



GE VERNOVA

ADMS

16.25.1

Release Notes

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Change History

Date	Change Description
Jul 3, 2023	ADMS 3.16.0.0: • This is a new document for ADMS 3.16.0.0.
Jul 21, 2023	ADMS 3.16.0.1: • Added 3.16.0.1. • Updated Known Issues. • Updated Common Vulnerabilities and Exposures.
Aug 31, 2023	ADMS 3.16.0.2: • Added 3.16.0.2. • Updated Known Issues.
Oct 19, 2023	ADMS 16.1.0: • Added 16.1.0. • Updated Known Issues. • Updated Common Vulnerabilities and Exposures. • Updated Compatible GE Software. • Updated Third-Party Software.
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Dec 20, 2023	ADMS 16.3.0: • Added 16.3.0. • Updated Known Issues. • Updated Common Vulnerabilities and Exposures. • Updated Third-Party Software.

Date	Change Description
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Feb 26, 2024	ADMS 16.5.0: • Added 16.5.0. • Added Test Information. • Updated Known Issues. • Updated Common Vulnerabilities and Exposures. • Updated Compatible GE Software.
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Apr 23, 2024	ADMS 16.7.0: • Added 16.7.0. • Updated Known Issues. • Updated Common Vulnerabilities and Exposures. • Updated Compatible GE Software.
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Date	Change Description
Sep 5, 2024	ADMS 16.11.0: - Added 16.11.0. - Updated Test Information. - Updated Common Vulnerabilities and Exposures. - Updated Third-Party Software. - Updated Prerequisites.
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Date	Change Description
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Oct 29, 2025	ADMS 16.22.0: • Added 16.22.0. • Updated Third-Party Software. • Updated Compatible GE Software. • Updated Known Issues.
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Date	Change Description
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1. ADMS 16.25.1

This chapter describes the incremental new features and defect fixes that were added since ADMS 16.25.0. The remainder of this document provides additional release-specific information that also applies to this release.

1.1. What's New in 16.25.1

No new features have been added in this release.

1.2. What's Changed in 16.25.1

The following changes have been made in this release.

1.2.1. Supported Software and Hardware Changes

This section describes changes to the system such as changes in system requirements, compatibility, or third-party software.

The following changes has been made in this release:

- Foundation - 25R13
- The Open Data Protocol (OData) version used by the DMS Data Service ([http://localhost:8801/dms/data/\\$metadata](http://localhost:8801/dms/data/$metadata)) has been upgraded to version 3.0 following a Microsoft-delivered NuGet package update. Clients using earlier OData versions must upgrade to version 3.0 to maintain compatibility.

1.3. What's Fixed in 16.25.1

This section lists the notable issues addressed in this release.

1.3.1. Archive

DEVICE_ID Mismatch in ORDERS, INCIDENT, and CALL Tables in Archiver

Grip ID: ETD-84213

Fix Version: 16.25.1

Severity: Medium

Issue: DEVICE_ID does not match in ORDERS, INCIDENT, and CALL tables in Archiver. ArchiveApp

throws exception, recorded in ArchiveNRTData log file, parent key not found while Incident Update. The deviceId for the fuse isn't present in the DEVICE table.

Solution: The Device_ID is missing from the device table which caused the issue. To handle this, added logic so that whenever an ArchiveIncidentUpdate message arrives for a Device_ID, the system populates the Device_ID in the device table.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Clearing Non-Outages Changes NRT NUMBER_OF_AFFECTED_CUSTOMERS from 0 to 1

Grip ID: ETD-84145

Fix Version: 16.25.1

Severity: Medium

Issue: When a non-outage is cleared in the thick client, the NUMBER_OF_AFFECTED_CUSTOMERS field in NRT changes from 0 to 1 at clearance. Prior to clearance, the value remains 0.

Solution: During data archival, no validation ensured that non-outage incidents retained the correct number of affected customers. Validation logic was added to ensure that, for non-outage incidents, the Number of Affected Customers field is set to 0.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Work Orders and Updates not flowing from ADMS to Workforce Management

Grip ID: ETD-84209

Fix Version: 16.25.1

Severity: Medium

Issue: Work orders and updates were not transmitted from ADMS to workforce management, preventing outage dispatch from ADMS. No system alerts were generated, and no errors were logged prior to restarting the CrewServer process on the active IF server.

Solution: The issue was caused by concurrent modification of a dictionary during iteration, which triggered an unhandled exception. The code was updated to create a copy of the dictionary using ToList(), and the iteration now runs on the copied collection to prevent concurrent modification errors.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

1.3.2. OMS

Customer Count Incorrectly Set to 1 During Non-Outage to Outage Conversion

Grip ID: ETD-84211

Fix Version: 16.25.1

Severity: Medium

Issue: When converting a non-outage to an outage downstream of a predicted incident, the customer count was incorrectly set to 1.

Solution: Updated the logic to re-calculate the customers count.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Partial Auto-Restoration at Multi-Phase Fuse Caused Cancelled Calls

Grip ID: ETD-84207

Fix Version: 16.25.1

Severity: Medium

Issue: Prediction at multi-phase fuse protecting multi-phase transformers partially auto-restored resulting in cancelled calls.

Solution: Improved logic for delta-wye 2 phase transformer.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Customer Calls Incorrectly Associated with Upstream Pending Clear Order

Grip ID: ETD-84171

Fix Version: 16.25.1

Severity: Medium

Issue: When a device is associated with multiple incidents (for example, both a pending clear incident and a new fuse predicted incident), the code fails to return the correct incident.

Solution: Updated the logic to return the newer incident in such cases, as the older one is already pending clear.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

EOMS Not Functioning Correctly with Large Customer Model

Grip ID: ETD-84121

Fix Version: 16.25.1

Severity: Medium

Issue: EOMS does not record Service Center correctly with a Large Customer. In some cases, the order lock cannot be obtained for editing, and the owner/organization field appears blank.

Solution: When creating an order for the Large Customer model, the ServiceCenter field was received as null. The binding occurred through ServiceCenterName, which was also null. Although the Service Center parameter contained valid values, it was not being used. The code was updated to first evaluate and prioritize ServiceCenterName. If ServiceCenterName is null, the system now retrieves the value from Service Center.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	

Issue Implication	Yes/No	Additional Information
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Invalid Task Results Sent Through API Halt Order Progression

Grip ID: ETD-84055

Fix Version: 16.25.1

Severity: Medium

Issue: ADMS processed task results received through the API even when the result was inconsistent with the task type. As a result, the associated order remained in an In Progress state.

Solution: The validation logic was updated to verify that the provided task status is applicable to the corresponding task type. If the status is not valid for the task, the system now returns a TASK_TRANSITION_INVALID exception, and the task status remains New in APF.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

1.3.3. Outage Assist

Merge Option Shown for Upstream Predicted Outages on Device Open

Grip ID: ETD-84176

Fix Version: 16.25.1

Severity: Low

Issue: When searching for existing incidents downstream of a device, the system returned all incidents affecting downstream service points without verifying whether the incident was truly downstream of the device. In some cases, upstream incidents were included in the results.

Solution: Additional validation was implemented to ensure that only incidents confirmed to be downstream of the device are returned.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	

Issue Implication	Yes/No	Additional Information
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Truck Type Column Shows None Under Crew Tab for Outage Assist

Grip ID: ETD-83909

Fix Version: 16.25.1

Severity: Medium

Issue: The Truck Type column displayed None under the Crew tab in Outage Assist when a vehicle was manually assigned to a crew.

Solution: The logic was updated to correctly retrieve and display the assigned vehicle type for the crew.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

1.3.4. DMS

Study Server Schematic View Missing Sections Present in Real Time

Grip ID: ETD-84053

Fix Version: 16.25.1

Severity: Medium

Issue: Significant sections of the surrounding stations that show up in RT are missing in study after initializing ST.

Solution: Updated Study initialization code to include geographically adjacent stations.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	

Issue Implication	Yes/No	Additional Information
ADMS Archive Schema	No	
User Interface	No	

1.3.5. DNAF

FISR Handle Miscoordination Documentation Unclear and Not Aligned with Observed Behavior

Grip ID: ETD-84175

Fix Version: 16.25.1

Severity: Low

Issue: The FISR Handle Miscoordination section in documentation is unclear on certain scenarios and results from testing revealed.

Solution: Clarify Miscoordination functionalities in FISR documentation.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

FISR Results Show Feasible Plan with Flow Violations

Grip ID: ETD-84185

Fix Version: 16.25.1

Severity: Low

Issue: FISR results show feasible plan with flow violation.

Solution: Improve the algorithm checking flow violation on bus tie breaker on restoration path.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

LVM Produces No Plan and Logs Do Not Clearly Indicate the Reason

Grip ID: ETD-84222

Fix Version: 16.25.1

Severity: Medium

Issue: The LVM does not produce any plan, and the reasoning is not clear from the result files.

Solution: Improved sensitivity calculation with bad load model so plan is generated in optimization engine. However, it is recommended to consolidate high volumes loads for better performance and fix zero values in three-phase loads in the model to have more realistic plan.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Incorrect Data in FISR Analyzed Tap Changing Transformers Section

Grip ID: ETD-84220

Fix Version: 16.25.1

Severity: Low

Issue: Incorrect logging in "Analyzed Tap Changing Transformers" Section in FISR result file.

Solution: Correct the log in "Analyzed Tap Changing Transformers" in FISR Results file

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

1.3.6. DOTS

OMSSIM Crashes When CIS_IVR_INTERFACE_ENABLED Is Set to False

Grip ID: ETD-84189

Fix Version: 16.25.1

Severity: Low

Issue: OMSSIM crash when CIS_IVR_INTERFACE_ENABLED is set to false

Solution: Handled Null exception code due to parameter _relayServerMsgHandler being passed as null when the CIS_IVR_INTERFACE_ENABLED flag is set to 0.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

1.3.7. User Interface

Bitmap Status Opposite of Device Status in Geographic View

Grip ID: ETD-84228

Fix Version: 16.25.1

Severity: Medium

Issue: In the geographic view, bitmaps assigned to the dmsTempSwitch object displayed an incorrect device state. When the temporary switch was open, the closed bitmap appeared, and when the switch was closed, the open bitmap appeared. This mismatch caused confusion about the actual operating state of the device.

Solution: Corrected the bitmap-to-status mapping for the dmsTempSwitch object. The geographic view now displays the correct bitmap for both open and closed states, ensuring the visual representation accurately reflects the actual device status.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

1.3.8. Tools

VoltageIsMeasuredLineLine Parameter Incorrectly Set to False

Grip ID: ETD-84127

Fix Version: 16.25.1

Severity: Medium

Issue: The VoltageIsMeasuredLineLine parameter was incorrectly set to "False" for internally generated stations created by SFBLITE, while the configuration file specifies it as "True".

Solution: Resolved an issue where the configured VoltageIsMeasuredLineLine parameter was ignored and the value defaulted to false. SFBLite now correctly applies the configured value for analog measurements.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

2. ADMS 16.25.0

This chapter describes the incremental new features and defect fixes that were added since ADMS 16.24.0. The remainder of this document provides additional release-specific information that also applies to this release.

2.1. What's New in 16.25.0

This section contains information about new features and functionality for ADMS 16.25.0.

2.1.1. ADMS integration with Enterprise Switching (ES) 1.0.0

ADMS is now integrated with new GridOS component Enterprise Switching, deployed within a GridOS cluster. This component has been certified and tested using individual clustered deployment.

Each component is deployed using its own manifest. When multiple GridOS components are deployed within the same cluster, the GridOS Foundation components shared across applications are consolidated and managed centrally by the services team.

2.1.2. Enhanced Step Separator for Switching Orders

The Switching Step Separator has been enhanced to improve the organization and readability of Switching Orders. Separators are now displayed as headers rather than footers, providing a clearer and more structured layout.

Users can configure and label separator headers through **Switching Order Client Configuration Editor > Switch Step tab > Switch Steps Separator**. Header text can be entered as custom text or selected from predefined options to support standardized workflows. These predefined options are fully editable, allowing users to add, update, or remove entries as needed.

Separator header text is included in all exported formats, including XLS, CSV, PDF, HTML, and Print Preview, ensuring consistency across documentation. (GRDDMSS3-2389)

Switching Steps									
☐	#	Nav	I/C	Step State	Device	Action	Phase	Tag Type	POLE
This is an example for custom text as a separator header.									
☐	1			EDIT	52530	OPEN	All		
☐	2			EDIT	52530	OPEN	All		

Switch Steps Separator

Switching Step Divider Text	Header
☒ A to B	Move Load from Moss Point Elder Ferry Brk 123 to Pascagoula Brk 432
☐ B to A	Move Load from Moss Point Pascagoula Brk 432 to Elder Ferry Brk 123

☒ Use Custom Text for Switching Step Separator

Add Delete

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	Yes	Screenshot included
ADMS Archive	No	
User Interface	Yes	Screenshot included
Demonstration Available	Yes	Available on request

2.1.3. Dynamic Device Support in Switching Order Add Step API

Version 2 (V2) of the Switching Order Web Service API has been introduced to support dynamic devices such as Cuts, Jumpers, and Temporary Grounds. This enhancement allows external systems to add switching steps with complete device specifications without creating or removing devices in the network model.

The updated API eliminates the need for manual intervention and provides a clean audit trail with full traceability of planned operations for temporary device management.

For more details, refer to the *Add a Switching Step Version 2* section of the *DMS Interface Server Programmer Guide*. (GRDDMSS3-3500)

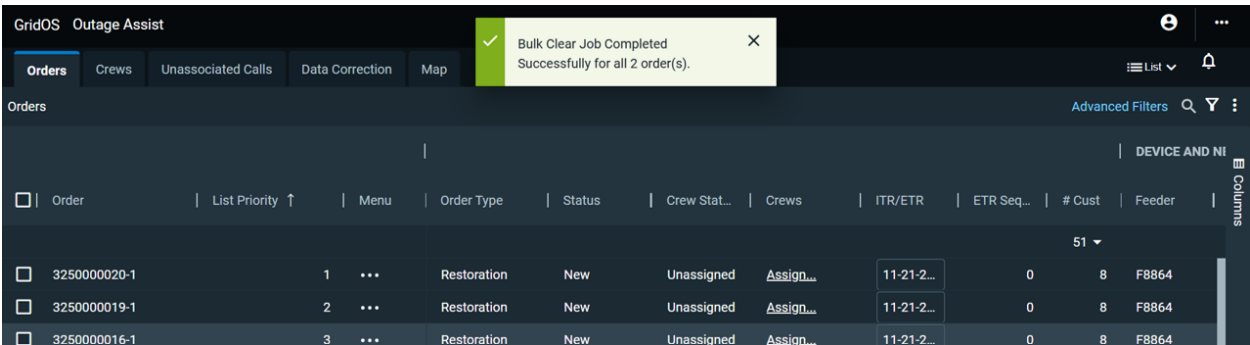
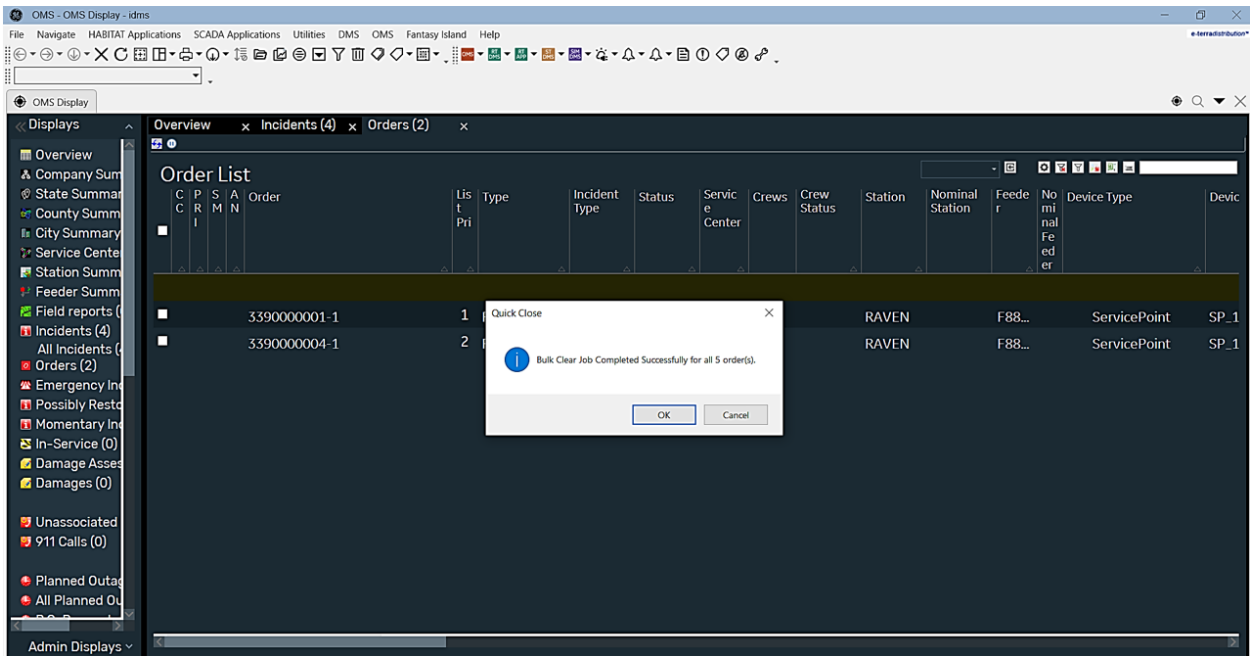
Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	Yes	Added V2 for creating step with dynamic devices POST /SWO/SwitchingOrders/AddStepV2
Configuration	No	
ADMS Archive	No	
User Interface	No	
Demonstration Available	No	

2.1.4. Bulk Closure of Restoration and Investigation Orders

Dispatchers can now close restoration and investigation orders in bulk from the Order List using **Quick Close** in both the ADMS and Outage Assist platforms. The system validates each selected order, skips locked orders, and automatically applies default values, including custom fields.

Custom fields are populated based on the device type using the QuickCloseInterruptType mapping, ensuring that only orders associated with supported devices are eligible for bulk closure.

For more information, see the *Multi-Select Quick Close of Restoration and Investigation Orders* section in the *Enterprise OMS User Guide* and the *Outage Assist User Guide*. (GRDDMSS3-3306 and GRDDMSS3-3307)



Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	

Feature Implications	Yes / No	Additional Information
API	Yes	Two new APIs support bulk order closure: • BulkCloseOrders • BulkCloseOrders?jobid=\${jobId}
Configuration	Yes	Screenshot included
ADMS Archive	No	
User Interface	Yes	Screenshot included
Demonstration Available	Yes	Available on request

2.1.4.1. Performance Summary

Test Environment

Performance testing was executed on a high-capacity server designed to simulate real-world and peak-load conditions:

- **CPU:** 16 vCPUs (AMD EPYC 7571, 2.5 GHz)
- **Memory:** 64 GB RAM
- **Architecture:** x86_64

What Was Tested

The **Bulk Closure of Restoration and Investigation Orders** feature was validated under varying load and concurrency scenarios to ensure reliability and scalability.

Test coverage included:

- Order volumes: 5K, 10K, 15K, and up to 30K orders
- Submission modes:
 - Sequential processing
 - Simultaneous multi-user submissions
- Execution methods:
 - Application-based execution (real user behavior)
 - Script-based execution (baseline comparison)

Key Performance Highlights

- **Consistent response times:**

Application-based bulk closure completed in approximately 2–3 seconds per order for batch sizes of 30, 60, 100 orders, even as order volume increased.

- **High-load stability:**

The system maintained stable performance across all tested loads (up to 15K orders) with no

observable degradation.

- **Strong concurrency support:**

Application-based bulk closure maintained an average processing time of 2–3.5 seconds per order for batch sizes of 30, 60, and 100 orders per user, even with multiple users performing bulk closures simultaneously.

- **Proven scalability at peak loads:**

At 30K orders with concurrent submissions, the feature continued to perform reliably, keeping average response times below 4 seconds per order.

- **Optimized application performance:**

Application-level execution demonstrated a significant performance improvement compared to script-based processing, reflecting real-world optimized behavior.

2.1.5. Enhanced Crew Manager User Interface

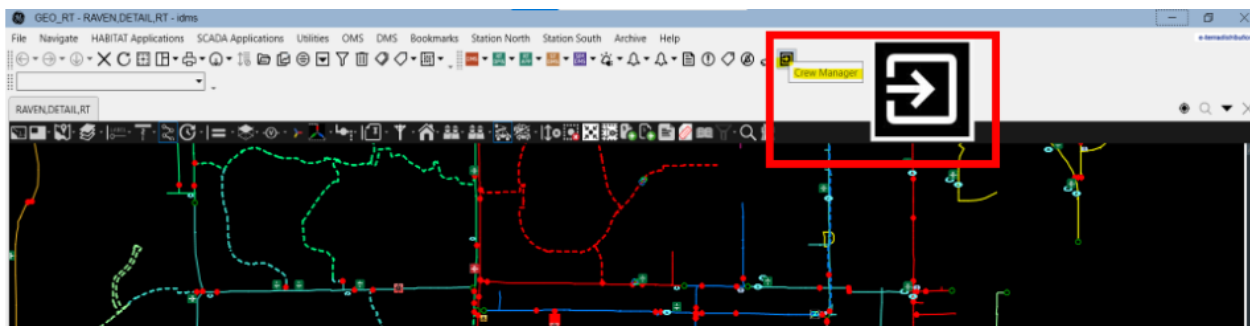
The Crew Manager user interface has been updated to simplify and streamline crew management operations. The interface supports managing up to 12,000 crews and simplifies key actions such as signing crews on or off duty, both individually and in bulk.

The Crew Manager window can be accessed from the browser toolbar icon or through the right-click context menu in the Crew Status Summary tab. The Crew Directory tab provides filtering capabilities to efficiently locate and manage crews.

The interface supports single and bulk check-in and check-out operations using buttons, icons, or the right-click context menu. Bulk operation limits are configurable, and all actions are logged for audit purposes. Users can also view and update Crew Division and Crew Area for multiple crews during sign-in operations.

Notifications are provided for all request types, allowing users to track operation status and better manage crew activities during high-volume events such as storms.

For more details, refer to the *Crew Manager* section of the *OMS User Guide*. (GRDDMSS3-E-774 and GRDDMSS3-E-792)



Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	Yes	Screenshot included
Demonstration Available	Yes	Available on request

2.1.5.1. Performance Summary

Test scenario

The Crew Manager UI was tested with 12,000 crews loaded, under active system load using event scripting, to validate real-world performance.

What was measured

Common crew actions were tested across multiple interaction methods:

- Single and bulk check-in / check-out
- Actions via button, right-click, and icon
- Bulk operations for 10 and 25 crews

Performance highlights

- The new Crew Manager UI consistently completed most single actions in ~5–6 seconds, even under load.
- Bulk actions (10 crews) completed in ~6 seconds, across all interaction methods.
- Bulk actions (25 crews) completed in ~7–11 seconds, showing predictable scaling with larger selections.
- Performance remained stable across different user interaction methods (button, right-click, icon).

Comparison with existing UI

- The new UI shows significant performance improvement over the existing Crew Status Summary view.
- Several actions that previously took ~30–40 seconds are now completed in under 10 seconds.
- Overall response time has been reduced by multiple times for bulk operations.

2.1.6. Hot Line Hold Process Automation

The Hot Line Hold (HLH) Process Automation enables end-to-end automation of HLH operations

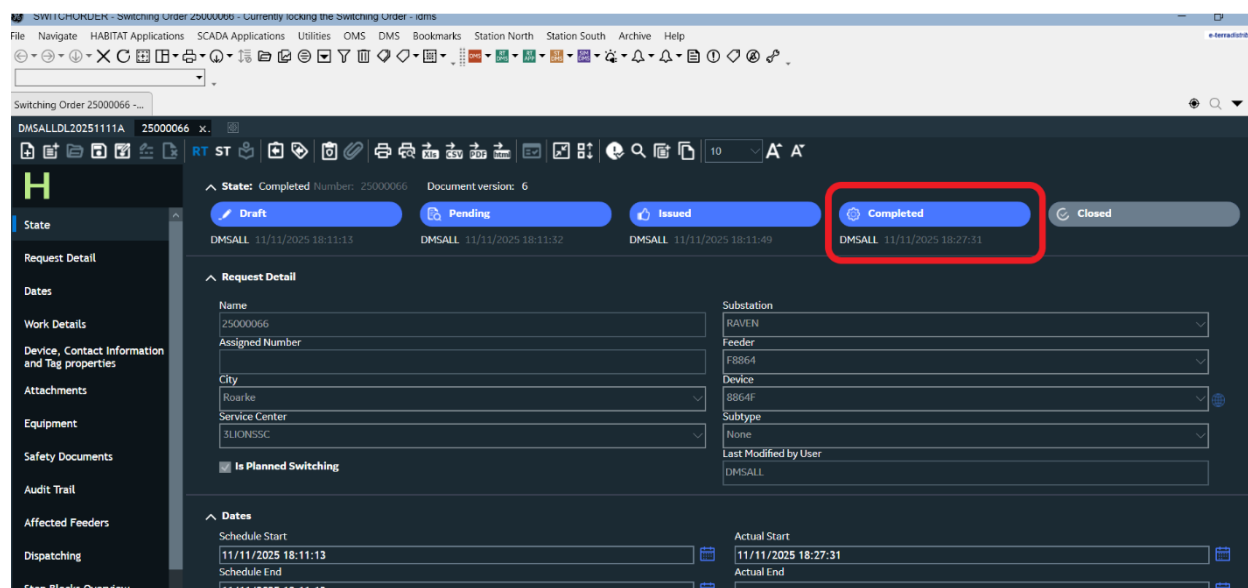
directly from the Geo View. Operators can initiate an HLH by selecting an asset, after which the system automatically identifies the first upstream protective device and pre-populates the required equipment details in the HLH document.

The HLH document supports the full lifecycle—Draft → Pending → Issued → Completed → Closed—allowing operators to save drafts, modify details, submit for validation, issue switching steps, and complete or close HLH activities. A new Area Details section has been introduced, enabling users to capture additional contextual information as needed.

Built-in system validations ensure consistency between the identified protective device and associated switching steps from creation through issuance. The solution supports automated execution with retry on failure, along with manual override options to facilitate coordination with field crews when required.

The system safely manages multiple tag dependencies on reclosers, ensuring the device remains disabled until all tags are cleared. Upon final closure, tags are automatically removed and auto-reclosing is re-enabled, with manual override support in case of execution failures.

This enhancement improves operational safety, reduces manual effort, and ensures consistent, auditable execution of Hot Line Hold processes. (GRDDMSS3-E-492)



Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	Yes	Screenshot included
Demonstration Available	Yes	Available on request

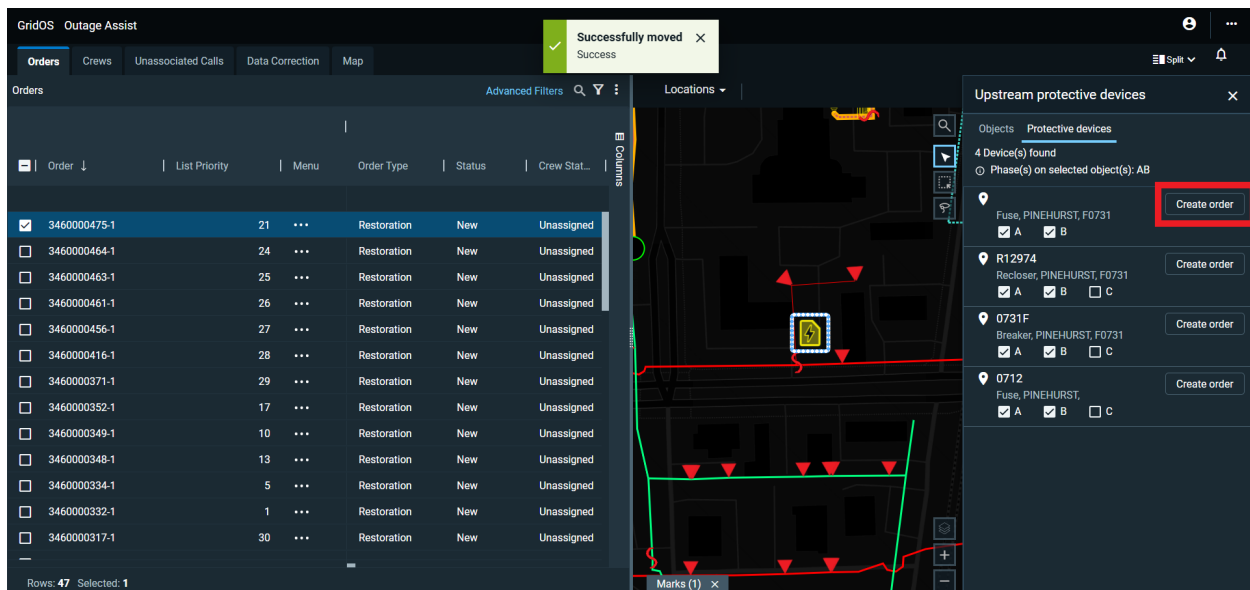
2.1.7. Upstream Protective Devices Panel Enhancements

The Upstream Protective Devices panel has been enhanced to improve operational efficiency when identifying common root causes of dense orders.

A new notification message has been added at the top of the Protective Devices tab, listing all phases from the Objects to provide better visibility for making informed decisions. For each device present phases for each upstream device will be listed.

The system will pre-select the phases listed in the notification, though users retain the ability to modify the selection if needed. Finally, users can now create orders for any upstream device on the selected phase(s). During order creation, the system will assess downstream for any existing orders and prompt users to merge them if necessary. (GRDDMSS3-3234)

For more information, see the *Upstream Protective Devices for a Single Order* section in the *Outage Assist User Guide*.



Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	Yes	Screenshot Included
Demonstration Available	Yes	Available on request

2.1.8. Enhanced Crew Assignment

The crew assignment pop-up has been enhanced to improve usability and efficiency. A configurable

parameter has been introduced to limit the number of crews loaded in the dropdown, defaulting to the top 100 crews. This ensures that the dropdown remains responsive. For access to the full list and to sort and filter for the most appropriate crew, the 'Advanced' filter is available. Additionally, the ability to 'search as you type' has been added to the dropdown field, allowing users to quickly find the crew they need without scrolling. (GRDDMSS3-3595)

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	Yes	Added configuration "maxCrewsToDisplayInAssignCrews" in the settings.json
ADMS Archive	No	
User Interface	No	
Demonstration Available	No	

2.1.9. Enhanced Geographic Display and Navigation for Substations

With the adjacency level set to 1, the geographic display now shows:

- The selected substation
- All substations connected by normally open switches
- The immediate parent and child substations

Additionally, the Navigation Builder tool now traverses through all parent and child stations of the main station, automatically including their respective parent and child stations. This provides a comprehensive view of all directly connected substations. (GRDDMSS3-3380)

For more information, see the *General Properties* section in the *Viewer Configuration Editor User Guide*.

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	No	
Demonstration Available	Yes	Available on request

2.1.10. ADMS Standardized Logging



Warning

The log format has been updated. Customer-developed tools, scripts, or integrations that rely on parsing the previous log format may be impacted and must be reviewed and updated to ensure compatibility with the new standardized format.

2.1.10.1. Logging Changes – Overview

To ensure system-wide consistency, clarity, and ease of operation, ADMS has adopted standardized logging. This change improves system coherence and accelerates issue resolution by making logs machine-readable, searchable, and human-friendly.

Standardized logging is implemented for the following ADMS applications:

CallServer, CrewServer, CrewSimulator, DmsInterfaceServer, NetDataServer, NetServer, NetServerCM, SwitchOrderServer, NetSrv, DnafSrv, StudyServer, OMS Interface Server.

All logs are formatted according to the specification below.

Standard Log Format

Field	Format / Details
timestamp	Local time format
logger	Identifier of the logging source
level	Logging severity in uppercase (ERROR, WARN, INFO, DEBUG, TRACE)
thread	(Optional) Thread name, where applicable
message	Free-form message text in English; concise and secure

Example Formatted Log Entries

```
02/05/2026 22:39:22.579 CallServer.CallServerFrm INFO Application version : 16.25.0.0 None
02/05/2026 22:39:22.580 CallServer.CallServerFrm INFO Initializing Call Server... None
02/05/2026 22:39:22.580 CallServer.CallServerFrm WARNING Message Security checking is disabled
None
```

2.1.10.2. Dynamic Logging

Dynamic logging filters log messages based on **environment variables** for specific modules such as OMS, DMS, and DNAF. Only messages at or below the configured log level are written to log files, significantly reducing log file size.

Dynamic Logging Behavior

1. Log files store only messages with a severity **less than or equal to** the log level set via environment variables.
2. If no environment variable is defined, the default log level is **INFO**, or the value defined in the configuration file.

- If no log level is defined in the configuration file, **INFO** is used by default.
 - Environment variables override configuration file settings.
3. Log messages are written to file only when their severity is less than or equal to the configured environment variable (for example, `CallServer_Log_Level`).
 4. Any change to the log level via environment variables is always logged.

Applications Supporting Dynamic Logging

Dynamic logging is currently supported for the following applications:

- NetServer
- Call Server
- Switch Order Server
- Crew Server
- Network Data Server
- NetDataReplicator
- DmsInterfaceServer
- OmsInterfaceServer
- NetviewerAdapter

Note

Log entries generated by **DLLs or shared libraries** are **out of scope** for dynamic logging.

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	No	
Demonstration Available	No	

2.2. What's Changed in 16.25.0

The following changes have been made in this release.

2.2.1. Supported Software and Hardware Changes

This section describes changes to the system such as changes in system requirements, compatibility, or third-party software.

The following changes has been made in this release:

- Forecasting - 3.11.0 & 16.25.0
- Gateway - 2.9.1 & 16.25.0
- Foundation - 25R12

2.3. What's Fixed in 16.25.0

This section lists the notable issues addressed in this release.

2.3.1. DMS

APF Random Crash

Grip ID: ETD-83877

Fix Version: 16.25.0, 16.24.0.1

Severity: Medium

Issue: APF randomly crash.

Solution: Added defensive code in the communication level.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

2.3.2. Archive

DEVICE_ID Mismatch Between ORDERS and INCIDENT Tables in Archiver

Grip ID: ETD-84090

Fix Version: 16.25.0

Severity: Medium

Issue: DEVICE_ID does not match in ORDERS and INCIDENT tables in Archiver.

Solution: The DEVICE_ID value was being updated in `archiveIncidentUpdate.FieldName`. However, in `ArchiveIncidentUpdate`, the corresponding field is named `OPENDEVICEID`. Due to this name mismatch,

the device ID was not updated. The field reference has now been corrected to OPENDEVICEID.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

2.3.3. EOMS

Icons Not Loading Correctly in Order Worksheet After Upgrade

Grip ID: ETD-84044

Fix Version: 16.25.0

Severity: Medium

Issue: Icons in the order worksheet are not loading correctly after upgrade.

Solution: The root cause of the issue is that the `gx-common.scss` file from the gx-ui package includes the following line: "`@import url(<a href='\"https://fonts.googleapis.com/css?family=Roboto:400,500,700;\"' class='\"bare\"'>https://fonts.googleapis.com/css?family=Roboto:400,500,700;\"</a>`". This import was causing icon loading issues. A post-build cleanup script (`post-build.js`) was implemented to remove the Google Fonts import from the compiled CSS after each build, ensuring it is no longer included.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

EOMS Stops Creating Orders After OMSMAIN Restart

Grip ID: ETD-84031

Fix Version: 16.25.0

Severity: Medium

Issue: EOMS failed to create new orders after the OMSMAIN service was restarted. The `ringBuffer.tailSequence()` value remained unchanged after initialization. A large volume of database events during the OMS server restart caused an invalid `tailSequence` value, leading to

inconsistencies with `HazelcastInstance.ringBuffer` event indexing.

Solution: The index is now read directly from the database instead of `HazelcastInstance.ringBuffer`, ensuring accurate and consistent event indexing.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

WMDBSYNC Gets Stuck in WMADDCREW

Grip ID: ETD-84135

Fix Version: 16.25.0

Severity: High

Issue: WMDBSYNC sometimes gets stuck when processing the WMADDCREW entry.

Solution: Before inserting or updating an `OrderId`, the system now checks for an existing assignment with the same `OrderId` and updates or merges it instead of inserting or blindly reassigning it. Deletes and missing rows are also handled idempotently to ensure retries or merged updates do not result in errors.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

2.3.4. OMS

Reference ID Not Added for Non-Outage Incidents

Grip ID: ETD-84146

Fix Version: 16.25.0

Severity: Medium

Issue: The Reference ID was not added for non-outage incidents and was not visible in the Customers tab.

Solution: When creating a non-outage incident, the Reference ID was missing due to incomplete data binding in the UI. The required mapping for the Reference ID has now been implemented and verified for both outage and non-outage incidents.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Incident Not Created When Customers Are De-Energized

Grip ID: ETD-83865

Fix Version: 16.25.0

Severity: Medium

Issue: Incidents not created at each connecting Network Protector when downstream customers were deenergized in the secondary network.

Solution: Improved incident prediction logic to create incident at Network Protector for such case.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Opening Second Phase on a Multiphase Lateral With Upstream Predicted Outage

Grip ID: ETD-84096

Fix Version: 16.25.0

Severity: Medium

Issue: The logic when deciding if a customer should be added to a new incident is missing an edge case.

Solution: Fixed the logic when deciding if a customer should be added to a new incident when there is already existing incident that affects this customer. In the case that the existing incident is a predicted incident and the new one is a confirmed, the customer should be added to the new confirmed incident.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Customers Tab Column LAST in OA and Incidents List AreaNoteETR Displaying 1969

Grip ID: ETD-84107

Fix Version: 16.25.0

Severity: Medium

Issue: Date and time columns showed 1969 when the values are not set or initialized.

Solution: Show empty when date and time values are not set.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Order Not Updated When Open Task Is Added to Confirmed Outage

Grip ID: ETD-84014

Fix Version: 16.25.0

Severity: Medium

Issue: Adding an open task to a confirmed outage does not update the order for the upstream device or create a new order. Instead, a new incident is created.

Solution: Allowed merging outage incidents into confirmed single-customer master incidents by adding condition and updating facility and device details in order worksheet during merge.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Archive Stalls When Deleting Restoration Steps

Grip ID: ETD-84088

Fix Version: 16.25.0

Severity: Medium

Issue: Archive stalling due to an error when attempting to delete some restoration steps.

Solution: Multiple ArchiveDbContext instances being created within a single TransactionScope, causing Entity Framework to escalate to distributed transactions.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

NetServer Crash Due to Unhandled Exception

Grip ID: ETD-83993

Fix Version: 16.25.0

Severity: Medium

Issue: Netserver crash due to unhandled exception and crash dump generated.

Solution: Multi-threading defensive code was not added earlier.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Incorrect Information Shown in Crew Selection Dialog

Grip ID: ETD-84131

Fix Version: 16.25.0

Severity: Low

Issue: The crew selection dialog box display shows incomplete/incorrect information.

Solution: The root cause of the issue is that the `gx-common.scss`

file from the gx-ui package includes the following line: "@import url(https://fonts.googleapis.com/css?family=Roboto:400,500,700);"". This import was causing icon loading issues. A post-build cleanup script (post-build.js) was implemented to remove the Google Fonts import from the compiled CSS after each build, ensuring it is no longer included.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Links Not Greyed Out When User No Longer Has Lock

Grip ID: ETD-84064

Fix Version: 16.25.0

Severity: Medium

Issue: Previous indicators that showed a user doesn't have a lock are no longer greyed out.

Solution: The root cause of the issue is that the `gx-common.scss` file from the gx-ui package includes the following line: "@import url(https://fonts.googleapis.com/css?family=Roboto:400,500,700);"". This import was causing icon loading issues. A post-build cleanup script (post-build.js) was implemented to remove the Google Fonts import from the compiled CSS after each build, ensuring it is no longer included.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Lead Crew Designation Dropping from Crew

Grip ID: ETD-83864

Fix Version: 16.25.0

Severity: Medium

Issue: Lead crew designation is intermittently removed from the crew in the grid tabular when the Assign button is clicked repeatedly in a short span.

Solution: Updated logic sets the existing lead crew's lead flag to false if it's different from the incoming lead crew.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

NetServer Crash During Crew Assignment from Click Application

Grip ID: ETD-83676

Fix Version: 16.25.0

Severity: Medium

Issue: Assigning a crew using the Click API could cause NetServer to crash.

Solution: Added defensive checks to safely handle potential null references.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Unexpected Repair Task Creation

Grip ID: ETD-83891

Fix Version: 16.25.0

Severity: Low

Issue: When a predicted stable order was created downstream of a confirmed breaker order, an unexpected repair task was generated.

Solution: An error returned by the OMS Server was not being forwarded by the OMS Interface Server to EOMS. The OMS Interface Server has been updated to forward the error, and EOMS has been updated to handle the error appropriately.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	

Issue Implication	Yes/No	Additional Information
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Incident Number Filter Does Not Work Correctly

Grip ID: ETD-83812

Fix Version: 16.25.0

Severity: Low

Issue: Filtering by cell value in the Incident Number column did not work correctly. When the = character was used in the filter cell, the selected incident was not displayed as expected.

Solution: The filtering logic has been updated so that incident numbers filtered using Filter on Cell Value are now correctly reflected in OMS.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

2.3.5. Outage Assist

Incident Could Be Cleared with a Future End Time

Grip ID: ETD-84116

Fix Version: 16.25.0

Severity: Medium

Issue: In OA, incidents could be cleared even when the end time was set to a future timestamp.

Solution: When setting the end time in OA, the Update Incident API is triggered. Previously, the endpoint did not validate future timestamps, and an issue was also identified in handling the "equal time" scenario. Both issues have been resolved by adding a new validation function in `DataEntryProcessing.cs` within the NetServer solution.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	

Issue Implication	Yes/No	Additional Information
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Duplicate Customers Displayed When Refreshing Customers Tab

Grip ID: ETD-84151

Fix Version: 16.25.0

Severity: Medium

Issue: In AG Grid, rowIdFields was configured to use accountNumber as the unique row identifier. However, accountNumber is not unique because a single account can be associated with multiple service points or meters, each with a different customerKey.

Solution: Updated rowIdFields from accountNumber to customerKey in the affected customers table configuration to ensure unique row identification.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Outage Assist Is Displaying 1969 Date in ETR Field

Grip ID: ETD-83910

Fix Version: 16.25.0

Severity: Medium

Issue: Outage Assist is displaying 1969 date in ETR field.

Solution: Display empty if ETR is not set.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Filter Menu Values Do Not Match Column Data on OA

Grip ID: ETD-83887

Fix Version: 16.25.0

Severity: Low

Issue: Filter menu values do not match column data on OA.

Solution: Update table json file so that the values match.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Bulk Actions Allow Operations on Orders Without Proper User Permissions

Grip ID: ETD-83770

Fix Version: 16.25.0

Severity: Medium

Issue: Permission check skipped for bulk action.

Solution: All selected items are checked for permission before the action is executed.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Casenote Failed to Update

Grip ID: ETD-84170

Fix Version: 16.25.0

Severity: Low

Issue: Cannot update Casenote in OA.

Solution: Need to add Casenote permission and grant access to users.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	

Issue Implication	Yes/No	Additional Information
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

OA Upstream Common Fails on Abnormal Network Topology

Grip ID: ETD-84047

Fix Version: 16.25.0

Severity: Low

Issue: When the network topology is unexpectedly switched between two substations, the DMS API service call to DNOM is unable to locate the device.

Solution: The payload has been updated to use the Nominal Station name, ensuring correct retrieval of upstream protective devices.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Crew Tab Column Widths Do Not Reset to Default

Grip ID: ETD-84169

Fix Version: 16.25.0

Severity: Low

Issue: When a column in the Crews tab was resized, selecting Return to Defaults did not restore the column widths to their default values.

Solution: The default column widths have been corrected in the `pref-init-oms` configuration to ensure proper reset behavior.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

2.3.6. Interface

Yellow Border Persists After ADMS Model Update

Grip ID: ETD-84120

Fix Version: 16.25.0

Severity: Low

Issue: After model update, the client RT session displays a yellow border.

Solution: Fixed client multi-threading data racing issue.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

2.3.7. Modeling

Station Sub-Fed Status Incorrect After GISA Conversion

Grip ID: ETD-83989

Fix Version: 16.25.0

Severity: Low

Issue: When a station marked as sub-fed via GISA is converted, the resulting XML incorrectly indicates that the station is not sub-fed. Additionally, the conversion process creates multiple added devices, including cases where devices are created directly on top of feeder reclosers, leading to confusion.

Solution: Converter looks for a Sub-fed flag before adding fake breakers.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

2.3.8. Internal Notifications

Email Notifications Triggered Twice for Affected Customer Subscriptions

Grip ID: ETD-83706

Fix Version: 16.25.0

Severity: Medium

Issue: When creating a subscription, selecting both **Notify on Incident Status Changes** and **Notify on ETR Changes** results in duplicate notifications when both events occur simultaneously.

Solution: Updated the logic to process records based on the latest journal time and added a unique key constraint in the `Outage_Event` table to prevent duplicate entries and eliminate duplicate notifications.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

2.3.9. SWO

EnableWebServices Must Be True for SWO Server Communication

Grip ID: ETD-84181

Fix Version: 16.25.0

Severity: Low

Issue: `ExtMobileSystem` calls `System.Net.WebRequest.CreateHttp(url)` to post messages and the url is only initialized when `EnableWebServices` is true. When the url is null, `WebRequest.CreateHttp()` throws exception that causes DMS Interface to crash.

Solution: `ExtMobileSystem` will not make a call `WebRequest.CreateHttp()` when `EnableWebServices` is false.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

SWO Adds Extra Switching Step on Boundary Devices

Grip ID: ETD-83845

Fix Version: 16.25.0

Severity: Medium

Issue: Duplicate switching steps were added to the switch order for boundary devices.

Solution: Duplicate switching steps have been removed from the switch order.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

2.3.10. Yambay

Mobile Client Is Not Updating the Current Out After Repair Task Is Complete

Grip ID: ETD-83885

Fix Version: 16.25.0

Severity: Medium

Issue: Mobile client is not updating the current out after repair task is complete.

Solution: Fixed the timing issue so data is up to date.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

2.3.11. Tools

Converter Crash in Certain Stations

Grip ID: ETD-83868

Fix Version: 16.25.0

Severity: Medium

Issue: Converter crashes when building certain models.

Solution: Added validation to ensure the measurement is correctly mapped to the DNOM object before processing.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

3. ADMS 16.24.0

This chapter describes the incremental new features and defect fixes that were added since ADMS 16.23.0. The remainder of this document provides additional release-specific information that also applies to this release.

3.1. What's New in 16.24.0

This section contains information about new features and functionality for ADMS 16.24.0.

3.1.1. Ground Switch Halos Disappear on SCADA RAG Switch

RAG switches are a special type of composite switch which has two states telemetered and controlled by SCADA (Open, Close) and one state that is not telemetered at all (Ground state, where a line crew has to go the field and manually change it to the Ground position). This ground position update is not visible to SCADA and currently can only be changed using the ADMS toolbar. The issue is that when SCADA data is resynchronized (measurement update, dynamics compression, and so on) the SCADA value of Open will override the dynamics and the ground status applied in the ADMS will go away. The loss of the ground status and associated ground halos is a safety concern.

To resolve this issue, a modeling update has been made to the ScadaValue field in the Composite Switch Model in the global file. A value of -1 now means the composite switch is in the manual state (for example, Ground), and any SCADA commands will have no effect on the composite switch state. The Converter validation logic has been updated to respect this new value. The ADMS server composite switch status processing has also been changed along with Viewer Configuration Editor updates to add menu options that allows a user to apply or remove the manual state. (GRDDMSS3-3416)

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	Yes	Requires an update to the <code>global.csv</code> file to properly support this device.
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	No	
Demonstration Available	Yes	Available on request

3.1.2. Dangerous Switching Indicator Added for Temporary Cuts

The Dangerous Switching indicator is currently available for temporary jumpers; however, this symbol is not available for temporary cuts. This feature supports the addition of the existing Dangerous Switching indicator for temporary cuts.

If there is a temporary cut which is in the open state and energized on both sides, the dangerous switching indicator should be shown if either the Dynamic Phase Mismatch or the Phase Angle Mismatch check is true. If one or both conditions are met, the ADMS client will draw the Dangerous Switching indicator on the cut to provide awareness to the user to not change the status of the cut from the open to closed state. The "Has Dynamic Phase Mismatch" and "Has Phase Angle Mismatch" fields have also been added to the cut object Analyst attributes popup. (GRDDMSS3-3381)

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	No	
Demonstration Available	Yes	Available on request

3.2. What's Changed in 16.24.0

The following changes have been made in this release.

3.2.1. Supported Software and Hardware Changes

This section describes changes to the system such as changes in system requirements, compatibility, or third-party software.

The following changes has been made in this release:

- Forecasting - 3.10.0 & 16.24.0
- Gateway - 2.8.4 & 16.24.0
- Foundation - 25R11

3.3. What's Fixed in 16.24.0

This section lists the notable issues addressed in this release.

3.3.1. DNAF

High Current Calculated for Jumpers Across Different Phases Regardless of State

Grip ID: ETD-84054

Fix Version: 16.24.0

Severity: Medium

Issue: When a jumper is placed across two different phases, BLA calculates a high current regardless of whether the jumper is in an open or closed state. A jumper connecting two different phases results in a high current calculation in both states.

Solution: Added support for single cross-phase temporary jumpers, with phase rotation now accounted for in the DPF engine.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

DNAF FISR Alarm Reason Text Truncation and Formatting Issues

Grip ID: ETD-84033

Fix Version: 16.24.0

Severity: Medium

Issue: DNAF FISR alarms FI237 and FI2310 were generating exception reasons with truncated characters. Additionally, the alarm text included unnecessary leading whitespace at the beginning of the Reason code, causing the alarm display to appear inconsistent and unclear.

Solution: The alarm text formatting was adjusted to remove leading whitespace and ensure the reason text fits correctly within the alarm display.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

LVM Minimize Demand and Plan Selection

Grip ID: ETD-84006

Fix Version: 16.24.0

Severity: Medium

Issue: Inconsistent behavior between LVM plan selection and DNAF Analyst Guide.

Solution: Adjusted the LVM plan selection truth table in the LVM logic to align with the Product Documentation.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Missing DNAF_LVM_RESET_CAP_TAP_COUNT Parameter in ADMS Installation Guide

Grip ID: ETD-84086

Fix Version: 16.24.0

Severity: Medium

Issue: The DNAF_LVM_RESET_CAP_TAP_COUNT parameter was not documented in the ADMS Installation Guide.

Solution: The ADMS Installation Guide was updated to include the DNAF_LVM_RESET_CAP_TAP_COUNT parameter.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

LVM Failsafe Uses Internal Power Flow and Ignores BLA/DPF

Grip ID: ETD-84052

Fix Version: 16.24.0

Severity: Medium

Issue: There was confusion regarding LVM failsafe behavior, as it was unclear whether voltage mismatch detection relied on standalone BLA/DPF status or internal power flow calculations.

Solution: Updated *DNAF Advanced Application Analyst Guide* with clarification that LVM failsafe operates independently of BLA/DPF, using internal UPF calculations without updating DPF status.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

3.3.2. DMS

Vault Bitmap Fails to Reflect Internal Device Abnormal Status (Elbows)

Grip ID: ETD-83932

Fix Version: 16.24.0

Severity: Medium

Issue: Vault was not showing the right icon.

Solution: Logic to select icon based on state is corrected.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

3.3.3. OMS

Call Priority Reset to None After NetServer Restart

Grip ID: ETD-83869

Fix Version: 16.24.0

Severity: Medium

Issue: The `ApplyCallsToDnomStation` process reset the call priority to None after a NetServer restart. When the snapshot file was created, call priority information was not saved for calls that were not marked as high priority, resulting in the loss of the original priority state.

Solution: Save the call priority correctly in the snapshot file, so when the server restart, all the call

information are restored correctly.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Map Stops Rendering After Order Dispatched to Crew

Grip ID: ETD-84039

Fix Version: 16.24.0

Severity: Medium

Issue: Mobility Gateway used Workorder Task or Service Point X-Y coordinates. While X-Y coordinates were not in Service Point due to customer model, they were also not in Workorder Task.

Solution: Instead of getting X-Y coordinates from Workorder Task or Service Point, Mobility Gateway will also use incident X-Y.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Undo Confirmation Fails to Recreate Service Point Incidents

Grip ID: ETD-84020

Fix Version: 16.24.0

Severity: Medium

Issue: When undo confirmation, the callSeconds on the affected customer didn't get set correctly so the leftover calls is not picked up when creating single customer incidents, fixed the callSeconds issue.

Solution: Reset the call seconds on the customer when undo confirmation, so that the remaining calls will be picked up during single customer creation.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	

Issue Implication	Yes/No	Additional Information
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Block Prediction Action Shows Permission Popup but Still Blocks Prediction

Grip ID: ETD-83465

Fix Version: 16.24.0

Severity: Low

Issue: Block prediction action did not honor permission check.

Solution: Code added in the client side to check user permission.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Converting Non-Outage Incidents Does Not Update Incident or Call Priority

Grip ID: ETD-83950

Fix Version: 16.24.0

Severity: Medium

Issue: When a non-outage incident was converted to another incident type, the system failed to update the incident's priority. As a result, the Incident Priority and Call Priority did not reflect the newly selected call information after conversion.

Solution: The logic was updated to recalculate the priority using the new call information prior to updating the incident record. This ensures the Incident Priority and Call Priority correctly reflect the new incident type and conditions after conversion.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Downstream Confirmed Outage Re-Numbers Existing Predicted Outage

Grip ID: ETD-84062

Fix Version: 16.24.0

Severity: Medium

Issue: The incident move up/move down and renaming was not documented in the *ADMS OMS User Guide*.

Solution: The *ADMS OMS User Guide* was updated to include guidance for incident move up and move down scenarios.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

3.3.4. Outage Assist

Outage Assist User Switch Actions Not Logged in Chronology

Grip ID: ETD-84022

Fix Version: 16.24.0

Severity: Medium

Issue: User-initiated switch actions performed from Outage Assist were not being logged in the chronology with the correct user information.

Solution: Enhance the chronology-log-service in Outage Assist to support SOMS mode.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

OA Upstream Common – Wrong Devices Included in UI and DMS API Results

Grip ID: ETD-83996

Fix Version: 16.24.0

Severity: Low

Issue: The trace upstream protected devices is returning additional devices which are not in the upstream path.

Solution: The trace logic has been updated to return only the protected devices that belong to the upstream path.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Bulk Actions Allow Unauthorized Operations on Orders

Grip ID: ETD-83770

Fix Version: 16.24.0

Severity: Medium

Issue: Bulk actions allowed users to perform actions on orders for which they did not have the required permissions.

Solution: Enhance the permissions API in OMS Interface Server to support both AOR and Organization permission check.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Incorrect Date Displayed for Crew ETA and Current ITR in OA

Grip ID: ETD-84109

Fix Version: 16.24.0

Severity: Medium

Issue: Crew's ETA and Current ITR showed 1969 when the value was empty.

Solution: When crew's ETA and ITR equals empty or Unix Epoch.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Incorrect End Time Displayed in Outage Assist After Editing

Grip ID: ETD-84108

Fix Version: 16.24.0

Severity: Low

Issue: In Data Correction, orders'endtime column showed the wrong value.

Solution: Update the display definition (json file) to use the value ENDTIME instead of CLOSEDTIME.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Only One Tag Visible When Multiple Tags Are Applied to a Device

Grip ID: ETD-84082

Fix Version: 16.24.0

Severity: Medium

Issue: In OA, when multiple tags placed on a device, only one tag was shown.

Solution: Fixed logic to get tag count.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

OA Upstream Common - Lasso Not Functioning for List Return

Grip ID: ETD-83944

Fix Version: 16.24.0

Severity: Low

Issue: The lasso tool frequently fails to locate upstream devices because the payload sent to the endpoint lacks required parameters.

Solution: When orders are selected with the lasso tool on the map, the tool filters out the required tracing parameters, causing the failure. Ensure the tool retains all necessary parameters before sending the request.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Custom Field Checkbox Not Updating Correctly in Outage Assist

Grip ID: ETD-84117

Fix Version: 16.24.0

Severity: Medium

Issue: In Outage Assist (OA), a custom field checkbox was not updating correctly even though the query returned the expected value. As a result, the checkbox remained unchecked when it should have been checked.

Solution: Updated the checkbox handling logic to correctly interpret boolean values returned by the query, ensuring the checkbox is displayed as checked when the value is true.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

3.3.5. Modeling

Model Exporter Exports Reclosers at Feeder Head for Feeder-Mapped SCADA Tags

Grip ID: ETD-84005

Fix Version: 16.24.0

Severity: Medium

Issue: The exported JSON model contains the SCADA tags of the upstream circuit breaker mapped to the feeders. The feeder recloser SCADA tags should be exported instead of upstream circuit breaker measurements when recloser is the feeder head device.

Solution: Added logic to export recloser SCADA measurements mapped to the feeder in the JSON Export when recloser is the feeder head device.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Incremental Change Not Picked Up at Tie Switch

Grip ID: ETD-83979

Fix Version: 3.15, 16.24.0

Severity: Medium

Issue: When MMA processes incremental changes for a substation, the substation may enter a queued state, during which the incremental changes are merged with the master file in the processing folder, and a new master file is then generated in the Input folder. If another incremental change arrives for the same substation, it should be merged with the latest master file available in the Input folder, not the one in the processing folder. Currently, the system merges it with the file in the processing folder.

Solution: While merging incremental substation file, latest master file from the Input folder is used when available.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

3.3.6. File Updater

File Updater Enters Hung State on Servers and Workstations

Grip ID: ETD-83833

Fix Version: 16.24.0, 3.15

Severity: Medium

Issue: File Updater enters an unresponsive state on servers and workstations.

Solution: Implemented changes to prevent race conditions, manage resource usage, and enhance reliability by rejecting large files, using dynamic timeouts, and isolating deserialization processes.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

3.3.7. History Correction

Data Correction Filter for End Time Does Not Work as Expected

Grip ID: ETD-84118

Fix Version: 16.24.0

Severity: Low

Issue: Data Correction Filtering failed to work with local time.

Solution: The code has been fixed, and Data Correction Filtering now operates correctly with local time.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

3.3.8. Interface

Incorrect Search Match Ranking in GIX/OA

Grip ID: ETD-84077

Fix Version: 16.24.0

Severity: Medium

Issue: The search results show incorrect rankings because sometimes one numerical string partially matches another numerical string, such as a meter number or a phone number.

Solution: Logic has been added to adjust the score based on the number of exact matches.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Lateral Boundary Device Labels Still Visible

Grip ID: ETD-83895

Fix Version: 16.24.0

Severity: Low

Issue: Boundary device labels between the backbone and lateral network do not follow the "non-main" zoom configuration. When zooming out beyond the configured range, the boundary device should hide along with its connected label, but the label remains visible. This happens even though the device is considered non-main and respects the backbone zoom threshold.

Solution: The issue has been resolved by adding an additional OR condition for the DMSrecloser logic. With this change, the connected label now correctly follows the non-main zoom configuration and hides as expected when the zoom level goes out of range.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

3.3.9. Yambay

Mobile Communication Status Not Updating in Order Worksheet

Grip ID: ETD-83801

Fix Version: 16.24.0

Severity: Medium

Issue: The MobileCommunicationStatus field was originally implemented as order-specific, but a recent feature enhancement intentionally changed it to behave as crew-specific. This design decision was later identified as incorrect, as the field must remain tied to individual orders. As a result, updates were being applied across all incidents assigned to a crew instead of only the current order.

Solution: Added explicit order-specific handling for MobileCommunicationStatus updates. This ensures the status is updated only for the current order and generates correct incident-level journal entries.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

3.3.10. ADMS Kit

Missing OmsInterfaceV1WebServiceClient.dll When Installing Interface Server Only

Grip ID: ETD-84093

Fix Version: 16.24.0

Severity: Medium

Issue: When installing only the Interface Server from the server kits, the file `OmsInterfaceV1WebServiceClient.dll` was missing from the server/bin directory.

Solution: The issue was resolved by updating the WiX project in the server kits to include `OmsInterfaceV1WebServiceClient.dll`.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

4. ADMS 16.23.0

This chapter describes the incremental new features and defect fixes that were added since ADMS 16.22.0. The remainder of this document provides additional release-specific information that also applies to this release.

4.1. What's New in 16.23.0

This section contains information about new features and functionality for ADMS 16.23.0.

4.1.1. Flexible Customer Search

Initially introduced for testing in 16.21.0 release, this feature is now fully functional and available for production use.

The Flexible Customer Search feature enables single-line input for customer name and address, supporting space-separated words in any order. The search also supports meter number, customer key, account number, and phone number.

The Call Entry interface now includes a single search box for customer data. The `CallEntryConfig.xml` file has been updated to remove field visibility options that are no longer applicable.

To use this feature, ensure that the Address field is configured in the Customer List Compiler. The feature is enabled by default and cannot be disabled starting with this release (16.23.0). (GRDDMSS3-E-490)

For more information, see *Search Tab* in the *Overview of Customer List Compiler Displays* chapter of the *Customer List Compiler User Guide*.

boardman lucy oak

Score	Name	Address		Phone Num...	Account Number	Meter Number	Service Point	Match
5.4	Boardman Lucy D	9419 OAK	SE	5679198	555146499	485671XE	SP_34457827	NAME1=BOARDMAN NAME2=LUCY ADDR2=OAK
4.9	Boardman Lucy R	9399 ELM	SE 104	5687580	555009973	3508500P	SP_33434442	NAME1=BOARDMAN NAME2=LUCY
4.9	Boardman Lucy M	8058 MOUNTAIN	W	5801893	555242076	280194WP	SP_17027731	NAME1=BOARDMAN NAME2=LUCY
4.9	Boardman Lucy G	7820 RIVER	PL	5612109	555191372	488974XL	SP_19821261	NAME1=BOARDMAN NAME2=LUCY
4.9	Boardman Lucy X	7215 MILL	N	5717672	555226335	340760YX	SP_34649406	NAME1=BOARDMAN NAME2=LUCY
4.9	Boardman Lucy K	9415 POPLAR	CT 1...	5602204	555222196	403496LN	SP_32795177	NAME1=BOARDMAN NAME2=LUCY
4.9	Boardman Lucy G	7178 VALLEY	NE 104	5729360	555130774	152660FK	SP_32843836	NAME1=BOARDMAN NAME2=LUCY
4.9	Boardman Lucy G	7185 RIVER	PL	5777619	555103069	498351QX	SP_34530493	NAME1=BOARDMAN NAME2=LUCY
4.9	Boardman Lucy K	9482 ELM	SW	5594374	555377073	296067KX	SP_155022907	NAME1=BOARDMAN NAME2=LUCY
4.9	Boardman Lucy E	7343 MILL	NW	5765196	555346010	163806CN	SP_13581153	NAME1=BOARDMAN NAME2=LUCY
4.9	Boardman Lucy Q	9581 POPLAR	E	5562747	555341337	365084HS	SP_19230028	NAME1=BOARDMAN NAME2=LUCY
4.9	Boardman Lucy M	7143 VALLEY	RD	5720361	555260516	425812NP	SP_33419456	NAME1=BOARDMAN NAME2=LUCY
3.5	Boardman Russ F	9397 OAK	PKWY	5954234	555033436	291468HM	SP_8981500	NAME1=BOARDMAN ADDR2=OAK
3.5	Boardman Ross W	9547 OAK	S	5728964	555147656	187467ZE	SP_34449041	NAME1=BOARDMAN ADDR2=OAK
3.5	Boardman Rick S	9132 OAK	ST	5726587	555234711	430600FT	SP_16566869	NAME1=BOARDMAN ADDR2=OAK
3.5	Boardman Jim T	9572 OAK	E	5870061	555309355	317367FT	SP_10687552	NAME1=BOARDMAN ADDR2=OAK
3.5	Boardman Chrissie	9343 OAK	CT 152	5722084	555371631	544225LM	SP_8519221	NAME1=BOARDMAN ADDR2=OAK
3.5	Boardman Jeff J	9336 OAK	CT 105	5631658	555204054	351095JX	SP_18376925	NAME1=BOARDMAN ADDR2=OAK
3.5	Boardman Victor X	9470 OAK	CT	5873266	555011676	408660PR	SP_13519003	NAME1=BOARDMAN ADDR2=OAK
3.5	Boardman Aaron M	9374 OAK	ST	5681852	555386976	260949MF	SP_11600915	NAME1=BOARDMAN ADDR2=OAK
3.5	Boardman Nathaniel	9417 OAK	CT	5589541	555216282	470267JL	SP_18931065	NAME1=BOARDMAN ADDR2=OAK
3.5	Boardman Tracey P	9640 OAK	S 125	5906102	555322015	550502JX	SP_9308806	NAME1=BOARDMAN ADDR2=OAK
3.5	Boardman Ben A	9210 OAK	ST	5925902	555131330	500402VQ	SP_9518146	NAME1=BOARDMAN ADDR2=OAK
3.5	Boardman David Y	9391 OAK	E	5773941	555375983	410392HK	SP_158865356	NAME1=BOARDMAN ADDR2=OAK
3.5	Boardman Bill H	9433 OAK	LN	5915771	555133368	315300GL	SP_12299539	NAME1=BOARDMAN ADDR2=OAK
3.5	Boardman Kenneth Y	9569 OAK	ST	5777604	555260566	403572IH	SP_34437843	NAME1=BOARDMAN ADDR2=OAK
3.5	Boardman Heath Q	9386 OAK	LN	5844773	555038398	538389WS	SP_15633421	NAME1=BOARDMAN ADDR2=OAK

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	Yes	Screenshot included
Demonstration Available	Yes	Click here for demonstration

4.1.2. Improved Filtering in Snapshot Viewer

A new filter option has been added to the Snapshot Viewer application. This option enables filtering of all data based on the entered text, in addition to the existing column filter.

For more information, see General Usage in the *Snapshot Viewer User Guide*.

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	No	
Demonstration Available	No	

4.2. What's Changed in 16.23.0

The following changes have been made in this release.

4.2.1. Supported Software and Hardware Changes

This section describes changes to the system such as changes in system requirements, compatibility, or third-party software.

The following changes has been made in this release:

- Forecasting - 3.10.0 & 16.23.0
- Gateway - 2.8.4 & 16.22.0
- Foundation - 25R09

4.3. What's Fixed in 16.23.0

This section lists the notable issues addressed in this release.

4.3.1. Archive

Archive App Crash Due to System.IO Exception

Grip ID: ETD-83615

Fix Version: 16.23.0

Severity: High

Issue: Archive App is crashing due to System.IO Exception

Solution: The new code allows both processes to write concurrently, preventing this race condition during rapid restarts. Addresses cross-process contention: `FileShare.ReadWrite` allows other ArchiveApp instances (from Procman restarts) to also write to the same file.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

4.3.2. Crew Management

Incident Name Not Displayed When Dispatching Crews

Grip ID: ETD-83829

Fix Version: 16.23.0

Severity: Low

Issue: When dispatching crews, whether multiple or individual, the GUID appears in the Dispatch Crew display under the Current Assignments output instead of the incident name, which is not helpful to users.

Solution: In the "Dispatch Crew to Incident" - Incident Form, initially the Incident GUID was appearing. We had modified the code in AssignCrewCtrl file in NetAddIns and now the Incident number is appearing on the screen as per the requirement.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Incidents Not Clearing When Multiple Mobile Crews Are Dispatched

Grip ID: ETD-83758

Fix Version: 16.23.0, 16.16.2

Severity: Medium

Issue: When an incident is assigned to multiple mobile crews, clearing the incident from one crew does not fully clear it. The incident remains active as a non-outage or stays in pending-clear or unassigned state instead of closing correctly.

Solution: When an incident is in a pending-clear state, its working status should not be affected by changes to crew status.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

4.3.3. DMS

Initializing Study Mode Does Not Load Proper Adjacent Stations

Grip ID: ETD-83871

Fix Version: 16.23.0, 16.18.0

Severity: Medium

Issue: Study Mode initialization had not been coded to support trans subfed stations at an electrical adjacency level greater than 1.

Solution: Added support in Study Mode code to handle adjacency greater than 1.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

DPF Incorrectly Changes Capacitor Status in Remote Mode

Grip ID: ETD-83804

Fix Version: 16.23.0

Severity: Medium

Issue: Due to a timing conflict between the SCADA event and the DPF run, the DPF uses an outdated value from the snapshot.

Solution: Improved the logic to prevent capacitor status conflicts caused by timing issues.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

LVM Triggering for Disabled and Locked Stations

Grip ID: ETD-83649

Fix Version: 16.23.0

Severity: Medium

Issue: LVM periodic execution is driven solely by a timer. During a model push, an LVM periodic solve can run before dynamics are applied, causing LVM to trigger for stations that are disabled or locked.

Solution: Updated the LVM trigger logic to coordinate with the model push process, preventing periodic solves from running until model loading and dynamics are fully applied.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

4.3.4. DNAF

Incorrect FISR Fault Indicator Measurement State

Grip ID: ETD-83971

Fix Version: 16.23.0

Severity: Medium

Issue: The Test_Mode bit on the fault indicator from SCADA is not taken into account when evaluating fault indicator changes before and after the FISR solution.

Solution: Check SCADA Test_Mode bit on the fault indicator when evaluating fault indicator changes before and after the FISR solution.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

LVM Over-Operates When Regulating Power Factor

Grip ID: ETD-83970

Fix Version: 16.23.0, 16.20.0

Severity: Medium

Issue: An unnecessary capacitor status change is being proposed by LVM

Solution: Enhanced the LVM power factor control logic to eliminate unnecessary capacitor status change.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Plan Manager Cannot Control Phase-Independent Caps

Grip ID: ETD-83813

Fix Version: 16.23.0

Severity: Medium

Issue: The capacitor phase-independent flag is not correctly serialized in the model, which caused a Plan Manager validation issue.

Solution: Ensure the capacitor's phase-independent flag is properly serialized in the model file.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	Need rebuild the model with new Converter from tool kit
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

LVM Plan Not Transferred to Plan Manager

Grip ID: ETD-83818

Fix Version: 16.23.0

Severity: Medium

Issue: When a station's LVM execution mode was changed from Advisory to Closed Loop, LVM generated a plan but it was not transferred to Plan Manager.

Solution: Improved DNAF handling of LVM plan updates by consolidating add/delete messages, enhancing log clarity when no valid plan is generated, and cleaning up multi-threaded logging sequences to prevent misinterpretation of plan processing.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	Need rebuild the model with new Converter from tool kit
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

4.3.5. OMS

Split Does Not Remove Disconnected Customers

Grip ID: ETD-83935

Fix Version: 16.23.0

Severity: Medium

Issue: When a customer is split from a merged non-outage incident that contains two or more calls, a new incident is created. However, the call used for the split remains on the original incident. This occurs because `callCustomer.CurrentIncident` is typed as `IOutageIncident`, causing `GetCurrentIncident` to return null, which prevents the call from being moved.

Solution: In `IncidentSplitting.cs`, when `GetCurrentIncident` returns null, the code now searches the incident library for a non-outage incident whose `nonOutageCall` customer key matches the call's customer key. That incident is then assigned as the call's current incident, and the call is moved to the newly created incident as intended.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	Yes	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Incident Counter Resets After OMS Switchover

Grip ID: ETD-84036

Fix Version: 16.23.0

Severity: Medium

Issue: After an OMS switchover performed without a manual copy, the incident count sequence restarts from 1. This results in duplicate incident names.

Solution: Added fix to make sure the `DailySequenceNumber` for the server is always kept up to date

for standby server.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Reference ID Not Assigned for Location-Based Calls

Grip ID: ETD-83998

Fix Version: 16.23.0, 16.18.0

Severity: Medium

Issue: The code logic differed between customer-based and location-based calls. For customer-based calls, the ReferenceId was assigned correctly. However, for location-based calls, the assignment of ReferenceId was missing, resulting in a blank value.

Solution: The missing logic for assigning ReferenceId to location-based calls was added. After this update, the assignment works consistently and as expected.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

ITRs Not Disabled When Creating a New Area Setting

Grip ID: ETD-83856

Fix Version: 16.23.0

Severity: Medium

Issue: When setting a different System Mode and disabling ITRs during the creation of a new Area Setting, the system calculates a new ETR instead of disabling the ITR. As a result, ITRs are not being properly disabled during Area Setting creation.

Solution: When OMS server configuration has OMS_CLEAR_EXISTING_INCIDENTS_ITR_ON_AREA_NOTE_ITR_DISABLED = 1 and Area Settings has ITR Disabled is checked, Area Settings will no longer allow ETR calculated regardless of Area Settings System Mode. Also, ITR for existing incidents will be disabled.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Data Replication Fails to Copy .dat Files Correctly

Grip ID: ETD-83923

Fix Version: 16.23.0

Severity: Medium

Issue: The issue was that when we switch from the active to the standby server, orders were appearing to be duplicated in the new active server. There were two issues. One in the Work Model system, we needed to check for orders that were in the Field Complete state. Second, the incident sequence number was restarting from 1 again.

Solution: Added the check for orders that are in the Field Complete state to prevent possibly duplicating those orders. Also, the incident sequence number was fixed to continue with the next number instead of starting over from 1 again.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Incident Custom Fields Truncated in NRT/Archive Database and ODATA

Grip ID: ETD-83746

Fix Version: 16.23.0

Severity: Low

Issue: Incident Custom Field are truncated in Archive/NRT Database when send through API.

Solution: Validation is handled for special characters before inserting in database. After validation, the custom field value is not truncating.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	

Issue Implication	Yes/No	Additional Information
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Split Restoration or Switching Steps Assigned to Incorrect Organization

Grip ID: ETD-83700

Fix Version: 16.23.0

Severity: Medium

Issue: Data synchronization gap where the organization value from the secondary incident order was lost, leaving the master order with its original (typically null or empty) organization value. Even if the organization had been set on the master order object in-memory, the lack of an explicit database persistence call meant the change was never saved.

Solution: Added organization synchronization code that extracts the organization value and updates the incident orders in the split event.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	Yes	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

4.3.6. SWO

SWO Recently Viewed List Disappears When Daily Log Requirement Is Unchecked

Grip ID: ETD-83994

Fix Version: 16.23.0

Severity: Medium

Issue: In the Switch Order Client Config, when **Needs a Daily Log** flag is unchecked, the recently viewed switch order list appears once and then disappears until the ADMS client is restarted.

Solution: The logic was updated so that when the **Needs a Daily Log** checkbox is cleared, the recently viewed switch-order list continues to persist. The list now remains visible after reopening the Switch Order window or restarting the client, even when the flag is unchecked.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	

Issue Implication	Yes/No	Additional Information
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Switch Order Client Connection Issues - Memory Spike

Grip ID: ETD-83836

Fix Version: 16.23.0

Severity: Medium

Issue: A memory growth issue was identified in the Switch Order Server during periods of high client activity. Frequent broadcast operations caused large object allocations to accumulate in the Large Object Heap (LOH), leading to memory usage reaching 4–5 GB and gradual performance degradation.

Solution: To improve memory management and reduce LOH growth, new environment-variable-driven configuration options were introduced to enable LOH Compaction in the Switch Order Server. These options allow automated compaction at configurable intervals and memory thresholds.

Variable	Description	Default
SWO_ENABLE_LOH_COMPACTION	Enables LOH compaction	false
SWO_LOH_COMPACTION_INTERVAL_MINUTES	Interval at which compacting is attempted	30
SWO_LOH_COMPACTION_MEMORY_THRESHOLD_MB	Memory threshold (MB) above which compaction is triggered	2000 (2 GB)
SWO_FORCE_LOH_COMPACTION	Forces compaction regardless of memory usage	0

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	Yes	New environment variables added
ADMS Archive Schema	No	
User Interface	No	

4.3.7. File Updater

File Updater Client Not Working with SecureFX 9.3 on Windows 11

Grip ID: ETD-83899

Fix Version: 16.23.0

Severity: Medium

Issue: The ADMS File Updater Client does not work with VanDyke SecureFX 9.3 on Windows 11. SecureFX 9.3 introduced changes to internal error messages, threading, locks, and protocol handling. The File Updater incorrectly interpreted these updated internal error codes as connection failures, which prevented normal operations.

Solution: Added a single error-analysis function covering all SecureFX versions to ensure consistent handling regardless of SecureFX version, fixed Exponential Backoff which reduces SFXCL calls by ~97%.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

4.3.8. Interface

Customer Tab Column Settings Reset After Closing Incident

Grip ID: ETD-83908

Fix Version: 16.23.0

Severity: Low

Issue: Reordered Columns under the Customer tab revert to the default order once the incident is closed.

Solution: Functionality has been added to the Customer tab under Incident; previously, it was not available.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Crew Vehicle Icon Incorrectly Displays at Incident Location

Grip ID: ETD-83907

Fix Version: 16.23.0

Severity: Medium

Issue: When an outage is dispatched to a crew, the yellow truck icon incorrectly appears at the incident location. This icon is intended to represent the crew's real-time GPS location, not the incident location.

Solution: The logic in CrewList.cs that updated vehicle positions based on crew assignments was removed. Vehicle locations are now updated only through incoming GPS data from external systems, ensuring the icon reflects the crew's real-time position.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

4.3.9. Yambay

ADMS Crew Updates and Additions Not Updating in Yambay Mobile Client

Grip ID: ETD-83863

Fix Version: 16.23.0

Severity: High

Issue: AddEmployee journal was not considered at the time this feature (Crew Sync with Mobile Enterprise) was implemented.

Solution: When OMS server loads AddEmployee journal and the journal is for the correct employee type, OMS server will initiate crew synchronization with Mobile Enterprise.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Mobile Communication Status Not Updating Correctly in Order Worksheet

Grip ID: ETD-83801

Fix Version: 16.23.0

Severity: Medium

Issue: MobileCommunicationStatus was changed from order-specific to crew-specific during a recent enhancement. This caused updates to apply to all incidents assigned to a crew instead of only the active order.

Solution: Restored MobileCommunicationStatus to order-specific behavior and ensured compatibility with both the original and recently introduced logic.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

5. ADMS 16.22.0

This chapter describes the incremental new features and defect fixes that were added since ADMS 16.21.0. The remainder of this document provides additional release-specific information that also applies to this release.

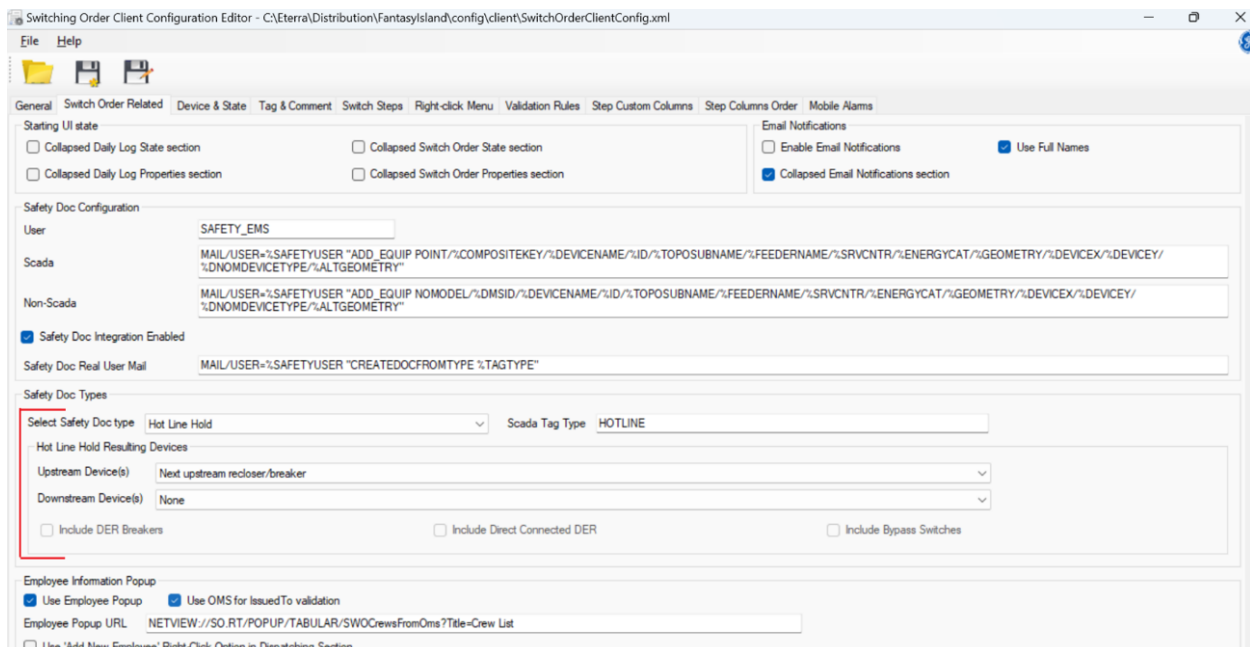
5.1. What's New in 16.22.0

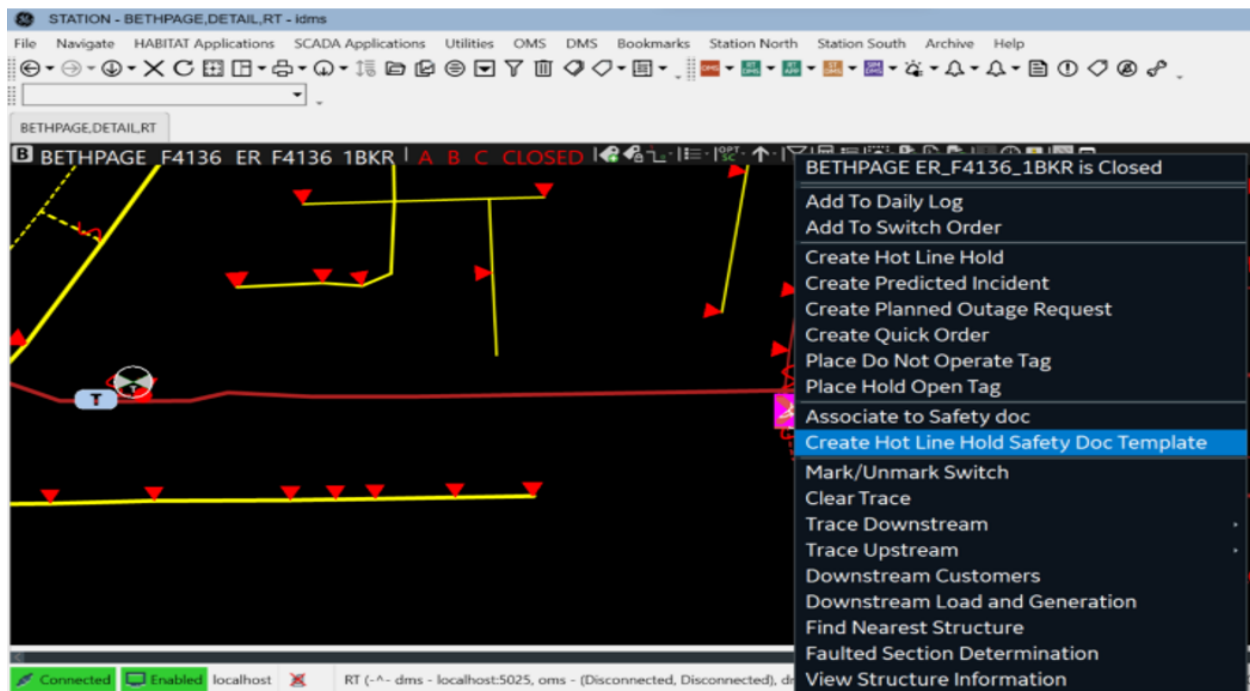
This section contains information about new features and functionality for ADMS 16.22.0.

5.1.1. Enhanced Device Addition for Safety Documents

Adding multiple devices to Clearance and Hot Line Hold (HLH) safety documents is now faster and more streamlined. With a single click, all relevant devices including upstream and downstream devices are automatically included, eliminating the need for manual selection. This improvement reduces configuration time and enhances operational accuracy.

To review the device selection logic used when creating Clearance and Hot Line Hold safety documents, navigate to Switch Order Editor Configurations and open the Switch Order Related tab. The new Safety Doc Types section displays the logic and rules applied for device inclusion. (GRDDMSS3-E-734 is available as an alpha release)





Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	Yes	Screenshot included
Demonstration Available	Yes	Click here for demonstration

5.2. What's Changed in 16.22.0

The following changes have been made in this release.

5.2.1. Supported Software and Hardware Changes

This section describes changes to the system such as changes in system requirements, compatibility, or third-party software.

The following changes has been made in this release:

- Forecasting - 3.9.0 & 16.22.0
- Gateway - 2.8.3 & 16.22.0
- Foundation - 25R09

5.3. What's Fixed in 16.22.0

This section lists the notable issues addressed in this release.

5.3.1. Archive

Archive Purge Command Failed Due to Constraint Violation

Grip ID: ETD-83830

Fix Version: 16.22.0

Severity: Medium

Issue: The purge logic in PurgingManager.cs deleted only DEVICE records directly associated with purged INCIDENTs, leaving orphaned DEVICE records that were no longer referenced by any INCIDENT after monthly batch purges. These orphaned records accumulated and were only removed during a full-range purge.

Solution: Updated the purge logic to identify and delete DEVICE records not referenced by any INCIDENT, SWITCHING_STEP, RESTORATION_STEP, or CALL, ensuring data integrity and consistent DEVICE counts during both monthly and full-range purges.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

5.3.2. OMS

Device Does Not Display in Geographic View on First Attempt

Grip ID: ETD-83894

Fix Version: 16.22.0, 16.21.0.1

Severity: Medium

Issue: When switching between searches for devices and structures, the initial Go to Geo action after returning to a device search displays a blank screen. The device loads correctly only after selecting Go to Geo a second time. The issue persists even if the search function is closed after each search.

Solution: Updated search logic to correctly locate the center point when navigating to a device.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Mismatch Between Crew Status and Incident Working Status

Grip ID: ETD-83841

Fix Version: 16.22.0

Severity: Medium

Issue: An edge case occurred when reassigning crews in ReassignCrewsForConfirmedClosedIncidents. If a downstream incident had the same device ID as the confirmed closed incident, the crew status could become inconsistent with the incident working status.

Solution: In ReassignCrewsForConfirmedClosedIncidents, make sure to reassign crew correctly when the downstream incident has the same device id as the original confirm closed incident.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Customers Marked Restored After Merging Fuse Orders

Grip ID: ETD-83832

Fix Version: 16.22.0

Severity: Medium

Issue: When a predicted incident was merged with a confirmed incident on the same device but with a different phase, some customers were incorrectly marked as restored.

Solution: Updated the logic to correctly handle restoration step merging and prevent customers from being automatically restored.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	

Issue Implication	Yes/No	Additional Information
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Restoration Order Created With Zero Customers on Multi-Phase Laterals

Grip ID: ETD-83769

Fix Version: 16.22.0

Severity: Medium

Issue: When creating a confirmed single-phase outage on a multi-phase lateral, some phase outages resulted in incidents with zero customers affected.

Solution: Updated the logic to prevent incident splitting when the new incident includes all customers from the existing one.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Unable to Restore Predicted Outage With Downstream Confirmed Outage

Grip ID: ETD-83684

Fix Version: 16.22.0

Severity: Medium

Issue: When a predicted outage had a downstream confirmed outage, repairing the task did not reduce the customer count to zero, causing the order to remain stuck.

Solution: Updated the logic to correctly handle restoration when a predicted outage has a downstream confirmed outage, allowing the incident to clear as expected.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Incident Priority Not Updated in NRT After Merge

Grip ID: ETD-83679

Fix Version: 16.22.0

Severity: Medium

Issue: When orders were merged, NRT did not update the incident priority to reflect the highest priority from all calls.

Solution: Updated the logic to ensure priority code changes are correctly applied and reflected in NRT after incidents are merged.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

NRT Device Table Incorrect for Underground Indicator on Jumpers

Grip ID: ETD-83560

Fix Version: 16.22.0

Severity: Low

Issue: The Device Table did not update correctly for the Underground_Indicator on jumpers. The IsOverhead value was incorrect when incidents were created from a jumper.

Solution: Updated the logic to remove an unnecessary conditional check, ensuring the IsOverhead value is correctly set in the Device Table for jumper incidents.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Incident Custom Fields are Truncated in NRT/Archive DB and NetDataServer ODATA

Grip ID: ETD-83746

Fix Version: 16.22.0

Severity: Low

Issue: Both NRT/Archive database and NetDataServer ODATA queries are truncating multi-line incident custom fields. Customer wanted to format the GUI for nice presentation and includes \r\n in the CrewServer UpdateIncident payload which is accepted and works just fine.

Solution: Created a method in NetListPrecompute.cs to replace all kinds of escape characters and called this method in the custom field value section.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Incident Updates Missing in NRT Database After Model Push

Grip ID: ETD-83828

Fix Version: 16.22.0

Severity: Medium

Issue: Few incident entries were skipped while archiving.

Solution: Enhanced error handling with chunked processing and row-by-row fallback using the new ExecuteBulkRecursive method to handle PK/DUP_VAL_ON_INDEX violations gracefully.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

NRT Incorrectly Marked Restoration Step as Momentary

Grip ID: ETD-83792

Fix Version: 16.22.0

Severity: Medium

Issue: Restoration step in NRT shows momentary = 1 even though the step duration is longer than 60s.

Solution: Refactored the logic to use endTime instead of cs.LastAMRRreportTime for duration comparison with ars.DeenergizedTime.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Predicted Outage Blocked by Confirmed Outage on Another Fuse

Grip ID: ETD-83831

Fix Version: 16.22.0

Severity: Medium

Issue: For Fuse that is modeled individually per phase, the energization checking was only checking branch status before.

Solution: For Fuse that is modeled individually per phase, the energization checking is updated to evaluate both the connecting node and the branches to ensure accuracy.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Undo Confirmation Lost Predicted Line Halo

Grip ID: ETD-83456

Fix Version: 16.22.0, 16.19.0

Severity: Medium

Issue: When undoing the confirmation of an incident that was created from a predicted one, the predicted halo is not displayed.

Solution: Updated the undo confirmation logic to re-associate service points and set PredictedStatic to true when a device-only predicted incident is undone, ensuring the prediction halo reappears.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	

Issue Implication	Yes/No	Additional Information
ADMS Archive Schema	No	
User Interface	No	

NetServerOMS Did Not Handle HA with K8s EOMS

Grip ID: ETD-83316

Fix Version: 16.22.0

Severity: Medium

Issue: It was discovered that the logic that creates the list of URLs for the WorkModel Service and establishes the initial end point with the WorkModel Service does not correctly handle some error conditions like when the URL is malformed or when the end point connection is successful, but it only returns an empty page.

Solution: Added the additional logic to check for and gracefully handle these error conditions with additional logging to inform the user.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Performance Slowdown in Crew Assignment

Grip ID: ETD-83933

Fix Version: 16.22.0, 16.21.0.2

Severity: Medium

Issue: In EOMS, assigning a crew experiences a significant delay.

Solution: 16.21.0.2 removed nclientID from the thick client to improve crew assignment performance; however, this solution can cause the system to lose information about the initiating user. 16.22.0 retains user information and changes the path of one specific query between the EOMS and OMS to eliminate the delay in crew assignment.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	

Issue Implication	Yes/No	Additional Information
User Interface	No	

5.3.3. DNAF

Primary Winding Tap Position Logged as Secondary in DPF Results

Grip ID: ETD-83849

Fix Version: 16.22.0

Severity: Low

Issue: In the DPF Results log, the Primary Winding Tap Position section was incorrectly logging the secondary winding tap position.

Solution: Updated DPF logging (Measured or Estimated) column now shows '--' for primary taps, as no measurement or estimation is required.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

AFR DNAF Solution Popup Not Retaining Default Problem Formulation

Grip ID: ETD-83834

Fix Version: 16.22.0

Severity: Low

Issue: The DNAF Solution Popup did not retain the default problem formulation after the user adjusted the dropdown and executed a solution.

Solution: The get objective functions for LVM, AFR, FISRCA, PLORS are modified by finding objective function from problem formulation index.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

LVM System Monitor Pulls Incorrect Problem Formulation for Similar Profiles

Grip ID: ETD-83820

Fix Version: 16.22.0

Severity: Medium

Issue: If LVM system monitor profile name is similar to another profile, problem formulations from another profile is used.

Solution: The issue is fixed by using appropriate string manipulation functions.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	Yes	LVM System Monitor Stations

DNAF Unresponsive When Receiving Dynamic Schedules

Grip ID: ETD-83917

Fix Version: 16.22.0

Severity: Medium

Issue: DNAF becomes unresponsive when a forecast posts more than 16,000 DERs with 24-hour schedules.

Solution: Improved dynamic schedule importing for high-volume datasets.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

5.3.4. DMS

Navigation Command Ignores NOGEOADJACENCY Parameter

Grip ID: ETD-83853

Fix Version: 16.22.0

Severity: Medium

Issue: After using the NOGEOADJACENCY parameter in a navigation command, subsequent navigation commands on the same viewport automatically included the parameter, preventing the expected data from being returned.

Solution: Updated the logic to ensure navigation commands respect the intended NOGEOADJACENCY behavior. As an alternative, Adjacency = 0 can be used to achieve the same effect.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

FISR Generates Plan With Identical Control Moves

Grip ID: ETD-83803

Fix Version: 16.22.0

Severity: Low

Issue: FISR generated a plan containing two identical control moves. This occurred during partial restoration when there were only voltage violations in a FISR loss-of-voltage solution.

Solution: Updated the logic to prevent generating identical control moves in partial restoration plans.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

FL Generates Excessive Results During Model Push

Grip ID: ETD-83795

Fix Version: 16.22.0

Severity: High

Issue: FL is triggered by non-zero fault current analogs with valid quality. Outdated fault current values can persist in SCADA field devices, causing unnecessary triggers during model push

operations that require SCADA measurement downloads.

Solution: Implemented a timestamp check to ignore fault current analogs with stale data.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

LVM Did Not Calculate Total Performance Index Correctly

Grip ID: ETD-83930

Fix Version: 16.22.0

Severity: Medium

Issue: LVM Performance Index is not calculated correctly in single step mode.

Solution: Logic change to make sure performance index is calculated in LVM single step mode.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

5.3.5. Modeling

PV and Wind SCADA Tags Not Exported in Model Exporter Tool

Grip ID: ETD-83823

Fix Version: 16.22.0

Severity: Medium

Issue: The EnergyResource types which were not under any DistTF are being skipped by ModelExporter.

Solution: Changed EnergyResource enumeration logic to select all EnergyResource objects.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	

Issue Implication	Yes/No	Additional Information
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

5.3.6. Yambay

Mobile Client Not Updating Current Out After Repair

Grip ID: ETD-83885

Fix Version: 16.22.0

Severity: Medium

Issue: Yambay Mobile Client did not update the Current Out value on the Outage and Information tabs after a repair task was completed. The customer count stayed at the original number instead of updating to 0.

Solution: The fix is querying incident for affected customers instead of relying on IncidentDbo due to its delayed updates.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

5.3.7. Tools

SnapshotViewer slow to export manuals.sav

Grip ID: ETD-83843

Fix Version: 16.22.0

Severity: Medium

Issue: Exporting a large file can take a long time

Solution: Performance has been improved by a better C# string manipulation logic

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	

Issue Implication	Yes/No	Additional Information
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

6. ADMS 16.21.0

This chapter describes the incremental new features and defect fixes that were added since ADMS 16.20.0. The remainder of this document provides additional release-specific information that also applies to this release.

6.1. What's New in 16.21.0

This section contains information about new features and functionality for ADMS 16.21.0.

6.1.1. Flexible Customer Search

A new flexible customer search is now available for testing. This enhancement simplifies the search process by supporting single-line input for customer name and address, accepting space-separated words in any order. The search also supports meter number, customer key, account number, and phone number.

Results are displayed after entering three characters and are ranked based on matched terms. To use this feature, ensure the Address field is configured in the Customer List Compiler and run the search from the Search tab. The feature is enabled by default and can be disabled by setting the DISABLEFLEXIBLESEARCH flag in the IDMS_FEATURES environment variable. (GRDDMSS3-2492 and GRDDMSS3-2639)

For more information, see *Search Tab* in the *Overview of Customer List Compiler Displays* chapter of the *Customer List Compiler User Guide*.

Customer List Compiler								
File Tools Help								
Conversion Dedicated Fields Data Schema Region/Management Area Schema View Customer Binary File Data Search								
boardman lucy oak								
Score	Name	Address	Phone Num...	Account Number	Meter Number	Service Point	Match	
5.4	Boardman Lucy D	9419 OAK	SE	5679198	555146499	485671XE	SP_34457827	NAME1=BOARDMAN NAME2=LUCY ADDR2=OAK
4.9	Boardman Lucy R	9399 ELM	SE 104	5687580	555009973	3508500P	SP_33434442	NAME1=BOARDMAN NAME2=LUCY
4.9	Boardman Lucy M	8058 MOUNTAIN	W	5801893	555242076	280194WP	SP_17027731	NAME1=BOARDMAN NAME2=LUCY
4.9	Boardman Lucy G	7820 RIVER	PL	5612109	555191372	488974XL	SP_19821261	NAME1=BOARDMAN NAME2=LUCY
4.9	Boardman Lucy X	7215 MILL	N	5717672	555226335	340760YX	SP_34649406	NAME1=BOARDMAN NAME2=LUCY
4.9	Boardman Lucy K	9415 POPLAR	CT 1...	5602204	555222196	403496LN	SP_32795177	NAME1=BOARDMAN NAME2=LUCY
4.9	Boardman Lucy G	7178 VALLEY	NE 104	5729360	555130774	152660FK	SP_32843836	NAME1=BOARDMAN NAME2=LUCY
4.9	Boardman Lucy G	7185 RIVER	PL	5777619	555103069	498351QX	SP_34530493	NAME1=BOARDMAN NAME2=LUCY
4.9	Boardman Lucy K	9482 ELM	SW	5594374	555377073	296067IK	SP_155022907	NAME1=BOARDMAN NAME2=LUCY
4.9	Boardman Lucy E	7343 MILL	NW	5765196	555346010	163806CN	SP_13581153	NAME1=BOARDMAN NAME2=LUCY
4.9	Boardman Lucy Q	9581 POPLAR	E	5592747	555341337	365084HS	SP_19230028	NAME1=BOARDMAN NAME2=LUCY
4.9	Boardman Lucy M	7143 VALLEY	RD	5720361	555260516	425812NP	SP_33419456	NAME1=BOARDMAN NAME2=LUCY
3.5	Boardman Russ F	9397 OAK	PKWY	5954234	555033436	291468HM	SP_8981500	NAME1=BOARDMAN ADDR2=OAK
3.5	Boardman Ross W	9547 OAK	S	5728964	555147656	187467ZE	SP_34449041	NAME1=BOARDMAN ADDR2=OAK
3.5	Boardman Rick S	9132 OAK	ST	5726587	555234711	430600FT	SP_16566869	NAME1=BOARDMAN ADDR2=OAK
3.5	Boardman Jim T	9572 OAK	E	5870061	555309355	317367FT	SP_10687552	NAME1=BOARDMAN ADDR2=OAK
3.5	Boardman Chrissie	9343 OAK	CT 152	5722084	555371631	544225LM	SP_8519221	NAME1=BOARDMAN ADDR2=OAK
3.5	Boardman Jeff J	9336 OAK	CT 105	5631658	555204054	351095JX	SP_18376925	NAME1=BOARDMAN ADDR2=OAK
3.5	Boardman Victor X	9470 OAK	CT	5873266	555011676	408660PR	SP_13519003	NAME1=BOARDMAN ADDR2=OAK
3.5	Boardman Aaron M	9374 OAK	ST	5681852	555386976	260949MF	SP_11600915	NAME1=BOARDMAN ADDR2=OAK
3.5	Boardman Nathaniel	9417 OAK	CT	5925902	555131330	470267JL	SP_18931065	NAME1=BOARDMAN ADDR2=OAK
3.5	Boardman Tracey P	9640 OAK	S 125	5906102	555322015	550502UX	SP_9308806	NAME1=BOARDMAN ADDR2=OAK
3.5	Boardman Ben A	9210 OAK	ST	5925902	555131330	500402VQ	SP_9518146	NAME1=BOARDMAN ADDR2=OAK
3.5	Boardman David Y	9391 OAK	E	5733941	555375983	410392HK	SP_158865356	NAME1=BOARDMAN ADDR2=OAK
3.5	Boardman Bill H	9433 OAK	LN	5915771	555133368	315300GL	SP_12299539	NAME1=BOARDMAN ADDR2=OAK
3.5	Boardman Kenneth Y	9569 OAK	ST	5777604	555260566	403572IH	SP_34437843	NAME1=BOARDMAN ADDR2=OAK
3.5	Boardman Heath Q	9386 OAK	LN	5844773	555038398	538389WS	SP_15633421	NAME1=BOARDMAN ADDR2=OAK

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	Yes	Screenshot included
Demonstration Available	Yes	Click here for demonstration

6.1.2. Multiple Prediction Rules

The OMS Server previously supported system-wide configuration of prediction rule parameters. With this release, prediction rules can be configured per feeder. This flexibility allows utilities to apply feeder-specific rules, leading to more accurate predictions. (GRDDMSS3-3270)

For more information, see *Configuring the Multiple Prediction Rule Sets* in the *Advanced Configuration of the ADMS Installation Guide*.

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	Yes	
ADMS Archive	No	
User Interface	No	
Demonstration Available	Yes	Click here for demonstration

6.1.3. Bounding box enhancements in Outage Assist

For details about using the bounding box to find common upstream devices or view orders in a list, see the *Outage Assist 16.21.0 Release Notes*. (GRDDMSS3-3224)

6.2. What's Changed in 16.21.0

The following changes have been made in this release.

6.2.1. Improvements to Existing Functionality

6.2.1.1. Switch Order Client Connection and Memory Management Improvements

Switch Order (SWO) server performance has been improved to prevent disconnections and excessive memory usage when saving switch orders. Log messages now provide better clarity, and the Find button is now disabled (greyed out) while a search is in progress.

6.2.2. Supported Software and Hardware Changes

This section describes changes to the system such as changes in system requirements, compatibility, or third-party software.

The following changes have been made in this release:

- Forecasting - 3.9.0 & 16.21.0
- Gateway - 2.8.3 & 16.21.0
- Foundation - 25R08

6.3. What's Fixed in 16.21.0

This section lists the notable issues addressed in this release.

6.3.1. NTP

Client Shows Subfed Station as Not Energized While Server Reports Voltage

Grip ID: ETD-83850

Fix Version: 16.21.0

Severity: Medium

Issue: Pass-through subfed station is not energized.

Solution: Fixed and improved sourceid assignment for pass-through subfed stations.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	

Issue Implication	Yes/No	Additional Information
User Interface	No	

6.3.2. Modeling

SFB No Longer Includes Backslash in SCADA Device

Grip ID: ETD-83807

Fix Version: 16.21.0

Severity: Medium

Issue: When SCADA device name contains back slash (for example, "69/13") the SFB assigns SCADA device name after dropping the slash.

Solution: Fixed the SFB validation code where SCADA name is read.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

6.3.3. OMS

ADMS Shows Seven-Hour Difference on Mobile ETA Update

Grip ID: ETD-83782

Fix Version: 16.21.0

Severity: Medium

Issue: ADMS shows wrong crew ETA.

Solution: Converting crew ETA from UTC to local time.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Area Settings UseParent Flag Not Working Correctly

Grip ID: ETD-83754

Fix Version: 16.21.0

Severity: Low

Issue: Incorrect use of Area Settings UseParent flag.

Solution: Fixed the code to build area's hierarchy in area settings.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Discrepancy Between CEA and OMS in Affected Customers

Grip ID: ETD-83753

Fix Version: 16.21.0

Severity: Medium

Issue: CEA Affected customer count included inactive customers.

Solution: Updated query filter to exclude inactive customers.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

NetServer Crash When Creating Snapshots

Grip ID: ETD-83736

Fix Version: 16.21.0

Severity: High

Issue: NetServer crashed when writing a new snapshot file.

Solution: Improved exception handling and enhanced logging were added to capture root causes and prevent future crashes.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

OMS Client Memory Utilization Reaches Maximum and Becomes Unusable

Grip ID: ETD-83675

Fix Version: 16.21.0, 16.20.0.1, 16.20.0

Severity: Medium

Issue: The OMS client became unresponsive after a few minutes of use because memory utilization reached 100 percent. The problem occurred due to structure drawing data not being cleared after each cycle.

Solution: The memory cleanup logic was updated to dispose of structure drawing data after each cycle. This prevents memory from maxing out and keeps the OMS client stable during extended sessions.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Work Model Service Server Returns HTTP 503 Error for EOMS API

Grip ID: ETD-83588

Fix Version: 16.21.0

Severity: Medium

Issue: Interface server often returns "HTTP 503: service unavailable" error for EOMS API.

Solution: Added an index on status in the orders table to speed up EOMS database queries.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	

Issue Implication	Yes/No	Additional Information
User Interface	No	

Outage Calls Linked to Order Not Cleared After Closure

Grip ID: ETD-83566

Fix Version: 16.21.0

Severity: Medium

Issue: When the flag OMS_CLEAR_OUTAGES_CALLS_FOR_MOMENTARY is set to 0, calls were not cleared correctly for non momentary incidents.

Solution: Fixed the logic to handle this flag when clearing non momentary orders.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

One Crew Appears Multiple Times on an Order in Same Status

Grip ID: ETD-83616

Fix Version: 16.21.0

Severity: Medium

Issue: When assigning a crew to an order, they are showing twice on the order with duplicate statuses.

Solution: Added a database constraint on the EOMS database so that the same crew can only have one active record for a specific order.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	Yes	Multiple attempts to assign the same crew to an order will only result in one assignment
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Set Predicted Not Rolling to Upstream

Grip ID: ETD-83168

Fix Version: 16.21.0

Severity: Medium

Issue: When customer is modeled as individual service point, single customer incident did not roll up correctly.

Solution: Fixed the logic related to service point lookup.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Netserver Crash in Affected Customers Query

Grip ID: ETD-83861

Fix Version: 16.21.0

Severity: High

Issue: Netserver crashes occasionally on startup or with a new customer file.

Solution: Added defensive logic to check for this condition.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

6.3.4. History Correction

Locate Transformer from History Correction Fails

Grip ID: ETD-83650

Fix Version: 16.21.0

Severity: Medium

Issue: The Locate command did not work because the "GEOGRAPHIC" parameter was missing from

the OMS display plugin.

Solution: The Tabular configuration was updated to use the geometry name, which is now dynamically picked up.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

6.3.5. Interface

NetViewerAdapter Crashes on Start with Unhandled Exception

Grip ID: ETD-83642

Fix Version: 16.21.0

Severity: Medium

Issue: NetViewerAdapter crashed due to duplicated settings related to landbase in viewerconfig.xml.

Solution: Handled the case when duplicated settings found.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Inconsistent Error Messages When Enrouting Crew to Multiple Orders

Grip ID: ETD-83539

Fix Version: 16.21.0, 16.17.0

Severity: Medium

Issue: In Outage Assist, inconsistent error messages occurred when routing a crew to multiple orders. In some cases, the error message in the worksheet disappeared too quickly.

Solution: Error handling was standardized across Outage Assist components. The same error message is now displayed in both the list view and the order worksheet view.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

6.3.6. Core

Hebrew Characters in Attribute Popups

Grip ID: ETD-83354

Fix Version: 16.21.0

Severity: Medium

Issue: Attribute popups would show corrupted characters for Hebrew strings.

Solution: Corrected the internal exchange between strings of different types.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

NetDataReplicator Stopped Working

Grip ID: ETD-83797

Fix Version: 16.21.0

Severity: Medium

Issue: With some error conditions, the NetDataReplicator application would exit rather than crash. In this case the application would not be restarted.

Solution: Removed the cases where the application would exit, and added additional error handling.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	

Issue Implication	Yes/No	Additional Information
ADMS Archive Schema	No	
User Interface	No	

6.3.7. DMS

Memory Spike in NetServ

Grip ID: ETD-83710

Fix Version: 16.21.0, 16.18.0, 16.16.2, 16.15.0

Severity: High

Issue: NetServ Memory Spike during manual dynamics synchronization.

Solution: The manual dynamic sync process has been optimized to handle large manual dynamic file syncs between one enabled NetSrv and three standby instances.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

6.3.8. EOMS

Unable to Set ETR When Disable ITR Flag Is Set in Area Settings

Grip ID: ETD-83559

Fix Version: 16.21.0

Severity: Medium

Issue: If the ITR disable flag is set in an area setting, you are no longer able to manually populate ETR data on any of the orders affected.

Solution: Handled null object when the default value of ETR Date is null.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	

Issue Implication	Yes/No	Additional Information
ADMS Archive Schema	No	
User Interface	No	

Cannot Enroute Crew After Arrival on Closed Order

Grip ID: ETD-83702

Fix Version: 16.21.0

Severity: Medium

Issue: Error message shows crew cannot be enroute because of a status conflict on a closed order. The issue happens because ADMS is not auto releasing the crew that was assigned in an order with Field Complete status.

Solution: Crews assigned to a FIELD_COMPLETE order will be auto released once the order is closed.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	Yes	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

6.3.9. DNAF

LVM Ignores ScVolt Deadband

Grip ID: ETD-83771

Fix Version: 16.21.0

Severity: Medium

Issue: LVM ignores ScVolt deadband in voltage setpoint functionality.

Solution: Updated the logic to check voltage setpoint deadband before and after range adjustment, to allow LVM to generate the meaningful plans.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	

Issue Implication	Yes/No	Additional Information
User Interface	No	

6.3.10. UI

Fault Bitmaps Matching Issue

Grip ID: ETD-83592

Fix Version: 16.21.0

Severity: Medium

Issue: When the status is changed, the fault bitmaps fail to match.

Solution: Added the missing else condition to match VCE.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

7. ADMS 16.20.0

This chapter describes the incremental new features and defect fixes that were added since ADMS 16.19.0. The remainder of this document provides additional release-specific information that also applies to this release.

7.1. What's New in 16.20.0

This section contains information about new features and functionality for ADMS 16.20.0.

- Find common upstream protective devices list from orders (GRDDMSS3-3309)
- Select orders in the list and show them on the map (GRDADMS-1759)
- Select orders on the map and show them in the list (GRDDMSS3-2504)
- Added functionality to prevent task status updates for orders that are already marked as complete.

See *Outage Assist 16.20.0 Release Notes* for details.

7.2. What's Changed in 16.20.0

The following changes have been made in this release.

7.2.1. Supported Software and Hardware Changes

This section describes changes to the system such as changes in system requirements, compatibility, or third-party software.

The following changes has been made in this release:

- Forecasting - 3.9.0 & 16.20.0
- Gateway - 2.8.2 & 16.20.0
- Foundation - 25R07

7.3. What's Fixed in 16.20.0

This section lists the notable issues addressed in this release.

7.3.1. Archive

Incorrect Fields in Archive Data Dictionary

Grip ID: ETD-83556

Fix Version: 16.20.0

Severity: Low

Issue: The Archive Data Dictionary incorrectly lists Dispatcher_Comments and Crew_Comments fields as present in the ORDERS table, even though these fields were removed from the actual database.

Solution: Removed Dispatcher_Comments and Crew_Comments fields from Data Dictionary.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Meter Ping Table Not Populated in Archive

Grip ID: ETD-83530

Fix Version: 16.20.0

Severity: Medium

Issue: The METER_PING table in the Archive was not populated when a meter was pinged through ADMS because the METERPING event was not journalized to CallJournal. In addition, the table does not contain the Response_Accuracy column, and the fields PINGREQUESTID, ROUTING, and METERID were instead recorded in the METERSTATUS journal entry.

Solution: Journalization was corrected to ensure that meter ping events are properly captured in the Archive.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Number of Current Customers Column Not Updated in Archive Database

Grip ID: ETD-83597

Fix Version: 16.20.0

Severity: Low

Issue: The number_of_current_customers column in the Incident table was not being updated correctly in the Archive (NRT) database due to a field name mismatch — the Archive expected CurrentCustomer, but CurrentCustomerCount was being passed instead.

Solution: Updated ProcessSingleCustomerRestoration to use CurrentCustomer instead of CurrentCustomerCount.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

7.3.2. OMS

Issue with Override Case Note Field in API

Grip ID: ETD-83554

Fix Version: 16.20.0

Severity: Medium

Issue: The API does not set the override case note correctly when the area note field is filled out, whereas the UI does.

Solution: Updated the logic to consider OverrideCaseNote is missing from API Side. Made changes in CreateUpdateAreaNoteDataEntryRequest method to consider OverrideCaseNote in NetDataEntryField when CustomerFeedbackOption == 1. Set the OverrideCaseNote to populate for CustomerFeedbackOption == 1 and not for CustomerFeedbackOption == 2.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Excessive Increase in OmsInputQueue Log on Standby OMS

Grip ID: ETD-83635

Fix Version: 16.20.0

Severity: Low

Issue: On a standby OMS, the OmsInputQueue log file was growing excessively due to repetitive OmsInitializeMeterMeasurement entries. This caused unnecessary log bloat and potential difficulties in log management.

Solution: The system now validates successful initialization before starting message processing. Messages are skipped if the server is not enabled, preventing excessive OmsInputQueue growth on redundant servers.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Standby OMS Server Not Reading Journals Properly

Grip ID: ETD-83609

Fix Version: 16.20.0

Severity: Medium

Issue: During failover testing, information discrepancies occurred between the main and standby servers, resulting in inconsistent incident replication and data misalignment after switching to the standby server.

Solution: The system now verifies that OmsDb is published before initializing the work model WebDb, ensuring proper sequencing and preventing initialization issues.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

NVA Memory Leak

Grip ID: ETD-83683

Fix Version: 16.20.0

Severity: Medium

Issue: Memory increase in NetviewerAdapter over time.

Solution: Updated NetViewer Adapter query and customer file processing.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Trouble Calls Column Not Updated in Archive Database

Grip ID: ETD-83583

Fix Version: 16.20.0

Severity: Medium

Issue: The Trouble_Calls column in the Incident table was not updated correctly in the Archive (NRT) database. This occurred because Trouble_Calls was not handled during incident updates in Archive, as the value was already available in journals.

Solution: Added handling for CALLCOUNT in the AddIncidentUpdate method to ensure the TROUBLE_CALLS column is updated correctly.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Incorrect AOR Assigned to Location or Unassociated Call

Grip ID: ETD-83653

Fix Version: 16.20.0

Severity: Medium

Issue: The incident.AOR value set in the CreateNonOutageIncident method was being overwritten in another method.

Solution: The system now preserves the existing incident.AOR value if it is already set and not empty. The value is only overwritten in the SetIncidentProperties method if it is not set or is empty.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	

Issue Implication	Yes/No	Additional Information
ADMS Archive Schema	No	
User Interface	No	

Undo Confirmation Not Working with Transformer Banks

Grip ID: ETD-83601

Fix Version: 16.20.0

Severity: Medium

Issue: After a predicted incident was created on a transformer bank, manually disconnecting the transformer bank and then clicking Undo Confirmation did not return the transformer bank to the closed state.

Solution: The transformer bank now re-energizes when Undo Confirmation is clicked.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Error with Past Start Time in Area Settings

Grip ID: ETD-83681

Fix Version: 16.20.0

Severity: Medium

Issue: Error when trying to set start time for an area setting in the past.

Solution: Adjust validation to allow setting start time in the past.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Ping Popup Pings Wrong Customer After Sorting

Grip ID: ETD-83692

Fix Version: 16.20.0

Severity: Low

Issue: The ping pop-up incorrectly pings the customer that was originally at the top of the list after sorting, instead of the intended customer.

Solution: The function returns sorted array index for sorted arrays and default index for unsorted arrays.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

OMS Crash Due to Affected Customer Query

Grip ID: ETD-83636

Fix Version: 16.20.0

Severity: High

Issue: OMS crashed when processing affected customer queries.

Solution: Added null checks to prevent application crashes when handling affected customer queries.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

7.3.3. DMS

Device Search Result Not Cleared After Unsuccessful Search

Grip ID: ETD-83668

Fix Version: 16.20.0

Severity: Medium

Issue: The Device Search pop-up retains previous results when searching for a non-existent device

after a successful search.

Solution: Fix the code to remove the first device results when searched for not found device.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Crosshatching and Feeder Info Missing for Abnormally Fed Feeder

Grip ID: ETD-83774

Fix Version: 16.20.0

Severity: Medium

Issue: Crosshatching missing on geographic when a feeder is abnormally fed while the other station is loaded in client.

Solution: The abnormally fed flag is now set correctly so crosshatching is displayed on the geographic view.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

7.3.4. DNAF

Island Status Not Updating After Convergence

Grip ID: ETD-83627

Fix Version: 16.20.0

Severity: Medium

Issue: When SCADA meas and model not matched, one KVA island is not converged.

Solution: Improved the algorithm so BLA will try to find a power factor closed to nominal PF, but still within the minimum and maximum allowed power factor in DCE, to calculate a reasonable KVA target.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Flag for Non-Conforming Load Not Set Correctly

Grip ID: ETD-83630

Fix Version: 16.20.0

Severity: Low

Issue: Non-conforming flag on load is not set correctly.

Solution: Set the flag in modeling tool (Converter).

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

LVM Triggering for Lateral Components

Grip ID: ETD-83707

Fix Version: 16.20.0

Severity: Low

Issue: The functionality works as expected, but logging is misleading.

Solution: Corrected the logging, so it is consistent with doc and software.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

7.3.5. Performance

Slow Map Navigation in ADMS Client

Grip ID: ETD-83628

Fix Version: 16.20.0

Severity: High

Issue: Client slowness navigating the map.

Solution: Reduce memory allocation in drawing logic.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

7.3.6. Yambay

Invalid Coordinates Sent to Mobile for Investigation Orders

Grip ID: ETD-83729

Fix Version: 16.20.0

Severity: Medium

Issue: Invalid coordinates sent to Mobile for investigation orders.

Solution: Correct coordinates are now used for investigation orders.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Incorrect Payload for Trouble Orders

Grip ID: ETD-83724

Fix Version: 16.20.0

Severity: Medium

Issue: Fields missing in payload for trouble orders.

Solution: Payload is populated correctly for trouble orders.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

7.3.7. Documentation

Missing Documentation for Initialize All Structures on Startup Toggle

Grip ID: ETD-83772

Fix Version: 16.20.0

Severity: Low

Issue: The documentation did not include information about the **Initialize All Structures on Startup** toggle in the VCE Structure tab.

Solution: The *VCE User Guide* has been updated to include the missing **Initialize All Structures on Startup** content.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

7.3.8. NRT

ArchiveApp Performance

Grip ID: ETD-83506

Fix Version: 16.20.0

Severity: Medium

Issue: The ArchiveApp user caused a performance issue. The RDBMS Install Guide included a section for creating an ArchiveApp user that was no longer needed.

Solution: The Creating an Archive Application User section has been removed from the *RDBMS Install Guide*.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

8. ADMS 16.19.0

This chapter describes the incremental new features and defect fixes that were added since ADMS 16.18.0. The remainder of this document provides additional release-specific information that also applies to this release.

8.1. What's New in 16.19.0

This section contains information about new features and functionality for ADMS 16.19.0.

- Upgraded third-party Angular version to 19 and UI component to 8.5. (GRDDMSS3-2903)

8.2. What's Changed in 16.19.0

The following changes have been made in this release.

8.2.1. Supported Software and Hardware Changes

This section describes changes to the system such as changes in system requirements, compatibility, or third-party software.

The following changes has been made in this release:

- Forecasting - 3.9.0 & 16.19.0
- Gateway - 2.7.5 & 16.19.0
- Foundation - 25R06

8.3. What's Fixed in 16.19.0

This section lists the notable issues addressed in this release.

8.3.1. DNAF

FISRCA Did Not Provide a Plan

Grip ID: ETD-83411

Fix Version: 16.19.0

Severity: Medium

Issue: FISRCA does not generate partial restoration for upstream device.

Solution: FISRCA is enhanced to produce partial plan for upstream device.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Incorrect Bus Voltage Limits in Reverse Flow

Grip ID: ETD-83596

Fix Version: 16.19.0

Severity: Medium

Issue: Logging caused confusion.

Solution: Improved bus voltage limits logging in result file.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

8.3.2. DMS

Tag Symbol on Geographic Not Reflected in DMS Client

Grip ID: ETD-83563

Fix Version: 16.19.0

Severity: Medium

Issue: When a device has one or more tags and is selected on the geographic view, the DMS client did not consistently display the correct symbol based on the tag type or the presence of multiple tags.

Solution: Updated the DMS client to show the appropriate tag symbol that matches the one displayed on the geographic view, accounting for tag type and multiplicity.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	

Issue Implication	Yes/No	Additional Information
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	Yes	

Numerical Logic Not Supported in DMS Queries

Grip ID: ETD-83582

Fix Version: 16.19.0

Severity: Medium

Issue: Existing queries do not support numeric comparison.

Solution: Numeric comparison support is added to the queries.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

DMS Search Tool Navigation Issue

Grip ID: ETD-83561

Fix Version: 16.19.0

Severity: Medium

Issue: Tags are duplicated between containers (switch cabinets, UI containers) and devices, when zoomed in.

Solution: Change the modeling of the internals to match the configuration.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	Yes	DNOM internal modeling
API	No	
Configuration	Yes	Container zoom level
ADMS Archive Schema	No	
User Interface	Yes	

8.3.3. Crew Management

Non-Outage Incident Cancel Produces Faulty WFM Message

Grip ID: ETD-83548

Fix Version: 16.19.0

Severity: Medium

Issue: When a Non-Outage incident is cancelled, the server sends out a cancel WFM message (which is correct) and followed by an update WFM, the second update WFM is wrong.

Solution: Removed the second wrong message.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

8.3.4. E-OMS

Orders Not Created Due to Invalid History Correction Entry

Grip ID: ETD-83607

Fix Version: 16.19.0

Severity: High

Issue: Orders were not created in ADMS/OA due to a malformed HISTORY_CORRECT entry.

Solution: Updated the logic to check for the '=' delimiter in the PARAMS field before parsing, preventing errors in the Work Model Adapter.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

8.3.5. Process Automation

Planned Outages Excluded from Automation

Grip ID: ETD-83536

Fix Version: 16.19.0

Severity: Low

Issue: Earlier PA partially (Auto transfer) supported for Planned switching type.

Solution: Planned outages are now completely excluded from process automation processing, regardless of auto-assign and auto-transfer rules.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

8.3.6. Modeling

Feeder Loss Not Flagged in Model Review

Grip ID: ETD-83474

Fix Version: 16.19.0

Severity: Medium

Issue: The Model Review log in the Model Manager application does not show updated counts of transformers, customers, or switches for feeders missing from the station external file due to a GIS issue.

Solution: Updated the logic to handle missing feeders so that the Model Review log displays information for those feeders.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Duplicate GUIDs Created for Measurement Points in SFBLite

Grip ID: ETD-83497

Fix Version: 16.19.0

Severity: Medium

Issue: When a station is reprocessed, Measurement Points with duplicate ID were written to the station internal xml.

Solution: Fixed the issue by clearing existing Measurement Points and adding a check before inserting a new measurement Id.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

SCADA PV Missing in ADMS Model Exporter JSON Output

Grip ID: ETD-83545

Fix Version: 16.19.0

Severity: Medium

Issue: ModelExport tool does not export all EnergyResources and exported EnergyResources were missing the "ScadaMeasurements" attribute.

Solution: Fixed the issue in the export logic and added the "ScadaMeasurements" attribute for all exported devices, including EnergyResources and Transformers.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

8.3.7. NRT

Planned Outage Customer Change Not Propagated to Database

Grip ID: ETD-83519

Fix Version: 16.19.0

Severity: Low

Issue: When editing the customer account number in a customer file; The system adds a new customer entry with the updated account number but fails to delete the existing customer record in the database.

Solution: Track planned outage customer changes using customer key to ensure accurate archiving and database updates when account number or premise ID changes. Since both premise ID and account number can be updated, but the customer key remains constant, you can always retrieve the latest values for premise ID and account number using the customer key.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Outage Incorrectly Converted to Non-Outage in NRT Incident Table

Grip ID: ETD-83586

Fix Version: 16.19.0

Severity: Low

Issue: In NRT table, the flag IsOutage is not set correctly.

Solution: Handles the case when converting so that the flag IsOutage has the right value.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

8.3.8. Performance

Improved Crew Assignment Responsiveness

Grip ID: ETD-83617

Fix Version: 16.19.0, 16.17.0.4

Severity: Medium

Issue: Crew assignment actions were slower than expected. After assigning a crew to a work order, there was a noticeable delay due to unnecessary data in the assignment message, which caused additional system processing.

Solution: The logic has been updated to exclude unnecessary data from the assignment message, reducing processing overhead and improving response time during crew assignment.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

8.3.9. OMS

Unable to Restore Predicted Order After OMS Main Bounce

Grip ID: ETD-83445

Fix Version: 16.19.0

Severity: High

Issue: Failures and delays transitioning devices to confirmed out or repaired (closed) were not detected by the EOMS in certain cases and task status was incorrect on the OMS order worksheet. This occurred most consistently immediately following an OMS restart and during heavy loading. The operation could not be retried, so the error was not recoverable from the order worksheet.

Solution: Rather than relying on the returned http message status, the EOMS now uses entries in the Incident Journal to definitively determine the correct device status following a command to change the device status. If the command fails, the task is left "In Progress" and the user has the option to retry the command from the order worksheet.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	Yes	User may now see check prediction and repair tasks transition through the "In Progress" state on the order worksheet

Incorrect OutageDeviceType Set During Investigation Order Creation

Grip ID: ETD-83591

Fix Version: 16.19.0

Severity: Medium

Issue: OutageDeviceType was incorrectly set to None (code 0) instead of ServicePoint (code 10) when creating an investigation order on a customer.

Solution: Updated the logic to set OutageDeviceType to None for non-outage incidents without a ServicePoint ID; otherwise, it is set to ServicePoint.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Incorrect Mapping of Affected Premises in NRT

Grip ID: ETD-83542

Fix Version: 16.19.0

Severity: Medium

Issue: In some cases, affected premises were incorrectly assigned to the final de-energized step in NRT, even when linked to a different device or restoration step. This caused discrepancies between NRT and the restoration sequence defined in the Order Worksheet or Update Incident.

Solution: Improved validation logic to ensure affected premises are only mapped to a restoration step if the customer is linked to the corresponding service point or device. This update ensures consistency between NRT and the defined restoration plan.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

8.3.10. UI

Blocking Tabular Comment Message Box

Grip ID: ETD-83537

Fix Version: 16.19.0

Severity: Medium

Issue: Tabular comment boxes in the ADMS client were blocking other edits. A message box appeared and prevented background work until it was manually closed.

Solution: Replaced the blocking message box with a non-blocking one that can be minimized, allowing continued work in the background.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Crash During PDF Export in Unattended Client

Grip ID: ETD-83541

Fix Version: 16.19.0

Severity: Medium

Issue: A crash occurred when DmsDisplay was accessed from a background thread after it had been disposed. Accessing the handle during the GetTopLevelControlWpf() call triggered an ObjectDisposedException.

Solution: Updated the logic to include disposal and handle validity checks within InvokeIfRequired, ensuring that DmsDisplay is accessed only on the UI thread and in a valid state.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Annotation Hover Fails to Show Comments

Grip ID: ETD-83606

Fix Version: 16.19.0

Severity: Medium

Issue: The comments doesn't show up when hover annotation.

Solution: Show the comments on UI.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

9. ADMS 16.18.0

This chapter describes the incremental new features and defect fixes that were added since ADMS 16.17.0. The remainder of this document provides additional release-specific information that also applies to this release.

9.1. What's New in 16.18.0

This section contains information about new features and functionality for ADMS 16.18.0.

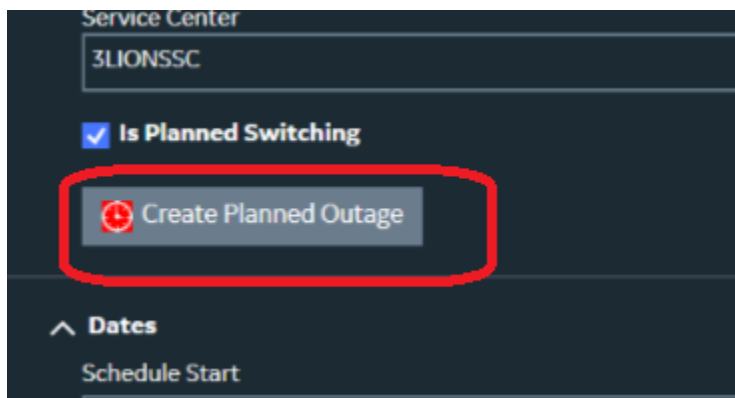
9.1.1. Enhanced Planned Outage Integration with Switching Orders

Planned outages can now be created with detailed customer interruption information from a switching order and its steps. This allows multiple interruptions to be recorded for a planned outage. These interruptions record the customers who are interrupted for the same period of time and provide a breakdown of the expected CMI for a planned outage.

To create detailed interruption records for a planned outage, all the steps on the switching order must first be executed in study mode with CBO enabled. This records the customers impacted by each step, and allows the detailed interruptions to be determined when the planned outage is created. The detailed interruptions recorded in this way are read-only, so the ability to add or remove customers is disabled.

When present on a planned outage, detailed interruptions are included in the results of the print and export functions, and are also made available to external systems via the Customer Experience Adapter. (GRDDMSS3-2525)

For more information, see *Creating a Planned Outage with Detailed Interruptions from a Switching Order* in the *Switching Operations User Guide*.



Update Customer Notification

VAIL 51115 Change To Selected Device Add Traced Customers

Customer List Notification Details Custom Fields Interruption Steps

Interruption Steps					
Interruption Id	Schedule Start Time	Schedule End Time	Duration	Affected Customers	CMI
1	18-05-2025 09:11:04	18-05-2025 11:11:09	02:00:05	296	36180
2	18-05-2025 09:11:04	18-05-2025 12:11:11	03:00:07	3	540
3	18-05-2025 13:11:13	18-05-2025 14:11:15	01:00:02	2	120

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	Yes	Customer Experience Adapter - PlannedOutageNotification now allows a list of Interruption objects containing the interruption details. Interruptions will include the following: InterruptionId, ScheduleStartTime, ScheduleEndTime, Duration, AffectedCustomersCount, CMI, PlannedOutageId, AffectedCustomersList, AffectedCustomersString
Configuration	No	
ADMS Archive Schema	No	
User Interface	Yes	Screenshot included
Demonstration Available	Yes	Click here for demonstration

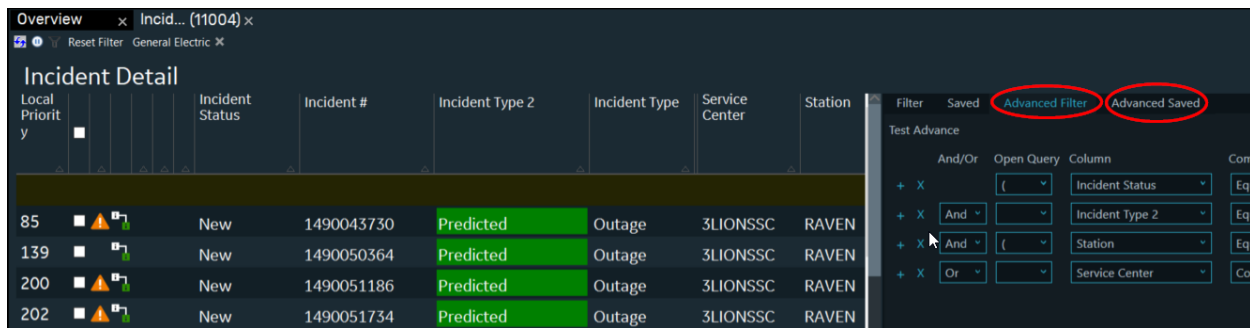
9.1.2. OMS Display Data Table Filter Panel Enhancements

Enhanced the filter panel in the OMS display data table with Advanced Filter functionality and introduced two tabs in the filter panel: **Advanced Filter** and **Advanced Saved**.

The Advanced Filter enables constructing filter queries using AND and OR operators, significantly improving data filtering and analysis capabilities. It allows applying multiple criteria simultaneously to refine search results and provides more granular control over data visibility.

Additionally, the Advanced Saved tab allows users to save customized filter queries for easy access and reuse, enhancing operational efficiency. (GRDDMSS3-2528)

See OMS Controls Overview section of the *OMS User Guide* for more details.

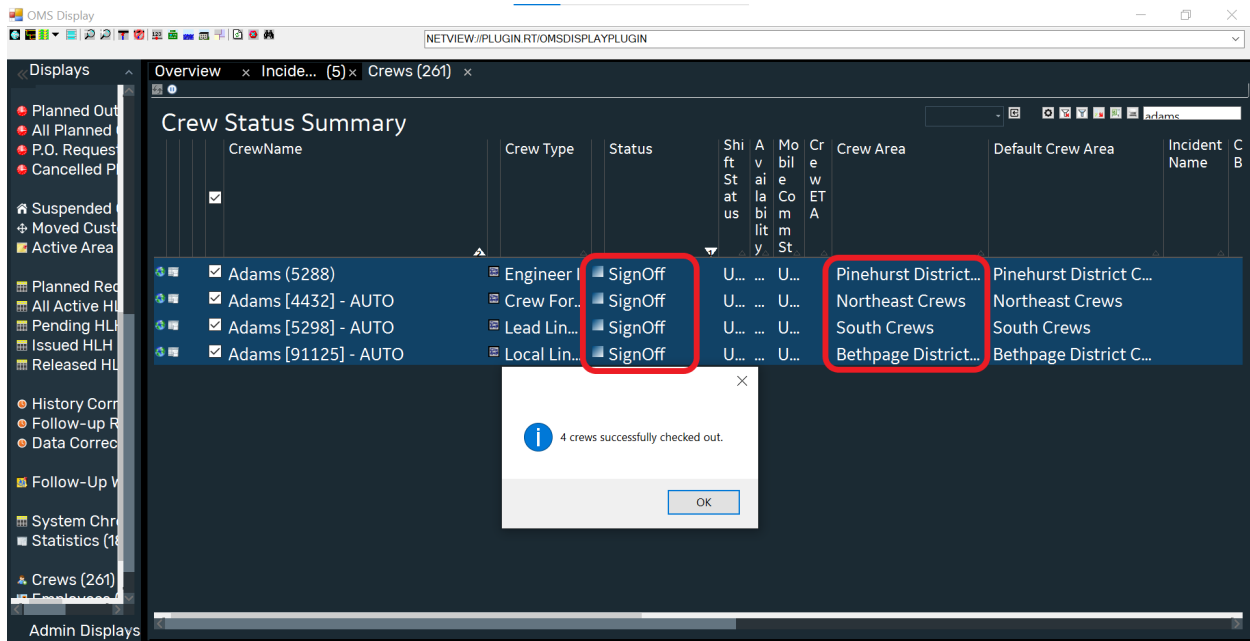


Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	Yes	Screenshot included
Demonstration Available	Yes	Click here for demonstration

9.1.3. Bulk Crew Sign On/Sign Off

The Bulk Crew Sign On/Sign Off feature simplifies the process of reassigning crews during shift changes and storm events. Dispatchers can now sign on or sign off multiple crews directly from the Crew Summary view without opening individual dialogs. A new interface allows managing crew assignments across organization levels and crew areas while maintaining visibility into active orders. Crew statuses and area assignments are updated automatically, and associated orders are unassigned when a crew is signed off. (GRDDMSS3-2568)

For more information, see Crew Status Bulk SignOn and SignOff section of the *OMS User Guide*.



Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	Yes	Screenshot included
Demonstration Available	Yes	Click here for demonstration

9.1.4. Push Crew Details from ADMS Crew Model to Mobile

Mobility Gateway automatically synchronizes crew model details with Mobile Enterprise anytime a crew is newly added, updated or deleted by:

- **Crew Modeling Tool** - Saved changes to `.crew` for any addition, update or deletion.
- **Crew Modeling API** - Any journaled crew model addition, update or deletion.
- **OMS Display - Crew Update Dialog** - Any journaled crew model addition, update or deletion. (GRDADMS-1435)

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	Yes	<CrewSynchronizationUri>crews</CrewSynchronizationUri> in MobilityGatewayAppConfig.xml
ADMS Archive Schema	No	

Feature Implications	Yes / No	Additional Information
User Interface	No	
Demonstration Available	Yes	Click here for demonstration

9.1.5. Composite Switch Support in OMS Order List and Worksheet

The OMS Order List and Worksheet now support composite switches, allowing users to view and manage orders associated with composite switches. This enhancement improves the visibility and management of complex switch configurations within the OMS interface.

In the Order List, the Device Type column will show **Composite Switch** for orders associated with composite switches. In the Order Worksheet, the Description field will also display **Composite Switch** when applicable. The Device Position field will also show the status of the composite switch. (GRDADMS-1817)

Overview x Incidents (1) x Orders (1) x 1750000002-1 x

Order List

	C C	P R	S M	A N	Order	List t s Pri	Station	Nominal Station	Feeder	Nominal Feeder	Device Type	Device	Phase
					1750000002-1	1	PINEH...		F07...		Composite Switch	73111	ABC

Incident 1750000002
1292 Main St, McDowell
73111_Preferred | F0731 PINEHURST (McDowell)

General

Facility

Customers

Damage Assessment

^ Device details

Device Name

73111

Description

Composite Switch

Phase

ABC

^ Operational state

Device position

Preferred

Voltage (kV) LL

13.20

- In the Coded Translation Tool, the OutageDeviceType has a new entry for 'Composite Switch'.
- Note that there is no UI changes for S-OMS related to this feature.
- There is no changes to the Rest APIs related to this feature.
- For the Incident and WorkOrder objects, there is a new property exposed. The name of the property is CompSwitchCurrentState. This property can be used to display in tabulars.

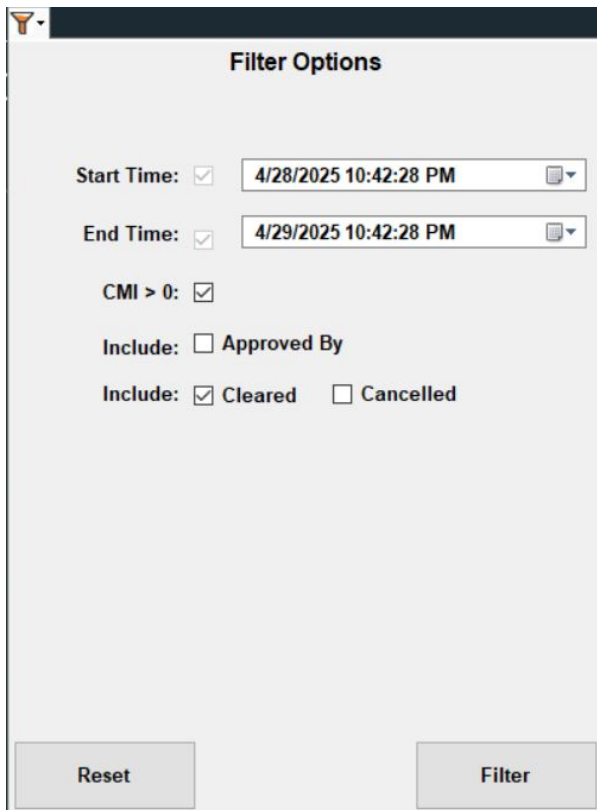
Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	Yes	Screenshot included
Demonstration Available	Yes	Click here for demonstration

9.1.6. Improvements

9.1.6.1. Enhancements to Data Correction

The following filters have been introduced or updated to improve performance and control the volume of data returned:

- **Start Time and End Time:** These filters are based on when the incident started and ended. Both are mandatory to help limit the number of records returned and address performance issues. The checkboxes are selected and disabled by default.
- **CMI > 0:** When selected (default), only records with a CMI value greater than 0 are displayed.
- **Include Approved By:** When selected, includes records that have been approved by a user. This option is unchecked by default.
- **Include Cleared:** When selected (default), displays only cleared incidents.
- **Include Cancelled:** When selected, displays cancelled incidents. This option is unchecked by default.



The screenshot shows a 'Filter Options' dialog box with a dark header bar containing a funnel icon. The dialog has a light gray background. It contains the following controls:

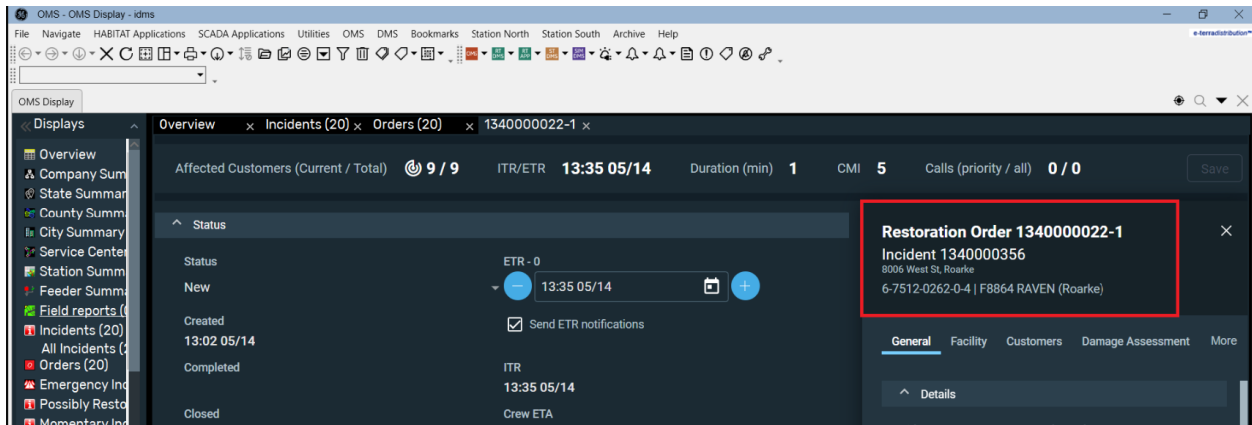
- Start Time:** A checkbox (checked) followed by a text field containing '4/28/2025 10:42:28 PM' and a calendar icon.
- End Time:** A checkbox (checked) followed by a text field containing '4/29/2025 10:42:28 PM' and a calendar icon.
- CMI > 0:** A checkbox (checked).
- Include:** A label followed by an unchecked checkbox and the text 'Approved By'.
- Include:** A label followed by a checked checkbox and the text 'Cleared', and an unchecked checkbox and the text 'Cancelled'.

At the bottom of the dialog are two buttons: 'Reset' on the left and 'Filter' on the right.

9.1.6.2. Improved Display of Order Information

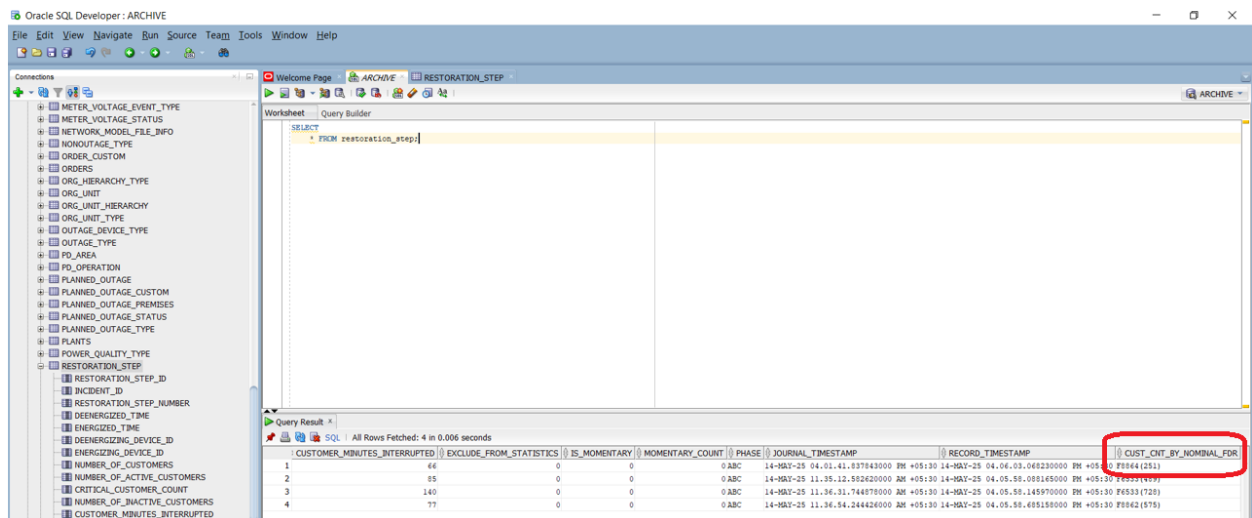
The order type and order name have been moved from the order header to the top of the incident side panel to improve visibility and support a more intuitive interface. This change is applicable for

both Outage Assist and Thick Client.



9.1.6.3. Database Update – Column Renaming for Size

The column `CUSTOMER_COUNT_BY_NOMINAL_FEEDER` in the `RESTORATION_STEP` table has been renamed to `CUST_CNT_BY_NOMINAL_FDR` to reduce byte size and improve efficiency for downstream applications that rely on fixed-length or size-sensitive data formats.



Note

This change is not backward-compatible. Any external systems, scripts, or integrations referencing the original column name will require updates to align with the new schema.

9.2. What's Changed in 16.18.0

The following changes have been made in this release.

9.2.1. Supported Software and Hardware Changes

This section describes changes to the system such as changes in system requirements, compatibility, or third-party software.

The following changes has been made in this release:

- Forecasting - 3.8.0 & 16.18.0
- Gateway - 2.7.3 & 16.18.0
- Foundation - 25R05
- Zitadel - Apisix Integration

9.2.1.1. Zitadel - Apisix

- Starting with this release, the primary authentication for ADMS is switched to Zitadel. Helm chart `outage-assist-values.yaml` and `adms-configuration` XML files are updated to use the Zitadel UAA endpoint.

This release tested with Zitadel as the primary authentication service for ADMS. UAA authentication may technically work, but are not tested.

9.3. What's Fixed in 16.18.0

This section lists the notable issues addressed in this release.

9.3.1. ADMS Interface

Investigation Orders Created via API Did Not Display Icons on the Map

Grip ID: ETD-83493

Fix Version: 16.18.0, 16.17.0.2

Severity: High

Issue: Investigation orders created through the API did not display icons on the geographic map and could not be navigated using the globe icon.

Solution: The system now derives X and Y coordinates from latitude and longitude only when both values are non-zero. If either value is zero or missing, the system uses the X and Y coordinates provided in the order data, bypassing coordinate conversion.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	

Issue Implication	Yes/No	Additional Information
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Limited GRPC Server Connections in DMSAPIServer

Grip ID: ETD-83336

Fix Version: 16.18.0, 16.17.0

Severity: High

Issue: DMS API Server supports only one GRPC server.

Solution: Enhance DMS API Server to support multiple GRPC servers.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Missing Error Message for Invalid LatestCallbackTime in AddCalls API

Grip ID: ETD-83339

Fix Version: 16.18.0

Severity: Low

Issue: When `WantsCallback` was true, the `LatestCallbackTime` field accepted invalid string values without triggering an error, leading to potential data inconsistencies.

Solution: Added validation to ensure `LatestCallbackTime` is present and in "hh:mm:ss" or "hh:mm" format when `WantsCallback` is true. An error is now returned for invalid input. Validation is skipped when `WantsCallback` is false.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

CEA Random Choice Sending Cancel Incidents

Grip ID: ETD-83459

Fix Version: 16.18.0

Severity: High

Issue: Cancelling an incident is not always processed correctly in Customer Experience Adapter (CEA).

Solution: Ensure all Cancel scenarios are processed correctly.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

9.3.2. APF

Loading a Saved Room Prevented Outage Confirmation

Grip ID: ETD-82979

Fix Version: 16.18.0

Severity: Low

Issue: After loading a saved room, users were unable to confirm an outage. This occurred because permissions were not reloaded correctly, resulting in lost or incomplete access.

Solution: Code added to handle correctly Netview commands when viewports are closed and reopened.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Scrollbar Unexpectedly Resets To Top In Grid Tabular Display

Grip ID: ETD-83489

Fix Version: 16.18.0, 16.17.0.2

Severity: Medium

Issue: When a user scrolls down in any Grid Tabular Display and waits for a few seconds, the scrollbar automatically returns to the top of the display.

Solution: Changes have been made to ensure the scroll position is properly cached by reaching the relevant part of the code as expected.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

9.3.3. AMI

Ping All Popup in AMI Returned Irrelevant Customers

Grip ID: ETD-83109

Fix Version: 16.18.0

Severity: Medium

Issue: The 'Ping All' popup in AMI used a wildcard search for the Customer List, which sometimes returned irrelevant customers.

Solution: Set `exactMatch` to true to ensure the search returns only exact match results, improving accuracy.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

AmiMessages Errors and Processing Fails

Grip ID: ETD-83535

Fix Version: 16.18.0

Severity: High

Issue: When encountering an invalid enumeration in AMI PingResponse, an exception occurs and is not processed correctly.

Solution: Return valid Status message to indicate the data element and location in the data being processed.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Meter Analog Details Not Displayed in Show Analogs Menu

Grip ID: ETD-83220

Fix Version: 16.18.0

Severity: High

Issue: The 'Show Analogs' menu did not display meter analog details for specific customers. This issue occurred because the required LoadID field was missing from the customer model files.

Solution: Identified that the absence of the LoadID field caused the display issue. Updated the documentation to specify that the LoadID field is mandatory for proper AMI association and to ensure meter analog details are displayed correctly.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

9.3.4. Archive

Archive PURGE Command Failed

Grip ID: ETD-82943

Fix Version: 16.18.0, 3.14

Severity: Medium

Issue: The Archive PURGE command failed when a foreign key constraint (FK_*_DEVICE) was violated. This occurred because child DEVICE records were still present and not properly filtered during purge operations across various incident-related sources (e.g., INCIDENT_CHRONOLOGY, SWITCHING_STEP, RESTORATION_STEP, CALL).

Solution: Updated the purge logic to more accurately identify and remove DEVICE records only when no valid references exist in related tables.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Duplicate Line Items Created Under Unique Custom Field IDs

Grip ID: ETD-83144

Fix Version: 16.18.0

Severity: Medium

Issue: Duplicate entries were created in the ORDER_CUSTOM table under the same custom field ID. This occurred when existing entries with the same order ID, field name, and field value were not properly removed before inserting new ones.

Solution: Updated the logic to removes any existing entry with the same information before adding a new one, preventing duplicates.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Incorrect End Time Displayed in NRT Incident Table

Grip ID: ETD-83403

Fix Version: 16.18.0

Severity: High

Issue: The NRT incident table incorrectly displayed an end time even when the client did not provide one. The field was previously set to a minimum date value, leading to misleading information.

Solution: Updated the logic to use 'null' instead of a default minimum date when no end time is provided by the client, ensuring accurate data representation.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Oracle NRT/Archive Upgrade Error

Grip ID: ETD-83427

Fix Version: 16.18.0, 16.17.0

Severity: Low

Issue: Oracle database upgrade was failing during the NRT/Archive upgrade process.

Solution: Updated the `UPGRADE_VERSION` parameter, which must be revised for every release to ensure archive upgrades are handled correctly.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Archive Log Error

Grip ID: ETD-83050

Fix Version: 16.18.0, 16.17.0.2

Severity: Low

Issue: Archive Upgrade logs are wrong.

Solution: Schema upgrade log was changed to represent the current version.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

9.3.5. Core

Stuck Tag Not Cleared from Boundary Device

Grip ID: ETD-83307

Fix Version: 16.18.0

Severity: High

Issue: Tags on boundary devices were occasionally not removed, even after typical recovery actions.

Solution: The fix ensures proper deletion of tags on both sides of a boundary device and clears unused dynamic tags during startup.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Configproxy Failed on Interface Call Server

Grip ID: ETD-83314

Fix Version: 16.18.0

Severity: High

Issue: ConfigProxy failed under load due to incorrect async usage in the `Timer_Tick` method.

Solution: Removed `async` from `Timer_Tick` to ensure proper synchronization during high-load scenarios.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Duplicate Incident Names

Grip ID: ETD-83376

Fix Version: 16.18.0, 16.17.0.1, 16.16.2

Severity: High

Issue: In some scenarios, duplicate incident names are found.

Solution: Code logic is changed to handle incident move down to prevent duplicate.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

9.3.6. Crew

Non-Evaluator Crews Incorrectly Set as Evaluators on Unassign

Grip ID: ETD-83418

Fix Version: 16.17.0, 16.18.0, 16.16.2, 16.17.0.1

Severity: High

Issue: Non-evaluator crews being set to evaluators on unassign.

Solution: When unassign crews non-evaluator crews are not changed to evaluators.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Crew Model Push Caused Customers to Appear Unconnected in NetServer

Grip ID: ETD-83469

Fix Version: 16.18.0

Severity: Medium

Issue: After a crew model push, customers appeared connected on the geo but were reported as unconnected in NetServer. New single-customer incidents were incorrectly shown as isolated.

Solution: Updated the logic to force a resync of `DnomServicePointIndex` after a crew model push, resolving the incorrect customer connectivity status.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Crew Updates Not Sent When Current Status Was Not Assigned or Dispatched

Grip ID: ETD-83576

Fix Version: 16.18.0, 16.17.0.3

Severity: Medium

Issue: Previously, the IncidentUpdate and CrewUpdate messages were sent from ADMS only if the current crew status was Assigned or Dispatched, and the new crew status was Dispatched. As a result, when the same crew was assigned to two orders and the current crew status was something other than Assigned or Dispatched, no update message was sent to WFM.

Solution: The logic has been updated to remove the dependency on the current crew status. Now, if the new Crew Status is Dispatched, and the crew is an MDT crew, and DISPATCHED is not included in the RELEASE_ALL_WORK_FROM_CREW_STATUS_CODES, ADMS sends the IncidentUpdate and CrewUpdate messages to the external system.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

9.3.7. DMS

Maximum Loading with PLOS

Grip ID: ETD-83317

Fix Version: 16.18.0

Severity: Medium

Issue: Max loading issue on PLOS plan.

Solution: Consider DG/DER in PLOS affected area to calculate correct max loading on PLOS plan.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

NTP Production Issue

Grip ID: ETD-83436

Fix Version: 16.17.0, 16.18.0, 16.16.2, 16.17.0.1

Severity: High

Issue: Miss NTP trigger for abnormal case.

Solution: Make sure NTP is triggered correctly.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Yellow Station Mismatch

Grip ID: ETD-83478

Fix Version: 3.15.6, 16.18.0

Severity: Medium

Issue: Polygon is not displayed correctly.

Solution: Improved client codes to support "z-order" polygon.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Incorrect IsOverhead Value for Jumper-Initiated Incidents

Grip ID: ETD-83355

Fix Version: 16.18.0

Severity: Low

Issue: The IsOverhead value in the realtime incident journal (archiveincident record) was incorrectly set to False when an overhead jumper became the device of record for an incident.

Solution: Updated the logic to ensure the IsOverhead value is correctly set to True when an overhead jumper is the device of record.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

9.3.8. DNAF

DPF Not Triggered by 2030.5 DER Status Change

Grip ID: ETD-83330

Fix Version: 16.18.0

Severity: Medium

Issue: DER status change from IEEE2030.5 Gateway does not trigger DPF.

Solution: Trigger DPF with DER connection status change from IEEE2030.5 when option is enabled.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

2030.5 DER LVM Eligibility Issue

Grip ID: ETD-83334

Fix Version: 16.18.0

Severity: High

Issue: DER in IEEE2030.5 Gateway is not eligible for LVM.

Solution: Make DER eligible if the status is connected and communication status is online from IEEE2030.5.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Incorrect Power Flow Directions While Toggling Phases

Grip ID: ETD-83350

Fix Version: 16.18.0

Severity: High

Issue: Wrong Power flow direction on de-energized phase

Solution: No power flow direction on de-energized phase.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

FISR Not Respecting Voltage Violations

Grip ID: ETD-83365

Fix Version: 16.18.0, 16.16.2, 16.17.0.1

Severity: Medium

Issue: FISR not check voltage violation on some restoration path.

Solution: FISR checks voltage violation on each restoration path.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	

Issue Implication	Yes/No	Additional Information
User Interface	No	

Incorrect Regulation Direction

Grip ID: ETD-83402

Fix Version: 16.18.0

Severity: Medium

Issue: FISR use incorrect flow direction.

Solution: FISR will check flow direction on device, not from Flow Direction status measurement when the "Enable Tap Emulation Logic for Telemetered Regulators and LTCs" is enabled.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

FISR Plan Moves and Setting Group Checks Logging Error

Grip ID: ETD-83413

Fix Version: 16.18.0

Severity: Low

Issue: Missing setting group logging.

Solution: Correct setting group logging in FISR result file.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

LVM/Plan Manager Did Not Issue Controls to Capacitors in Partial State

Grip ID: ETD-83422

Fix Version: 16.18.0

Severity: High

Issue: Plan manager device state check.

Solution: Improve Plan Manger pre-plan checks for device partial state case.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

DNAF and Forecast Server Hang

Grip ID: ETD-83423

Fix Version: 16.18.0, 16.17.0.2, 16.17.0, 16.16.2, and 16.15.0

Severity: High

Issue: The DNAF and Forecast servers entered a stalled state in a production environment. The DNAF server was restarted, and a Windbg trace was captured on the Forecast server prior to the restart.

Solution: Resolved an issue related to the use of a dangling pointer that caused the servers to hang.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	Yes	
ADMS Archive Schema	No	
User Interface	No	

LVM Completion Time is Incorrect

Grip ID: ETD-83477

Fix Version: 16.18.0

Severity: Medium

Issue: LVM completion time is not displayed correctly on UI.

Solution: Show correct completion time on UI.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	

Issue Implication	Yes/No	Additional Information
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Voltage Flattening Component

Grip ID: ETD-83512

Fix Version: 16.18.0

Severity: Medium

Issue: Ungrounded Y connection capacitor cause sensitivity issue in Heuristic LVM.

Solution: Use correct threshold for ungrounded Y capacitor in Heuristic LVM sensitivity calculation.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Meshed FL Cannot Find the Location of a Fault

Grip ID: ETD-83484

Fix Version: 16.18.0

Severity: Medium

Issue: The modeling node of device is incorrect.

Solution: Correct the node of device.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

9.3.9. E-OMS

Incorrect Error Message Shown When Enrouting an Active Crew

Grip ID: ETD-83039

Fix Version: 16.18.0, 16.17.0

Severity: Medium

Issue: When attempting to enroute a crew already marked as Onsite on another order, the system blocked the action but displayed an incorrect error message.

Solution: Updated the logic to display the correct message:

Crew [X] is working on order [1] and must be released from that order before it can be en routed to order [2].

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Data Correction Order Tab Frequently Comes Up Empty

Grip ID: ETD-83412

Fix Version: 16.18.0, 16.17.0.2

Severity: Medium

Issue: Clicking 'Reset Table to Default' did not restore column widths to their default values.

Solution: Set "width": 100 in the configuration to restore expected behavior.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Data Correction Rejected in ADMS Client

Grip ID: ETD-83575

Fix Version: 16.18.0, 16.17.0.3

Severity: Medium

Issue: The ADMS client rejected data corrections and displayed a pop-up message stating that the user ADMS did not have permission.

Solution: The code was updated to resolve the permission issue during data correction.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

9.3.10. FISR

FISR Triggering Not Respecting Recent Lockout

Grip ID: ETD-83448

Fix Version: 16.18.0

Severity: High

Issue: FISR triggering does not check the correct lockout measurement.

Solution: Improved triggering logic in checking old lockout measurements.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Enforce Station Backfeed Limit

Grip ID: ETD-83457

Fix Version: 16.18.0

Severity: Low

Issue: Incorrect backfeed violation.

Solution: Made code change to filter out 0.00 backfeed violation.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	

Issue Implication	Yes/No	Additional Information
ADMS Archive Schema	No	
User Interface	No	

9.3.11. Modeling

Converter Crash Due to Missing Configuration for Reclosers or Switches

Grip ID: ETD-83284

Fix Version: 16.18.0

Severity: Low

Issue: The converter crashed when reclosers or switches did not have corresponding configurations in the global file.

Solution: Removed the condition that checked for the obsolete Telemetered field in the global file.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

9.3.12. NRT

Energized Time Cleared but End Time Not Updated in NRT Incident Table

Grip ID: ETD-83447

Fix Version: 16.18.0

Severity: Medium

Issue: When the energized time was cleared from a restoration step, the end time was removed from the order but was not updated in the NRT incident table, resulting in inconsistent data.

Solution: Modified the logic to update the incident message when the energized time is cleared.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	

Issue Implication	Yes/No	Additional Information
ADMS Archive Schema	No	
User Interface	No	

Replication Issues in NRT Downstream Applications

Grip ID: ETD-83406

Fix Version: 16.18.0, 16.17.0.1, 16.17.0, 16.16.2

Severity: Medium

Issue: NRT downstream applications were unable to replicate data.

Solution: The column `CUSTOMER_COUNT_BY_NOMINAL_FEEDER` in the `RESTORATION_STEP` table has been renamed to `CUST_CNT_BY_NOMINAL_FDR` to reduce byte size and improve efficiency for downstream applications that rely on fixed-length or size-sensitive data formats.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Archive Stopped by NRTDataLoader Due to Crew Custom and Area Note Custom Fields

Grip ID: ETD-83426

Fix Version: 16.18.0

Severity: Low

Issue: The archive process was halted by NRTDataLoader due to issues with Crew Custom and Area Note Custom fields. The failure was caused by mismatched case sensitivity in data saved to a child table with a foreign key relationship to the parent table.

Solution: Updated the logic to remove uppercase characters when saving to the child table, ensuring compatibility with foreign key constraints and preventing archive failures.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Fixed Issue With END_TIME Not Resetting In NRT Incident Table

Grip ID: ETD-83472

Fix Version: 16.18.0, 16.17.0.2

Severity: Medium

Issue: When a user reopens a device, the END_TIME in the NRT Incident table is not set to null.

Solution: Changes have been made to set EndTime to null when its value is reset.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

9.3.13. OMS

Unnecessary Merging of Isolated Incidents

Grip ID: ETD-83315

Fix Version: 16.18.0

Severity: Medium

Issue: The system attempted to merge two isolated incidents, resulting in warning messages in the netomssserver.log. This behavior caused performance degradation and generated unnecessary log entries.

Solution: Removed isolated incidents at the beginning of the process to prevent unnecessary merging logic. This fix improves performance and eliminates redundant log entries later in the code.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Resolved Timeout Issue In Data Correction Tabular View

Grip ID: ETD-83495

Fix Version: 16.18.0, 16.17.0.2

Severity: High

Issue: The Data Correction tabular view failed to display results due to a query timeout after 100 seconds, resulting in an empty display.

Solution: The change is to restrict the returned dataset (limit 10,000 records) with the end date of the associated incidents.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Orders Not Created Due to Loop in Incident Processing

Grip ID: ETD-83271

Fix Version: 16.18.0, 3.14

Severity: High

Issue: Orders were not created because the process entered a loop when the related incident could not be found.

Solution: Added cleanup logic to remove the incident from the underlying NetList when it is deleted, preventing the loop.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Set Predicted Not Rolling to Upstream

Grip ID: ETD-83168

Fix Version: 16.18.0, 16.17.0

Severity: Medium

Issue: A downstream Predicted Stable (single customer) Restoration Order is not rolling to an upstream Predicted (fuse) Restoration Order when the action is done to "Set Predicted".

Solution: Code in the OMS Server is fixed to handle rolling upstream correctly.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Orders Recorded Unrelated Chronology

Grip ID: ETD-83321

Fix Version: 16.18.0

Severity: High

Issue: Chronology events were getting recorded even after the order moved to the Pending Clear state.

Solution: Updated the logic to stop recording chronology events once the order moves to the Pending Clear state.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Several Momentary Incidents Created

Grip ID: ETD-83340

Fix Version: 16.18.0, 16.17.0, 16.16.2

Severity: High

Issue: Several momentary incidents created instead of sustained outage incident.

Solution: Correct API is used to that check all phases independently.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	

Issue Implication	Yes/No	Additional Information
User Interface	No	

Invalid Restoration Step During Outage Merge

Grip ID: ETD-83346

Fix Version: 16.18.0

Severity: Medium

Issue: Merging active outages into a pending clear order created an invalid restoration step. The restoration step retained the master incident's merged date and time, resulting in negative CMI.

Solution: Updated the logic to set the energized time to match the de-energized time when it is later than the last restoration step's energized time.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

ArchiveApp Crash During Area Note Processing

Grip ID: ETD-83450

Fix Version: 16.18.0, 16.16.2

Severity: High

Issue: ArchiveApp crashed while processing area notes due to unhandled logic when adding chronology entries.

Solution: Updated the logic to defer chronology entry handling by collecting entries before processing.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Storm Assist Customers Tab from Within Incident is Displaying "00" and "01" Instead of MAC, URG, VIT, MED

Grip ID: ETD-83460

Fix Version: 16.18.0, 16.16.2

Severity: Medium

Issue: Critical customer columns display "00" and "01" instead of MAC, URG, VIT, MED.

Solution: OA code changed to handle coded translation.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Calls Before Start Time Included in Momentary Incidents

Grip ID: ETD-83461

Fix Version: 16.18.0, 16.16.2

Severity: Medium

Issue: Called received prior to device opening time are pulled into a momentary incident.

Solution: Added the checking of the code time versus the momentary incident start time.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Incorrect Calls Included in Momentary Incidents

Grip ID: ETD-83467

Fix Version: 16.18.0, 16.16.2

Severity: Medium

Issue: Calls that occurred before the incident start time or before device open were incorrectly associated with some momentary incidents.

Solution: Updated the logic to prevent updating the incident ID, first call time, and call count fields for such calls.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

DNAF Reprocessed All Stations After Midnight Compression or Restart

Grip ID: ETD-83470

Fix Version: 16.18.0

Severity: Medium

Issue: DNAF did not load previously solved data after midnight dynamics compression or system restart, causing all stations to be reprocessed unnecessarily.

Solution: Disabled the option by default, restoring previous behavior during midnight dynamics compression.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Mandatory Custom Order Field Causes Order Symbol Persistence When Moving Up

Grip ID: ETD-83444

Fix Version: 16.18.0, 16.17.0.2

Severity: Medium

Issue: Configuring the custom order field as mandatory in the OmsServerConfig leads to issues when moving up orders. Specifically, the original order symbol does not disappear from the map and retains the same order number as the new one.

Solution: Bypass the mandatory field check when the DeleteIncident message comes from the incident journal - because the OMS has already performed all necessary checks and deleted the incident.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	

Issue Implication	Yes/No	Additional Information
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Outage Calls Created Non-Outage Orders After Crew Model Updates

Grip ID: ETD-83518

Fix Version: 16.18.0, 16.17.0.2

Severity: High

Issue: Outage calls were incorrectly generating non-outage orders after changes were made using the Crew Modeling tool. This occurred because the system did not reinitialize the mapping between service points and stations, resulting in mismatches such as: ServicePoint Station Name : does not match with DNOM Station Name.

The issue was temporarily resolved only by restarting OMSMain.

Solution: Updated the mapping logic to force reinitialization of the `DnomServicePointIndex` whenever a new model (for example, crew or customer) is applied.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Momentary Outage Start Time Did Not Match Device Operation Time in Alarm Viewer

Grip ID: ETD-83392

Fix Version: 16.18.0, 16.17.0, 16.16.2

Severity: High

Issue: Start time for momentary outages do not match device operation time shown in Alarm Viewer.

Solution: Moved the momentary incident processing outside of the `highntplatenacy` skip and moved the processing of `currentcustomersout` to BEFORE the momentary incident processing in `OmsEngine`. Also changed some logic within the momentary incident processing to look at incident `EndTime` as opposed to `PendingClear` status b/c the `PendingClear` was being set later on in `OmsEngineTimer`.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

9.3.14. Order Worksheet

Crew Assignment from Order Worksheet Was Slow

Grip ID: ETD-83357

Fix Version: 16.18.0

Severity: High

Issue: Crew assignment from the Order Worksheet took 30 seconds or more to complete.

Solution: Improved crew assignment performance from the Order Worksheet. The fix leverages local caching of crews, crew areas, and incidents to speed up query processing.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Order Worksheet Facility Tab Shows Incorrect Device Description

Grip ID: ETD-83481

Fix Version: 16.18.0

Severity: High

Issue: Order worksheet under → Incident Panel (Side panel), On that label we should show device type info, instead of All or None.

Solution: Description was showing wrong information. Now it shows deviceType on the current opened order. (Fuse, Switch, Distribution Transformer, ...)

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	

Issue Implication	Yes/No	Additional Information
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

9.3.15. Performance

NetserverRO High CPU Usage

Grip ID: ETD-83364

Fix Version: 16.12, 16.17.0, 16.18.0, 16.16.2

Severity: High

Issue: NetServer ROs show high CPU usage.

Solution: Removed redundant UpdateCallAmiCounts().

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

9.3.16. Study

Study Server Crashing

Grip ID: ETD-83446

Fix Version: 16.18.0, 16.17.0.2

Severity: Medium

Issue: The Study Server was crashing due to a dangling pointer.

Solution: Resolved the dangling pointer usage that caused the server to crash.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	

Issue Implication	Yes/No	Additional Information
User Interface	No	

9.3.17. UI

ADMS client crash

Grip ID: ETD-83368

Fix Version: 3.14, 3.15.6, 16.18.0

Severity: High

Issue: Switch Order component of ADMS client crash when updating switch orders.

Solution: Added protective fix to prevent crashes to Switch Order client component.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	Yes	Crash fix in Switch Order

9.3.18. Work Model

Extra Commas Displayed in Crew Column for Single Assignment

Grip ID: ETD-83164

Fix Version: 16.18.0

Severity: Medium

Issue: Unassigning crews does not remove comma in crew column in Orders Table.

Solution: Removed comma in CrewName when crews are unassigned.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Invalid Crew Assignment Impacts Order Processing

Grip ID: ETD-83482

Fix Version: 16.18.0, 16.17.0.2

Severity: High

Issue: An invalid crew was assigned to an order, resulting in delays in creating the order. This led to a failure in completing the switching step for the close task, as the crew could not be found in the system, causing continuous retries and stopping the order creation process.

Solution: Enable the crew state transition to complete even when the crew is not found in the OMS. This change ensures that the crew status can transition effectively, accommodating situations where the crew may have been deleted from the model or an invalid crew was assigned via the API.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

NetData Server Memory Leak Causing CPXNETDATA Crashing

Grip ID: ETD-83291

Fix Version: 16.18.0

Severity: High

Issue: CPXNETDATA crashing seemingly randomly. NetDataServer goes unregistered and effectively crashes.

Solution: Make sure that the method for handling CfgProxy is not reentrant, clean up the socket connections correctly so that the next timer will work and support for reading more than 255 bytes.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

Unable to Select Power TF Symbols for Some Stations on Schematic View

Grip ID: ETD-83185

Fix Version: 16.18.0

Severity: High

Issue: Cannot select some Transformers on the Schematic View.

Solution: Converter sets the geometry bounds to include power transformer placements.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	Yes	DNOM compiler fix
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

9.3.19. Order Worksheet

Incident Panel Did Not Display Full Content When Scrolled

Grip ID: ETD-83463

Fix Version: 16.18.0

Severity: Medium

Issue: The content box in the side panel was cut off, preventing the full information from being visible.

Solution: Increased the height of the side panel by updating the CSS from height: `calc(100% - 233px);` to height: `calc(100% - 315px);`.

Issue Implication	Yes/No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive Schema	No	
User Interface	No	

10. ADMS 16.17.0

This chapter describes the incremental new features and defect fixes that were added since ADMS 16.16.2. The remainder of this document provides additional release-specific information that also applies to this release.

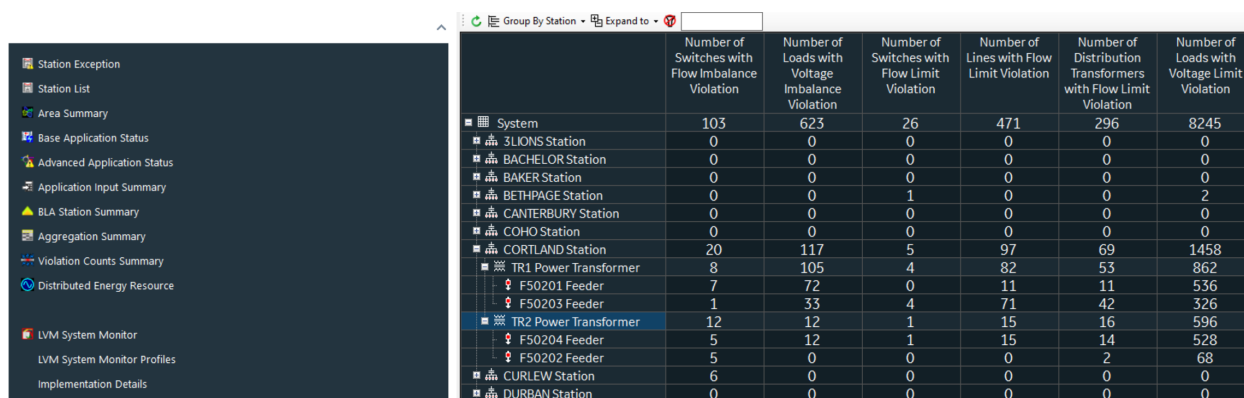
10.1. What's New in 16.17.0

This section contains information about new features and functionality for ADMS 16.17.0.

10.1.1. Violation Counts Summary

A new Violation Counts Summary view provides a consolidated display of loading, voltage, and imbalance violations. The summary is organized by feeder, power transformer, substation, and area, and shows violation counts for switching devices, conductors, distribution transformers, and customer loads. (GRDADMS-1282)

See the DNAF violation counts summary display section of the *DNAF User Guide* for more details.



	Number of Switches with Flow Imbalance Violation	Number of Loads with Voltage Imbalance Violation	Number of Switches with Flow Limit Violation	Number of Lines with Flow Limit Violation	Number of Distribution Transformers with Flow Limit Violation	Number of Loads with Voltage Limit Violation
System	103	623	26	471	296	8245
3LIONS Station	0	0	0	0	0	0
BACHELOR Station	0	0	0	0	0	0
BAKER Station	0	0	0	0	0	0
BETHPAGE Station	0	0	1	0	0	2
CANTERBURY Station	0	0	0	0	0	0
COHO Station	0	0	0	0	0	0
CORTLAND Station	20	117	5	97	69	1458
TR1 Power Transformer	8	105	4	82	53	862
F50201 Feeder	7	72	0	11	11	536
F50203 Feeder	1	33	4	71	42	326
TR2 Power Transformer	12	12	1	15	16	596
F50204 Feeder	5	12	1	15	14	528
F50202 Feeder	5	0	0	0	2	68
CURLEW Station	6	0	0	0	0	0
DURBAN Station	0	0	0	0	0	0

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	Yes	Screenshot included
Demonstration Available	Yes	Click here for demonstration

10.1.2. Auto-Complete for Manual Switches

The Switching Order application now supports automatic completion of switching steps involving

manually controlled SCADA devices. Operators no longer need to update the device status in the SCADA display and then return to complete the step. When enabled, the application updates the device status and completes the corresponding step automatically.

This functionality can be configured through the Switching Order Client Configuration Editor based on utility requirements. (GRDDMSS3-2039)

See the switch steps tab section of the *Switching Order Client Configuration Editor User Guide* for more details.

Switch Steps

☐ SD Colum...

☒ Warn about problems before issuance of ...

☒ Auto-fill completion time for SCAD...

☒ Disable 'Complete Step Confirm' popup

☒ Auto Complet...

☒ Auto-complete Non-Scada steps from SW...

☒ Copy the issued To Field whe...

☐ Disable 'Issue Step Confirm' popup

☒ Disable Comment editing after completion

☒ Auto complete Non Controllable S...

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	Yes	Added Auto complete Non Controllable Scada points from SWO in RT mode in Switching Order Client Configuration Editor.
ADMS Archive	No	
User Interface	Yes	Screenshot included
Demonstration Available	Yes	Click here for demonstration

10.1.3. Redesigned Order Worksheet

This release introduces a redesigned Order Worksheet to improve dispatcher efficiency by streamlining access to frequently used fields and minimizing scrolling. The updated layout prioritizes critical information—such as crew comments and resolution details—enabling faster completion of key workflows. The new design is available in both the thick client and Outage Assist.

Key enhancements include a reorganized Comments & Instructions section, with tabs for Crew Comments, Dispatcher Comments, and Instructions arranged in a logical order. A visual indicator highlights tabs containing content.

The Resolution section now includes System Affected, Cause, and Sub-Cause fields, relocated for improved clarity. Utilities can configure custom fields in any section and choose whether they appear at the top or bottom. Additional updates include displaying up to three tasks in the Tasks section, expanding the Resolution section by default, and refining layout density to reduce navigation time. (GRDDMSS3-2694)

See the Main Panel section of the *EOMS User Guide* and the Custom Fields tab of the *OMS Server Configuration Editor User Guide* for more details.

^ Instructions & Customer Notes

Case Note

Power has been restored

● Crew Comments

Dispatcher Comments

Instructions

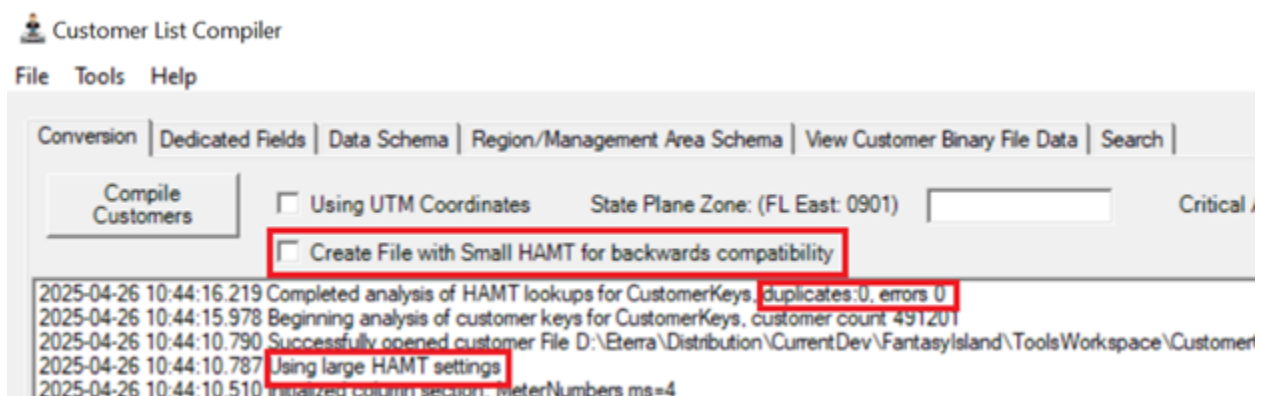
Crew Assigned

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	Yes	Screenshot included
Demonstration Available	No	

10.1.4. Smaller HAMT for Backwards Compatibility in Customer List Compiler

A new option in the Customer List Compiler allows creating customer files with a smaller Hashed Array Mapped Trie (HAMT) for compatibility with older client, server, and interface applications. By default, a larger HAMT is used.

See the Conversion Tab section of the *Customer List Compiler User Guide* for more details.



Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	Yes	Screenshot included
Demonstration Available	No	

10.1.5. Flexible Customer Search

A new customer search feature is available for testing. This flexible search allows single-line input using the CustomerName and Address dedicated fields, accepting names and addresses as space-separated words in any order. It also supports searches by meter number, customer key, account number, and phone number. Enable the feature by setting the `ENABLEFLEXIBLESEARCH` flag in the `IDMS_FEATURES` environment variable. Ensure the Address field is configured in the Customer List Compiler. Use the Search tab to test the feature. Results appear after entering three characters and are ranked based on matched terms. (GRDDMSS3-2492)

10.2. What's Changed in 16.17.0

The following changes have been made in this release.

10.2.1. Supported Software and Hardware Changes

This section describes changes to the system such as changes in system requirements, compatibility, or third-party software.

The following changes has been made in this release:

- Foundation - 25R03

10.3. What's Fixed in 16.17.0

This section lists the notable issues addressed in this release.

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
Archive	ETD-83301	New Coded Translation Editor Items are not Archived	Medium	Archive - Coded Translation Data

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
Crew	ETD-83348, 969859	Crew Model Push Deletes Calls	High	OMS Server - Core processing
DMS	ETD-83002, 917586	When 3 or more boundary switches are closed ADMS shows false loops.	Medium	NTP Loop Determination
DMS	ETD-83153	BLA doesn't stop when it reaches the threshold.	Medium	BLA Island Convergence Check
DMS	ETD-83173	Tracing to switch fails to stop on a single phase fuse inline on a three phase.	Medium	Client Tracing
DMS	ETD-83189, 936504	Station disappears from geographic viewport during CRC check	High	Client Station Map Synchronization
DMS	ETD-83289, 956341	ADMS allows LineSwitch to be closed via WFM when step restored from a SCADA Recloser incident	High	Interfaces - WFM Processing
SWO	ETD-83090, 950094	SO, SWO, DIS TrustedClient component not communicating with different users	Medium	Switch Order Server Email Notifications
SWO	ETD-83152, 932360	Switch Orders - Misleading CBO for a Close/Energerizing Step	High	Switch Order Server CBO processing
OMS APF	ETD-83202, 950094	APF and Client Crashing when opening OW	Medium	APF Client Web Browser
SWO	ETD-83282, 969429	Study Mode SWO doesn't show de-energized customer count popup warning	Medium	Switch Order Server CBO processing
OMS	ETD-83356, 972953	NetServer crashes multiple times with NullReferenceException	High	Interfaces - WFM message processing
DNAF	ETD-83068, 927283	FISR CA does not generate any plans for substations	Medium	FISRCA Plan Generation
DNAF	ETD-83155, 945374	RLD value does not change correctly (SCADA and DNAF)	High	RLD Calculation triggering
DNAF	ETD-83176, 936677	LVM Invalid Cap Delta Voltages	Medium	Heuristic LVM
DNAF	ETD-83200, 936676	AFR won't finish solving.	Medium	AFR Network Selection
DNAF	ETD-83203, 938058	LVM Reg Margin Insufficient	Medium	LVM Network Selection
DNAF	ETD-83215, 941232	FISR Plan State Shows as "Feasible" with no Violations when 3-Wire/4-Wire Check Fails	Medium	FISR switching check for 3-wire/4-wire
DNAF	ETD-83244, 944424	DPF Log shows Effective Override High and Low values for local capacitors switched	Medium	DPF logging

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
DNAF	ETD-83256, 945372	DPF Back Calculation Voltage Angle	Medium	DPF Back Calculation
DNAF	ETD-83266, 946793	FISR Alarm- Undefined reason Contact DMS support code=0	Medium	Cross phase Jumper logic in FISR
DNAF	ETD-83268, 946791	Line flow limit violations	Medium	Limit Monitoring
DNAF	ETD-83281, 951940	Update Documentation for "Maximum Number of DPF Result Files Created for Any One Station" in DCE	Low	Documentation
DNAF	ETD-83287, 954192	Tap Changer manual entry for the secondary winding of a local transformer	Medium	DPF logging
DNAF	ETD-83328, 967395	DNAF Hang During Midnight Compression	High	Null Pointer
DNAF	ETD-83345, 969550	DermKafkaAdapter error with LVM 2030.5 DER dispatches	High	DERMS Kafka Adapter data serialization
E-OMS	ETD-83218, 944044	Order worksheet calls disappearing after merge	Medium	Order worksheet
E-OMS	ETD-83235, 943514	Order worksheet flickering	Medium	Order worksheet
Grid Tabular	ETD-83300, 958725	ACKNOWLEDGEDTIME variable does not work in conditional test of grid tabular	Medium	Client - Order List
NRT, OMS	ETD-83219, 944829	Same customer on multiple incidents (in NRT - not in memory)	High	NRT
OMS	ETD-83103, 919482	OMS Hierarchy Level Filters Go Blank	Medium	OMS Client
OMS	ETD-83139, 917613	Cannot create follow-up Maintenance Order from Investigation/Restoration Order	Medium	Client - Order List
OMS	ETD-83340, 969207	Several momentary incidents created for Recloser instead of sustained outage incident.	High	OMS Server - Core processing
OMS	ETD-83168, 934155	Set Predicted not rolling to upstream	High	OMS Server - Core processing
OMS	ETD-83225, 941385	Momentaries coming in as Confirmed Incidents	High	OMS Server - Core processing
DMS	ETD-83147, 932361	Not Able to Change Symbol to Custom Bitmap for Primary Meter Load	Low	ADMS Client - Geographic display
OMS	ETD-83344	Affected Customer Count Not Updating	High	OMS Server - Core Processing
OMS, Outage Assist	ETD-82919, 917453	Having to Restart OMS ROs for Synchronization	Medium	OMS Server - Core Processing

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
Order-Worksheet	ETD-83341	Total customers out shows 0 on all order worksheets	High	Order worksheet
EOMS	ETD-83093, 916997	Arriving crew on one order is arriving crew on all orders they are dispatched to	Medium	Order worksheet
EOMS	ETD-83175, 934154	Unable to Close Restoration Order	Low	Order worksheet
EOMS	ETD-83293, 957530	NetDataServerRO Increase In Memory Consumption Affects Data Correction Performance	Medium	NetDataServer RO Memory
EOMS	ETD-83323, 965761	Users unable to open order worksheets	Medium	Order Worksheet
Study	ETD-83073, 945377	Study Mode on ADMS Client Computers fails to start.	High	Study Initialization
UI	ETD-83070, 941284	The ADMS Client was still crashing	High	Client logging component
UI	ETD-83265, 946794	APF crash	High	Client logging component
UI	ETD-83275	Hover over a transformer causes .NET unhandled exception	Medium	Client Null Pointer
EOMS	ETD-83295	OMS Display - Tabular Hierarchy Filter Goes Blank After a Period of Time	Low	Client - OMS Display Plugin

11. ADMS 16.16.2

This chapter describes the incremental new features and defect fixes that were added since ADMS 16.16.1. The remainder of this document provides additional release-specific information that also applies to this release.

11.1. What's New in 16.16.2

No new features have been added in this release.

11.2. What's Changed in 16.16.2

The following changes have been made in this release.

11.2.1. Supported Software and Hardware Changes

This section describes changes to the system such as changes in system requirements, compatibility, or third-party software.

The following changes has been made in this release:

- Forecasting - 3.8.0 and 16.16.2
- Foundation - 25R01

11.3. What's Fixed in 16.16.2

This section lists the notable issues addressed in this release.

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
Archive	ETD-82991	Call added to archiver with no details when customer split from outage	Low	Call journal-Archive App processing
Archive	ETD-83041	Deleting Adhoc Crew Crashing Archiver	Low	Crew journal-Archive App processing
Archive	ETD-83050	Archive Upgrade logs wrong?	Low	Version update after Database upgrade in Archive Database

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
Archive	ETD-83071	OMS Archiver database is having errors affecting various interfaces from getting accurate status	High	Switch Order, Incident Switch Steps processed by Archive App
Archive	ETD-83134, 927640	Customers appearing on restoration steps for multiple devices	Medium	OMS Server - Incident processing
Core	ETD-83127, 925075	Abnormal halos when accessing the ADMS with Partial Model View turned off	High	Thick client - Geographic display
Crew	ETD-83022, 899305	EOMS Issues with Backward Flow Crew Model Status Update	High	EOMS Order Worksheet
DMS	ETD-83006	Marking a device from a tabular	Medium	Thick client - Geographic display loading
DMS	ETD-83123, 922659	DSC System Level Default Plan Execution	High	ADMS client
DMS	ETD-83125, 927282	LVM not using all eligible devices in Mixed Mode	Medium	Mix-mode LVM solution area
DMS	ETD-83130, 925087	Structure Layer Annotation won't render if placed outside circuit DNOM extents	Medium	Thick client - Geographic display
DMS	ETD-83147, 932361	Not Able to Change Symbol to Custom Bitmap for Primary Meter Load	Low	Thick client - Geographic display
DNA	ETD-83160, 932453	Meshed FL cannot find the location of a fault.	Medium	Meshed Fault Location
DNAF	ETD-83058, 914140	LOV Restoring from De-energized Feeders	High	Loss of Voltage FISIR network selection
DNAF	ETD-83059, 914144	AFR finding violations not in the switchable network selection	Medium	AFR network selection
DNAF	ETD-83067, 914138	Station Outage closed into a fault	High	Station Outage Event FISIR triggering
DNAF	ETD-83133, 927281	FISIR Partial Plan Logic Flaw Reported by PPL	Medium	FISIR plan generation
DNAF	ETD-83297	DNAF Server Hang because of thread crash	High	DPF Network Selection
Documentation	ETD-83129	Documentation defect: When crew server to send IncidentUpdateWfm msg to WFM	Low	Interface WFM Documentation
EOMS	ETD-83111, 902454	Two queries perform for the same Incident	High	EOMS Order Worksheet

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
EOMS	ETD-83175, 934154	Unable to Close Restoration Order	Low	EOMS - eventhandler
EOMS	ETD-83194	GraphQL Queries cancelled - Performance Order Worksheet Issue	Medium	EOMS Order Worksheet
Interface	ETD-82473	AmiMessages Errors and Processing Fails	Low	Call Server API
Interface	ETD-83084	Traced Load and Generation Popup Showing Large Currents	Medium	Documentation
Interface	ETD-83115, 922606	Inconsistent CEA IncidentUpdateNotification message generation for partial restoration incident update	High	Customer Experience Adapter (CEA) configuratio
Interfaces	ETD-82995	Search can't find automated crews	Medium	Netviewer Adapter crew search
NTP	ETD-83110, 954998	Parent substation feeders were shown abnormal when they were looped in child substation	Medium	NTP abnormal loop determination
OMS	ETD-83026	Restoration orders recreated right after cancelled due to non-outage	Medium	OMS Server-Incident processing
OMS	ETD-83028, 902007	OMSMAIN restarted on the enabled PROD server with a null reference exception on customer	Medium	OMS Server Components
OMS	ETD-83031, 937112	Data Customer information not populating on replicated server	Medium	NetDataServer - Odata
OMS	ETD-83052	Planned Outage Callback number formatting not working	Medium	Planned outage validations for email and phone number are corrected
OMS	ETD-83093, 916997	Arriving crew on one order is arriving crew on all orders they are dispatched to.	Medium	EOMS - eventhandler
OMS	ETD-83105, 925076	Area Note ETR does not apply to an incident after a user makes a manual ETR change in 16.15 and greater	Medium	OMS Server - ETR processing
OMS	ETD-83113, 920504	OMSMain Crash from order merge	High	OMS Server - Incident Processing
OMS	ETD-83119	WFM Complete for SCADA recloser moving to Pending Clear without Closing Device	Medium	Interface WFM Processing
OMS	ETD-83136	Incidents are not being sent to SS9 when a MDT crew is assigned (no incident update message is sent to the crew server)	High	Interface WFM Processing
OMS	ETD-83151, 931955	"IsOverhead" value incorrect when incident created from Cut	Medium	OMS DNOM - Interface
OMS	ETD-83156, 936604	NetviewAdapter Midnight Crash	Medium	NAV data serialization

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
OMS	ETD-83196, 936836	unhandled exception for null reference in set affected customers for incident	High	OMS Server - Query
OMS	ETD-83197, 936905	Net Data Server Crash	High	NetDataServer - processing Incident journal message
OMS	ETD-83202, 950094	APF and Client Crashing when opening OW	Medium	
OMS	ETD-83206	NtpCombined IslandsSolved Message Performance Issue with certain Station	High	Server OMS processing
OMS	ETD-83208	Dot Net Exception When Selecting Overflow button from Client Toolbar on Geographic	High	Thick client - toolbar
OMS	ETD-83225	Momentaries coming in as Confirmed Incidents	High	OMS Server-Incident processing
OMS	ETD-83238, 943704	Data - Customer energized time incorrect with both active and pending clears	Medium	NetDataServer - OData
OMS	ETD-83239, 944520	Last Open Device not updating for Step Restored Incidents after Momentaries from upstream Breaker	Medium	OMS Server - Incident Processing
OMS	ETD-83259, 950077	Prediction using meter events is rolling up slower then customer calls	High	OMS Server - Incident Processing
OMS-Crew	ETD-83082	Outage Call isn't creating outage incident when same customer receives non-outage call within the simmer time	Medium	OMS Server-Incident processing
OMS-Crew	ETD-83099, 919435	Unassigned or Signing off a crew does not always release all work	High	Crew Assignment
OMS-Crew	ETD-83273, 950115	Crew returns to service center when released with multiple orders assigned	Medium	Crew Assignment
Order Worksheet	ETD-83048, 914937	Consumes Full laptop/browser memory when OW left open	Medium	Order worksheet leaks
Outage Assist	ETD-83062, 917003	OA - Assets/settings.json from preference service path	Low	Outage Assist Deploy
Performance	ETD-83132, 927354	NVA high memory consumption	High	Server and Netviewer Adapter dynamics
Process Automation	ETD-82989	Process Automation inconsistent logging related to truck zones	Low	Process Automation - Logging
Process Automation	ETD-83021, 891480	Process Automation auto-transfer erroneous actions in production	High	Process Automation - Auto Transfer

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
Process Automation	ETD-83035, 899304	Auto-transfer default org not working with X/Y 0,0	Medium	Process Automation - Auto transfer
Process Automation	ETD-83042, 912196	Process Automation doesn't Auto-Assign when an Auto-Transfer rule is matched	High	Process Automation - Auto Assign
Process Automation	ETD-83045	Process Automation Config Editor incorrectly rejects rule as duplicate	Medium	Process Automation - Configuration
Process Automation	ETD-83066, 914550	Process Automation unassigns a manually assigned crew and assigns a different crew	High	Process Automation - Crew Assignment
Process Automation	ETD-83190, 937685	Process Automation matching crews incorrect truck zone	Medium	Process Automation - Crew Assignment
Process Automation	ETD-83199, 941550	Process Automation Calculating Incorrect Truck Zone	Medium	Process Automation - Crew Assignment
Process Automation	ETD-83224, 941578	PA Crash 2 hours after auto-dispatch	High	Process Automation - Auto Dispatch
SA	ETD-82924	When creating a task for an upstream device for an already open device, two Create Tasks windows popup	Low	Order work sheet - Task creation
Switching Orders	ETD-83137, 941865	.NET Issue Opening Switching Orders	Medium	Switching Order UI
SWO	ETD-83090	SO, SWO, DIS TrustedClient component not communicating with different users	Medium	Notification Publisher
UI	ETD-82748, 714448	Left-clicking does not unselect previously selected rows	Low	OA - Tabular display
UI	ETD-82997	Order symbol blocks tag visibility	High	Thick client - Geographic display
UI	ETD-83015, 914849	Customizing Order's Table Filter Panel leads to OA Unresponsiveness	High	OA - Order table
UI	ETD-83085, 926286	Memory Leak Leading to Client to Crash	High	Thick Client - Schematic View
UI	ETD-83094, 917169	Unhandled exception on structures toolbar drop down when viewport not maximized	Low	DMS - Viewer Toolbar

12. ADMS 16.16.1

This chapter describes the incremental new features and defect fixes that were added since ADMS 16.16.0. The remainder of this document provides additional release-specific information that also applies to this release.

12.1. What's New in 16.16.1

No new features have been added in this release.

12.2. What's Changed in 16.16.1

The following changes have been made in this release.

12.2.1. Supported Software and Hardware Changes

This section describes changes to the system such as changes in system requirements, compatibility, or third-party software.

The following changes has been made in this release:

- Gateway - 2.7.2 and 16.16.1
- Scada - 3.6.2
- Foundation - 25R01

12.3. What's Fixed in 16.16.1

This section lists the notable issues addressed in this release.

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
Alarms, DNAF	ETD-83053, 916998	Alarm text message for some FISR alarms have changed	Medium	Alarm text
Core	ETD-83047, 914145	DEM 16.8 Large ISD Log Files Over 100 MB	Medium	ISD logging
DMS	ETD-82586, 727829	When Confirming Device In, the Confirmation Popup is Missing Device Information	Medium	OMS - Incident Processing
DMS	ETD-82774, 872607	Crew model validation schema incorrect	Medium	OMS - Crew Model loading

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
DMS	ETD-82838, 816836	CBO does not indicate customers being dropped when there is downstream loop	Medium	NTP CBO Check
DMS	ETD-83003, 885821	Upstream trace when multiple paths back to source are available	High	Client NTP tracing
DMS	ETD-83006, 891101	Marking a device from a tabular	Medium	Client- Locate device on Geographic
DMS	ETD-83027, 913524	Hung DnafSrv processes	High	NTP scheduling when server restart
DMS	ETD-83072, 914137	Plan Execution Lock Not Showing in LVMSM Profiles	Low	LVM System Monitor Profile Display
DNAF	ETD-83001, 885822	Recloser on disabled feeder was incorrectly set as eligible for reconfiguration	High	FISR device eligible check
DNAF	ETD-83029, 914147	FISR Fails to Generate a Partial Restoration Plan	Medium	FISR Plan generation logic
DNAF, Interface	ETD-83064, 914139	DNAF GRPC server binds localhost ip instead of 0.0.0.0 ip on the server for port 50055	High	DNAF Server GRPC component
EOMS	ETD-82369, 900620	Slow performance on the first time you open an order	Medium	Order Worksheet
FISR	ETD-82959, 866801	"Check Flow After Make Before Break Switching" Causing a 5 Minute Delay on FISR Implementation Event/Alarm	High	FISR Plan Manager
Grid Tabular	ETD-82952, 866855	GridTabular Dynamic Filtering Scroll Bar Resetting to Top	Low	Grid Tabular UI
Internal Notifications	ETD-83004, 881734	Internal Notifications Subscription Data corrupted during pod restart	High	Internal Notifications - Deployment issue
Modeling	ETD-82771, 857586	ADMS reports primary side outages at secondary network level, rather than feeder level as expected for downtown secondary network	High	OMS prediction engine
Modeling	ETD-82932, 864463	Customer Compiler obfuscation feature failing	Low	Customer list compiler
OMS	ETD-82613, 715693	Incident not Created during Performance Test in OMS	High	Customer list compiler
OMS	ETD-82867, 864625	Non-outage isolated incident being created with ServPoint OutageDeviceType	Medium	OMS - Incident Processing
OMS	ETD-82868, 857577	Deenergized Lines Journal not created for transformer with secondary lines	Low	OMS prediction engine
OMS	ETD-82891, 857437	Multiple Incident for the same device	High	OMS Server - Core Processing

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
OMS	ETD-82972, 872272	OMSMAIN Crash Loop	High	OMS DNOM accessor
OMS	ETD-83010, 891931	Unhandled exception causing ADMS client froze\crash when user worked an outage	Medium	Client Switch Order Container
OMS	ETD-83026, 898802	Restoration orders recreated right after cancelled due to non-outage	Medium	OMS Server - Core Processing
OMS	ETD-83032, 895504	No horizontal scroll bar at the bottom of the Incident Detail page	Low	Incident Processing
OMS-Crew	ETD-83082, 915844	Outage Call isn't creating outage incident when same customer receives non-outage call within the simmer time	Medium	OMS Server - Core Processing
Performance	ETD-82968, 872141	Incident creation continuing after performance test end	High	OMS - Incident Processing
UI	ETD-82794, 738012	Permanent Jumpers disappear at high zoom level	Medium	Client - Automatic feeder schematics
UI	ETD-82828, 853694	Crew status "assigned" delay	Medium	OMS OrderWorkSheet
UI	ETD-82997, 879095	Order symbol blocks tag visibility	High	Client - Geographic
DMS	ETD-82975	Toggle phase issue	Low	Client - Phase Toggling Popup
SWO	ETD-82955	Switch Order: When adding multiple steps via one click, order is reversed.	High	Client Switch Order Container
Crew	ETD-82948	ISEVALUATOR property not updating when Crew Type changed in UI	Medium	OMS Client - Crews Tabular
UI	ETD-82847	AFS schematic displays does not respect the VCE configured visibility options	Medium	Client - Automatic feeder schematics
OMS	ETD-83119	WFM Complete for SCADA recloser moving to Pending Clear without Closing Device	Medium	OMS - WFM processing

13. ADMS 16.16.0

This chapter describes the incremental new features and defect fixes that were added since ADMS 16.15.0. The remainder of this document provides additional release-specific information that also applies to this release.

13.1. What's New in 16.16.0

No new features have been added in this release.

13.2. What's Changed in 16.16.0

The following changes have been made in this release.

13.2.1. Supported Software and Hardware Changes

This section describes changes to the system such as changes in system requirements, compatibility, or third-party software.

The following changes has been made in this release:

- Gateway - 2.7.1 and 16.16.0
- Forecasting - 3.7.0 and 16.16.0
- Foundation - 24R13

13.3. What's Fixed in 16.16.0

This section lists the notable issues addressed in this release.

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
Crew	ETD-82948, 865977	ISEVALUATOR property not updating when Crew Type changed in UI	Medium	Client - Crew tabular
DMS	ETD-82889, 857439	Hung DnafSrv and ForecastSrv processes in 16.13	High	DMS, DNAF
DMS	ETD-82905, 857434	eTerra crashing with structure layer annotation originating inside extents but going outside	High	Client - structure drawing
DMS	ETD-82912, 859538	LOADORBUSHASHIGHVOLTVIOLATION and LOADORBUSHASLOWVOLTVIOLATION fields always return True	High	DMS - Query DMS Interface server

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
DMS	ETD-82975, 870538	Toggle phase issue	Low	Client - CBO dialogs
DNAF	ETD-82903, 866802	CLFISR triggered off a Manual Operation (Emergency drop) of a feeder breaker, and the plan implemented even after disabling the feeder for CLFISR	High	DNAF - FISR
DNAF	ETD-82963	Non-telemetered regulators going to +16/-16	High	DNAF - Distribution power flow
DOTS	ETD-82900, 857438	TDL3 cannot be reset on CLTRPDOTSRQ81 or RALACDOTSRQ81	Medium	DOTS - Real time - Breaker lockout
FISR	ETD-82931	FISR LOV stops on "measurement utilized for triggering network applications flag was not enabled".	High	DNAF - FISR - Loss of Voltage
Grid Tabular	ETD-82833, 816862	SchedulePointsStatic_GridDisplay does not work in GIX due to duplicate field in query	Medium	NetView GridTabular
Interfaces	ETD-82870, 853388	CallServer does not send CallV2 LatLong to OMS	High	Call Server Interface API
NTP	ETD-82946	CBO message when energizing	High	Network topology Check before operating message
OMS	ETD-82688	OMS - When Merging Incidents the highest Incident Status is not kept	Medium	OMS User Guide has been updated
OMS	ETD-82730, 716360	Advanced crew selection does not work with full FPL crew model	Medium	OMS Worksheet
OMS	ETD-82766, 722384	UI Performance issue with measurement tabular filtering	High	OMS GridTabular performance
OMS	ETD-82803, 738159	Blocked Prediction Ignores Permission Check	Low	OMS Server components
OMS	ETD-82851, 855884	Area Setting Repeating sending Auto Clear Momentaries on same incident	Low	OMS Server logging
OMS	ETD-82859, 853339	Archive Timeout Exception Information Needed	Medium	Additional Logging for Archive App Timeout
OMS	ETD-82861, 853630	NVADAPTER Crash	Medium	Outage Assist back end
OMS	ETD-82888, 864468	Slow Incident Refresh due to TroubleDescriptionTranslated	High	OMS server - query performance
OMS	ETD-82899, 857436	Start time of an order backdating	Medium	OMS Server - call handling
OMS	ETD-82915, 860382	OMS Display Plugin - Client Data values not adding up	Low	Incident Processing

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
OMS	ETD-82945, 865339	OMSMAIN crash due to System.NullReferenceException	Medium	OMS Server - NTP Processing
OMS	ETD-82964, 866785	Slow Merge of Single Customer Incidents	High	OMS server - timer processing
OMS	ETD-82998	OMS Main crashing when loading Hurricane Milton dynamics	High	OMS server - Dynamics loading
OMS, UI	ETD-82938, 864465	UI date selection Issue In History Correction's Edit Restoration Step	High	Client - OMS Date picker dialog

14. ADMS 16.15.0

This chapter describes the incremental new features and defect fixes that were added since ADMS 16.14.0. The remainder of this document provides additional release-specific information that also applies to this release.

14.1. What's New in 16.15.0

This section contains information about new features and functionality for ADMS 16.15.0.

14.1.1. File Updater Support for Digitally Signed Files

The File Updater has been enhanced to provide improved security around the files it transports. It now supports digital signing of files by the server and the subsequent verification of signed files by clients. To use this feature an appropriate certificate must be installed on both server and clients.

See the configuration section of the File Updater Installation Guide for more details.

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	Yes	Via File Updater Configuration Editor.
ADMS Archive	No	
User Interface	No	
Demonstration Available	Yes	The demo will be available upon request.

14.1.2. Enhanced Switching and Outage Planning

Added an option to create a planned outage from the switching order window, allowing users to record the scheduled time, track de-energized or energized customers for each switching operation, and utilize this data to generate interruption steps under the planned outage for precise customer notifications with CMI details.

See the Creating a Planned Outage from a Switching Order section of the Switching Operations User Guide for more details. (GRDADMS-1287)

See the General Tab section of the Switching Order Client Config Editor User Guide for more details.

Other Settings

6

Max Number of Tabs

☐ Confirm Existing Daily Log Usage

☒ OMS Link Enabled

☒ Allow Planned Outage From SWO

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	Yes	Added Allow Planned Outage From SWO in Switching Order Client Config Editor.
ADMS Archive	No	
User Interface	Yes	Screenshot included.
Demonstration Available	Yes	Click here for demonstration.

14.1.3. Internal Notifications Roles & Permissions

14.1.3.1. Roles and Permissions

New roles and permissions have been introduced to provide enhanced control and flexibility in managing access within the Internal Notifications application.

14.1.3.2. Enhanced Security Features

- **Istio Enabled:** Istio has been enabled for all Internal Notification services to improve security and service management.
- **Credential Management:** All application credentials have been migrated to Kubernetes secrets, ensuring secure storage and access.

For more details, refer to the *Internal Notifications Installation Guide*.

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	Yes	Retrieve the client secret value from the Kubernetes secret to configure the client credentials for the CoreNotificationServer.
Configuration	Yes	Refer <i>Internal Notifications Installation Guide</i> for changes in Configuration.
ADMS Archive	No	
User Interface	No	

Feature Implications	Yes / No	Additional Information
Demonstration Available	No	

14.2. What's Changed in 16.15.0

The following changes have been made in this release.

14.2.1. Supported Software and Hardware Changes

This section describes changes to the system such as changes in system requirements, compatibility, or third-party software.

The following changes has been made in this release:

- Gateway - 2.6.4 and 16.15.0
- Forecasting - 3.7.0 and 16.15.0
- Foundation - 24R12

14.3. What's Fixed in 16.15.0

This section lists the notable issues addressed in this release.

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
Archive	ETD-82884, 854384	Merged Order Causes Duplicate Restoration Steps in Archiver	High	Archive Data Integrity
Archive, OMS, Order-Worksheet	ETD-82824, 747240	Splitting restoration or switching steps breaks archive	Critical	Archive
Core	ETD-82798, 730729	New Tag Types in EMS was not showing on DMS Toolbar after upgrading to 16.x	High	Toolbar helper function
Core, Modeling	ETD-82538, 698420	TAG inserted on Station internals high-side is not displayed on geographic	High	Client - Geographic Display
Core, UI	ETD-82461, 640591	Placing mobile substation on non-geographic display does not appear without manual refresh of APF	Medium	Client display call up
DMS	ETD-82590	state icon for fault indicator missing in popup	Medium	Event Scripter - UI
DMS	ETD-82586, 727829	When Confirming Device In, the Confirmation Popup is Missing Device Information	Medium	Confirm Toggle Switch UI

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
DMS	ETD-82617, 729710	"Data Storm" that happened during peak HELENE roughly 1030 events/alarms	Medium	DMS Dynamics
DMS	ETD-82753, 730115	Breaking the Loop with tie to Open Action	Medium	NTP CBO Check
DMS	ETD-82779, 720366	APF Crashed due to Duplicate Structure Id	Medium	Client - Geographic Display
DMS	ETD-82838, 816836	CBO does not indicate customers being dropped when there is downstream loop	Medium	NTP CBO Check
DMS	ETD-82877, 852808	APF crash on SCE production system: 2 different users experienced APF crash while working on SWO	High	Client - Geographic Display
DMS	ETD-82885	Dangerous Condition Icon showing up improperly when opening device and de-energizing tap. This seems to occur on 3 phase, C phase, AC phase and BC phase devices.	Medium	Query function for Client
DMS	ETD-82889, 857439	Hung DnafSrv and ForecastSrv processes in 16.13	High	DPF result file handler
DMS	ETD-82895	CBO Issue on a Looped network, impacted by status of another pole switch on a different feeder	High	NTP CBO check
DNAF	ETD-82708, 730035	DEM 16.8 Dual Prod ReadOnly DNAF Hang	Medium	DNAF Dynamics
DNAF	ETD-82827, 816837	AFR doesn't generate plan	Medium	AFR Plan Generation
DOTS	ETD-82900, 857438	TDL3 cannot be reset on CLTRPDOTSRQ81 or RALACDOTSRQ81	Medium	Simulator lockout reset
E-OMS	ETD-82334, 605232	Cancelling of incident leaves bad data occasionally	Medium	OMS Worksheet
E-OMS	ETD-82780, 729694	An extra close task is created when working a manual switching order	Medium	OMS Worksheet
Grid Tabular	ETD-82763, 737439	gendpf_griddisplay not working in GIX	Medium	Grid Display
Interface	ETD-82739	Status message not sent when LatestCallbackTime is wrong format CallMessages	Medium	Call Server Interface API
Interface	ETD-82813, 747044	UpdateCrewStatus API updates crew status even if the CrewId Field is empty/blank	Medium	Crew Server Interface API
Interface	ETD-82814, 747048	UpdateIncident API Status Code is not working as intended	Medium	Crew Server Interface API
Internal Notifications	ETD-82549, 698160	Internal Notifications Processable Entity Error Logging	Low	Internal Notifications - Users
Internal Notifications	ETD-82575, 698081	Passwords not secure in values file for Internal Notifications	High	Internal Notifications - Security

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
Internal Notifications	ETD-82800, 730959	Internal Notification issue with Helm Chart	Low	Internal Notifications Deploy
Modeling	ETD-82697	Model Manager service crash	Medium	
NRT	ETD-82786, 729807	Core/NRT/Reporting - Restoration Details are inconsistently missing phase information	High	OMS Server - Restoration Steps Processing
OMS	ETD-82650, 730536	AMI interface Power Restore Generating Trouble Tickets	High	OMS Server - Incident Processing
OMS	ETD-82764, 747050	Momentary Orders not created	Medium	OMS Server - Incident Processing
OMS	ETD-82816, 853570	Restoration and switching steps are not being recorded properly	Medium	OMS Server - Incident Processing
OMS	ETD-82820, 816670	Breaker momentaries coming in as outages.	High	OMS Server - Incident Processing
OMS	ETD-82899, 857436	Start time of an order backdating	Medium	OMS Server - Incident Processing
OMS	ETD-82964, 866785	Slow single customer incident roll-up processing	High	OMS Server - Incident Processing
OMS, UI	ETD-82938, 864465	UI date selection Issue In History Correction's Edit Restoration Step	High	Client - History Correction
OMS, Work	ETD-82809, 816838	APF Client does not Connect to HTTPS EventHandler	Low	OMS Worksheet
OMS-Incidents	ETD-82258, 855352	Wrong ETR/ITR reported when area note deleted	Medium	OMS Server - Incident Processing
OMS-Incidents	ETD-82862, 850163	Nonoutage customer split from merged confirmed incident is in both incidents and loses called status	Medium	OMS Server - Incident Processing
Order-Worksheet	ETD-82778, 745380	RO users can select drop down menus when requestIncidentLock = "READONLY"	Medium	OMS Worksheet
Order-Worksheet	ETD-82836, 823604	Outage Assist allowing users to complete tasks on SCADA Devices	Medium	OMS Worksheet
Outage Assist	ETD-82699, 728942	Outage Assist Layers - Similar Layer Names cannot use Spaces	Low	OA UI
Outage Assist	ETD-82916, 859516	Comment added in editable cell gets copied over to incorrect order	Low	OA UI

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
Process Automation	ETD-82826, 818952	Auto-dispatch not respecting Crew Assigned Max Incident Count	Medium	Process Automation Rules
SWO	ETD-82762, 715960	DMS Switch Order Issue Button locks during execution	Medium	Client Switch Order Container
UI	ETD-82552, 689942	Customer Count in Lines Analyst Attributes different with Trace Down Customer	Medium	Line Query Helper
UI	ETD-82720, 714594	ADMS Client (IDMS_GIT_315_20240719.1) Crashing	Medium	Client Model Load
UI	ETD-82766, 722384	UI Performance issue with measurement tabular filtering	High	Client - Grid Tabular
UI	ETD-82807, 739830	.NET Unhandled Exception	Medium	APF Crash - Create/Move/Update Annotations

15. ADMS 16.14.0

This chapter describes the incremental new features and defect fixes that were added since ADMS 16.13.0. The remainder of this document provides additional release-specific information that also applies to this release.

15.1. What's New in 16.14.0

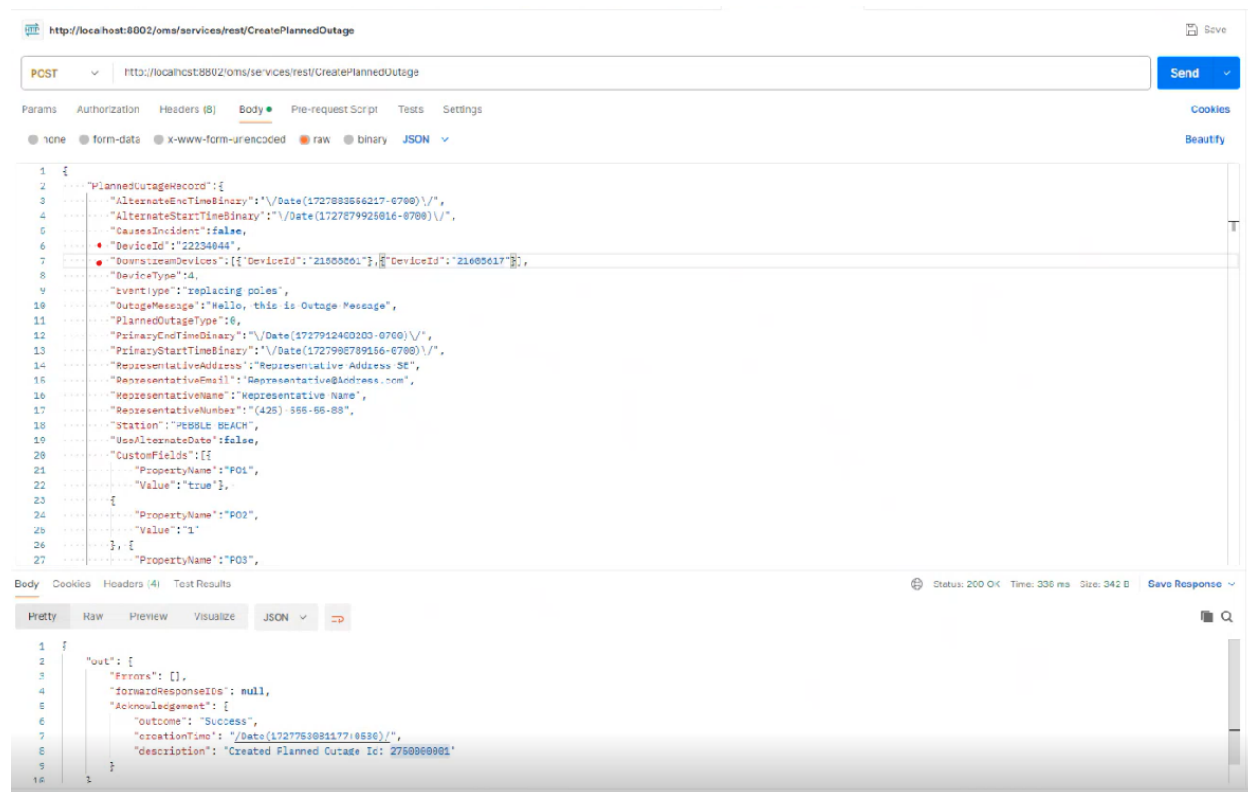
This section contains information about new features and functionality for ADMS 16.14.0.

15.1.1. Support Feeder Sections for Planned Outages

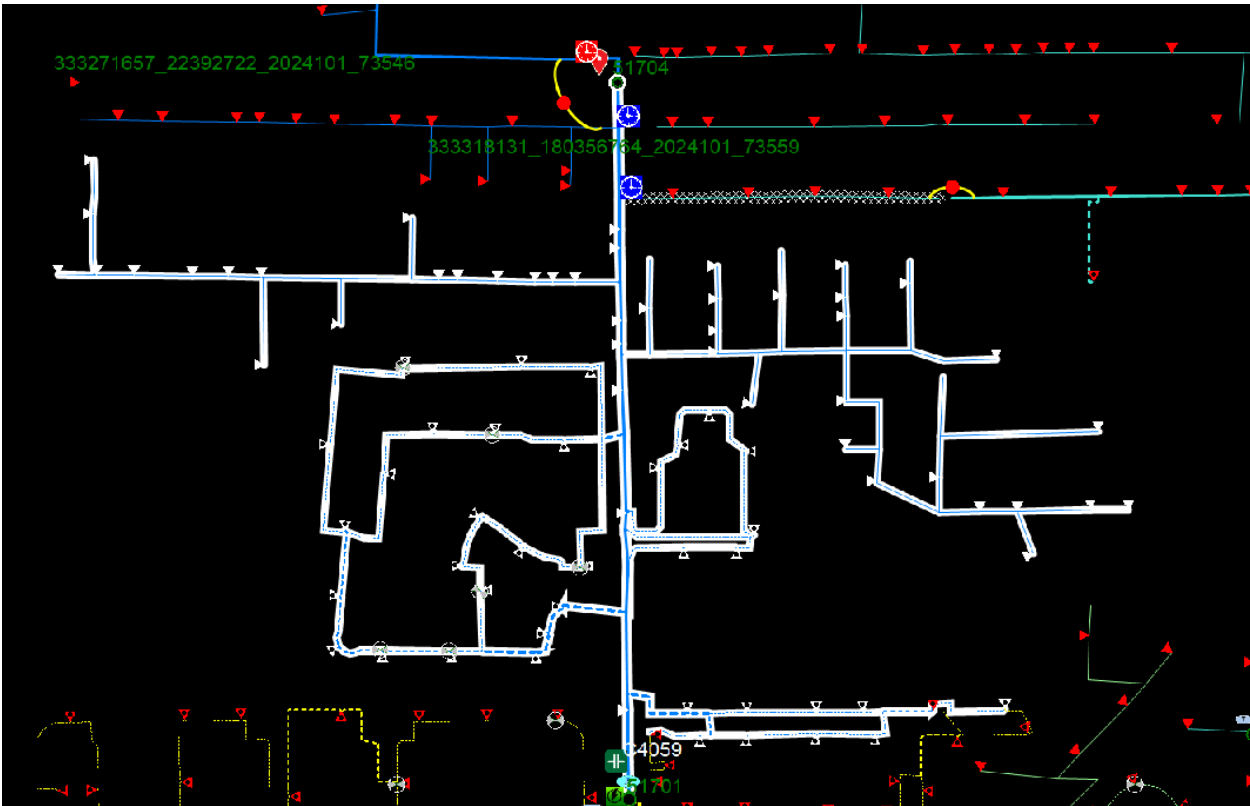
The planned outage system now supports outages that affect only part of a feeder, where downstream sections remain energized by alternate feeders.

Previously, a planned outage was defined by a single device, affecting all downstream customers. With this update, a planned outage can be defined by multiple devices, including one upstream device and optional downstream devices. The outage will only affect the section of the feeder between the upstream and downstream devices.

The CreatePlannedOutage API now includes a new field, DownStreamDevices. (GRDDMSS3-2333)



Blue icons are now displayed near delimiting devices. Delimiting devices and their associated customers are not part of the outage, as they are backfed through external devices such as jumpers.



Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	Yes	DownStreamDevices
Configuration	No	
ADMS Archive	No	
User Interface	Yes	Screenshots included
Demonstration Available	Yes	Click here for demonstration

15.1.2. Mobile Enterprise Updates Crew Shift and Availability Statuses

Mobility Gateway now supports setting CrewShiftStatus and CrewAvailabilityStatus. CrewShiftStatus is set in response to the NotifyCrewStatus message and the precise behavior is configurable to handle the fixed values received from Mobile Enterprise. CrewAvailabilityStatus can be populated through the NotifyCrewAvailability message, which performs a direct passthrough of the received value based on coordination of coded translations between ADMS and Mobile Enterprise. In addition, the Crew Shift Status and Crew Availability Status fields can now be updated

manually from the Crew Update Dialog. (GRDADMS-1662)

Crew Update Dialog - Crew Name : Lewis [5270] - AUTO

Primary Employee: Y Jeff Lewis Make This Employee Primary Create Temp Crew

Current crew employees

ID	NAME	TEMPCREWNAME	RADIONUMBER	CREWAREA	DEFAULTCREWAREA	EMPLOYEE
FI2098	Dave S Panasky		4168	Northeast Crews	Northeast Crews	Lineman
FI1816	Peter C Barber		4096	Northeast Crews	Northeast Crews	Apprentice

Properties

Name	Value
Active	True
Auto Dispatch	True
Automatic Dispatch Id	5270
Company Name	
Contractor	False
CreatedBy	DMSALL
Crew Area Id	CL1051
Crew Area*	Northeast Crews
Crew Availability Status	On Break
Crew Division*	DispatchNorth
Crew Shift Status	On Shift
Crew Status	Signed On
Crew Type	Lead Lineman
Default Crew Area Id	CL1051
Default Crew Area*	Northeast Crews
Default Location X	951394
Default Location Y	790487
Department	
Evaluator	False
Foreign	False
Home Area	Northeast Crews
Id	FI1960
Is Adhoc	False
MDT	True

Available employees

ID	NAME	CREWNAME	TEMPCREWNAME	RADIONUMBER	CREWAREA
FI1841	Aaron X Cappucci	Thompson [5274] - AUTO		4104	Northeast Crews
FI1842	Adam V Briscoe	Ratzlaff (4245)		4105	Northeast Crews
FI2093	Al I Owen	Spears [5272] - AUTO		4166	Northeast Crews
FI1903	Alex Vincent McCormick			4018	Northeast Crews
FI2739	Alicia I Beaudette				Northeast Crews
FI1846	Alvin U Davey	Druhan [5273] - AUTO		4107	Northeast Crews
FI2077	Arne L Coulborn	Ratzlaff (4245)		4162	Northeast Crews
FI1832	Ben E Perkizas	Atkinson (4247)		4102	Northeast Crews
FI1976	Bill Y Persons	Persons [5275] - AUTO		4260	Northeast Crews
FI1973	Bob Jon Thompson	Thompson [5274] - AUTO		4259	Northeast Crews

Filter: Split Crew

Employees Vehicles

OK Cancel Apply

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	Yes	Mobile Enterprise endpoint
Configuration	Yes	Setting CrewShiftStatus in response to NotifyCrewStatus messages requires setting <CrewShiftStatusMap> in MobilityGatewayAppConfig.xml.
ADMS Archive	No	
User Interface	Yes	Screenshot included
Demonstration Available	Yes	Click here for demonstration

15.2. What's Changed in 16.14.0

The following changes have been made in this release.

15.2.1. Supported Software and Hardware Changes

This section describes changes to the system such as changes in system requirements, compatibility, or third-party software.

The following changes has been made in this release:

- Gateway - 2.6.4 and 16.14.0
- Forecasting - 3.6.0 and 16.14.0

- Foundation - 24R10
- Angular version to 18

For more information, refer to [Third-Party Software](#).

15.3. What's Fixed in 16.14.0

This section lists the notable issues addressed in this release.

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
Archive	ETD-82711, 714598	ArchiveApp Crashes "System.ArgumentException: An item with the same key has already been added"	High	Archive Processing
Client	ETD-82550	Client Memory consumption increases due to Structures	High	Client Structure Initialization
Core	ETD-82627, 706728	NRT backed-up - up to 20 hours	Medium	Processing real time journal data
Crew	ETD-82490, 674270	Crew signed off reverts to Available status	Low	Crew Processing
Customer Model Compatibility	ETD-82770, 719045	Customer List Compiler binary cannot be read	Medium	Customer List Compiler
DMS	ETD-82636	Structure layer objects do not render if there is only a single entry	Medium	Client Geographic Display
DMS	ETD-82777, 720343	CBO on Device Toggle Not Showing Notifications Properly	Low	CBO
DNAF	ETD-82605, 697887	Flow limit and Loading percentage values are not updated after the DPF is run	Low	Query Helper functions
DNAF	ETD-82622, 705631	FISR can't see Fault Indication from SCADA	High	FISR Triggering
DNAF	ETD-82651, 716016	Sanity Check on Measurement Time stamps	Medium	FISR Triggering timestamp check
DNAF	ETD-82722	Two issues found when the LVM Voltage Safety Check Auto Disabling Check is enabled	Medium	LVM Volt Safety Check
DNAF	ETD-82741	AFR doesn't generate plan, involving subFed station devices	Medium	AFR network selection
DNAF	ETD-82760	FLISR Run more than 3 mins for certain tripping case	High	FISR de-energized load processing.
DNAF	ETD-82781, 730085	Meshed FL	Medium	Meshed Fault Location fault targets check

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
E-OMS	ETD-82369, 611160	Slow performance on the first time you open an order	Medium	OMS Order Worksheet
E-OMS	ETD-82677, 714059	Crew status time can't be updated to past time via calendar icon in order worksheet	Low	OMS Order Worksheet
E-OMS	ETD-82730, 716360	Advanced crew selection does not work with full crew model	Medium	OMS Order Worksheet
Interface	ETD-82594, 705609	GET method of switching order creation and updates is not aligned with same POST abilities	Low	DMS Interface API processing
Interface	ETD-82706, 709516	DMS Interface Server JSON Formatting for Double Quotes	Low	DMS Interface API processing
Internal Notifications	ETD-82575, 698081	Passwords not secure in values file for Internal Notifications	Medium	Internal Notifications
NTP	ETD-82599, 697883	Conductor is removed rather than de-energized if the upstream device is partially open	Low	Converter static topology processing
OMS	ETD-82438, 689973	Manually throwing over a throw over device is causing an order to be created	Medium	OMS NTP processing
OMS	ETD-82568, 691722	Downgrade Predicted Stable order to Open device is also creating orders for all customers who called	Medium	OMS NTP event trigger
OMS	ETD-82613	Incident not Created during Performance Test in OMS	High	OMS call processing
OMS	ETD-82822, 747034	Storm Replay NetServer Performance Issue	High	OMS parallel recloser incident processing
OMS-Incidents	ETD-82556, 690822	OMS Display Plugin Overview Values Don't Tie to Incident Detail Values	Medium	Client - OMS Display Plugin
Outage Assist	ETD-82336, 629055	Creating and Executing Open task generates another order	Medium	Outage Assist UI
Process Automation	ETD-82618, 698948	Process Automation is transferring/assigning the order even though Auto Transfer/Assign is disabled by Area Setting	Medium	Incident Processing, Area Setting
Process Automation	ETD-82769, 722402	Process Automation - Scenario where order doesn't get assigned until order worksheet is opened	Medium	Crew Processing, AutoDispatch
Study	ETD-82731, 738027	Study is Not Saved for Full Model Network for the existing ST Network	Medium	Study server client
SWO	ETD-82600, 697876	Place CUT in Study SWO and the Device State Information Pop-up is wrong for Switches/Comp Switches - incorrect messages issued in Study SWO	Medium	Switch Order CBO Study Mode logic
UI	ETD-82582, 708225	Substation Internal View Inconsistencies	Medium	Client drawing

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
Work	ETD-82745	NetROServer crash due to exception "Collection was modified; enumeration operation may not execute"	Medium	Area Settings

16. ADMS 16.13.0

This chapter describes the incremental new features and defect fixes that were added since ADMS 16.12.0. The remainder of this document provides additional release-specific information that also applies to this release.

16.1. What's New in 16.13.0

This section contains information about new features and functionality for ADMS 16.13.0.

16.1.1. ADMS Coded Translations In Mobile Enterprise

The ADMS definition of Coded Translations dictates the values that users can select (for example, cause and sub-cause). When a field user selects values from Coded Translations, these values must match with those recognized by ADMS to ensure the message is properly communicated.

To ensure consistency between ADMS and Mobile Enterprise, Mobility Gateway reads the complete Coded Translations configuration and sends it to Mobile Enterprise. (GRDADMS-1688)

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	Yes	Mobile Enterprise endpoint
Configuration	Yes	These settings are part of the <i>Mobile Enterprise Web Service Client</i> plugin definition in the <code>MobilityGatewayAppConfig.xml</code> configuration file. • <code><CodedTranslationXMLPath></code> • <code><CodedTranslationsUri></code>
ADMS Archive	No	
User Interface	No	
Demonstration Available	No	

16.1.2. Managing the Suspension of Orders via Mobility Gateway

Mobility Gateway can automatically suspend and unsuspend an order without operator intervention. With appropriate configuration, order suspension and unsuspension can be performed based on crew status changes initiated by Mobile ADMS. When automatic suspension is carried out, the suspension time is set and the suspension reason can be set to a configured value. (GRDADMS-1746)

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	Yes	Mobile Enterprise endpoint
Configuration	Yes	These settings are part of the <i>Mobile Enterprise Web Service Client</i> plugin definition in the <i>MobilityGatewayAppConfig.xml</i> configuration file. • <SuspendOrderOnCrewRelease> • <ReleasedSuspensionReason> • <SuspendOrderOnCrewSuspension> • <StoppedSuspensionReason> • <CrewStatusesToUnSuspendOrder>
ADMS Archive	No	
User Interface	No	
Demonstration Available	Yes	Click here for demonstration

16.1.3. Annotation Placement API now supports Device ID and Device Type

The Distribution Interface Server Annotation Placement API has a limitation where the device ID must be provided to associate the annotation with a device. When the new optional device ID and device type parameters are supplied, the devices link for the new annotation will be populated.

When using the API to add annotations, the Status bitmap in the Annotation Summary display will not be populated automatically, as this is a UI-dependent feature. Subsequent editing and saving of the annotation will populate the Status bitmap in the Annotation Summary display.

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	Yes	Distribution Interface Server Service API for Annotation
Configuration	No	
ADMS Archive	No	
User Interface	No	
Demonstration Available	No	

16.2. What's Changed in 16.13.0

The following changes have been made in this release.

16.2.1. Supported Software and Hardware Changes

This section describes changes to the system such as changes in system requirements, compatibility, or third-party software.

The following changes has been made in this release:

- Gateway - 2.6.3 and 16.13.0
- Forecasting - 3.6.0 and 16.13.0
- Foundation - 2024-09

16.3. What's Fixed in 16.13.0

This section lists the notable issues addressed in this release.

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
Archive	ETD-82382,612776	PST Code (Area Settings) not recorded correctly in the Archive/NRT	Medium	Area settings changes to reflect in NRT database
Archive	ETD-82465,675998	Invalid Area-Note ETR being set in Archive as some default string	Medium	Area settings changes to reflect in NRT database
Crew	ETD-82558,689768	Crew advancing incorrect incident on delayed UpdateCrewStatus message	Medium	OMS Server Crew Processing
DMS	ETD-82249,138344	180 degrees rotation for DISTTF in model creates black discrepancy in the DISTTF bitmap in UI	Low	Thick client - Geographic Display
DMS	ETD-82455,645683	Drop down in the feeder filter disappears and does not allow user action	Medium	Thick client-Toolbar
DMS	ETD-82534,689950	Non Load Break Switch de-energizing Capacitor, DERs & Gens	Medium	CBO check
DMS	ETD-82391	Future tie in loop swaps in unrelated switch in one-line loop diagram	Medium	Converter static NTP processing
DMS	ETD-82632	Regulator RLD Status Points Not Updating	Medium	Dynamics
DMS	ETD-82548,690157	Structure Attributes not translated in Right-Clicks	Medium	DMS Structure Right Click Menu
DMS	ETD-82738	Incorrect warning regarding de-energizing customers for Jumpers	Medium	CBO check
DNAF	ETD-82528,689955	DPF Results Logging Error when reverse energized	Medium	DPF logging
DNAF	ETD-82529,689958	DPF Application Voltage Regulation Logic Error When RegulationControl.csv is Built in the Model	Medium	DPF Advanced Regulation

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
DNAF	ETD-82701	Dual Prod DNAF Issue/Hang Event	High	Fault Location
DNAF	ETD-82567, 697884	DPF diverged case and BLA KVA Target Values Wrongly Calculated	Medium	BLA KVA allocation
DNAF	ETD-82653, 705633	The Dynamic phase mismatch notification doesn't list all the abnormal phase connection	High	DNAF server CBO
DNAF	ETD-82674	Study Dnaf Solution PopUp doesn't take given Evaluation Time	Low	Study mode DPF timestamp
DNAF	ETD-82714	PLOS will solve for a de-energized device	Low	PLOS status change
DNAF, FISR	ETD-82554, 697882	FISR Auto Trigger by Fault Event Problems	High	FISR triggering fault placement
DNAF, FISR	ETD-82676, 705635	FL halos back after dnaf server restart, after passing 1 hr	Medium	FL LST file serialization
EOMS	ETD-82580, 692196	Can't Fully Scroll Down on Order Worksheet	Medium	OMS - Order Worksheet
EOMS	ETD-82598, 696559	Unassociated Call Appears Twice in Customer List	Medium	OMS - Order Worksheet
EOMS-Orders	ETD-82591,698099	-2 orders are getting created randomly	Medium	Order Name format
Interface	ETD-82544, 697563	WmsModelMessages Processing Issues	Low	Crew Modeling API
Internal Notifications	ETD-82574	Internal Notification documentation not clear	Medium	The Internal Notification Installation Guide has been updated
OMS	ETD-82695	FISR Dynamic Eligibility broken	Medium	FISR device eligibility check
OMS-Incidents	ETD-82569, 698401	Resetting predicted stable after move up removes incident	Medium	OMS Server-Incident Processing
Process Automation	ETD-82717, 712227	Process Automation dispatching crews that are not covering truckzon	Medium	OMS server with Process automation
SWO	ETD-82519, 689766	Messages don't popup from Study Switching Order	Medium	Switch Order CBO Study Mode logic
SWO	ETD-82600, 697876	Place CUT in Study SWO and the Device State Information Pop-up is wrong for Switches/Comp Switches - incorrect messages issued in Study SWO	Medium	Switch Order CBO Study Mode logic
UI	ETD-82539, 683229	UI Performance issue with measurement tabular filtering on Value	High	Grid Tabular Filter Panel

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
UI	ETD-82436, 632937	Blank UI row on the tasks menu	Low	OMS - Order Worksheet
UI	ETD-82461	Placing mobile substation on non-geographic display does not appear without manual refresh of APF	Medium	Client mobile substation processing
UI	ETD-82552, 689942	Customer Count in Lines Analyst Attributes different with Trace Down Customer	Medium	NTP customer count calculation
UI	ETD-82611, 700050	Order Worksheet UI has unused space	Low	OMS - Order Worksheet
UI, WebFG	ETD-82582	Substation Internal View Inconsistencies	High	Thick client - Geographic Display
Work Model	ETD-82625	No Outage Action does not Cancel or Clear the Order	Low	The EOMS User Guide has been updated

17. ADMS 16.12.0

This chapter describes the incremental new features and defect fixes that were added since ADMS 16.11.0. The remainder of this document provides additional release-specific information that also applies to this release.

17.1. What's New in 16.12.0

This section contains information about new features and functionality for ADMS 16.12.0.

17.1.1. Enhanced Process Automation Application

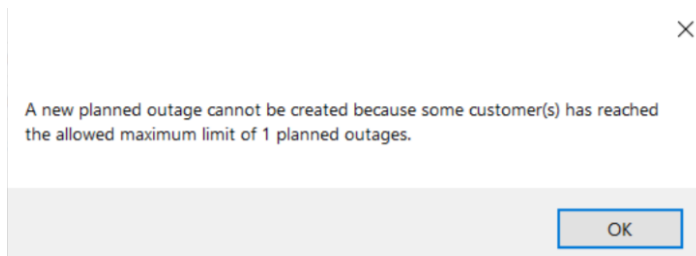
This release introduces new properties to the Process Automation configuration file (ProcessAutoConfig.xml) to support crew assignments based on truck zones. (ADO-674323)

By default, the `<TruckZoneAreaPath></TruckZoneAreaPath>` parameter is not defined. To define this parameter, navigate to `<TruckZoneAreaPath>`
`%IDMS_ROOT%/models/RT/TruckZone.geojson</TruckZoneAreaPath>`. (GRDDMSS3-2453)

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	Yes	Updated ProcessAutoConfig.xml with <code><TruckZoneAreaPath></TruckZoneAreaPath></code>
ADMS Archive	No	
User Interface	No	
Demonstration Available	Yes	Click here for demonstration

17.1.2. Planned Outage Validation

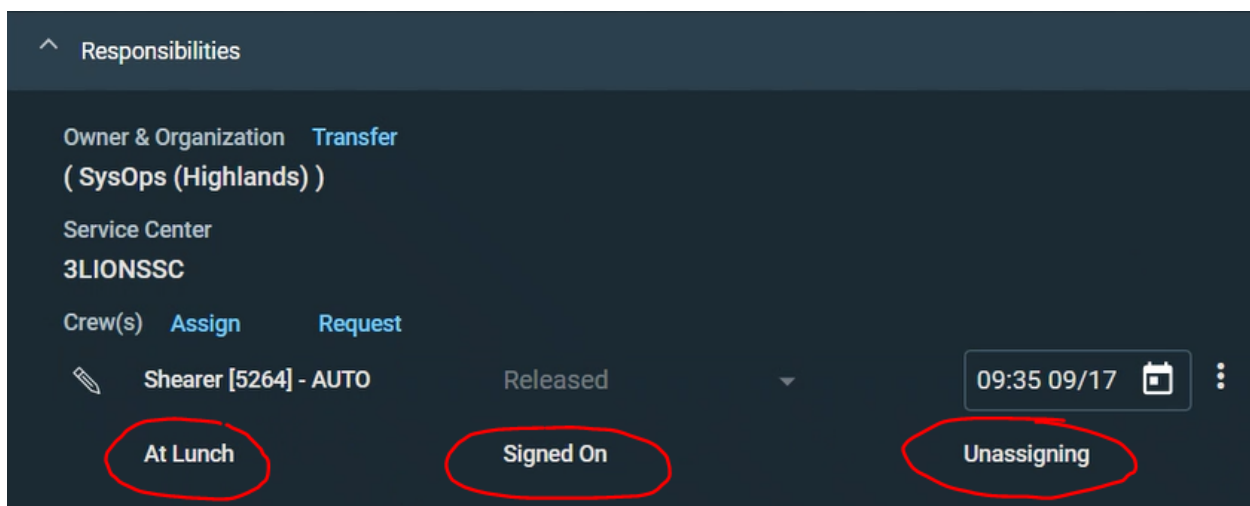
The system now ensures that customers are not affected by multiple planned outages within a specific date range. Previously, a new planned outage could be created even if a customer was already part of an ongoing outage. With this enhancement, the system prevents overlapping outages for the same customer and displays an alert if the limit is exceeded. (ADO-564131, GRDADMS-1289)



Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	Yes	NUMBER_OF_DAYS_PRIOR_TO_PLANNED_OUTAGE_CREATION ALLOWED_NUMBER_OF_PLANNED_OUTAGES
ADMS Archive	No	
User Interface	Yes	Screenshot included
Demonstration Available	Yes	Click here for demonstration

17.1.3. Enhanced Crew Status Visibility

Crew statuses, including Shift Status, Mobile Communications Status, and Crew Availability Status, are now displayed directly in the Order Worksheet and grid order tabular. On the Order Worksheet, these statuses appear in the Responsibilities section next to the assigned crew members. In the Order List grid, only the lead crew's statuses are displayed. This enhancement improves decision-making efficiency by allowing dispatchers to access crew information without needing to open the Crew Details panel. (ADO-607262, GRDADMS-1717)



OverviewIncidents (1)Orders (1)2570000001-1

Order List

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Order

List Pri

Type

Incident Type

Status

Service Center

Crews

Crew Status

Crew Shift Status

Crew Mobile Comm Status

Crew Availability

Station

Nominal Station

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2570000001-1

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Restoration

Outage

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Dowley [5253] - AUTO

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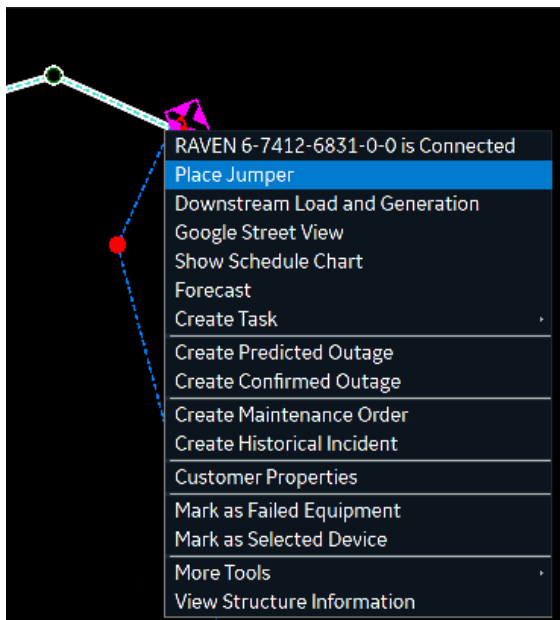
Available

RAVEN

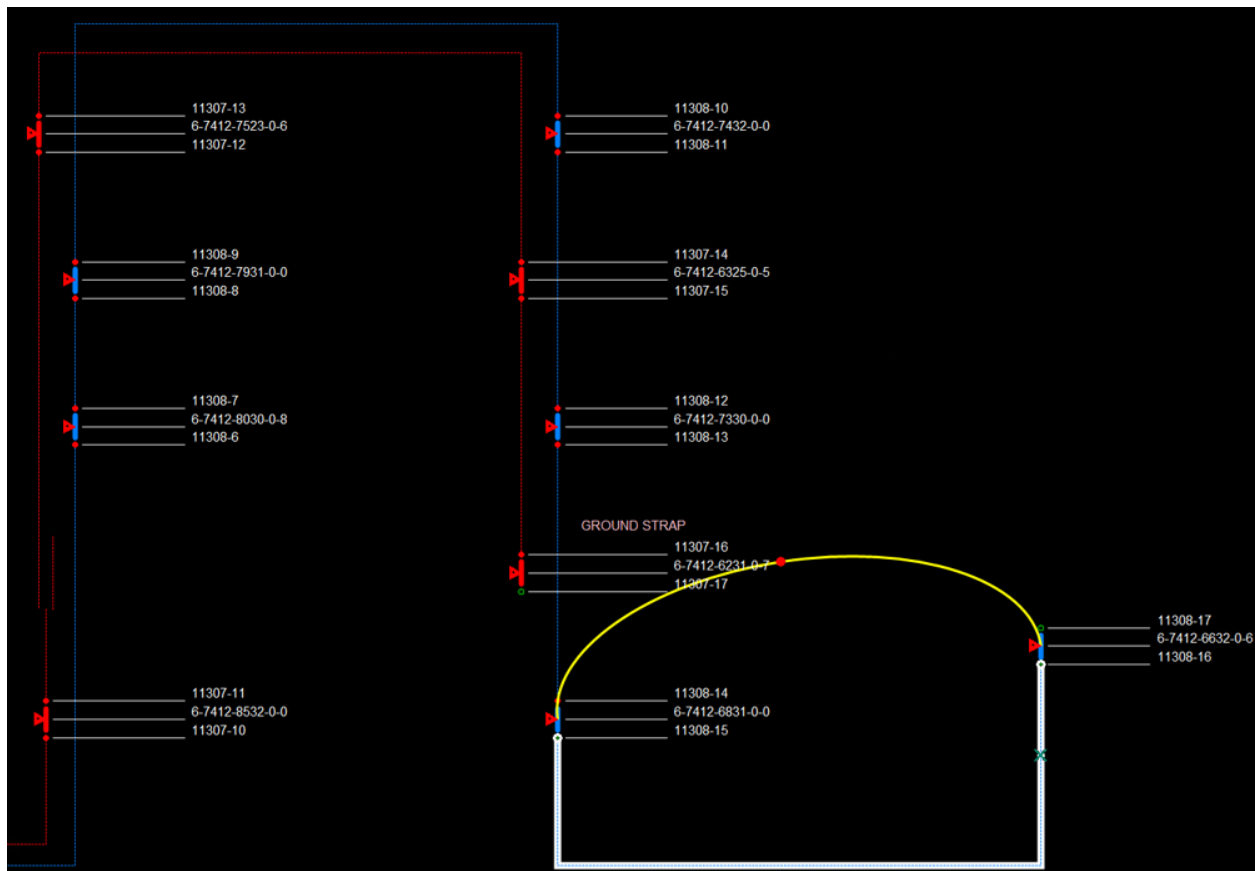
Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	Yes	Updated OmsDisplayPluginConfig.xml with <ShowCrewStatusesInCrewDisplay></ShowCrewStatusesInCrewDisplay>
ADMS Archive	Yes	Added the 3 crew status columns in Order and CrewAssignments table
User Interface	Yes	Screenshot included
Demonstration Available	Yes	Click here for demonstration

17.1.4. New Jumper Capabilities

It is now possible to place a temporary jumper on a distribution transformer, in addition to a line. Either end of the jumper, or both, can now be connected to a transformer. The jumper connects to the high side of the distribution transformer. (ADO-582109, GRDDMSS3-1994)



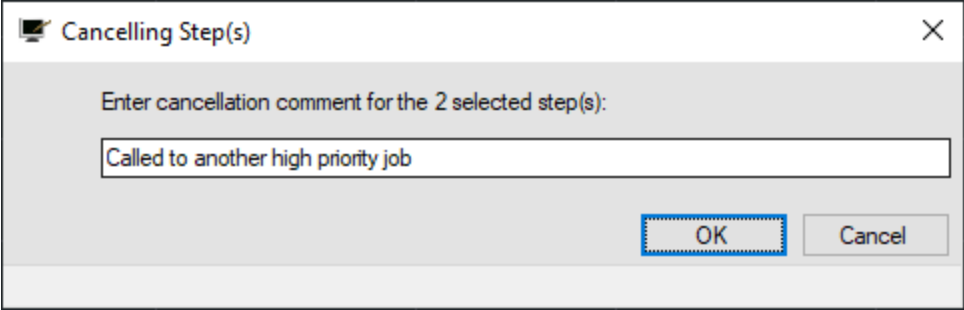
Any jumpers placed on an underground transformer will be shown on the corresponding URD loop schematic.



Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	Yes	Requires changes to ViewerConfig.xml to add toolbar or right click buttons, to enable the feature. See the delivered sample ViewerConfig.xml for examples.
ADMS Archive	No	
User Interface	Yes	Screenshots included
Demonstration Available	Yes	Click here for demonstration

17.1.5. Enhanced Switch Order To Allow Cancelling Multiple Steps

It is now possible to select multiple Issued steps and click the Cancel option. A new popup will appear so that the user operator can indicate a note for cancelling all selected steps.



All steps are cancelled and the note introduced is updated for each step in the comment column.
(ADO-647940, GRDDMSS3-2431)

☐	#	Nav	I/C	Step State	Device	Action	Issued To	Comment	
☐	0.1			CANCELLED	6-7803-4902-0-8	DISCONNECT	Crawford [5251] - AUTO	[STEP CANCELLED] Called to another high priority job	
☐	0.2			CANCELLED		TRANSFER OF OPERATIONS RESPONSIBILITY	Crawford [5251] - AUTO	[STEP CANCELLED] Called to another high priority job	

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	Yes	Screenshots included
Demonstration Available	Yes	Click here for demonstration

17.2. What's Changed in 16.12.0

The following changes have been made in this release.

17.2.1. Supported Software and Hardware Changes

This section describes changes to the system such as changes in system requirements, compatibility, or third-party software.

The following changes has been made in this release:

- Gateway - 2.6.1 and 16.12.0
- Forecasting - 3.6 and 16.12.0
- Foundation - 2024-08

17.3. What's Fixed in 16.12.0

This section lists the notable issues addressed in this release.

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
Crew Server	ETD-82501, 674254	Dispatching Crew from Crew Server API fails to Dispatch Crew in Thick Client	Medium	Crew processing logic
Crew, Crew Management	ETD-82242	Netsrv (DMSMain) does not get enabled role after a failover to Standby when cfgproxy is enabled with mTLS	Medium	Crew processing logic
Crew, Crew Management	ETD-82424, 62942	Crew "CurrentIncident" not updating to highest CrewStatus	Medium	Crew processing logic
Crew, Crew Management, OMS	ETD-82448, 637068	Crews Tabular Unable to show All Incidents Assigned to Crew	Medium	Client Crew Tabular
Crew, OMS	ETD-82498, 674240	Netserver Crash on Crew File Update	Medium	Crew Model Update
DMS	ETD-82406	SWO Viewport not refreshed during Room Loading	Medium	Switch order window loading
DMS	ETD-82456, 640587	Incorrect CBO Notification with Tie Switch	High	NTP CBO Check
DMS	ETD-82464, 644236	Data syncing issue between SCADA 3.5 and DMS 16.9 (for example, Measurements not updating in DMS Tabulars, original reported issue - DFT display values were always showing 0.00)	High	DMS fail-over
DMS	ETD-82480, 665432	Phase Angle Mismatch Decoration doesn't show up for jumper.	Medium	Fast Dynamics
DMS Interface Server	ETD-82491	ADMS Tag modification resets X/Y to 0,0	Medium	DMS server TAG processing
DMS Server	ETD-82246, 638518	NTP Not Solving in RT	Medium	DMS Server NTP processing
DNAF	ETD-82507, 674717	DNAF performance from Optimized LVM	High	LVM Triggering
DNAF	ETD-82520	2030.5 DER LVM Eligibility Should not Check SCADA Measurements	Medium	LVM device eligibility
DNAF	ETD-82313, 622661	Wrong DPF Feeder Flow computed values for open spare feeders	High	Converter
DNAF	ETD-82404, 644237	FISR not restoring all feeders during concurrent events at PPL	Medium	FISR Triggering logic DNAF backend
DNAF	ETD-82418, 622662	FISR De-energized and Restored Customer Counts	Medium	FISR Network reduction algorithm

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
DNAF	ETD-82452, 638519	Maximum Number of DNAF Result Files created for any one substation	Low	DNAF logging
Documentation	ETD-82543	Incorrect figure shown in DNAF Advanced Applications Analyst Guide	Low	Updated the DNAF Advanced Applications Analyst Guide
Documentation	ETD-82375	On-Line Documentation - Does not allow to search for server configuration items	Low	Documentation
E-OMS	ETD-82346, 611161	Some tasks are not being transferred to new order when moving up	Medium	Work Model backend
Interface	ETD-82415, 628926	DmsInterfaceServer failure messaging issue	Low	DMS Interface server
Internal Notifications	ETD-82388, 615596	Pop-up appearing as 'no records found' but customer searchable for affected customer type subscription	Low	Internal notifications backend
Modeling	ETD-82560, 689911	Incorrect tftype was used in transformer model	Medium	Converter
NTP	ETD-82468	De-energized sections not energizing when tied to energized source	High	DMS NTP backend
OM Server	ETD-82504	Arrived crew does not move to order until netserver restart	Medium	Incident processing logic
OMS	ETD-82322, 604852	Data Correction not showing orders in production	Medium	NetDataServer performance
OMS	ETD-82516, 676190	Devices Not Closing when Step Restoring an Incident from Wfm	Medium	OMS Incident processing logic
OMS	ETD-82517, 676197	Incident Restoration Step Info Updating Incorrectly when Step Restoring an Incident from Wfm	High	Incident processing logic
OMS	ETD-82422, 644153	End Time not being consistently set on Investigation orders	Medium	API Error propagation
OMS	ETD-82413	Update Incidents - All Affected Customers not shown	Medium	Updated the DMS User Guide
OMS	ETD-82442, 645510	Evaluator assignments being moved to crew column when incidents are merged	Medium	Crew assignment processing
OMS	ETD-82478, 646045	FEEDER and STATION tables not updating with new SERVICE_CENTER_ID	High	Archive app
OMS, Archive	ETD-82385, 612662	Queries deadlock on CALL table while NRT is updating it	High	Archive App processing real time journal data
OMS, Archive	ETD-82367, 612639	Archive purge script times out for moderate amounts of data	High	Purge data from archive database
OMS, E-OMS	ETD-82398, 628889	Create Close Task popup displaying wrong status for phases	Medium	Task creation for Orders

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
OMS, SWO	ETD-82470	Playing SWO in Study Mode can Send commands to RT DNOM	High	Switch Order Server backend
Order Work	ETD-82561	Organization Type on Transfer Organization Popup not consistent	Low	OMS Order Worksheet Display
Order Worksheet	ETD-82580, 692196	Can't Fully Scroll Down on Order Worksheet	Medium	OMS Order Worksheet Display
Study	ETD-82595	Study Mode savecase not loading in Geographic when retrieving from study mode	Medium	Study Server
SWO	ETD-82443, 631932	Planned Request "APPROVE" action is throwing exception in ADMS	High	SWO server backend
SWO	ETD-82454, 663748	SWO implementing steps when already in desired state in study mode	Medium	Switch Order server backend
Tagging	ETD-82446, 674565	Slowness of Tagging Creation and Delete on ADMS 16.9	High	Tag creation and deletion
Tagging	ETD-82361, 607341	Support for 31 tags	High	DMS server and UI
Tagging	ETD-82460	Tag prevents open of a composite switch	High	Composite switch processing
Tagging	ETD-82466	Unpredictable lag in dmsQuery API for Tags	High	IST Tag processing
Tools	ETD-82434, 629693	Netconf file overwrites settings	High	Netconf file
UI	ETD-82191, 644151	Order status changed to field complete without restoring customers	Medium	Order work sheet processing logic
UI	ETD-82217, 638518	DNAF server hangs after running for few hours	High	DNAF server edge case fix
UI	ETD-82329, 612775	Annotation Geometry is changing while using POST request method to update the status	Low	DMS Interface
UI	ETD-82348, 629702	Tracing color options repeating in right-click menu	Low	Viewer configuration editor
UI	ETD-82380, 629072	Boundary Switch not triggering NTP when de-energized	High	Client NTP triggering
UI	ETD-82389, 632209	Fuse/line/transformers of pending clear order still shown as predicted on geographic display	Medium	Incident processing logic

18. ADMS 16.11.0

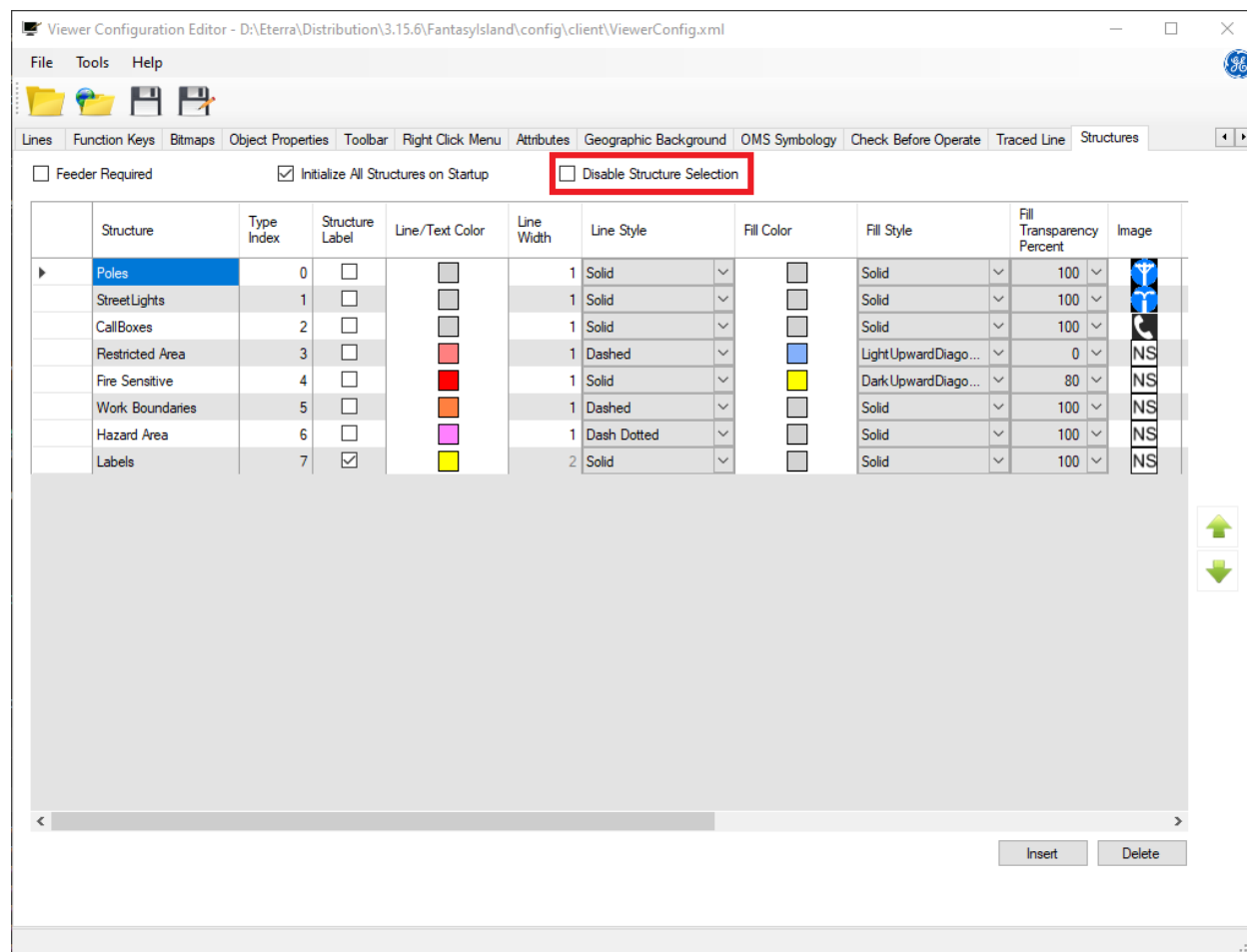
This chapter describes the incremental new features and defect fixes that were added since ADMS 16.10.0. The remainder of this document provides additional release-specific information that also applies to this release.

18.1. What's New in 16.11.0

This section contains information about new features and functionality for ADMS 16.11.0.

18.1.1. Enhanced Structure Selection Control

Added a **Disable Structure Selection** checkbox to the Structures tab in the Viewer Configuration Editor. When selected, this option prevents structure placements from being selectable in the various ADMS views. By default, this checkbox is unchecked, maintaining the previous standard functionality where structures are selectable. (ETD-82101)



Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	Yes	Screenshot included
Demonstration Available	No	

18.2. What's Changed in 16.11.0

The following changes have been made in this release.

18.2.1. Supported Software and Hardware Changes

This section describes changes to the system such as changes in system requirements, compatibility, or third-party software.

The following changes has been made in this release:

- Gateway - 2.6 and 16.11.0
- Forecasting - 3.6 and 16.11.0
- Foundation - 2024-07

18.3. What's Fixed in 16.11.0

This section lists the notable issues addressed in this release.

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
Crew Management	ETD-82425, 629704	CrewStatus not retaining highest status when crew is dispatched to multiple incidents	Medium	Crew Assignment Handling for Multiple incidents
DBSync	ETD-82297, 605247	In DBSync, the CrewName on CrewAssignments entity, is erroneously being marked as Required	High	Removed required flag from the CrewName field.
DMS	ETD-82356, 612974	Slow to initialize DMS in client with many structures	Medium	Client initialization
DMS	ETD-82371	CBO gives "Load Break" warning when "Making Load"	Medium	CBO topology check

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
DNAF	ETD-82359	FISR Plan Generation Fails When Breaker Pri/Alt Meastype is Modeled in DMS	Medium	FISR plan generation logic
DNAF	ETD-82363, 614812	FISR PRV Forming Incorrect Protection Zones	Medium	FISR post-PRV processing
DNAF	ETD-82373, 611593	PLOS shouldn't solve for station that the device is not part of.	Medium	PLOS outage component processing
DNOM database	ETD-82112, 604976	DNOM SQL database does not well support localization	Low	Loading data into DNOM database tables
DOTS	ETD-82341	Erratic Behavior in SIM server (NetServerSIM)	Medium	DER Dynamics
EOMS	ETD-82255, 580187	RO users can still modify Order Worksheet when requestIncidentLock = "READONLY"	Medium	Added additional mode checks for read only.
EOMS	ETD-82305, 605234	Momentary Orders Not Converting to Restoration Orders	Medium	Added order type conversion code for this specific situation.
Order Worksheet	ETD-82286, 611165	Orders Disappearing from Order List	Medium	Fixed a sensitivity to specific text characters in input fields
OMS	ETD-81914, 578978	Incorrect Updates to Crew's Status & Current Incident	Medium	Crew Assignment Handling for Multiple incidents
OMS	ETD-82243, 580776	Customer energized time incorrect with both active and pending clears	High	Fixed incorrect energized time by adding incident name check
OMS	ETD-82264, 604911	Ghost Calls - incident customers with a call time after incident end time are "orphaned".	High	Fixed Ghost call issues
OMS	ETD-82269, 612972	Incident type is created 'Single customer' instead of "Non-outage" when Non-Outage CallType call is submitted	Medium	Added Validation for Non Outage Action
OMS	ETD-82276, 138787	Unknown closed device ID in restoration step for Momentary incident	Medium	Momentary Incidents
OMS	ETD-82294, 604971	AreaSettings API is not updating Archive/NRT correctly	Medium	Update of Area settings through API
OMS	ETD-82311, 604972	Incidents not rolling up after first of three phases predicted out	Medium	Incident creation and rolling up to upstream device

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
OMS	ETD-82330, 139088	ADMS is creating orders with (-2, -3 or -4) when trying to keep the order with the same number	Medium	Order number generation logic
OMS	ETD-82335, 604622	Ping and Scan AMI windows are blank with Avista Data models	Medium	Ping and Scan AMI actions
OMS	ETD-82353, 612525	Downgrade Predicted Stable order to open downstream device gets the order stuck	High	Updated for Predict Stable Logic When Downgrading Incident
OMS	ETD-82374, 608753	Sorting coded translation columns with images is not working and giving exceptions	High	Handled the null exceptions by adding visibility condition
OMS	ETD-82390, 613215	NVA Crash Loop	Low	Need install latest C++ lib
OMS, SWO	ETD-82470	Playing SWO in Study Mode can Send commands to RT DNOM	High	Implementation of Switching steps in SWO
SIM	ETD-82429	Event Scripter not connecting to RT	Low	Documentation
Switch Order	ETD-77382, 581666	Multi-Select Issuing Switching Steps results in wrong warning	Medium	CBO check from switch order
Switch Order	ETD-82218, 612756	Switch Order opened switch order as user NETCLIENT instead of user ID	High	Opening a Switch order through logged in user
Switch Order	ETD-82357, 607237	SWO CBO "X" doesn't abort implementation	High	CBO check from switch order
Switch Order	ETD-82358, 608967	SWO Save popup "X" doesn't abort	High	Closing of switch order view port
Switch Order	ETD-82378, 611328	Viewport - Maximum Viewport behavior issues	High	Navigation from switch order view port

19. ADMS 16.10.0

This chapter describes the incremental new features and defect fixes that were added since ADMS 16.9.0. The remainder of this document provides additional release-specific information that also applies to this release.

19.1. What's New in 16.10.0

This section contains information about new features and functionality for ADMS 16.10.0.

19.1.1. Process Automation Application

The Process Automation application has been improved with the followings:

- Ensured Auto-Assignment is based on appropriate Organization Rules
- Allowed Auto-Assignment of orders that are In Progress
- Prevented Auto-Transfer of orders that are manually Transferred using DoNothing Transfer action
- Added additional attributes, such as Call Priority and Incident Working status, to Auto-Assign and Auto-Transfer rules.

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	No	
Demonstration Available	No	

19.1.2. Enhanced SWO Plan Editing Capabilities

The SWO plan was previously modified only by control room operators in the secured zone. With the latest changes, users from the corporate zone can now modify the rejected SWO plan. These users are typically the ones submitting the Switching Request for the same SWO plan. (ADO-137436, GRDDMSS3-1964)

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	Yes	Added AllowEditRejectSWO=false

Feature Implications	Yes / No	Additional Information
ADMS Archive	No	
User Interface	No	
Demonstration Available	No	

19.1.3. Enhanced Device Navigation Setting in Switch Steps Tab

A Use `KEEPPLAYERSETTINGS` checkbox has been added to the **Switch Steps** tab in the **Device Navigation** section of the Switching Order Client Configuration Editor. By default, this checkbox is selected, maintaining the existing behavior. When unchecked, navigating to geographic views will revert all layer settings to their default state, including Laterals, Non-Reconfigurable Sections, Future, Labels, and so on. (ADO-138489, GRDDMSS3-2366)

Switching Order Client Configuration Editor - D:\Eterra\Distribution\3.15.6\FantasyIsland\config\client\SwitchOrderClientConfig.xml

File Help

General Switch Order Related Device & State Tag & Comment **Switch Steps** Right-click Menu Validation Rules Step Custom Columns Step Columns Order Mobile Alarms

Switch Steps

☐ SD Column Visible ☒ Warn about problems before issuance of a step ☒ Auto-fill completion time for SCADA devices ☒ Disable 'Complete Step Confirm' popup
☒ Auto Complete Steps ☒ Auto-complete Non-Scada steps from SWO in RT mode ☒ Copy the Issued To Field when Copying a Step ☐ Disable 'Issue Step Confirm' popup
☒ Disable Comment editing after completion

Device Navigation

Default Locate: Geographic
Default Zoom: 10
Zoom level for Containers: 100
Zoom level: 10
Zoom level: 100

☐ Show PIP for Non-Container Objects
☐ Show PIP for Container Objects
☐ Scada Displays Not Available
☒ Use geo navigation for URD devices ☒ **Use KEEPPLAYERSETTINGS**

Isolation Switch Navigation URL: FGCMDDISPLAY/VIEWPORT=GEO_%DYN/URL"/%DYN/VIEW/DETAIL/%STATIONNAME?L
Ground Navigation URL: FGCMDDISPLAY/VIEWPORT=GEO_%DYN/URL"/%DYN/VIEW/DETAIL/%STATIONNAME?L
Customer Navigation URL: FGCMDDISPLAY/VIEWPORT=GEO_%DYN/URL"/%DYN/VIEW/DETAIL/%STATIONNAME?L

State Name Localization
Open State Name:
Close State Name:

Employee List for Issued To field
☒ Use Employee File
Employee list file path: %IDMS_ROOT%\config\server\SWOEmployee.txt

Switch Step Grid Colors

Real-time

Edit (Even Rows) Aa
Edit (Odd Rows) Aa
Selected Aa
Issuing Aa
Issued (Manual) Aa
Issued (State Feedback) Aa
Issued (State Matches) Aa
Completed Aa
Cancelled Aa
Read-Only Cell Aa
Error Aa
Validate With No Change Aa
Validate With Change Aa
Validate With Warnings Aa

Study Mode

Edit (Even Rows) Aa
Edit (Odd Rows) Aa
Selected Step Aa
Implementing Aa
Implemented Aa
Read-Only Cell Aa
Error Aa

Manual Step Action/State Type
Action/State Type: NODEVICEACTIONS

Completion Time-Out
Time to wait for step completion, seconds: 10

Hover Over
Tooltips visibility mode: Show always

Safety Instructions
☒ Enable Safety Instructions
Safety Instructions File Path: %IDMS_ROOT%\config\client\SWOMultilineInstructionsList.txt

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	

Feature Implications	Yes / No	Additional Information
Configuration	No	
ADMS Archive	No	
User Interface	Yes	Screenshot included
Demonstration Available	No	

19.1.4. Config for Op Center with a Value Higher in the AOR Hierarchy

Configuration that will update all outgoing messages to send a different level of the organizational hierarchy. This configuration is for deployments in which service centers do not form the lowest level of the organizational hierarchy. (ADO-136365, GRDDMSS3-2947)

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	Yes	WFM_ORG_LEVEL_FOR_OP_CENTER parameter in NetServerOmsConfig.txt
ADMS Archive	No	
User Interface	No	
Demonstration Available	No	

19.1.5. Field Damage Assessment support in Mobility Gateway.

Enable a field assessor using Mobile Enterprise application to create a component-based damage assessment message that can be sent to ADMS via Mobility Gateway. (GRDDMSS3-1954)

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	Yes	Added 'FieldDamageAssessmentOnly' flag to 'MobileEnterpriseWebServiceClient' Plugin in MobilityGatewayAppConfig.xml
ADMS Archive	No	
User Interface	No	
Demonstration Available	No	

19.1.6. Support for Mutual TLS (mTls) in DMS Interface Server.

Add support and configuration to allow mutual TLS connections to the DIS Interface Server. (ADO-605633)

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	Yes	Added the following to <code>DmsInterfaceServerConfig.xml</code> • 'CertSubjectName' Certificate client Subject(s) (CN). • 'RevocationMode' Certificate revocation mode (NoCheck 0, Online 1, Offline 2). • 'EnableTrustedClientDualSSL' Enable mTls for Trusted Clients • 'TrustedClientCertSubjectName' Certificate client Subject(s) (CN) for Trusted Clients • 'TrustedClientCertStore' Certificate store for Trusted Clients • 'EnableDualSSL' Enable specific Trusted Client configured for mTls
ADMS Archive	No	
User Interface	No	
Demonstration Available	No	

19.2. What's Changed in 16.10.0

The following changes have been made in this release.

19.2.1. Supported Software and Hardware Changes

This section describes changes to the system such as changes in system requirements, compatibility, or third-party software.

The following changes has been made in this release:

- Gateway - 2.6 and 16.10.0
- Forecasting - 3.6 and 16.10.0
- Foundation - 2024-06

19.3. What's Fixed in 16.10.0

This section lists the notable issues addressed in this release.

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
Archive	ETD-82214, 138258	ArchiveApp Purge throws 'child record found' violation	Medium	Purge cleared or cancelled incidents from Archive database
DMS	ETD-81950, 131041	Multiple Consoles unable to search, remove tags, or toggle assets - DMS Main Hang	Medium	Dynamics synchronization
DMS	ETD-82213, 138137	Client Nav Performance using LOCATEOMSCOMP	Medium	DMS client code
DMS	ETD-82247	Structure layers names drop down do not show full name	Medium	Structure name display from Toolbar
DMS	ETD-82254	Incorrect info displayed Composite switches nominal stations/feeders fields	Medium	Query helper function
DNAF	ETD-82216, 138219	FISR Fails to Restore Customers Upstream of a Fault (Breaker and VCR LO)	Medium	FISR network selection
DNAF	ETD-82289, 138673	FISR triggered and placed a fault based on a lockout that was 20 minutes old	Medium	FISR Triggering
DNAF	ETD-82292	FL Result file is empty & 0Kb	Medium	FL solution reporting
Documentation, Process Automation	ETD-82271, 138510	Process Automation Auto-Assign for service points	High	Considered Single Customers incidents as well
DOTS, UI	ETD-82237, 138352	snapshotconverter from browser does not work when launching from a client workstation	Low	Snapshot converter - Added Dynamics path, Event Scripts Path configuration
E-OMS	ETD-82164, 135845	(OA and ADMS) search customers in large outage hangs client	Medium	Order Worksheet - Customer search performance improvement
E-OMS	ETD-82297, 605247	In DBSync, the CrewName on CrewAssignments entity, is erroneously being marked as Required	Medium	EOMS failover in Windows deployments
OMS	ETD-82071, 133327	OA - Downstream Order Takes the Same Organization as Upstream Order	Low	Orders - Organization Assignment

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
OMS	ETD-82166, 135995	Two restoration orders on same customer	Medium	OMS Journaling code
OMS	ETD-82180, 138208	The same de-energized customer is part of two incidents.	Low	Incident processing
OMS	ETD-82204,138259	NRT - All calls are not displayed in the Affected Premises table	Medium	Writing customer calls related data to Archive database
OMS	ETD-82236	Predicted fuse order reopened in Map with new switching step by ADMS after order auto cleared by AMI Power-In event	High	used PredictedStable property to prevent switching step creation
OMS	ETD-82238, 138422	Excessive "Creating New Incident because of energization Delay in Incident"	Low	Incident creation
OMS	ETD-82245, 138343	ADMS created new SNCs on customers without receiving a new call/ami out	Medium	OMS Processing ..
OMS	ETD-82264	Ghost Calls - incident customers with a call time after incident end time are "orphaned".	High	setting SingleCustomerS immerPeriod for the new call so that new incident gets create
OMS	ETD-82318	"Operation is taking longer than expected" message is appeared after Blocking Outage Prediction	Medium	Port Configuration
SWO	ETD-82089, 134808	Multi-Select Issuing Switching Steps out of order	Medium	SWO Client logic for Multi-Select.
SWO	ETD-82257,138421	ST Switch Order does not implement RECL/ground steps	High	Implementation of switching step in ST mode from switch order
UI	ETD-81940	After traces are cleared, trace color 1 is used instead of color 0	Low	Client drawing
UI, Tagging	ETD-79213, 135802	Tagging does not work for DMS-Only composite switches	High	Server logic for composite switch

20. ADMS 16.9.0

This chapter describes the incremental new features and defect fixes that were added since ADMS 16.8.0. The remainder of this document provides additional release-specific information that also applies to this release.

20.1. What's New in 16.9.0

No new features have been added in this release.

20.2. What's Changed in 16.9.0

The following changes have been made in this release.

20.2.1. Supported Software and Hardware Changes

This section describes changes to the system such as changes in system requirements, compatibility, or third-party software. The following changes has been made in this release:

- Gateway - 2.5.4 and 16.9.0
- Forecasting - 3.5.1 and 16.9.0
- Foundation - 2024-05

20.3. What's Fixed in 16.9.0

This section lists the notable issues addressed in this release.

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
Archive, NRT	ETD-82178, 137571	NRT Performance Issues with Customer Models	High	Writing NRT data to Archive database
Core	ETD-82223, 138136	NetServer crashing while updating Crew model	High	OMS Server, crew model update
Core	ETD-82229, 138216	Dynamics backlog and DMS main server (NetserverCM) slowness due to DYN\$J TapSave msg dynamics	High	Dynamics processing
EOMS	ETD-82166, 135995	Two restoration orders on same customer	Medium	OMS server, Incident processing

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
IN	ETD-82273	Internal Notification black bar covers up elements in top right of UI	Medium	UI alignment issue
Mobility	MOB-435, 134557	Payload for tag details has an incorrect creationTime	High	Mobility Gateway
OMS	ETD-82180, 138208	The same de-energized customer is part of two incidents	Low	OMS Server, incident processing
OMS	ETD-82245, 138343	ADMS created new SNCs on customers without receiving a new call/ami out	Medium	OMS Server, incident processing related to customer call time
OMS	ETD-82224, 138218	DER missed checking three phase kw and kvar measurements	Medium	Converter
UI	ETD-81837, 137449	Power Flow Direction Indicator	Medium	Client, Analyst Output Attribute popup, Flow diagram

21. ADMS 16.8.0

This chapter describes the incremental new features and defect fixes that were added since ADMS 16.7.0. The remainder of this document provides additional release-specific information that also applies to this release.

21.1. What's New in 16.8.0

This chapter contains information about new features and functionality for ADMS 16.8.0.

21.1.1. Automatic Release Of Crew's Assigned Work

With the ADMS 16.8.0 release, a new configuration-driven feature has been introduced. When auto-release is enabled and a crew's status changes to one of the configurable auto-release trigger values (default Unavailable or Sign Off), the crew is automatically released from all assigned incidents. (ADO-134437, GRDDMSS3-1899)

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	Yes	RELEASE_ALL_WORK_FROM_CREW RELEASE_ALL_WORK_FROM_CREW_STATUS_CODES
ADMS Archive	No	
User Interface	No	
Demonstration Available	No	

21.1.2. DMS Interface Server (DIS)

DMS Interface Server (DIS) support for Mutual TLS (ETD-81853)

A new configuration in DIS will allow enabling mutual TLS (mTls) between DIS and client applications. When configured, client applications will be able to present a client certificate for authentication. The Server will validate that the client is from a valid issuer by configuration. (GRDADMS-1801)

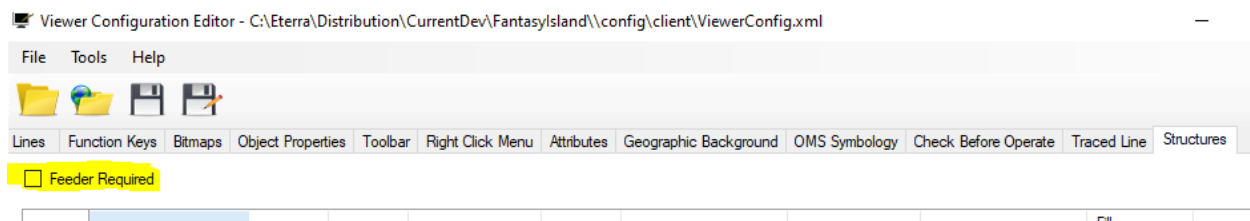
Certificate client Subject (CN) or Issuer (CN) in mutual tls connection If CertSubjectName not defined and SecureWebService is true only Server Side TLS is enabled
<CertSubjectName></CertSubjectName>

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	

Feature Implications	Yes / No	Additional Information
Configuration	Yes	
ADMS Archive	No	
User Interface	No	
Demonstration Available	No	

21.1.3. Viewer Configuration Editor

A new option has been added in the Structure tab of the Viewer Configuration Editor named "Feeder Required". This checkbox was added to optimize the performance of the Structure Layer drawing. If the checkbox is checked, then every structure must have a feeder id associated with it. It is not enforced by default for backward compatibility.



Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	Yes	Screenshot included
Demonstration Available	No	

21.2. What's Changed in 16.8.0

The following changes have been made in this release.

21.2.1. Supported Software and Hardware Changes

This section describes changes to the system such as changes in system requirements, compatibility, or third-party software. The following changes has been made in this release:

- Foundation 2024-04 version is used
- Forecasting 3.5.1 & 16.8.0
- Gateway 2.5.3 & 16.8.0

Refer to [System Requirements and Software Compatibility](#)

21.3. What's Fixed in 16.8.0

This section lists the notable issues addressed in this release.

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
AMI	ETD-82137, 134706	"Call Ping Popup - Single Customer" not working with Address values containing Commas	High	Updated code if Customer Model contains commas
Archive	ETD-82219, 137542	Minimum permission set needed to run archive for archive user in SQL Server Database	Medium	Database
Core, UI	ETD-82101	Performance degradation of client with full structure layer on	High	Client drawing code; structure serialization
DMS	ETD-82117, 134653	Recloser Device Search Inconsistent	Medium	DNOM search index
DMS	ETD-82133, 135801	Display URL command changed to all uppercase when called from DMS toolbar	Medium	Client web navigation
DMS	ETD-82145, 135834	CBO does not appear in study mode when isolating a downstream device	Medium	NTP CBO check
DMS	ETD-82192, 137443	DMS Crash during storm restoration	Medium	NTP outage island processing
DNAF	ETD-82125, 134583	Inconsistent BLA status	Low	Not a defect; no code change
DNAF	ETD-82135, 134665	FISR not implementing plans in closed loop, citing voltage violations, even with 'Calculated Voltage Violation Limit Violation not worse' checked.	Medium	FISR violation check
DNAF	ETD-82150, 135704	DNAF Process restarts at mid night and sending out team switch control commands	High	Measurement processing
DNAF	ETD-82157, 135590	Crowded and un-readable text on LVMSM Webpage	Low	LVMSM UI
DNAF	ETD-82190, 137444	DNAF Crash on LOV	Medium	LoV network selection
E-OMS	ETD-81884, 128101	Client unresponsive when multiple order worksheets are open	Medium	Oms Display plugin querying of Order data
E-OMS	ETD-81914, 128099	Enrouting Crew to second order changes incident working status of the first	Medium	Crew Assignment processing on Orders

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
E-OMS	ETD-82154, 135756	Assigning a crew to a manually created task is auto arriving them	Medium	Workorder service checks incident status instead of order status
FISR	ETD-82162, 135596	Parent Station doesn't restore the child station during FISR	Medium	FISR LoV sub-fed network selection
FISR	ETD-82163, 135699	Child station won't trigger FISR, if parent energizing feeder is disabled.	Medium	LoV triggering
FISR	ETD-82200, 137484	Child's FISR Triggering issue	Medium	FISR sub-fed triggering
Interface	ETD-79191, 134812	API not searching filtering correctly	Medium	Querying the calls through Oms Interface Server API
Internal Notification	ETD-82182, 137505	Corenotes Payload too large error	High	Core Notification Server sends station wise device data as payload
Modeling	ETD-81808	Composite switch momentarily changes to "preferred" after abnormal energization is removed	Medium	NTP Composite switch processing
NRT	ETD-82159, 135809	NRT Call Record Journal Timestamp Not Updated By ArchiveApp	Medium	ArchiveApp - Update Call Table
OMS	ETD-81965	"Unknown" closed device ID in restoration details of throw-over switch activation order	Medium	OMS Server - Incident processing
OMS	ETD-82038, 137447	Delay After Energization Timer Not Working Correctly (leads to double counting of customers)	Medium	OMS Server - Incident Processing
OMS	ETD-82056, 133949	Investigation Order created from U-Call shows one affected customer in order worksheet	Medium	OMS Server - Incident processing
OMS	ETD-82096, 134944	Missing switching step when ami_in comes in	Low	Single Customer incident creation through Power In and Power Out of Customer.
OMS	ETD-82113, 134184	"Create Crew" from Grid Tabulars resulting in Blank Popup	Medium	Online crew modeling

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
OMS	ETD-82114, 134156	NetServer and NetDataServer exit when customer model pushed by File Updater	High	NetServer reading the newly pushed Customer model file
OMS	ETD-82118, 134842	During the 4/10 storm OMS systems lags keeping the NTP delta high on processed and received messages	High	OMS Server - NTP Processing
OMS	ETD-82147	Non-Outage IncidentUpdate Crew Server msg POST Duplicates Calls	Medium	OMS Server - Incident Customer Association
OMS	ETD-82188, 137244	NetServer Crash - Enumeration operation may not execute	Medium	OMS Server Query Engine
OMS	ETD-82225	Continuous netserver.exe aborts due to dictionary insert	High	NTP processing during topology change on phase level
OMS	ETD-82198	IncidentStatus oData not updating correctly on WFM UpdateIncident	Critical	OMS Server - Incident status updates.
OMS, OMS-Crew, Outage Assist	ETD-81984, 132264	OA - Crew Icons for Multiple Assigned Crews Overlapping	Medium	OMS Server - Spread out crew locations
OMS-Incidents	ETD-82116, 136409	Cannot merge non outage to predicted outage from tabular	Medium	Merge incidents from UI
SWO	ETD-80206, 135807	Pop-up is not shown to user while performing actions on Composite switch in a switching order	Medium	NTP CBO check for composite switch from Switch Order
UI	ETD-81837, 137449	Power Flow Direction Indicator	Medium	Client drawing code on attribute popup
UI	ETD-82111, 134183	Boundary switch Marked in the geographic view is not shown as marked in AFS schematic	Medium	Client Automatic Feeder Schematics
UI	ETD-82136, 135803	Tabular display auto-refresh clears the dynamic filter	Medium	Client - tabular dynamic filter
UI	ETD-82149, 135800	Group Control: With browser window maximized, group control scroll-bar at the bottom for sideways scroll is not visible	Medium	Client - Group Control Display

22. ADMS 16.7.0

This chapter describes the incremental new features and defect fixes that were added since ADMS 16.7.0. The remainder of this document provides additional release-specific information that also applies to this release.

22.1. What's New in 16.7.0

No new features have been added in this release.

22.2. What's Changed in 16.7.0

The following changes have been made in this release.

22.2.1. Deprecated Features

There are no deprecated, removed, or changed features in this release.

22.2.2. Supported Software and Hardware Changes

This section describes changes to the system such as changes in system requirements, compatibility, or third-party software. The following changes has been made in this release:

- Foundation March-2024 version is used
 - PDI is dropped
 - Agro CD/Helm charts is now used for installation
- Habitat 5.12.1
- Scada 3.6.1
- APF 4.4 SP2
- Browser 4.4 SP2
- Forecasting 3.5 & 16.7.0
- Gateway 2.5.1 & 16.7.0

Refer to [System Requirements and Software Compatibility](#).

22.3. What's Fixed in 16.7.0

This section lists the notable issues addressed in this release.

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
Archive	ETD-82018, 132296	Cancelled SWO step ends up in switching_step_incident table	Medium	Switching Orders
Core	ETD-82008, 131058	Temp Ground on a line not able to remove and giving a message that Cannot found TempGround	Medium	Dynamic Module
DMS	ETD-82036, 133584	ADMS Operator Client crashing while performing testing on this recently received kit- 315_20240222.2	High	ADMS Client
DNAF	ETD-81937	Meter loads under primary metering points not being recognized by Power Flow	Medium	Converter
DNAF	ETD-81971	Lockouts older than 60 seconds impacting triggering causing incorrect triggers	High	FISR Triggering
DNAF	ETD-82002, 130930	LVM plan execution terminated in fast execution operation mode	High	Plan Manager
DNAF	ETD-82015, 131349	LVMSM Profiles display causes APF to hang	High	ADMS Client
DNAF	ETD-82016, 131350	DNAF Calculation popup refreshing during DSDR mode	High	ADMS Client
DNAF	ETD-82039, 133246	BLA/DPF is showing odd behavior for certain sets of measurements	High	BLA
E-OMS	ETD-82043, 133307	Advanced Restoration Steps doesn't store incident ID when opened second time	Medium	ADMS Client
Installation	ETD-81996, 133709	NetServerRO needs to be added to idms.appdef	Low	Applications to run under Procman Control
Modeling	ETD-59713	Converter error messages are missing in the log file	Medium	Converter
Modeling	ETD-81929, 132131	Customer Compiler Tool doesn't complete while running in batch mode	High	Modeling Tools - Customer Compiler
Modeling	ETD-81997, 130786	Encryption Tool's Grant Access button is not granting key access to other user	Medium	Encryption Tool (MMA Interface)
Modeling, Tools, Tools-Modeling	ETD-82020	Model Manager is Hanging - Suspected on User Preferences	Medium	Modeling Tools - Model Manager
OMS	ETD-81664, 124095	OA - Performance Issue when clicking between open Order Worksheets	Low	Outage Assist
OMS	ETD-81886	Oldest trace not cleared before reusing color on new trace	Low	Tracing
OMS	ETD-81894, 127991	Incident Management - Trip and Reclose actions on breaker script fails to show steps with second and third run	Medium	Update Incident dialog - Switching Steps
OMS	ETD-81917, 128252	Viewer Configuration Editor Lines Tab Preview Issues	Low	Viewer Config Editor - Lines tabular

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
OMS	ETD-81945, 128903	Data Correction Tabular doesn't show Dispatcher and Crew Comments	Low	Data Correction Tabular
OMS	ETD-82027, 132263	Meter Analysis not triggering pings from ADMS	Low	Order Worksheet - Meter Analysis tab - Documentation
OMS	ETD-82029, 132297	Area Setting LastUpdate time gets updated on model push	Medium	Incident Journal - AREANOTEUPDATE entries
OMS	ETD-82031, 132135	Structure symbol Icons don't resize according to the zoom level matching structure sizes configured in in Object Properties	Low	Viewer Config Editor - Structure tab
OMS	ETD-82041, 133253	Unable to change Area Note start time	Medium	Area Settings
OMS	ETD-82044	NetServer Crashed during Affected Customers Query (further occurrence)	High	NetServer
OMS	ETD-82057, 133050	ADMS stops displaying new orders when there is an order with non-existent crew	Medium	Order and Incident lists
OMS	ETD-82073, 133214	"IsOverhead" value incorrect when incident created from Jumper+Cut	Medium	Real Time Journal
OMS	ETD-82075, 133311	Storm Assist - Suspended order does not auto remove crew	Low	Order List
OMS	ETD-82076, 133275	Open Delta Transformers customers list contains all customers on the feeder	Medium	Map - RMB - Call Ping Popup options
OMS	ETD-82086, 133601	NetServer crash due to System.NullReferenceException (AddRowsToAffectedCustomersQuery)	Critical	NetServer
OMS	ETD-82090	Unplanned failover on PROD OMS which caused by process failure on enable server and standby server was not able to take over became disable, the BUCC server became new enable with a number of fake old incident appeared in outage portal as a result	High	NetServer
OMS	ETD-82097, 133947	ADMS is setting Start Time with End Time of the closed order	Medium	Incident Start Time
OMS	ETD-82099, 133946	VCE Convert BMP to PNG Color Conversion Broken	Medium	Viewer Config Editor - Conver BMP images to PNG
OMS	ETD-82105, 133969	Mutex issue when customer model pushed by file updater	High	File Updater
Outage Assist	ETD-81958, 134032	OA: AOR- Order transfer organization issue	Low	ADMS Client, Outage Assist - Order List - Transfer Organization

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
Outage Assist	ETD-82123, 134646	OA - Cause not auto-updating when Incident Page is open	Medium	Outage Assist - Incident Worksheet
SA	ETD-82103, 134116	Storm Assist can update locked orders though right click menu	Medium	Outage Assist - Order List
Tools	ETD-81991, 130323	Event-Scripter Automatic Navigation doesn't work	High	DOTS
Tools	ETD-82088	Converter Throws an Exception for Customer Data	Medium	Converter
UI	ETD-73689, 128731	Group Control: No quality status/tag status available for the devices which show up in Group Control UI	High	ADMS Client
UI	ETD-82067, 133330	Slowness in ADMS application performance when working with graphics.	High	ADMS Client
UI	ETD-82068, 133216	Temporary Device doesn't exists notification is shown in CBO popup	Medium	CBO Popup
UI, APF	ETD-82119, 134582	Client crash after deploying new network models	Medium	ADMS Client
Mobility	MOB-429	Mobility Gateway should not send Mobile payloads where orderStatus is "IN_PROGRESS" and incident id is null	High	Mobility Gateway
Mobility	MOB-430	When orderStatus is "FIELD_COMPLETE" and incident id is null and deviceType is null, remove the incident section from the payload and provide a value 'UNKNOWN' for deviceType	High	Mobility Gateway
Mobility	MOB-432	Payloads for Trouble Orders does not have geoLocation coordinates anymore	High	Mobility Gateway

23. ADMS 16.6.0

This chapter describes the incremental new features and defect fixes that were added since ADMS 16.5.0. The remainder of this document provides additional release-specific information that also applies to this release.

23.1. What's New in 16.6.0

No new features have been added in this release.

23.2. What's Changed in 16.6.0

This section includes changes to existing features, as well as deprecated or removed features.

23.2.1. Deprecated and Removed Features

There are no deprecated, removed, or changed features in this release.

23.2.2. Potential Upgrade Requirements

This section describes changes in the system that may impact your end user workflows, particularly if customizations have been made in your system. You should review these changes and make appropriate adjustments as part of your upgrade process to ensure that all workflows are preserved.

23.2.2.1. Netviewer Adapter Configuration File Update

After upgrading to ADMS 16.6.0, users must change the TileServiceCompression setting to gzip (or none) in the NetViewerAdapterConfig.xml file.

The config definition looks like this:

```
<!-- Compression to apply to published tiles & configurations. -->
<!-- Valid values: "br" (Brotli compression), "gzip" (Gzip compression), and "none" (no compression) -->
<!-- Note that Brotli compression does not work in some versions of .NET -->
<TileServiceCompression>gzip</TileServiceCompression>
```

23.2.3. Supported Software and Hardware Changes

This section describes changes to the system such as changes in system requirements,

compatibility, or third-party software.

With the 16.6.0 release, there has been a migration from .NET Core 6 to .NET Core 8. As a result, installation of .NET Core 8 is necessary to ensure compatibility and optimal performance with the application.

Refer to section [Prerequisites](#).

23.3. What's Fixed in 16.6.0

This section lists the notable issues addressed in this release.

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
Archive, OMS	ETD-81999, 130729	Incident Chronology Entries Not Writing to RealTime Incident Journal	Medium	OMS Chronology entry
Core	ETD-81976	Excessive logging on independent follower DMS	Medium	Follower ADMS Main and DNAF Servers
DMS	ETD-81887, 128232	Manual entries for Primary Taps on The Independent Follower	Medium	Tap Changer Manual Entries
DMS	ETD-81950, 131041	Multiple Consoles unable to search, remove tags, or toggle assets	High	Main Server core improvements of handling of sending dynamics messages to Standby Server and processing of SCADA measurement messages.
DMS	ETD-81967, 130292	Right-click options predict and move up not functioning with dependent DMS Follower integrated	High	OMS Server Incident Processing
DMS, UI	ETD-81724, 125050	Slow Initial Loading of Geographic/Schematic View in Browser because of Geojson Files	Medium	Client Structure Information Loading
DNAF	ETD-80770	DNAF performance issue: while triggering manual DPF solve. DPF solve does not start (queued DPF runs)	High	DNAF Manual Triggering
DNAF	ETD-81824, 128477	DNAF Server in production - memory spiking to 28GB	High	DNAF Server
DNAF	ETD-81931, 128516	No Measurements in Triggering File	High	FISR
DNAF	ETD-81959, 130033	Capacitor State Mismatch between SCADA and DNOM	High	DPF, LVM, and UI
DNAF	ETD-81990, 130189	Tag Type missing from the LVM Device Exclusion List display	Medium	LVM Device Exclusions List Display
DNAF	ETD-82009, 131057	Inconsistent DNAF calculation popup after LVM Profile change	Medium	DNAF Calculation Pop-up Display
DNAF	ETD-82000, 131063	DPF Backcalc with zero voltage values	High	Power Flow

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
DNAF	ETD-82011, 131065	LVM Profile Names do not allow "%" and "."	Medium	LVM System Monitor Profiles
E-OMS	ETD-81865, 126937	Work model DB sync writes null values	Medium	Work Model DB Sync
E-OMS	ETD-81888, 127670	Work Model DBSYNC failed to replicate order data with NULL custom field value	Medium	Work Model DB Sync
E-OMS, OMS	ETD-81926, 128749	Validation Rules Stop Working when Passed on Previous Order	Medium	Incident Clearing Validation logic
E-OMS	ETD-81944, 130357	OMS_INCIDENT_LOCKOUT_TIMER not working	Medium	Work Order Loader
FISR	ETD-81852, 127158	FISR respecting reach violation on station outside of switching area	Medium	FISR Solution
History Correction	ETD-81964, 129925	Order Custom Field Name values getting stored inconsistently	Low	Chronology Message Configuration
Internal Notifications	ETD-81877, 127696	Internal Notifications - popup spelling issue	Medium	Internal Notifications UI
Internal Notifications	ETD-81993, 130368	Corenote Certificate Error Incorrect Logging message	Medium	Internal Notifications logging
Modeling	ETD-81839	Power flow and voltage discrepancies in switching devices station internals/externals	Medium	DMS Main server query processing
Modeling	ETD-81910, 131766	Unexpected De-energization in Study Mode	Medium	Interface Modeling
OMS	ETD-81010, 127891	Ping popup for Single Customer & single service point shows up empty	Medium	OMS Server Query Processing engine
OMS	ETD-81761, 124424	Unassociated calls showing as double in the order worksheet	Medium	Work Model Data Provider
OMS	ETD-81838, 128247	Error popup when there are many historical incidents in history correction	Medium	Oms Display Plugin Data Service connector
OMS	ETD-81861, 128104	NULL End Time for orders in Data Correction (caused CMI to grow)	Medium	Incident Processing Code
OMS	ETD-81863, 127426	OMS - New "Trouble Clear" incidents are created each night, shortly after midnight or when models pushed to abnormally feeding stations.	Medium	Incident Processing Code
OMS	ETD-81873, 128102	No Outage on an order is not always excluding cmi	Medium	OMS Server Incident Processing Code
OMS	ETD-81895, 128500	Predicted fuse order reopened with new switching step by ADMS sometime after order auto cleared by AMI Power-In event	High	OMS Server Incident Processing
OMS	ETD-81904, 127990	AOR Not Assigned for Single Customer Outages	High	Network Topology Processing code
OMS	ETD-81920, 128796	Incident Priority not updated when Investigation and Maintenance Orders are completed	Low	Incident Processing code

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
OMS	ETD-81932, 128524	Duplicate incidents being created on the same service point	High	Incident Processing code
OMS	ETD-81953, 129373	Call Aggregation Not respecting set rules	Medium	Oms Configuration Server tool
OMS	ETD-81962, 129541	Customer Correction empty service point	High	Customer Correction Popup
OMS	ETD-81973	OA: Show "Laterals" Shows Labels As Well	Low	Net Viewer Adapter
OMS	ETD-81980, 130537	Auto cleared momentary incident and order status discrepancy	Medium	OMS Server Query Engine
OMS	ETD-81982, 130839	AMI PRN event restores customers but do not create 'closed' Switching Step	Medium	Incident Processing code
OMS	ETD-81994, 130799	Call lingers around even after order is closed	High	OMS Server call handler processing
OMS	ETD-81995, 130344	Restored orders showing as predicted on map	High	OMS Server Incident Processing
OMS	ETD-82010	AMI messages from Old restored Orders getting associated to new orders	Medium	OMS AMI Handler
OMS	ETD-82035	OA - Server Name Displayed in Chronology	Medium	Server User Name processing
Outage Assist	ETD-81930, 128456	OA: Globe icon on facility details does not work for Maintenance Order	Low	OMS Order Worksheet
Outage Assist	ETD-81933, 128750	OA - Device Search Failing for Reclosers and Fuses	High	Outage Assist UI
Outage Assist, SA	ETD-81969	SA cannot print to pdf order parameter that come from coded translation editor	High	Outage Assist UI
Outage Assist	ETD-81979	OA - Transformer Labels not displaying on Map	Medium	NetViewAdapter tile processing
Outage Assist	ETD-82001	OA - sending double ping requests when pinging from the map	Medium	Outage Assist UI
SIM	ETD-81883, 128231	MPP 3-phase close is broke, 3-phase open works	High	Measurement Modeling
UI	ETD-80153	Tabular Export csv encoding	High	Grid Tabular Client processing
UI	ETD-81870, 128103	Order worksheet chronology scroll down bar has an infinite scroll	Low	Order Worksheet
UI	ETD-81882, 128455	Group Control: Add "Last Operation Time" for the controlled devices to reflect when it was last operated.	Medium	Group Control UI
UI	ETD-81946, 129372	Moving annotation on the screen clears trace marks	Low	Client Drawing

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
UI	ETD-81981, 130237	Temp Ground&Fault Char to be increased in proxysrv.almmsg for removed and PCname are not pointed to right PC for Ground and Perm Fault	Medium	Alarm Messages

24. ADMS 16.5.0

This chapter describes the incremental new features and defect fixes that were added since ADMS 16.4.0. The remainder of this document provides additional release-specific information that also applies to this release.

24.1. What's New in 16.5.0

This section contains information about new features and functionality for ADMS 16.5.0.

24.1.1. Process Automation Application

24.1.1.1. Allow Reprocessing of Orders for Auto-Assignment

The Process Automation application has been enhanced to reconsider orders under the following conditions:

- Any changes in the values of IncidentType, SystemMode, NonoutageType, DeviceType, AffectedCustomersCount fields.
- Until the orders match with the specified Time Range, Date range.
- Availability of crews with configured crew statuses. (ADO-123662, GRDDMSS3-1851/1878)

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	No	
Demonstration Available	Yes	Click here for demonstration

24.1.2. Process Automation Configuration Editor

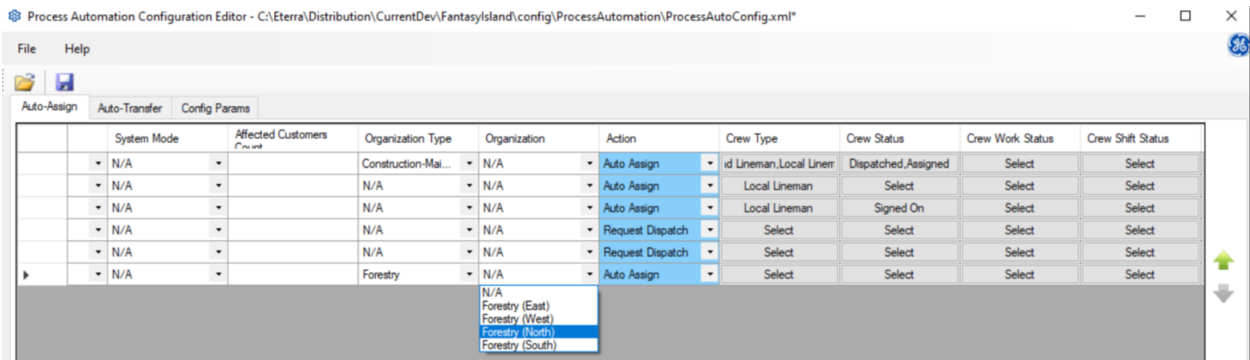
24.1.2.1. Auto-Assign Tab (EOMS)

Two new columns have been added to Auto-Assign tab. (ADO-123661, GRDDMSS3-1853)

Organization Type: Organization type as defined on the Organization tab in the OMS Server Configuration Editor. This matches the mapped organization where the order belongs to.

Organization: Organization as defined on the Organization tab in the OMS Server Configuration

Editor. This matches the organization where the order belongs to.



Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	Yes	Screenshot included
Demonstration Available	Yes	Click here for demonstration

24.1.3. Improved Drag and Drop Functionality for Switching Order

The drag and drop feature for switching order has been improved. It now automatically adds the device as a new step at the bottom, preventing accidental edits to existing steps and improving the user experience. (ETD-81797)

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	No	
Demonstration Available	Yes	Click here for demonstration

24.1.4. Check Before Operate for Composite Switches

Implemented the Check Before Operate functionality for composite switches when accessed from Switch Order. (ADO-123583, GRDDMSS3-1887)

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	No	
Demonstration Available	Yes	Click here for demonstration

24.2. What's Changed in 16.5.0

There are no deprecated, removed, or changed features in this release, nor are there any changes to supported software or system requirements.

24.3. What's Fixed in 16.5.0

This section lists the notable issues addressed in this release.

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
Archive	ETD-81645, 125041	Incorrect Start Time in Device Feeder Table	Medium	Archive Stored Procedure
Archive	ETD-81720, 124387	Archive tables (EMPLOYEE, VEHICLE and CREW) data found empty	High	NetDataServer NRT processing
CallServer	ETD-81773	Documentation - Call Server CallMessages requires UpTo in Response	Low	CallServer schema made UpTo field as not nullable
Core	ETD-81486, 124249	Loop fed stations not showing as de energized (No white halo) 0kV with angle in RT mode	Medium	Topology with Looped Sub-fed Stations
Core, SCADA-DMS	ETD-81780	DMS to Scada Energization Point Defaulting to Suspect when ISD Site Cycles	Medium	DMS main server
Core	ETD-81808	Composite switch momentarily changes to "preferred" after abnormal energization is removed	High	Composite Switch Energization Handling
Core	ETD-81835	Dangerous Switching Condition Incorrectly Shown on Geographic	Medium	DMS core. Added a new way of phase angle mismatch check
Crew Management , OMS	ETD-81804, 125835	Dispatch Crew is Inconsistent on Incident Multiselect	Medium	OMS server Crew Management logic
DMS	ETD-81600, 127068	ADMS network, schematic and one-line views on SIM Display	High	Client Automatic Feeder Schematics
DMS, DNAF	ETD-81834	LVM Infeasible LP case	Medium	DNAF LVM Module

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
DMS	ETD-81850	LOCATE netview command selects annotations instead of devices	Medium	Client Annotation selection
DMS, Interface Server	ETD-81860	DMS Data Service: Abnormal are not fetching updated values on request	Medium	DMS Interface Server
DNAF	ETD-80619	DNAF/NTP triggering too much	Low	DPF Result File Processing
DNAF	ETD-81758, 124032	FISR LOV Fails to Generate a Plan when Breaker Measurements are Suspect (Fault Placed on Breaker Zone)	Medium	FISR Loss of Voltage Triggering
DNAF	ETD-81828	FISR Log: Station Default Problem Formulation	Low	FISR Result File
DNAF	ETD-81830, 125705	FISR doesn't take default Problem Formulation	Medium	FISR Solution Configuration
DNAF	ETD-81840	Incorrect limit 1 for distribution transformers	Medium	Power Flow Limit Monitor
DNAF	ETD-81848	FISR Auto-Triggers and drops bolt but does not recognize the bolt	Medium	FISR Triggering
Documentation	ETD-81781, 125830	Running Customer Loader in a redundant configuration	Medium	Updated distribution_customer_list_compiler_user_guide
E-OMS	ETD-81621, 124289	ETR Field Not Displaying in Order Worksheet in Status Section	Medium	Order Worksheet
E-OMS	ETD-81787	Downgrading a predicted tx outage did not create a completed task on new order	Medium	Incident processing move down logic
E-OMS	ETD-81888, 127670	Work Model failed to replicate order data with NULL custom field value	Medium	Work Model Db Sync
FISR	ETD-81679	When ISD link is repaired between SCADA and DMS FISR is able to fire on "old" faults	Medium	FISR Fault Triggering
Interface	ETD-80582	CyberArk ddls should be delivered under server/bin, not under InterfaceSDKSamples	Medium	Interface kit
OMS	ETD-80485, 124385	Closed task is added after complete a Check Prediction task with No Outage	Medium	OMS Predicted Incident switching Step handling
OMS	ETD-80981, 124906	Open Geographic and Customer Tracing Performance Issues	Medium	DMS client NTP processing improvements
OMS	ETD-81479	Incident at Circuit Breaker being created with early Start Time	High	Incident processing StartTime logic
OMS	ETD-81509	Switchorder stops accepting new connections.	Medium	Switch Order Socket connection logic
OMS	ETD-81759, 124374	Incident Location Description, State, County, and Municipality not populated when Confirmed Order created within a predicted extent	Medium	Incident processing Split Incident logic

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
OMS	ETD-81791, 125766	Customer count is duplicated when merging incidents in History	Medium	History Correction Merge Incident logic
OMS	ETD-81803, 125445	History Correction Filter Resets on Incident List Update	Medium	History Correction filtering logic
OMS	ETD-81818, 126361	History Correction tab network field reverts back to default network of "Zachary" immediately, when user selects a different network and then apply any other filter criteria	Medium	History Correction filter logic
OMS	ETD-81820, 126067	Symbol Icons from the structure layer won't resize when changing Bitmap Size	Low	Client drawing of Structure layer
OMS	ETD-81827	OA - Reader/View Only user is able to modify/write	High	Outage Assist Read Only logic
OMS	ETD-81847	No call type nor incident type when converting Non-outage order to Outage order	Medium	Incident Processing logic
OMS	ETD-81856, 126656	Isolated Incidents defaulting to Pending Clear Incident Status	High	Incident Processing logic for checking pending clear
OMS	ETD-81863, 127426	OMS - New "Trouble Clear" incidents are created each night, shortly after midnight or when models pushed to abnormally feeding stations.	Medium	Incident processing logic for cancelling an incident
OMS	ETD-81898, 127915	Call Aggregation Not working for Investigation orders	Medium	Documentation update
OMS	ETD-81903, 127985	NetServer Crashed during Affected Customers Query	High	Extra check added to prevent OMS server crash
Outage Assist	ETD-81892, 127854	Outage Assist text alignment issues with longer strings	Medium	Outage Assist UI
SWO	ETD-81786	Switch Order - if SWO is locked, User2 expects to see the message "Locked By User1" in the "State" section of the SWO	Medium	SWO lock processing logic
SWO	ETD-81797, 125851	Switch Order drag and drop changes existing step device without warning user	Medium	SWO UI Drag and Drop logic
UI	ETD-81309, 124911	AFS callup command fails to locate a line device	Medium	AFS client callup logic
UI	ETD-81795, 125236	Structure Information window does not behave as other information popup	Medium	Client Structure Information popup logic
UI	ETD-81837	Analyst Output: Power Flow Direction Indicator	Medium	Client Analyst Output drawing
UI	ETD-81866, 127066	Navigation from SCADA OneLine to DMS Geographic freezes the client	High	Client processing logic

25. ADMS 16.4.0

This chapter describes the incremental new features and defect fixes that were added since ADMS 16.3.0. The remainder of this document provides additional release-specific information that also applies to this release.

25.1. What's New in 16.4.0

This section contains information about new features and functionality for ADMS 16.4.0.

25.1.1. Process Automation Application

25.1.1.1. Evaluate Conditions for Process Automation Application (EOMS)

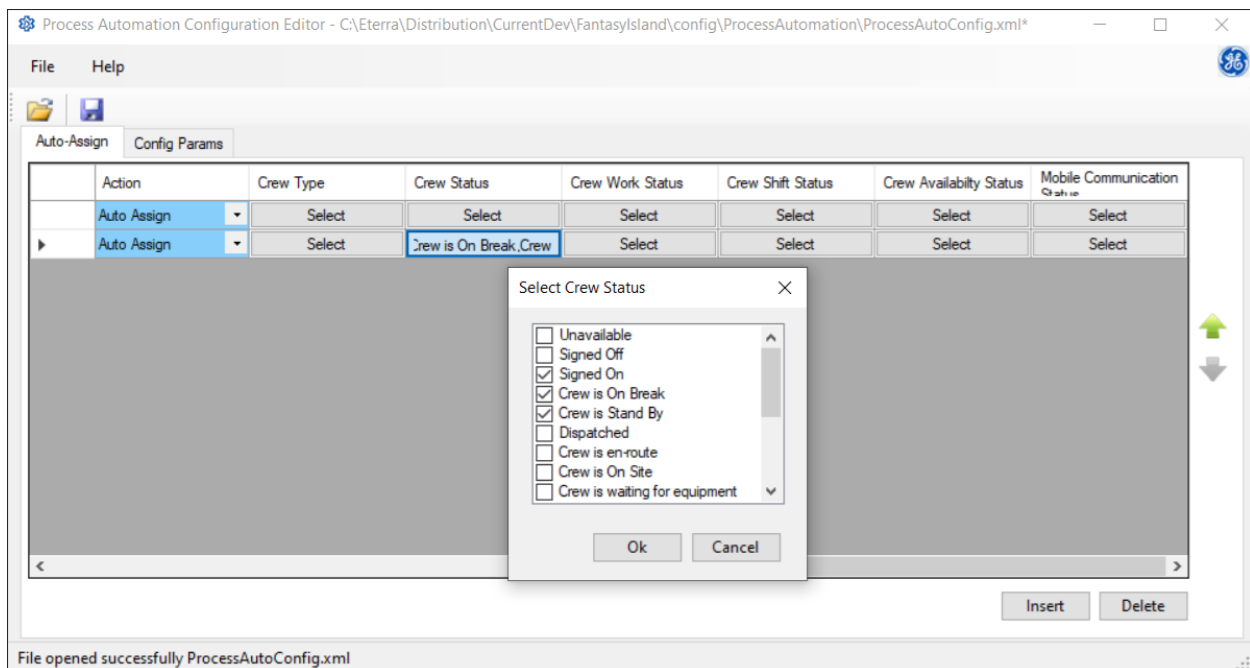
The Process Automation application has been enhanced to re-consider orders with an OrderStatus of New or Suspended and a SummaryCrewStatus of Unassigned or Released until the orders are deleted from Orders tab of the OmsDisplayPlugin. (ADO-123870, GRDDMSS3-1850)

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	No	
Demonstration Available	Yes	Click here for demonstration

25.1.2. Process Automation Configuration Editor

25.1.2.1. Auto-Assign Tab

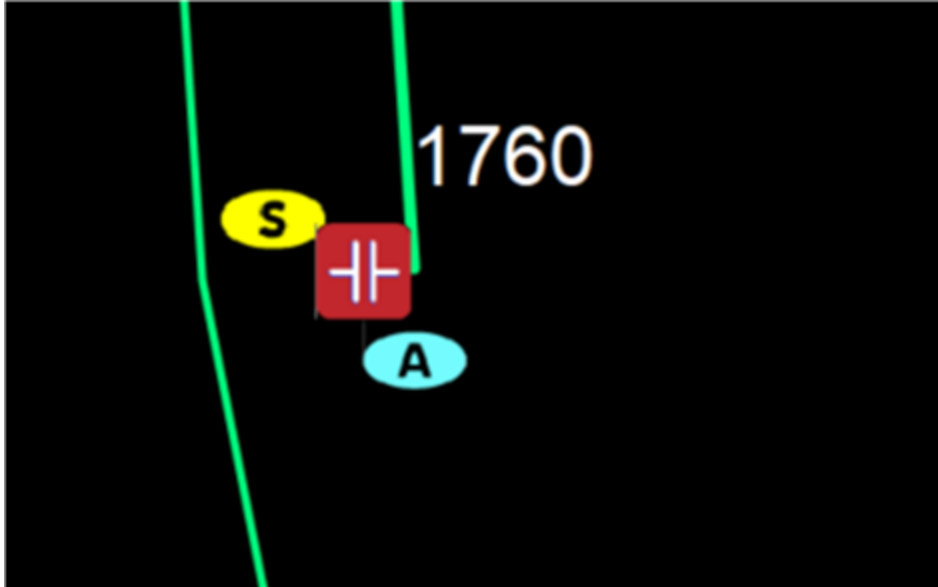
The Crew input parameter columns on the Auto-Assign tab has been enhanced to allow the selection of multiple crew statuses. These columns include Crew Type, Crew Status, Crew Work Status, Crew Shift Status, Crew Availability Status, and Mobile Communication Status. (ADO-123873)



Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	Yes	Screenshot included
Demonstration Available	Yes	Click here for demonstration

25.1.3. Suspect State of SCADA Capacitor

The SCADA capacitor will show a Suspect State when SCADA reports a quality of Suspect for a capacitor, as shown in the image below with [image3]. (ETD-81762, GRDADMS-1802)



Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	Yes	Screenshot included
Demonstration Available	No	

25.2. What's Changed in 16.4.0

There are no deprecated, removed, or changed features in this release, nor are there any changes to supported software or system requirements.

25.3. What's Fixed in 16.4.0

This section lists the notable issues addressed in this release.

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
Archive	ETD-80158	Build Version is 1.0.0.00000 in Archive logs (call, nrt, oms, and so on). Should be updated to reflect current ADMS/ArchiveApp version.	Low	ArchiveApp UI
Core	ETD-78931	Three phase recloser status measurement are not showing up in the measurement tabular.	Medium	DMS Query Engine

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
Core	ETD-80136	Updating Customers.bpcl crashed OMS Main and NetDataServer.	High	NetServer, NetDataServer, CallServer, DMS Client (Loading of Customer file)
Core, OMS	ETD-81659	Non-outages converted to outages on service point incidents confirm out not working properly.	Medium	NetServer (OMS processing)
Crew Management	ETD-81539	Crewserver not processing "comment:" field with some special characters and sets timestamp for next GET to 0001-01-01	Medium	SWO Web Service, Crew Server Web Service
DMS	ETD-81142	Trace Between function doesn't include taps with hypernode.	Medium	DMS NTP tracing
DMS	ETD-81403	Boundary CompositeSwitch current state changes after model update (Old Status are not being removed from Dynamics).	High	DMS – State preservation of Composite Switch
DMS	ETD-81415	DNAF crashes with the newer currentDev kit.	Medium	DNAF LVM module
DMS	ETD-81495	Future layer not working correctly.	High	DMS – Future Layer Processing
DMS, DNAF, Tools	ETD-81651	DNAF App logs doesn't show DEC version correctly (1.0.0.000).	Low	DNAF Server APP log output
DMS, DNAF	ETD-81799	DMS Main crash with DNAF high memory (>100G) usage.	High	DNAF LVM
DNAF	ETD-81505	Active Management inverters are not using measurements to determine KVAR output for LVM eligibility.	High	DNAF PowerFlow Shunt Processing
DNAF	ETD-81669	Fault not placed if a device status measurements are modeled as multi-phase point (MPP).	High	DNAF FISR Triggering
E-OMS	ETD-81468	OA user loses lock after client user opens and closes an order worksheet.	Low	Client - EOMS Order Lock Processing
E-OMS	ETD-81500	Losing lock no longer grays out order worksheet selections.	Medium	UI- Disable Button
Interface	ETD-80951	Lack of evidence in logging that Netserver OMS retrieving UAA ID and UAA secret from CyberArk when secret Custom store is defined in configuration for AOR permission manager UAA endpoint.	High	NetServer Logging
Modeling	ETD-81591	Substation Editor swaps power transformer nominal voltages after placing voltage points.	Medium	Substation Editor
OMS	ETD-80207	OMS - Current customer out count is incorrect, when a predicted stable single customer incident is merged to an upstream confirmed fuse incident.	High	NetServer - OMS Merge processing
OMS	ETD-81036	Incorrect Restoration Step "Time Off" & "Phase" Values For Partial Restoration	High	NetServer – OMS Processing

Component (Module)	Issue Number	Description	Severity	Applicable to ADMS Component
OMS	ETD-81257	OMSMAN (NetServer) crashed during Performance Test (CallHandler).	Medium	NetServer Cancelled Incident chronology processing
OMS	ETD-81347	High Priority Calls in OMS Server Config not working.	Medium	Oms Server Configuration Editor
OMS	ETD-81542	Netserver in crash loop.	High	NetServer - OMS Query Engine
OMS	ETD-81571	Deleting an Area Setting after Expiration Removes Storm Name.	Medium	NetServer - AreaNote Processing
OMS	ETD-81751	OMSMAN (NetServer) crashed after Performance Test (WorkOrder).	Medium	NetServer - WorkModel processing
Outage Assist	ETD-80779	Outage Assist Capacitor Bitmap doesn't match the capacitor status.	Medium	NVA - Processing of Capacitor Dynamics
Outage Assist	ETD-81436	Outage Assist tag symbology contains squared numeration.	Low	NVA - Tag decoration query
Outage Assist	ETD-81718	OA Data Correction missing device and device type on elbows.	Medium	UI - Data Correction Table
Process Automation	ETD-81595	Auto-Transfer not working for some Organization types.	Medium	ProcessAutomation EOMS Query Service
SWO	ETD-81006	Switchorder time validation issue.	Medium	SWO Server side validation
SWO	ETD-81191	Switch Order Schedule Start/End Time is shifted by 1 hour in the PDF exported.	Medium	SWO - Export to PDF
SWO	ETD-81316	Regressive Behavior in Custom 'ListBox' Field of Switch Order.	High	SWO Client WPF form
SWO	ETD-81477	Switchorder steps are not unselected when using return to normal.	Medium	SWO Client processing
Tagging	ETD-81618	COMPHASMULTTAG field is being populated with True even with just one Tag.	Medium	DMS - Query Module for the Device with Multiple Tags
Tools	ETD-81798	Converter hangs for long time (>2minutes) after station conversion.	Low	Converter tool performance improvement

26. ADMS 16.3.0

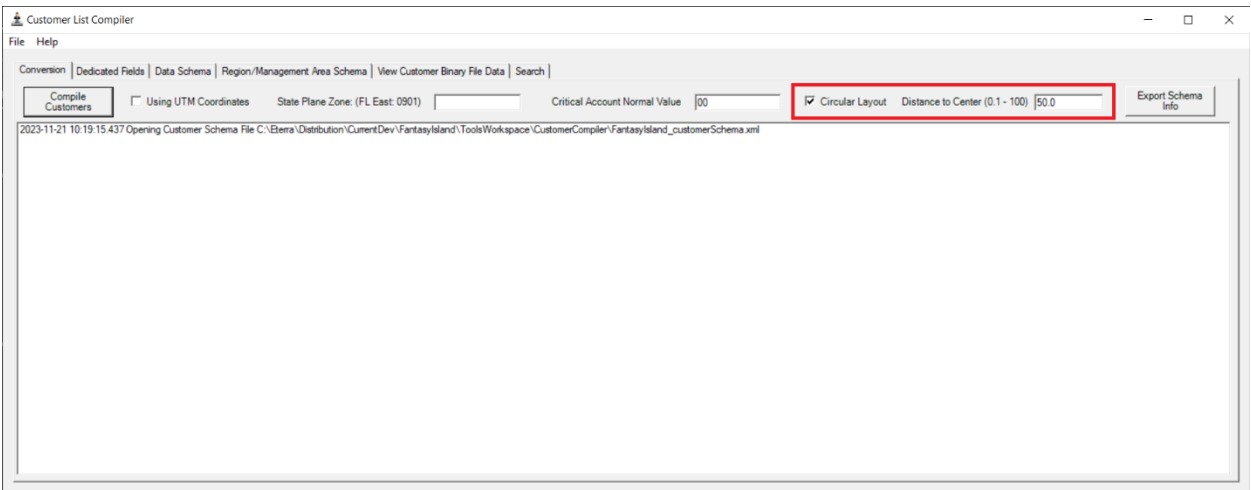
This chapter describes the incremental new features and defect fixes that were added since ADMS 16.2.0. The remainder of this document provides additional release-specific information that also applies to this release.

26.1. What’s New in 16.3.0

This section contains information about new features and functionality for ADMS 16.3.0.

26.1.1. Circular Layout

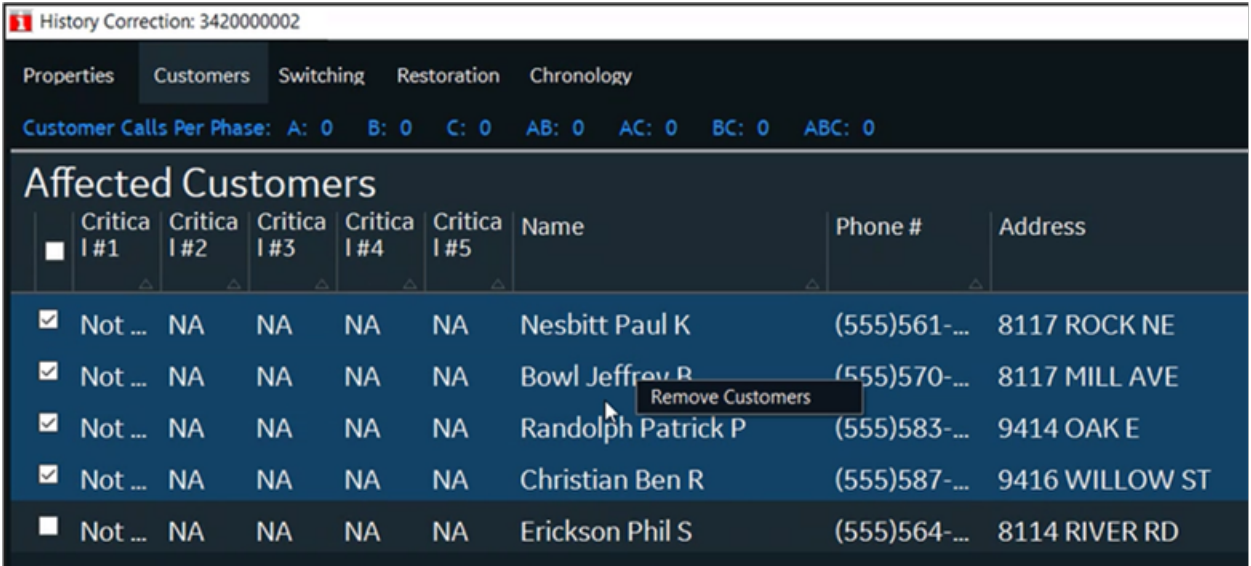
The Customer List Compiler has been enhanced to enable the spreading of customer positions during the compilation process. This new option can be enabled before compiling and will only apply to customers sharing the same X and Y coordinates. The distance to the center of the circle can be adjusted from 0.1 to 100. (ETD-80110)



Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	Yes	
ADMS Archive	No	
User Interface	Yes	Screenshot included
Demonstration Available	Yes	Click here for demonstration

26.1.2. Exclude Customer(s) from Outage

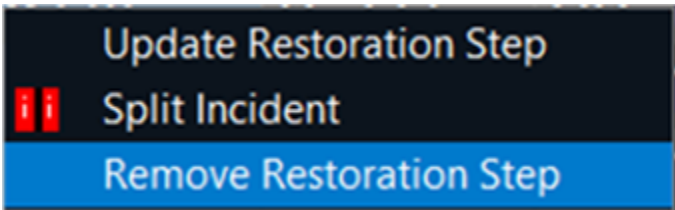
The History Correction window has been enhanced to allow the exclusion of specific customers from the outage scope. A new option, Remove Customer(s), is accessible through a right-click on the customer in the History Correction Customers window. (ETD-79843)



Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	Yes	Screenshot included
Demonstration Available	Yes	Click here for demonstration

26.1.3. Restoration Options in Data Correction

The restoration step during data corrections has been enhanced to allow an option to split and cancel the specific restoration step on an outage. A new option, Remove Restoration Step, is accessible through a right-click on the restoration step in the Data Corrections Restoration Tab. (ETD-79844)



Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	Yes	Screenshot included
Demonstration Available	Yes	Click here for demonstration

26.2. What's Changed in 16.3.0

There are no deprecated, removed, or changed features in this release, nor are there any changes to supported software or system requirements.

26.3. What's Fixed in 16.3.0

This section lists the notable issues addressed in this release.

Component	Issue Number	Description	Severity
Archive, OMS	ETD-81042	Archive DB Switching Step Incident Table Not Updating 'nrt'	Medium
Core, DNAF	ETD-80771	DMS tracing failed on lines with the path cannot be resolved error	High
DMS	ETD-81028	Grey halo obscures labels behind future layers on DMS geographic screen	Medium
DMS	ETD-81225	NOP states is not being preserved in ADMS after modeling push	High
DMS	ETD-81291	Using trace function on alt fed sections of line changes de-energized lines to energized	High
DMS	ETD-81333	Study mode CBO Pop-up doesn't follow the ST NetConf Pop-up BG Color, all other pop-up follows	High
DMS, DNAF	ETD-81280	Meshed FL Fails to Trigger Consistently – Long Delay Between Fault and DNAF Trigger	High
DNAF	ETD-80656	After running some time with ADMS clients, yellow border is showing up with GeoDisplays.	High
DNAF	ETD-80895	SubFed Stations Increase Feeder Numbers that Rejected by Application	Medium
DNAF	ETD-81244	Meshed FL Fails to Trigger Consistently – No Triggering Inside DNAF	High
DNAF	ETD-81457	DNAF Port Disconnected	High
DNAF, Modeling	ETD-81161	Adding fields: County, LocationDescription, Municipality, State; to Substation object doesnt work in DNOM	Medium
E-OMS	ETD-80536	Damage assessment value not reflected in tabular or print summary page	Medium

Component	Issue Number	Description	Severity
E-OMS	ETD-80964	CMI is inconsistent in Data Correction	High
E-OMS	ETD-81375	Missing horizontal scroll bar on Crew Advanced Tab	Low
Modeling	ETD-81151	Associating SCADA Measurements to Composite Switches	Low
Modeling	ETD-81314	Composite Switch Reference Objects	Medium
OMS	ETD-80483	Service Point Incidents auto-created after Scada Incident Clear	High
OMS	ETD-80721	No_Prediction_of_Good_Qual_Devices Configuration Not Respected	Medium
OMS	ETD-81199	Other trouble tickets reset and are no longer displayed from "Other Trouble" tabular	High
OMS	ETD-81465	Tracing up on a de-energized section show line back energized	Medium
OMS, UI	ETD-80646	Splitting Pending Clear Incident - No Warning Message/Popup	Low
OMS-Crew	ETD-80824	Unable to update crew status via Crew Update Dialog popup	Medium
OMS-Incidents	ETD-81266	History Correction tab network field reverts back to default network, when any user progresses an incident	High
Outage Assist	ETD-81474	Globe icon on facility details does not work for SNC and Investigation Order	Low
UI	ETD-81214	No text on buttons in area setting when opened again	Low

27. ADMS 16.2.0

This chapter describes the incremental new features and defect fixes that were added since ADMS 16.1.0. The remainder of this document provides additional release-specific information that also applies to this release.

27.1. What's New in 16.2.0

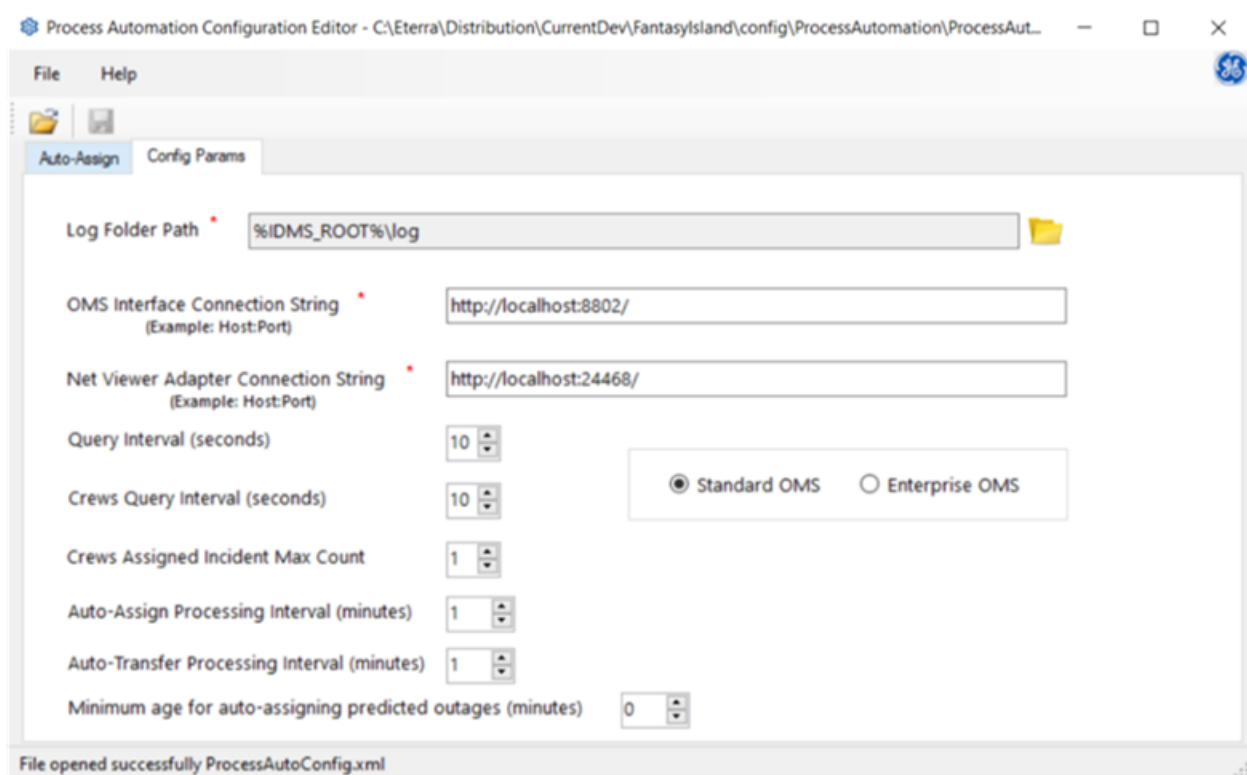
This section contains information about new features and functionality for ADMS 16.2.0.

27.1.1. Process Automation Configuration Editor

27.1.1.1. Configuration Parameter Tab

Configuration Parameter tab has been enhanced and added a field Minimum age for auto-assigning predicted outages (minutes).

Minimum age for auto-assigning predicted outages (minutes): Time period in minutes to process non-stable predicted outages for auto-assignment. The minimum value configured should be more than 0 for processing and the age of predicted outages should be over the configured value for crew assignment. (ETD-81016)

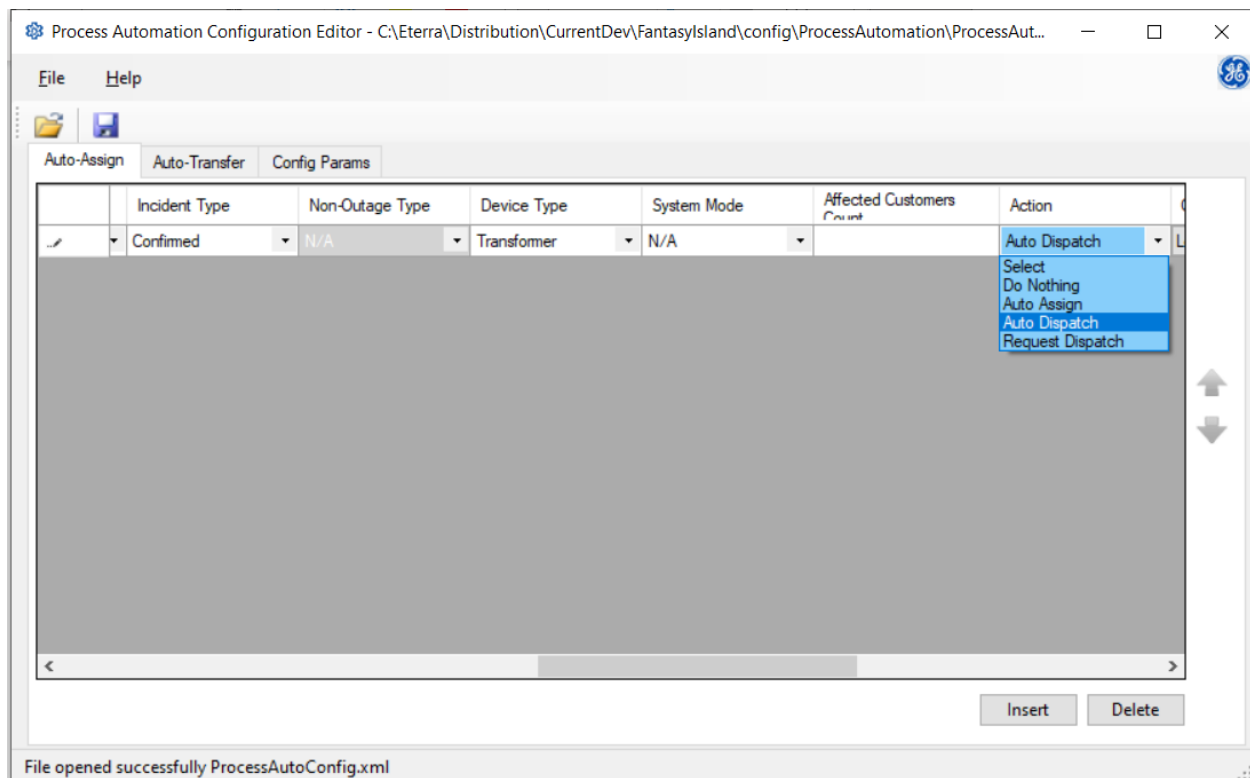


Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	Yes	Screenshot included
Demonstration Available	Yes	Click here for demonstration

27.1.1.2. Auto-Assign Tab

Auto-Dispatch: The Process Automation application dispatches a crew to the incident or order.

The Process Automation application tries to dispatch the nearest crew among crews whose type, status, and crew area based on service center match the rule. The application tries to allocate number of incidents or orders to crews based on configured max count. If the application does not find any crew that matches the selection criteria, the application does not dispatch any crew. (ETD-79473)



Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	

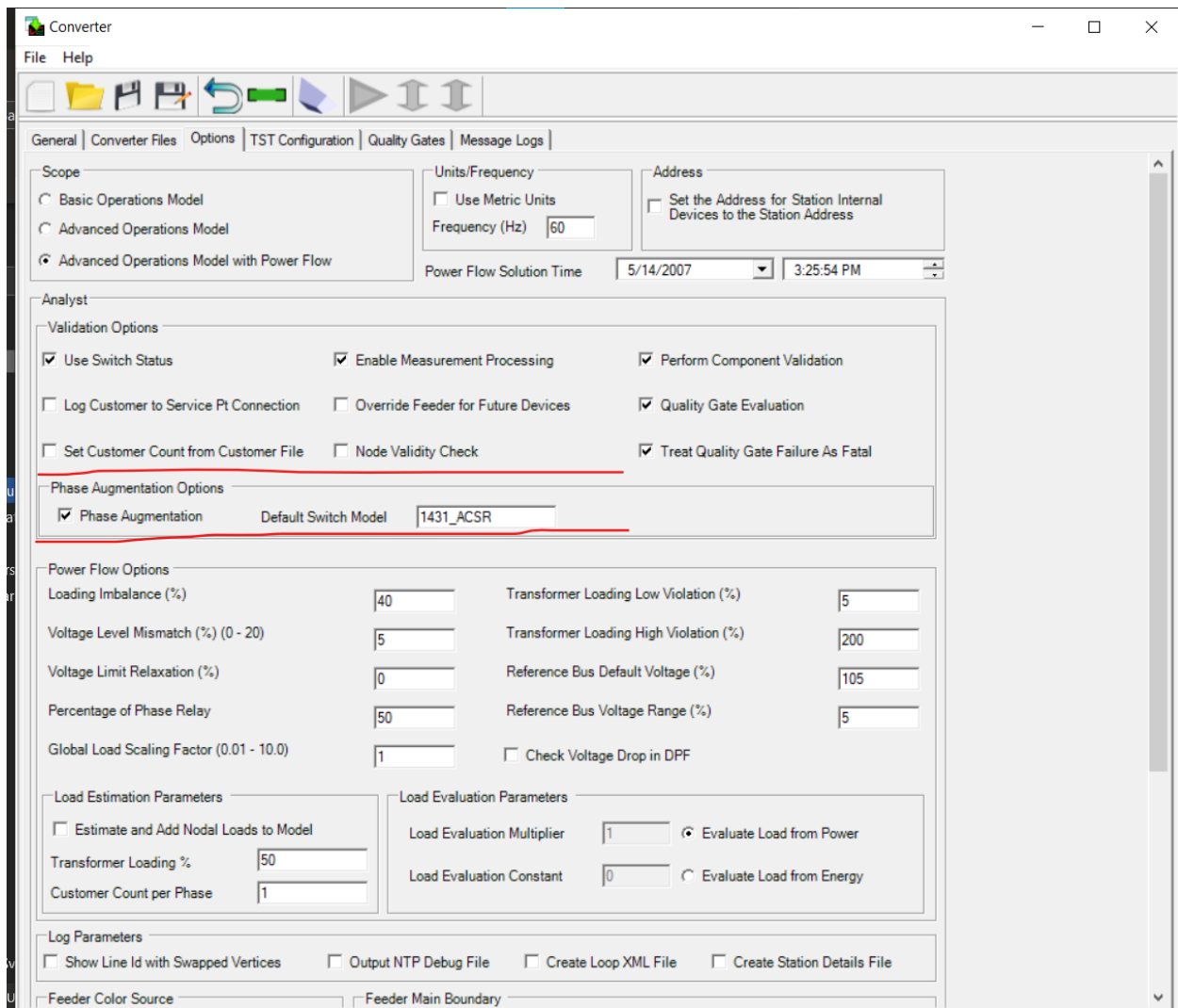
Feature Implications	Yes / No	Additional Information
ADMS Archive	No	
User Interface	Yes	Screenshot included
Demonstration Available	Yes	Click here for demonstration

27.1.2. Phase Augmentation

Phase augmentation adds jumpers to bridge missing downstream phases for switching devices and regulators.

- **Phase augmentation:** Determines whether jumpers will be added to bridge missing downstream phases.
- **Default Switch Model:** Displays the default assigned Switch Model for jumpers.

This feature does not apply for fuses that are part of a Underground Residential Loop (URD Loop) or any loop structure. (ETD-80110)

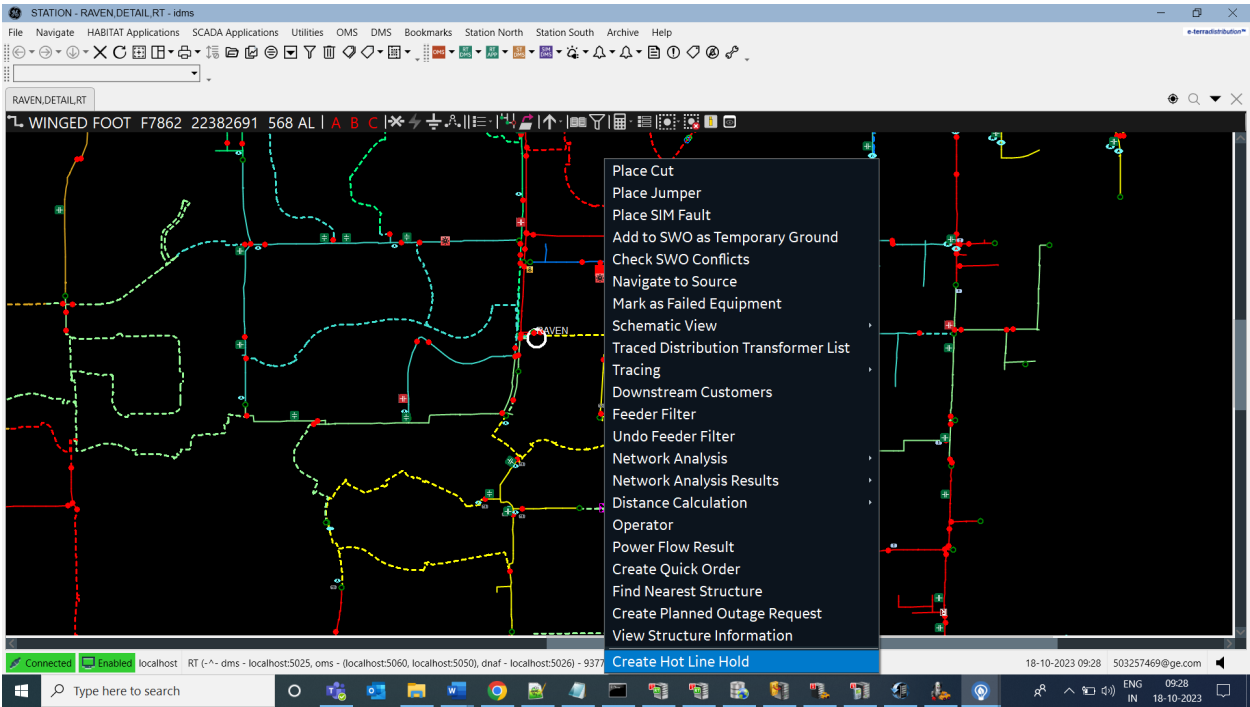


Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	Yes	
ADMS Archive	No	
User Interface	Yes	Screenshot included
Demonstration Available	No	

27.1.3. Enhanced Asset Safety Workflow

When creating a Hot Line Hold (HLH) or Energize Work Permit (EWP) Safety Document for an asset location, the system now seamlessly traces upstream to the protection device. This automatically adds the device to the switch order, creating two predefined switching steps in the safety document. Additionally, users have the flexibility to add these steps manually if needed. (ETD-

79841)



Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	Yes	Screenshot included
Demonstration Available	Yes	Click here for demonstration

27.1.4. Switching Order Enhancements

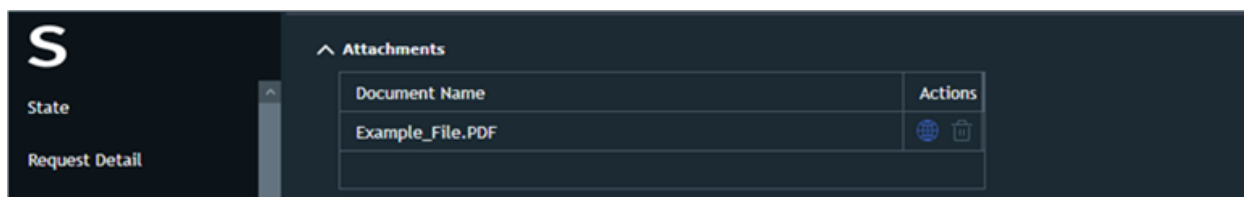
The following features were added:

1. New Attachments Section.
2. Updated Planned Request to be able to add multiple Switching Orders.
3. Implemented Open By record in Audit Trail (ETD-79816)

27.1.4.1. Attachments Section

Added the capability to attach documents and files to switch orders and other switching documents, such as planned requests, daily logs, and HLH.

Operators can now add external documents to the switching order. The files are listed in the new Attachments section and will be opened with the default windows application to read the document, based on the file extension.



Operators must ensure the attached file is in a shared location and accessible to all clients.

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	Yes	Screenshot included
Demonstration Available	No	

27.2. What's Changed in 16.2.0

There are no deprecated, removed, or changed features in this release, nor are there any changes to supported software or system requirements.

27.3. What's Fixed in 16.2.0

This section lists the notable issues addressed in this release.

Component	Issue Number	Description	Severity
Archive	ETD-78940	Incorrect 'Restoration_step_number' and 'Restoration_step_Id' columns data in Switching step incident table	Medium
Core, DNAF	ETD-78329	After DMS Process Recycle, DMS and SCADA Stopped Syncing	High
Core, DNAF	ETD-80620	Fault indicator and locked out status update issue if measurements are modeled as multiphase	High
DNAF	ETD-78294	A phase of Regulator missing voltage violation calculation in LVM	High
DNAF	ETD-80640	DNAF Port Disconnected	High
DNAF	ETD-80648	LVM result log is randomly missing daily	High
DNAF	ETD-80825	Station Outage Logic allows FISR Disabled Feeders to Operate	High
DNAF	ETD-80866	Incorrect back calc voltages on Canton 115KV	High

Component	Issue Number	Description	Severity
DNAF	ETD-80930	Back Calculation Selecting Incorrect Voltages	Medium
DNAF	ETD-81281	When system is updated, old tags can cause FISR to disable for the entire Feeder	High
DNAF, FISR	ETD-78387	Event Summary reports FISR Status every 3 hours	Low
DNAF, FISR	ETD-79278	FISR solving after re-enabling the test station.	Medium
DNAF, FISR	ETD-80593	Dual Production: FISR doesn't restore the whole child station when an LOV happens on parent.	High
DNAF, FISR	ETD-80850	Fault indicators won't show up in FISR triggering file	Medium
DNAF, Modeling	ETD-75337	Trace Down for a specific phase is not tracing all transformers with that phase	Medium
DNAF, SCADA-DMS	ETD-80641	DEMW ISD suspect or stale data issue	High
DOTS	ETD-81037	Multi-phase point lockout status not resetting if breaker is manually opened and closed	Medium
E-OMS	ETD-79613	EOMS - Cannot Close Out Abnormally Restored Orders	Low
E-OMS	ETD-79668	Order Worksheet not displaying user id for who has it locked	Medium
E-OMS	ETD-80531	ADMS Client stops responding when updating the System Contributing	Medium
E-OMS	ETD-80612	Populating End Time for an Order Marks task as No Outage	Medium
E-OMS	ETD-80616	Update Restoration Step End Time defaults to year 1753	Low
E-OMS	ETD-80673	AMI Power On is setting check prediction task to No Outage	Medium
E-OMS	ETD-80674	Customer who called on order worksheet missing info	Medium
E-OMS	ETD-80675	Incident/order state transitions very slow on FPL systems	High
Interface, UI	ETD-77653	Label for temp grounds still appears in unexpected location in 3.15	Medium
Modeling	ETD-80140	ADMS Model Exporter for DERMs network model (duplicates DER)	High
Modeling	ETD-80839	Customer Merge Tool is creating blank lines for customer records that are marked as deleted	Low
NRT, OMS	ETD-80226	Looping caused Excessive Journal Creation leading to Delayed Data in NRT	High
OMS	ETD-69293	FISR point not copying to safety document when used from daily log	Medium
OMS	ETD-74355	Incidents are not populating correctly in the AFFECTED_PREMISES table	Medium
OMS	ETD-78121	Closed device ID UNKNOWN in restoration step and no of restored customers are zero in switching step	High
OMS	ETD-79819	Current Incident Name column under Crew tab not showing accurate information	Low
OMS	ETD-80057	Splitting device from switching or restoration tab, with AOR as the Organization	Medium

Component	Issue Number	Description	Severity
OMS	ETD-80330	Merging Outages Opens/Confirms Device and Sets Crew Arrived/Onsite	Low
OMS	ETD-80520	Statistics Tabular not updated - SAIDI, SAIFI, CMI	Low
OMS	ETD-80622	AOR for primary meter customer incidents set to DEFAREA	High
OMS	ETD-80642	Customers out value delta between overview and incident tabular	High
OMS	ETD-80685	Restoration Steps appear incorrect when cut is removed	Medium
OMS	ETD-80720	Breaker Prediction after Step Restoration results in Dropped Incident	High
OMS	ETD-80765	Show/Hide Function Performance Issues	Medium
OMS	ETD-80793	"Create and Execute" option in "Create Task" pop-up not working as expected	Low
OMS	ETD-80798	Adding "enter" + capital F in Crew comments causing client error and breaks chronology	Medium
OMS	ETD-80847	Missing parameters from NetServerOmsConfig_localhost for OIS_UAA	Medium
OMS	ETD-80853	Dependent DMS Follower produces different incident counts	High
OMS	ETD-80868	MDT Crew icon not showing near incident when manually assigned and merged	High
OMS	ETD-80870	After failover OMS server is stuck repeating this message in log and you cannot do any OMS functions on the client.	Medium
OMS	ETD-80939	Historical Incident Times are Incorrect	Medium
OMS	ETD-80977	Data - Customer PriorityCall flag not set	Medium
OMS	ETD-80984	Can't run Process Automation from PROCMAN	Medium
OMS	ETD-81144	"DNOM object is not a CDmsSwitch"	Medium
OMS	ETD-81150	Duplicate customer on incidents	High
OMS, Outage Assist	ETD-80820	TX merges and restoration does not restore customers; different variety to ETD-77521	Medium
OMS-Incidents	ETD-80684	OMS - When a predicted incident on a fuse is created while there is an existing trouble cleared incident present on the same fuse, the newly created predicted incident immediately goes to trouble clear	High
Outage Assist	ETD-79485	SA-Not all Critical Customer Types are showing as Diamonds in Storm Assist	Medium
Outage Assist	ETD-80401	OA - Completion Time gets updated after Maintenance Task Creations	Low
Outage Assist	ETD-80547	OA missing customer ping halo for "undetermined" ping status	Medium
Outage Assist	ETD-80797	OA - crew selection basic popup called up from Orders tabular - "select crew" drop down list wrong	Medium
Outage Assist	ETD-80828	OA - Evaluator Crews Status can't be changed from the OA Incidents Tabular	Medium

Component	Issue Number	Description	Severity
Outage Assist	ETD-80838	OA - Duration miscalculated on print summary report	Medium
Outage Assist	ETD-80848	OA - ETR Issues in Incident	Medium
SA	ETD-80854	Outage Assist deployment should provide option to use other storage classes than default "local-path" class	High
SWO	ETD-80988	When adding regulators to SWO the SWO client hangs and has to be closed and reopened	High
SWO	ETD-81188	SWO hung when adding certain stations	Medium
Tools	ETD-80224	Event-Scripter exception error while opening geo view navigation	High
Tools	ETD-80998	Snapshot viewer unhandled exception when loading 16.1 dynamics files	High
Tools-Configuration, Tools-Modeling	ETD-79285	Substation File Builder - Bus element association	Medium
UI	ETD-77847	Structure Layer Toolbar - multiple issues	Medium
UI	ETD-79885	Multi Select of Orders for Geographic not working	High
UI	ETD-80319	Single customer ping without AMI notification should be consistent	Low
UI	ETD-80559	Not possible to change the color of an existing network tracing	Medium
UI	ETD-80584	ADMS client increasing memory consumption with SCE model while trying to open multiple geographic displays	High
UI	ETD-80698	Merging orders from Map VS Orders Table	Medium
UI	ETD-80986	Can't open OMS plugin with APF/Browser 4.4 SP1	High
UI	UIUX-6179	Saved filter is not applied when reselected from 'personal' list	Low

28. ADMS 16.1.0

This chapter describes the incremental new features and defect fixes that were added since ADMS 3.16.0.2. The remainder of this document provides additional release-specific information that also applies to this release.

Note

The product version numbering has changed. The prefix '3' is no longer included in the version numbering for ADMS. The version numbers for all the documents will be updated in the next monthly release.

28.1. What's New in 16.1.0

This section contains information about new features and functionality for ADMS 16.1.0.

28.1.1. Process Automation Configuration Editor

28.1.1.1. Auto-Assign Tab

A new column has been added to Auto-Assign tab.

Affected Customers Count: The maximum number of customers impacted by an incident up to which automatic dispatching will be used. For example, if you set the Affected Customers Count to 100, any incident affecting up to 100 customers will be automatically dispatched for resolution. However, larger outages will be processed manually. (ETD-79480)

Process Automation Configuration Editor - C:\Eterra\Distribution\CurrentDev\FantasyIsland\config\ProcessAutomation\ProcessAutoConfig.xml*

File Help

Auto-Assign Auto-Transfer Config Params

	Type	Order Status	Incident Type	Non-Outage Type	Device Type	System Mode	Affected Customers Count	Action
▶		N/A	Single Customer	N/A	N/A	N/A	20	Auto Assign
		N/A	Confirmed	N/A	Transformer	N/A	30	Auto Assign
		N/A	Predicted	N/A	N/A	N/A		Auto Assign
		N/A	Non Outage	N/A	N/A	N/A		Request Dispatch
		N/A	Isolated	N/A	N/A	N/A	5	Request Dispatch
		N/A	Momentary	N/A	N/A	N/A		Do Nothing

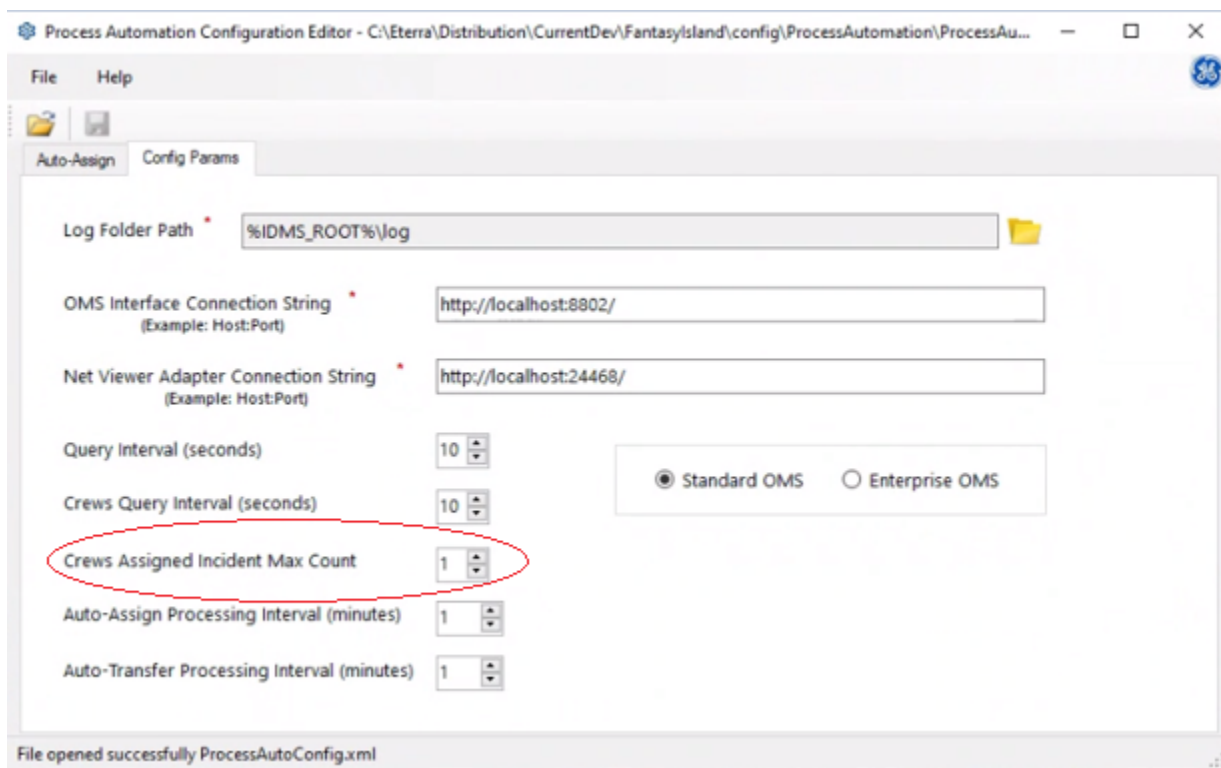
Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	

Feature Implications	Yes / No	Additional Information
ADMS Archive	No	
User Interface	Yes	Screenshot included
Demonstration Available	No	

28.1.1.2. Configuration Parameter Tab

Configuration Parameter tab has been enhanced to define and manage the maximum number of incidents a crew can handle.

Crews Assigned Incident Max Count: The maximum allowable incidents count for any crew. (ETD-79479)



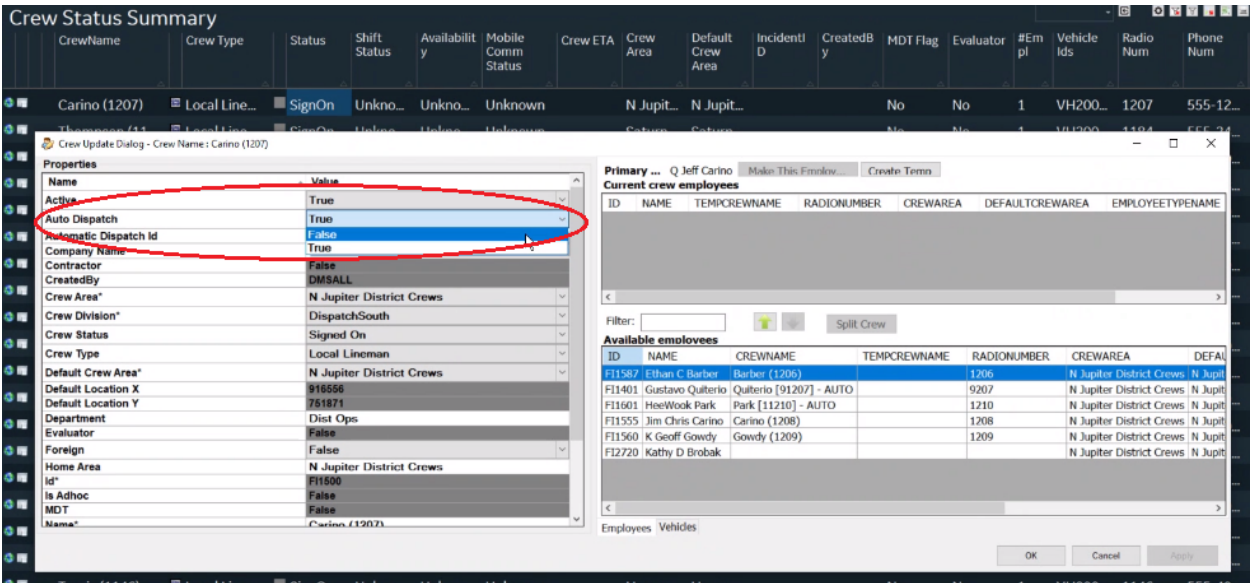
Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	Yes	Screenshot included
Demonstration Available	Yes	Click here for demonstration

28.1.2. Process Automation Application

A new parameter Auto Dispatch has been introduced into the ADMS Crew Model and integrated into the Crew Properties section within the Crew Update Dialog to control crews for the auto dispatch process for the Process Automation application.

28.1.2.1. Crew Update Dialog

Auto Dispatch: Controls the crews for auto dispatch process. (ETD-79471)



Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	Yes	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	Yes	Screenshot included
Demonstration Available	Yes	Click here for demonstration

28.1.3. Handling Inactive Customers

This feature is related to how the OMS Engine handles inactive customers. The following configuration setting is introduced in NetServerOmsConfig_localhost.txt

INCIDENT_PENDING_CLEAR_EXCLUDE_INACTIVE_CUSTOMERS

The above setting controls when the incident status becomes pending clear.

- If set to 1, when incident CurrentCustomer (which is active customer count) becomes 0, the

incident becomes pending clear.

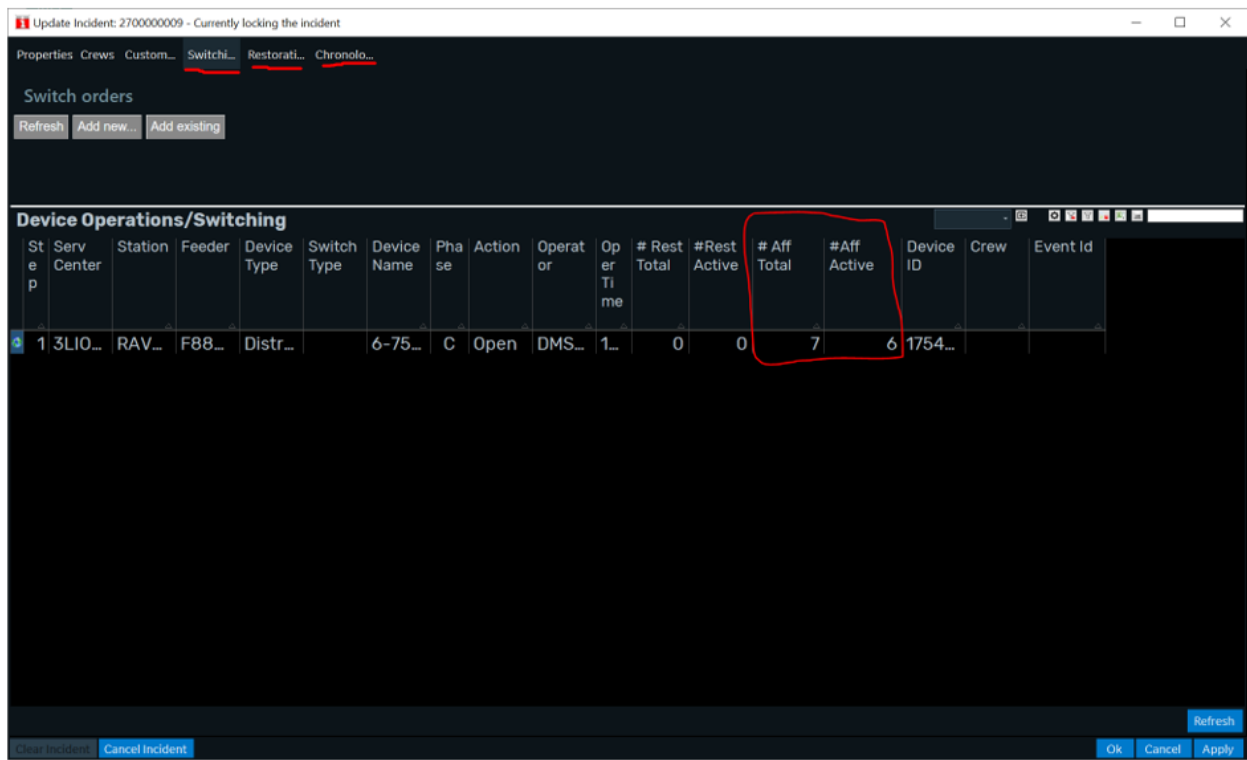
- If set to 0, when incident TotalCurrentCustomer (which includes both active and inactive customers) becomes 0, the incident becomes pending clear.
- Default value is 1.

There are two new properties exposed at incident class level.

- TotalAffectedCustomersCount: The total affected customer count, which includes inactive customers, if any.
- TotalCurrentCustomer: The total current customer count, which includes inactive customers, if any.

The following UI screens are updated in the Update Incident Popup.

- Switching Steps Tab: shows both total count and active customer count.
- Restoration Steps: shows both total and active customer count.
- Chronology: shows only the active customer count for related journal messages.



Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	Yes	NetServerOmsConfig_localhost.txt
ADMS Archive	No	

Feature Implications	Yes / No	Additional Information
User Interface	Yes	Screenshot included
Demonstration Available	Yes	Click here for demonstration

28.1.4. Exporting Data from Tabular Displays

When exporting tabular data to CSV, there is a new option for encoding. Users can choose Ascii or UTF8.

In netconf.xml file, there is a new config setting GridTabularCsvExportAscii.

- True = use Ascii
- False = use UTF8
- Default value is False

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	Yes	netconf.xml
ADMS Archive	No	
User Interface	No	
Demonstration Available	No	

28.2. What's Changed in 16.1.0

This section includes changes to existing features, as well as deprecated or removed features.

- Crew auto-assignment algorithm auto dispatches the crew based on crew area instead of service center. (ETD-79468)

28.3. What's Fixed in 16.1.0

This section lists the notable issues addressed in this release.

Component	Issue Number	Description	Severity
DMS	ETD-76484	3.15 open UI container details popup doubles client memory	High
AMI	ETD-72443	"Ping All Customers on Transformer" only pings first ServicePoint, not all.	Medium
ADMS_Work	ETD-79740	Outage Assist user who left an order worksheet open perpetually keeps taking back the lock even after getting unlocked with the ADMS Incident Locking Management	Medium

Component	Issue Number	Description	Severity
Core	ETD-79669	Existing grounding condition not visible	High
Core	ETD-80404	Old Status of Composite Switch is keeping in the Dynamic	High
Core	ETD-80621	ADMS is not keeping the changed value for Tap Position Overridden	Low
Core, Modeling	ETD-80455	SWO lookup failed	Medium
Core, UI	ETD-80505	Server side Feeder halo (FHAL) measurement values to SCADA for loop & parallel not respecting client (VCE) priority setting	High
Crew	ETD-79808	Need to add the ability to disable <CaseNotesUri>CaseNotes</CaseNotesUri> coming from the OMS server for the V2 messages	Medium
DMS	ETD-79700	Future layer not working correctly	High
DMS	ETD-80443	Multiple Consoles unable to search, remove tags, or toggle assets	High
DMS	ETD-80577	DMS Server sending dynamics to non-dynamic DMS clients	High
DMS	ETD-80594	DPF log: "Non-zero current flow measurement detected on a deenergized device" message on Energized Devices.	Medium
DMS, Tagging	ETD-80630	Unable to operate switch in follower after restrictive tag is removed	Medium
DNAF	ETD-70809	LOV Station Outage not consistent	Medium
DNAF	ETD-79733	LVM Indirectly Confirmation of Capacitor Operation is not Working Correctly	Medium
DNAF	ETD-79818	DER still showing disconnected with no disconnection reason	High
DNAF	ETD-80446	3 Wire/ 4 Wire CBO notification appears while doing switching in Delta - Wye network	High
DNAF	ETD-80454	Incorrect Date Time shown in Abnormal Summary for Composite Switches	Medium
E-OMS	ETD-79679	Splitting device from switching tab creating multiple tasks	Medium
E-OMS	ETD-79781	Some fields in the order worksheet are still editable when locked	Medium
E-OMS	ETD-79809	Search function in the customers tab of an order worksheet does not filter	Medium
FISR	ETD-79912	Parent Disabled, Child Enabled, LOV on parent outside child zone.	Medium
Grid Tabular	ETD-79788	Comments not typable on tabular	Medium
Interface	ETD-79191	OIS API not searching filtering correctly	Medium
Interface	ETD-80327	Orphans are being created in CE when Other Trouble Incidents are being Cancelled	High
Interface	ETD-80352	Event Date Time is not specified while notifying network events to external systems	Low
Modeling	ETD-79611	Workstations missing customer model_MMA to update the upload folder in atomic way	High
Modeling	ETD-80205	Unable to host Model Manager Rest Service	Medium

Component	Issue Number	Description	Severity
Modeling, Schematics	ETD-79994	NOP Missing for Telemetered and primary Feeder CB during upload new mod file	High
NRT	ETD-80596	ETR is cleared in the incident record when incident is cleared.	High
OMS	ETD-74738	Splitted order status not changing to field complete automatically	Medium
OMS	ETD-77892	After merging a non-outage/investigation order to a outage/restoration order for the same customer, the Order Worksheet does not display both of the calls	Medium
OMS	ETD-79131	Can't acquire lock on incident popup for primary meter incidents and certain isolated incidents.	High
OMS	ETD-79497	Order End Time is Different timezone (+7)	Low
OMS	ETD-79670	OMS Data Server Daily Snapshot using old data	High
OMS	ETD-79686	Incident working status does not stay assigned when assigning a crew to multiple incidents	Medium
OMS	ETD-79739	NetdataReplicator error over replicating	Medium
OMS	ETD-80157	Update Incident Window Blocks Other Windows	Medium
OMS	ETD-80216	IncidentLocationDescription does not appear in Data Correction tabular	Low
OMS	ETD-80329	Momentaries not autoclearing	Medium
OMS	ETD-80331	Incident clearance validation rules do not stop incident from being cleared	Medium
OMS	ETD-80438	Call Aggregation stores row number of coded translation editor item but uses key value in computation	Medium
OMS	ETD-80476	oData AreaNoteCode incorrect for incidents created before the Area Note	Medium
OMS	ETD-80672	Discrepancy in call replication	Low
OMS	ETD-80786	No new incident is created when we Split Multiple Customers from an Incident	Medium
OMS	ETD-80834	When clicking 'confirm device in / out' buttons, update incident window switches focus to oms display plugin	Low
OMS-Incidents	ETD-80213	Incident details tab is not showing customer calls by phase when using Event scripter for customers on clean dynamics	Medium
SWO	ETD-80403	No Tooltip for Start/Stop Recording Buttons in Switching Order (Study Mode)	Medium
SWO	ETD-80517	Switch Order State displays "Approved" state instead of "Validated" when the document is validated	Medium
SWO	ETD-80633	UIUX: Add space between geographic navigation and Alternate navigation	Medium
Tools	ETD-79461	De-energized while building the model with ADMS 3.15.5 converter	High
UI	ETD-79532	De-Energize line are showing greenish shadow for underground cable	Medium

Component	Issue Number	Description	Severity
UI	ETD-79671	Issue selecting order from geographic	Medium
UI	ETD-79785	Dynamic Schedules Chart is not show properly- Different DateTime format	Medium
UI	ETD-79884	Tags inside cabinets/vaults only visible from far zoom	High
UI	ETD-80153	Tabular Export csv encoding	High
UI	UIUX-6162	Saved filters in advanced crew assign tab not saving	Medium

29. ADMS 3.16.0.2

This chapter describes the incremental new features and defect fixes that were added since ADMS 3.16.0.1. The remainder of this document provides additional release-specific information that also applies to this release.

29.1. What's New in 3.16.0.2

No new features have been added in this release.

29.2. What's Changed in 3.16.0.2

There are no deprecated, removed, or changed features in this release, nor are there any changes to supported software or system requirements.

29.3. What's Fixed in 3.16.0.2

This section lists the notable issues addressed in this release.

Component	Issue Number	Description	Severity
ADMS_Archive	ETD-79440	Archive upgrade for 3.16.1 from 3.15.5 shows up as "Upgrade to 3.15" instead of 3.16 version	High
ADMS_Core	ETD-79271	After DMS Process Recycle, DMS and SCADA Stopped Syncing	High
ADMS_DNAF	ETD-78538	FL fails to solve with known values when the fault is downstream of a recloser that is in SCADA	Medium
ADMS_DNAF	ETD-78562	FISR Incorrectly Stating All Customers are Restored	High
ADMS_Interface	ETD-78290	Custom Fields of Converted Planned Incident are not populated in OMS Interface Server query	Medium
ADMS_Kit	ETD-79210	End Time populates with incorrect (past) end time, causing negative CMI count	High
ADMS_Modeling	ETD-77613	Two cases of Model Manager GUI missing updates	Low
ADMS_Modeling	ETD-79295	Model Manager Trace logs are created using machine local time but in UTC format	Low
ADMS_OMS	ETD-79087	Ghost calls revive automatically now at random times. Previously occurred after failover/switchover, customer binary update, crew file update	Medium
ADMS_OMS	ETD-79512	Ghost calls revive automatically now at random times. Sanity checks implementation	Medium
ADMS_Tools	ETD-77539	Model Manager Reprocess from Upload History Issue	Medium
ADMS_Tools	ETD-79084	Crew Modeling Tool Row Limitation for Crew Files	High

Component	Issue Number	Description	Severity
ADMS_Tools	ETD-79155	Converter is not logging any warning for disconnected fault indicator	Low
ADMS_UI	ETD-79055	SCADA Tabular Misspelling	Low
Archive	ETD-79379	Archive upgrade dacpac fails from 3.9 DB to 3.15.5 DB.	High
Core	ETD-78955	Duplicate incident created on rolup	High
Core	ETD-78552	New ADMS halo measurement not getting updated in SCADA without manual trigger	Medium
Crew Management	ETD-79309	Incorrect Crew is loaded upon selection & vehicles wrongly associated	Medium
DMS	ETD-79105	FL exact location determination	Low
DMS	ETD-79140	SIM: Protective device, lockout and Reclosing Cycle	Medium
DMS	ETD-79581	Back Calc not finding measurements	Medium
DNAF	ETD-79081	Load should only require dynamic real schedule for DPF	High
DNAF	ETD-79088	Static PF and dynamic real power for DERs	Medium
DNAF	ETD-79208	Network Analysis: SCC Failed - Power Flow Not Converged	High
DNAF	ETD-79286	Regulator's RLD State stay good quality in SCADA after DNAF server is down	High
DNAF	ETD-79288	Forecast server hang	High
DNAF	ETD-79289	LVMSM Implementation Does Not Complete	High
DNAF	ETD-79330	LVM Execution Mode does not Persist	High
DNAF	ETD-79334	LVM Failsafe Overrides the Locked Fixed Voltage Reduction Problem Formulation	Medium
DNAF	ETD-79335	Station's active LVM problem formulation shows empty in DSC	Medium
DNAF	ETD-79380	Incorrect Measured Real Power in Aggregation Summary Display	Medium
DNAF	ETD-79381	LVM Pre-Implementation State Check does not work	Medium
DNAF	ETD-79416	LVM Result Log is wrong for Fix Voltage Reduction Problem Formulation	Medium
DNAF	ETD-79659	Fails to Trigger Consistently – DNAF Not Calling DNA	High
FISR	ETD-78454	FISR doesn't restore the whole child station when an LOV happens on parent.	Medium
FISR	ETD-79121	DNAF Plan Cancel feature not usable	Low
FISR	ETD-79150	Old FISR plans not being cleared	Medium
History Correction	ETD-79343	OMS - When creating a split incident from a historical incident, the cause, sub cause and remarks are not populating on the properties tab of newly created historical incident	High
Internal Notifications	ETD-79154	CoreNotificationServer is not running successfully in Procman	High
OMS	ETD-76598	Check Prediction Task "No Outage" not Defaulting End Time to Start Time	High

Component	Issue Number	Description	Severity
OMS	ETD-76656	Positive Ack received from WFM is not updating the incident working status	Medium
OMS	ETD-79059	Switch cabinet internals have several Other Trouble incidents and Vehicle icons	High
OMS	ETD-79061	CI and CM tables are not functioning correctly in ADMS which is affecting Reporting, due to Incident identical name	High
OMS	ETD-79119	ETR & sequence #s updated in perpetuity	High
OMS	ETD-79129	NetSrv will not start if config has a blank line at top	Low
OMS	ETD-79211	Chronology Log captures inaccurate states	Medium
OMS	ETD-79428	Crew Symbol does not move with Order Symbol in FI	Medium
OMS	ETD-79451	Cannot mass cancel Orders by Order command nor default install	Low
OMS	ETD-79658	Master CallType is overwritten	Low
OMS-Crew	ETD-79057	Create Crew is Creating a Temporary Crew	Medium
OMS-Incidents	ETD-79672	System is generating incidents with a call time in the past when the customer was involved in a previous outage	High
SWO	ETD-78753	Initial default active element to be the Review button for a new SWO	Low
Tools-Configuration, Tools-Modeling	ETD-79287	The Reactor element association in SSE is not interpreted correctly when running the SFB application	Medium
UI	ETD-79110	Fault Location Manual Entry Tabular Filter.	Medium
UI	ETD-79568	Reader/View Only user is able to modify/write to Incident from List view	High
WebFG	ETD-79348	APF/ETB/ADMS client unresponsive when navigating to (or "near") substation whose model is out of sync	High

30. ADMS 3.16.0.1

This chapter describes the incremental new features and defect fixes that were added since ADMS 3.16.0.0. The remainder of this document provides additional release-specific information that also applies to this release.

30.1. What's New in 3.16.0.1

No new features have been added in this release.

30.2. What's Changed in 3.16.0.1

There are no deprecated, removed, or changed features in this release, nor are there any changes to supported software or system requirements.

30.3. What's Fixed in 3.16.0.1

This section lists the notable issues addressed in this release.

Component	Issue Number	Description	Severity
ADMS_Archive	ETD-78227	Unable to run the Archive Upgrade script between Builds of the Same Release	Medium
ADMS_Core	ETD-78596	New DMS log files are not created after failover as expected	Low
ADMS_Core	ETD-78977	NetData Replicator process stops replication upon encountering an exception	High
ADMS_DNAF	ETD-79040	Meshed FL fails when auto-throwover switches change during station initialization.	High
ADMS_DNAF	ETD-78936	FL application displays "Failed - Internal" status when A-C-G fault placed on recloser R01163 Springfield LA, however, FL halo appears correctly	Medium
ADMS_DNAF	ETD-78667	Internal bus limits don't get adjusted with limit adjustment	Medium
ADMS_DNAF	ETD-78563	FL DNA Algorithm Not Predicting Location	Medium
ADMS_DNAF	ETD-78407	Reason do not appear in Alarm Viewer, when FL Failed to complete due to phase target were incorrectly set	Medium
ADMS_DNAF	ETD-78384	Fault Location halo doesn't respect the limits set for boundary by fault targets	Medium
ADMS_DNAF	ETD-78374	Non-radial FL does not find fault location for the faults close to a switch	High
ADMS_DNAF	ETD-78281	Cannot set high voltage limit for cap bank 38393543	High
ADMS_DNAF	ETD-77859	FL doesn't solve for the first time after starting DNAF	Medium

Component	Issue Number	Description	Severity
ADMS_Interface	ETD-78520	Custom Fields not in PO payload when from PR	Medium
ADMS_Interface	ETD-78431	Adapter performance is heavily impacted by processing missing transactions	High
ADMS_Interface	ETD-77714	Tracing was slowdown when adjacent set to 1 or higher	High
ADMS_Interface	ETD-77473	AMI Power on Messages are not setting the restoration times	Medium
ADMS_Modeling	ETD-78713	CrewImportSchema.xsd to be updated	Medium
ADMS_Modeling, ADMS_Tools	ETD-78157	SFBLite issues tested in IDMS_GIT_DEV_20230511.1	Medium
ADMS_Modeling	ETD-76945	Model Quality Stat Logger does not account for Switching Devices	Medium
ADMS_OMS	ETD-78312	OMS - When creating a historical incident with remarks on an energized device, historical incident is being created without any remarks	Medium
ADMS_OMS	ETD-78101	Modified flag not saving on restoration steps in data correction	Medium
ADMS_OMS	ETD-77521	TX merges and restoration does not restore customers	Medium
ADMS_Tools	ETD-78515	Model Manager Validation Summary - Message, Explanation, and Solution Values Duplicating	Medium
ADMS_Tools	ETD-78511	Model Manager Client Not Opening or Buffering	Medium
ADMS_Tools	ETD-75429	Upload History Tab not Populating for Model Manager Client	Medium
ADMS_UI	ETD-78246	Blue line flashes sporadically over incident tabular	Low
ADMS_Work	ETD-77234	Device name for no-devices order is empty	Low
Archive, NRT	ETD-78785	3.12 Archive Database - Network Daily Outage Reports Discrepancies	Medium
DMS	ETD-78870	Haloing between Sim and RT not showing after breaker is open from SCADA	High
DMS	ETD-78512	Add SCADA Sanity check for start and end times - Momentary Incident on SCADA devices in History Correction is expanding End Time to 5 hours in the future	High
DNAF	ETD-79026	LVMSM cannot initiate	High
DNAF	ETD-78770	Meshed FL doesn't trigger with kits after June 16th.	High
DNAF	ETD-78561	Inconsistency between switch lockout status and its SCADA measurements	Medium
DNAF	ETD-78521	DPF periodic trigger issue	Medium
DNAF	ETD-78333	Fault Halo on gets cleared on a three phase segment for a single phase fault even if the faulted phase is de-energized	High
FISR	ETD-78523	Subfed triggering Issues.	Medium
FISR	ETD-78514	The Fault triggering log on parent, doesn't have information on why it stopped, when the fault is on Child or parent disabled.	Medium
IDMS UI	ETD-78295	Pull box missing from cabinet SG0096	High
IDMS UI	ETD-77849	Structure layer rendering - Issues with drawing line strings	Medium

Component	Issue Number	Description	Severity
Interfaces	ETD-78854	Click tickets are not sending geographic location (city & zip code) for Riverlake network. Snow Ticket created INC 11401492	High
NRT, OMS	ETD-77612	NRT - CREW_ACTIVITY Table - INCIDENTASSIGNMENT_STATUS_CODE not correctly update when the Incident is reassigned using the Dispatch / SS9 application	Medium
OMS	ETD-78752	Font size modification required for Filters in the incident screen, so they are visible	Low
OMS	ETD-78665	Customer Correction wrong service point	Medium
OMS	ETD-78492	Damage Assessment Messages not Showing in Chronology Log	Medium
OMS	ETD-78490	Order Worksheet Affected Customers Count Inconsistent	Medium
OMS	ETD-78433	Planned Outage not retaining status of having a switch order linked in Grid Tabular	High
OMS	ETD-78432	OMS Issue-Disconnect	High
OMS	ETD-78163	Incorrect Incident Restoration Step Data	High
OMS	ETD-78134	ADMS Restoration Steps are incorrect after clearing incident with momentaries	High
OMS	ETD-77826	Crew Server - Converting Incident to Non-Outage results in IncidentUpdate Dropping Customer List	Medium
OMS	ETD-76592	Service Point Check Prediction Task "Confirmed Closed with Power" Can't Progress Order	Medium
OMS-Incidents	ETD-78747	(Grouping/Rollup) Multiple incidents being created when customers (AMI or calls) are reporting in on an confirmed outage	High
SWO	ETD-78233	TrustedClient options do not send messages	Low

31. ADMS 3.16.0.0

31.1. Release Highlights in 3.16.0.0

This document provides release-specific information for the ADMS 3.16.0.0 product. It covers the incremental changes since ADMS 3.15.

The following major improvements have been implemented for ADMS 3.16.0.0.

Distribution Network Analysis Functions (DNAF)

FISR Triggering LOV

Some customer feeder breakers do not have all the required SCADA measurements for FISR to be triggered when the breaker is opened due to a downstream fault. This function triggers the loss of voltage FISR and restores power to customers downstream of the faulted section. (ETD-75736)

Sub-Fed Station Support for FISR and LOV

Prior to the 3.16.0.0 release, the FISR fault and loss of voltage triggers did not function properly for all sub-fed station configurations, and some customers have a lot of these configurations in their network which hinders their ability to take the full advantage of the FISR application.

DNAF Decoupling – Pre containerization

As part of a multi-phase effort to decouple the network applications, additional functionality and components were developed including the following:

- Support for other DNAF including PRV, SCC, LVM, AFR, and PLOS was added
- New model serializer/deserializer was added
- Development started on DNAF database (Operational Data Store or ODS) (ETD-75754)

Outage Management (OMS)

- Options for Setting ETRs when Splitting Outages
- Enhanced Configuration of Summary Displays
 - Including the ability to summarize based on any incident working status value
- Critical Customer Icon for Non-outage Incidents and Orders
- Improved Handling of Assignments to Signed Off Crews
- Hazardous Customer Filter for Customer/Call Cards (E-OMS)
- Enhanced Data Content for Customer/Call Cards (E-OMS)
- Enhanced Crew Search (E-OMS)
- Incident Name as an Option for Order Naming (E-OMS)
- Creation of Default Damage Assessment Location (E-OMS)

- Added Columns for Auto-transfer Rules in Process Automation (E-OMS)

Enterprise Outage Management (EOMS)

Enterprise OMS Version Number and Release Notes

Enterprise OMS no longer uses a different version number structure, instead it is considered as part of the ADMS 3.16.0.0 product.

Subsequently, separate Enterprise OMS release notes are no longer delivered; all relevant Enterprise OMS information is provided here in the ADMS 3.16.0.0 Release Notes.

Further incorporation of Enterprise OMS documentation into ADMS documentation will be done in a future release.

31.2. What's New in 3.16.0.0

This section contains information about new features and functionality for ADMS 3.16.

31.2.1. Distribution Management System

31.2.1.1. Hazardous Customer Filter for Customer/Call Cards

Introduced in 3.15.4

The Hazardous filter is a new block filter added to Customer/Call Cards that checks for the presence of text in the Hazardous property of customers. This feature provides a quick and easy way for Operators to identify potential hazards for crew members. (ETD-77569)

1020000008-1 x

Save

Incident 1020000008

No Location Description

SP_34749018 | F3540 DURBAN (Widgeon)

General Facility Customers Damage Assessment More v

Calls
1

High Priority
0

Critical
0

Informative
0

Hazardous
1

Search

Berry Phil Y
7148 ROCK NE

Lights Out
No Call Time
543114OQ
555-573-1505

"No Comment"

☐ Informative

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	Yes	Screen shot included
Demonstration Available	Yes	Click here for demonstration

31.2.1.2. Improved Transformer Bank Search and Navigation

Introduced in 3.15.3

The transformer bank is searchable in both "analyst search" and "device search". When a distribution transformer is searched, the results will also display the associated transformer bank, which can be navigated. A simple tool bar and attributes display have been added for the transformer bank. When opted-in to the viewer configuration editor, the transformer bank label

displays all the distribution transformer IDs in multiline. (ETD-75731)

Feature Implications	Yes / No	Additional Information
Modelling (DNOM, BPCL, Crew)	Yes	
API	No	
Configuration	Yes	viewerConfig.xml
ADMS Archive	No	
User Interface	Yes	
Demonstration Available	Yes	Click here for demonstration

31.2.1.3. Customize Port

Introduced in 3.15.1

ADMS Sample Reports Setup has been enhanced to customize the port. During installation, the system will check whether the port the user entered is available for installing it on that port (ETD-72113).

ADMS Sample Reports (CustomBuild) Setup

Provide Data Source information

Page 1 of 3

Data Source Settings

Please select SQL Server Reporting Service Instance to deploy reports to

SSRS

Report Server Port 80

☐ Install Sample Reports Oracle

☒ Install Sample Reports Microsoft SQL Server

< Back Next > Cancel

Feature Implications	Yes / No	Additional Information
Modelling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	Yes	
User Interface	Yes	Screen shot included
Demonstration Available	No	

31.2.2. Distribution Network Analysis Functions

31.2.2.1. FISR Triggering Loss of Voltage

Introduced in 3.15.5 and 3.16.0.0

Some customer feeder breakers do not have all the required SCADA measurements for FISR to be triggered when the breaker is opened due to a downstream fault. This function triggers the loss of voltage FISR and restores power to customers downstream of the faulted section. (ETD-75736)

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	Yes	viewerConfig.xml
ADMS Archive	No	
User Interface	Yes	
Demonstration Available	No	

31.2.2.2. Sub-Fed Station Support for FISR and LOV

Introduced in 3.15.5 and 3.16.0.0

Prior to the 3.16.0.0 release, the FISR fault and loss of voltage triggers did not function properly for all sub-fed station configurations, and some customers have a lot of these configurations in their network which hinders their ability to take the full advantage of the FISR application.

The enhanced FISR triggering code considers the child and parent station dependently in handling the following cases:

- When a fault or loss of voltage occurs in the child station, the parent station is not impacted. Only the child station will see the fault or loss of voltage condition and trigger the FISR application.
- When a fault or loss of voltage happens in the parent station, both the parent and the child station(s) are impacted. (ETD-75722)

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	No	
ADMS Archive	No	
User Interface	No	
Demonstration Available	No	

31.2.3. Outage Management System

31.2.3.1. Incident Name as an Option for Order Naming

Introduced in 3.15.4

IncidentName can now be used as the Order Name. When the new {IncidentNumber} template parameter is used on the Order Name tab of the OMS Server Configuration Editor, the Incident Name is used to set the Order Name. This allows you to easily cross-reference incidents and orders when using different user interfaces. (ETD-77565, ETD-77566, and ETD-77568)

To use IncidentName in the Order Name:

1. Open the OMS Server Configuration Editor.
2. Go to the Order Name tab.
3. Remove all the existing text from the *Order name format* text box.
4. Add "{IncidentNumber}" to the Order name format.
5. Save the configuration file.
6. Restart the OMS Server.

Note

Other templates can be used in addition to "{IncidentName}" but, we suggest using it with no other templates.

Caution

If "{IncidentName}" is used, you must disable the Create Related Maintenance Order option on the Order List. To do so, change the value of the OMS_RELATED_MAINTENANCE_ORDER_ENABLED parameter to 0 in the NetServerOmsConfig_localhost.txt file.

Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	

Feature Implications	Yes / No	Additional Information
API	No	
Configuration	Yes	NetServerOmsConfig_localhost.txt
ADMS Archive	No	
User Interface	No	
Demonstration Available	Yes	Click here for demonstration

31.2.3.2. Options for Setting ETRs when Splitting Outages

Introduced in 3.15.1

The user has three options when using the "Split" function from a tabular list (Restoration Step, Switching Step, Customer/Call) to specify where the new incident's ETR should come from:

- Copy the originating/parent's ETR to the new incident
- Calculate the new ITR from the standard ITR Matrix if the ETR had not been set
- Don't automatically set an ETR so that the user can set it manually afterwards (ETD-75884)

Feature Implications	Yes / No	Additional Information
Modelling (DNOM, BPCL, Crew)	No	
API	No	
Configuration	Yes	NetServerOmsConfig
ADMS Archive	No	
User Interface	No	
Demonstration Available	Yes	Click here for demonstration

31.2.3.3. Critical Customer Icon for Non-outage Incidents and Orders

Introduced in 3.15.3

The non-outage incidents with critical customers are now marked with the critical customer icon and specified in the critical customer list. (ETD-75720)

Note

For this behavior to work properly the WFM_NONOUTAGE_INCLUDE_PRIORITY_AND_CUSTOMER parameter must be set to 1 in NetServerOmsConfig_localhost.txt file.

Feature Implications	Yes / No	Additional Information
Modelling (DNOM, BPCL, Crew)	No	
API	No	

Feature Implications	Yes / No	Additional Information
Configuration	Yes	NetServerOmsConfig_localhost.txt
ADMS Archive	No	
User Interface	Yes	
Demonstration Available	Yes	Click here for demonstration

31.2.3.4. Improved Handling of Assignments to Signed Off Crews

Introduced in 3.15.5 and 3.16.0.0

When an incident is assigned to a Signed Off crew (such as MDT, Non-MDT, Evaluator), the crew's Current Status now remains as Signed Off. However, the Incident Status will indicate that the crew has been assigned to the incident. The crew's Current Status will change to Assigned only when the crew's status is changed to Sign-On or Available. (ETD-77489)

31.2.3.5. Handling of Inactive Customers

Introduced in 3.15.5 and 3.16.0.0

Handling of Inactive Customers has been enhanced in the following ways: (ETD-77441)

- Inactive customers are associated with incidents automatically
- Inactive customers are displayed in affected customer lists
- Inactive customers are left out of affected customer counts on outages, restoration steps and switching steps
- Inactive customers to be left out of CMI calculations

i Note

Calls from inactive customers are still involved in the prediction engine

31.2.3.6. Setting ETR When Splitting Incidents

Introduced in 3.15.5 and 3.16.0.0

Various enhancements have been made related to how ETR is set when splitting incidents.

Previously:

- When splitting an incident from a device, ETR is calculated.
- When splitting a customer from an incident, ETR is the same as the existing incident.

In this release, there are now three options you can choose:

- Copy the parent incident's ETR

- Re-calculate ETR
- Unset ETR

Additionally, there are two new configuration settings in the OMS Server Configuration file:

- DEFAULT_ETR_OPTION_FOR_SPLIT_INCIDENT_BY_CUSTOMER = KEEP
- DEFAULT_ETR_OPTION_FOR_SPLIT_INCIDENT_BY_DEVICE = NEW

The command to split is:

```
NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/COMMAND/CREATE/MANUALINCIDENT?CUSTOMERKEY=&RID=%CUSTOMERKEY&
```

You can specify the new parameter, which has one of the following three values:

- ETR=KEEP
- ETR=NEW
- ETR=UNSET

For example:

```
NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/COMMAND/CREATE/MANUALINCIDENT?CUSTOMERKEY=&RID=%CUSTOMERKEY&ETR=KEEP
```

31.2.3.7. Filter by Feeders

Introduced in 3.15.5 and 3.16.0.0

From the incidents/orders lists, you can now choose the feeders to which the incidents/orders belong to. With this change, both the nominal and as-operated feeder information is considered.

31.2.4. EOMS

31.2.4.1. Enhanced Crew Search

Introduced in 3.15.5 and 3.16.0.0

The Crew Selection dialog that is launched from the Map or Order List now has a name-based search option. There is now a text box above the "Select Crew" dropdown which allows you to type part of the name of a crew to limit the options available in the dropdown. The search is case-insensitive and is performed against all the components of the crew name. (ETD-77735)

1. Select the filter on the right to restrict the overall list of crews.
2. Filter the list of crews further by typing part of the crew's name in the text box that contains the background text "Type crew name to search".
3. Select the crew from the filtered list in the scrollable dropdown.

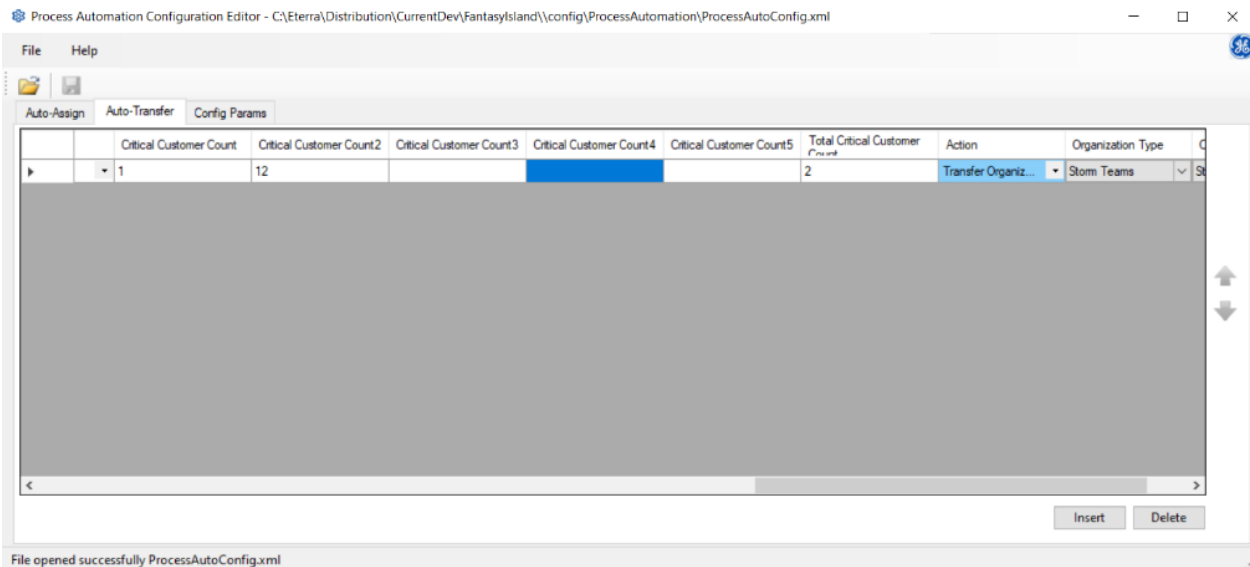
31.2.4.2. Added Columns for Auto-transfer Rules in Process Automation

Introduced in 3.15.3 and 3.15.4

The Auto-Transfer tab of the Process Automation Configuration Editor has been enhanced with new columns that can be used when defining auto-transfer rules. (ETD-75848, ETD-75849, ETD-75850)

The added columns are as follows:

- Critical Customer Count 1 – 5: The minimum number of critical customers of each critical type (1-5) for the transfer rule to apply to an order. Critical types are defined as coded translation items.
- Total Critical Customer Count: The minimum number of all critical customers of types 1-5 for the transfer rule to apply to an order. Critical types are defined as coded translation items.
- Priority Code: The Priority Code that an incident must have for the transfer rule to apply to its order. The Priority codes are defined by the Priority_Code coded translation item.
- Description: The Description (derived from the first call) that an incident must have for the transfer rule to apply to its order. The Descriptions are defined by the Description coded translation item.
- Condition1: The Condition1 value (derived from the first call) that an incident must have for the transfer rule to apply to its order. The Condition1 values are defined by the Condition_1 coded translation item.
- Condition2: The Condition2 value (derived from the first call) that an incident must have for the transfer rule to apply to its order. The Condition2 values are defined by the Condition_2 coded translation item.



Feature Implications	Yes / No	Additional Information
Modeling (DNOM, BPCL, Crew)	No	
API	Yes	

Feature Implications	Yes / No	Additional Information
Configuration	Yes	ProcessAutoConfiguration.xml
ADMS Archive	No	
User Interface	Yes	Screen shot included
Demonstration Available	Yes	Critical Customer Columns - Click here for demonstration Priority Code and Call Type Columns - Click here for demonstration

31.2.4.3. Handling of Orphaned Orders

Introduced in 3.15.5 and 3.16.0.0

Orphaned Orders are orders that show as active (not closed or cancelled) but which have been left in the system without a reference to an incident. While this is an unexpected situation, this feature has been added to allow such orders to be removed from the Order List if it does occur.

When an attempt is made to open the Order Worksheet for an orphaned order, an alert pop-up will be presented to the user to allow the order to be closed or cancelled. The text on the popup reads as follows:

The incident for order <OrderName> has been removed from the real-time database. It isn't expected for this order to still to be in your list so you can choose to close it or cancel it and the worksheet will be closed.

Buttons are presented to close or cancel the order and pressing either one of these will force the order into the corresponding state, removing it from the Order List. It will not be possible to open the worksheet for such an order.

31.2.4.4. Creation of Default Damage Assessment Location

Introduced in 3.15.5 and 3.16.0.0

Previously, to record component-based damage assessment information on the Order Worksheet, a Damage Assessment Location had to be manually created for the corresponding incident. This is no longer the case. When first adding damage assessment information on the Order Worksheet, a default damage assessment location is created automatically (if one has not already been created). (ETD-75768)

31.2.4.5. Enhanced Data Content for Customer/Call Cards

The customer/call cards displayed on the Order Worksheet have been enhanced to include the following data fields:

-

For customers:

- Account Number
- Customer Type
- Hazardous Customer (shown in red if present)
- Critical Types (shown as a tooltip text over the critical (diamond) icon)

- For calls:

- Description
- Condition_1
- Condition_2



Pratt Billie B (555292470)

Residential 7538 RIVER NE

Lights Out

Wires Down

Pole Down

Pole/Pole

21:53 06/08

447288AW

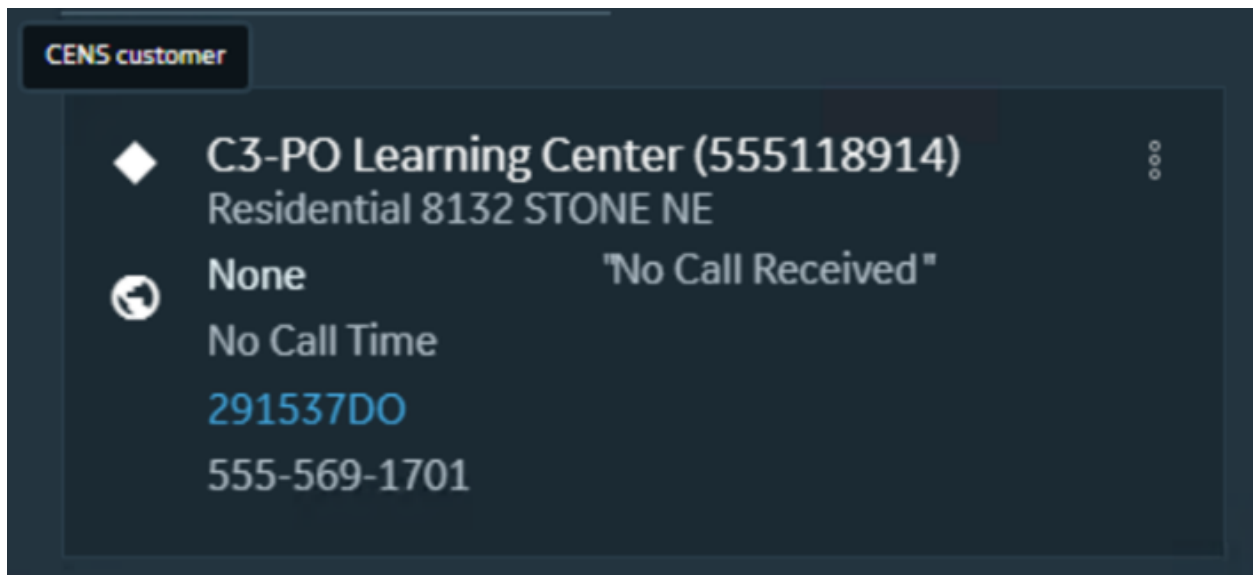
555-575-2076

Vicious dog roaming in the
back yard

"No Comment"

☐ Informative





31.3. What's Changed in 3.16.0.0

This chapter includes changes to existing features, as well as deprecated or removed features.

31.3.1. Deprecated and Removed Features

There are no deprecated, removed, or changed features in this release.

31.3.2. Potential Upgrade Requirements

This section describes changes in the system that may impact your end user workflows, particularly if customizations have been made in your system. You should review these changes and make appropriate adjustments as part of your upgrade process to ensure that all workflows are preserved.

31.3.2.1. OMS CrewServer Configuration File Update

There is a change in the OMS CrewServer configuration file (CrewServerConfig.xml) that will affect what value is populated in the REST outbound WFM messages; IncidentUpdateV2 & CrewUpdateV2.

The config definition looks like this:

```
<!-- Determines whether the Incident "Id" or "Name" (the default) is sent in the -->

<!-- IncidentUpdate.IncidentId and CrewUpdate.IncidentId message fields. -->

<PostIncidentIdOrName>id</PostIncidentIdOrName>
```

This configuration controls what value, either the Incident.GUID (aka Incident.ID) or the Incident.Name, is populated into the outbound IncidentUpdateV2.IncidentId & CrewUpdateV2.IncidentId fields. In earlier versions of OMS we populated the outbound IncidentId field with the Incident.Name. This was incorrect and limited the WFM's ability to track the messages when Incident names change, etc. Each outbound message has an IncidentName field which will always be populated with the Incident.Name.

If your WFM interface assumes the Incident.Name will be populated in the outbound IncidentId field, you should change this config to "Name".

31.3.3. Supported Software and Hardware Changes

This section describes changes to the system such as changes in system requirements, compatibility, or third-party software.

Refer to chapter [System Requirements and Software Compatibility](#).

31.3.4. Database or Backend File Changes

This section describes changes to backend system files, such as database tables or configuration files.

Table 1. EOMS Database Changes

Table	Column	Comment
TASKS	SWITCH_STEP_NUM_ID	Added to address issues with the creation of duplicate tasks (ETD-75823).

31.4. What's Fixed in 3.16.0.0

This chapter lists the notable issues addressed in this release.

Component	Issue Number	Description	Severity
ADMS	ETD-76632	Cannot Merge Nonoutage to Single Customer Outage	Low
ADMS_Archive	ETD-77140	Affected Customer is removed from Pending Clear Incident	Medium
ADMS_Archive	ETD-77288	ArchiveApp server used to move order journal files to a back-up directory, archived. Added logic to copy the file and then delete the file as well as a system requirement to check for success. If there is failure, the system logs the error and continuous working.	High
ADMS_Archive	ETD-75698	Archive app crash during archiving work order data	High
ADMS_Archive	ETD-76766	Exclude_From_Statistics is not reflecting outage_type_code	Medium
ADMS_Archive	ETD-77398	Archive App Crash due to Foreign Key db error caused by "Create Historical Incident"	Medium

Component	Issue Number	Description	Severity
ADMS_Archive, ADMS_OMS	ETD-77626	AS_OPERATED_STATION in NRT not populating for certain orders/incidents	Medium
ADMS_Archive, FISR	ETD-73988	FISR LOV: Bad outputs - not archived and FISR log mixups - when multiple solutions triggered at same time	High
ADMS_Core	ETD-75501	Upstream Trace Scada is Skipping Controllable Device	Medium
ADMS_Core	ETD-75703	Feeder level is not showing up in Study Manual Input Parameter Entry	Medium
ADMS_Core	ETD-75959	ADMS Inactive User Appearing on Incident Chronology	Medium
ADMS_Core	ETD-77268	New Query Field for Future Devices in Station	Medium
ADMS_Core	ETD-77911	When merging downstream predicted incidents from a confirmed incident a restoration time is triggered.	High
ADMS_Core	ETD-78105	Popup de-energization notifications are not correct for TIE Composite Switch	Medium
ADMS_Core	ETD-78159	Load scaling factor in SIM doesn't inherit properly for the sub-elements at station level.	Medium
ADMS_Core	ETD-78207	ISD with ISD Logs on failover appends to an old C ISD file, not showing proper enabled status	Low
ADMS_Core	ETD-78297	Dependent follower not honoring permissions in 3.15.4	High
ADMS_Core, ADMS_UI	ETD-77454	Tasks could not be created from the OMS Order Worksheet in the Kubernetes / Foundation deployment. Note that this fix was an API change which requires that the core ADMS and Enterprise OMS be upgraded together.	High
ADMS_DNAF	ETD-73488	DNAF Configuration Transformer Zero Seq Impedance value not impacting calculation	Medium
ADMS_DNAF	ETD-73607	Fault Location Application Triggered Multiple Times	Medium
ADMS_DNAF	ETD-75460	For manually triggered DNAF plan execution (FISR, FSD, AFR), system log to mention the name of the initiating user	Medium
ADMS_DNAF	ETD-75579	Device is Eligible in Analyst but Ineligible in FISR Log	Medium
ADMS_DNAF	ETD-75605	Non-radial Fault Location solves only one fault when faults happen simultaneously on more than on feeder	Low
ADMS_DNAF	ETD-75641	DOTS Slow Performance of Non-Radial Fault Location in Study Mode	High
ADMS_DNAF	ETD-75659	BRT DOTS - Faults on different laterals downstream of the same SCADA device gives wrong predicted fault locations	High
ADMS_DNAF	ETD-75676	FL Halos will disappear by dynamic compression	High
ADMS_DNAF	ETD-75945	Wrong Fault Location and Fault Currents	Medium
ADMS_DNAF	ETD-75949	FL doesn't solve for the first time after starting DNAF.	Medium
ADMS_DNAF	ETD-76787	FL fails to find any location after starting DNAF.	High
ADMS_DNAF	ETD-76935	SCC calculated values are not in the range of FL	Medium
ADMS_DNAF	ETD-76965	Incorrect on/off level and limit of temperature control type capacitor reported in DPF and LVM result file	High

Component	Issue Number	Description	Severity
ADMS_DNAF	ETD-77006	BRT DOTS Single Phase to Ground Faults fails to solve when a valid solution is expected	High
ADMS_DNAF	ETD-77036	BRTDOTS: Non - Radial FL doesn't solve when a fault happens in the abnormally energized section	High
ADMS_DNAF	ETD-77448	Manual Enable Checkbox in DSC doesn't reflect AUTO_OFF status	Low
ADMS_DNAF	ETD-77475	DNA FL Not Predicting Fault Locations	Medium
ADMS_DNAF	ETD-77548	FISR Scenario Not Producing Plan	High
ADMS_DNAF	ETD-77640	FISR Plan Shows Incorrect Number of Customers Restored	High
ADMS_DNAF	ETD-77642	LVM does not generate correct plan similar to ETD-75789 and ETD-75790	Medium
ADMS_DNAF	ETD-77689	FISR Stating No Deenergized Branches in Valid Scenario	High
ADMS_DNAF	ETD-78150	LTC/Regulator/Capacitor daily operation count not resetting	Medium
ADMS_DOTS	ETD-74270	RCL and CB reclose cycle and lockout issues	Low
ADMS_DOTS	ETD-76737	Event scripser return to normal is not resetting fault currents in SCADA/RT	Medium
ADMS_DOTS	ETD-77636	Event Scripser Not Using CTE Keys for Customer Call Type	Medium
ADMS_Interface	ETD-75879	CoreNotificationServer not running successfully in Procman	Medium
ADMS_Interface	ETD-77179	ANPS Adapter does not support UAA auth	Medium
ADMS_Interface	ETD-77188	ANPS Adapter sets as CreateAreaNote upon receiving an update	Low
ADMS_Interface, ADMS_UI, UI	ETD-77019	Unable to export to PDF right-to-left language text in a SWO	Medium
ADMS_Interface, ADMS_UI, UI	ETD-77057	Operator/Analyst Attributes pop-up malfunctioning with /RIGHTTOLEFTLAYOUT	Medium
ADMS_Kit	ETD-76771	OMS - When a predicted incident is denied down, one of the resulting single customer incident(s) briefly goes to trouble status before returning to unassigned status	High
ADMS_Modeling	ETD-70963	MMA Crashed Due to Unhandled Exception in Application Log	Medium
ADMS_Modeling	ETD-76522	Conversion point regulators generating Voltage level Errors.	Medium
ADMS_Modeling	ETD-77216	Model Manager Service Client Stops processing after ME timeout	Medium
ADMS_Modeling	ETD-77394	Customers being marked as Special Accounts incorrectly	Low
ADMS_Modeling	ETD-78135	Adhoc - No limits for parameters from global.csv file from converter	Medium
ADMS_OMS	ETD-74225	ADMS - AMI counts are not updating in the incident tabular, number displayed is not correct and does not reflect the number of AMI calls received in the ADMS	Medium
ADMS_OMS	ETD-75220	Momentary Orders are not created	Low
ADMS_OMS	ETD-75394	Open device voltage assignment in OMS incidents	Medium
ADMS_OMS	ETD-75662	OIS/NVA search no longer working	High
ADMS_OMS	ETD-76827	SA 2.1 - Unable to create incident from an unassociated call in storm assist	High

Component	Issue Number	Description	Severity
ADMS_OMS	ETD-76962	Workmodel journal files archived too soon on enabled OMS server	High
ADMS_OMS	ETD-76962	Added DataArchivedIntervalDays parameter in the ArchiveAppConfig.xml file, which controls the number of days the ArchiveApp will wait before moving and old journal file to the Archive directory. Default value = 1 day.	High
ADMS_OMS	ETD-76997	OMS - In AMDS 3.12, when confirming in a non scada recloser or a transformer, incident is reverting back to predicted if there are inactive customers downstream of the device	High
ADMS_OMS	ETD-77315	Negative Duration/CMI after restoring outage via AMI API	Medium
ADMS_OMS	ETD-77490	Incident name has 11 digits while configured for 10	Medium
ADMS_OMS	ETD-77523	Customers being counted on multiple outages	Medium
ADMS_OMS	ETD-77838	Incident is not properly cancelled from Data Correction tab	Medium
ADMS_OMS, OMS	ETD-75162	Setting OMS_NEW_INCIDENT_DELAY_AFTER_ENERGIZATION to a low number removes customers when pending clear	Medium
ADMS_OMS	ETD-78226	"Cuts" incidents are showing inconsistent number of calls and customers.	High
ADMS_SWO	ETD-76929	SWO: Navigation to Schematic or Scada view from Switching Steps Labels should be set to zoom based	Low
ADMS_Tools	ETD-74782	Model Manager ISSG and SSG File Synchronization	Medium
ADMS_Tools	ETD-75593	Process Automation Configuration Editor Pointing to Wrong Folder	Medium
ADMS_Tools	ETD-77380	Grid Tabular Shortcut icon opens 2 gird tabulars	Low
ADMS_UI	ETD-74382	Newly commissioned devices do not appear in Automatic Feeder Schematic	High
ADMS_UI	ETD-76594	Toolbars are replicating	Medium
ADMS_UI	ETD-76805	Telemetered Fault Indicator shows A symbol instead of T	High
ADMS_UI	ETD-76861	ADMS netview slowness and taking more memory	Medium
ADMS_UI	ETD-77169	Locate normal feeder in SCADA display is not working when the line is de-energized	Low
ADMS_UI, DMS	ETD-76000	3.15 client disconnects from server during model updates	Medium
ADMS_UI, UI	ETD-76612	Annotation combined Hebrew tooltip must match combined Hebrew comment	Low
ADMS_Work	ETD-77554	Chronology tab: Messages for one scenario is not reflecting under Chronology tab	Low
ADMS_Work	ETD-77568	Disabled "Create Related Maintenance Order" in the UI. Added OMS_RELATED_MAINTENANCE_ORDER_ENABLED parameter in the NetServerOmsConfig_localchost.txt.file. Disabled = 0, Enable = 1.	Low
ADMS_Work	ETD-77732	Field Clear Device Level: Separate AMI IN and Crew Usage into two different configs	Low
AMI	ETD-72443	"Ping All Customers on Transformer" only pings first ServicePoint, not all.	Medium

Component	Issue Number	Description	Severity
AMI	ETD-72444	"Scan AMI Analogs - Multiple Customers" doesn't display results (or even customer)	Medium
Archive	ETD-74998	Fields LAST_UPDATED_BY and CLEARED_BY in INCIDENT table are not getting populated when user updates the related incident in the order worksheet	Medium
Archive	ETD-75161	Unable to Upgrade Archive DB in Oracle from 3.9 to 3.12	Medium
Archive	ETD-75591	ArchiveApp PURGE is Deleting Active Incidents and Orders Related Data	High
Archive	ETD-76872	Archive App new user script permission issue	Low
Archive	ETD-77367	ArchiveApp archives multiple instances of restoration steps	High
Crew	ETD-77958	crew model validation schema incorrect	Medium
DMS	ETD-76754	Customer Search not working with multiple entries	High
DMS	ETD-77286	Standby follower server syncs its Dynamics and manuals with the leader not its redundant pair during failover	High
DMS	ETD-77658	Block Un/Block Requires the user to click off the device and then back on to it to update status	Low
DMS	ETD-78123	ISAUTOMATEDSWITCH Query Not Populating Correctly for Boundary Devices	Medium
DMS	ETD-78204	Study Savecase Not Retrieving Manuals Correctly	Medium
DMS, DNAF	ETD-77462	RLD points in SCADA are not match with those in DMS measurement tabular	High
DNAF	ETD-74612	Peterson coil modelling on a side earthing transformer is not supported	Medium
DNAF	ETD-74711	FSD feeder ranking is not correct according to provided formula	Low
DNAF	ETD-75458	The result summary tabular is wrong	Low
DNAF	ETD-75462	Proposed switching steps order to be optimized	Medium
DNAF	ETD-75464	FISR Device Eligibility on Dynamic vs Nominal Feeder	Medium
DNAF	ETD-75692	FISR Plans in the FISRPLANSTATUS_GRIDDISPLAY Window reset to "Not Executed" at Midnight	High
DNAF	ETD-77011	Power Flow CBO: Overloaded Power Transformers are not listed on CBO pop-up	Medium
DNAF	ETD-77024	Meshed FL Solution is way off in a special scenario in looped condition.	High
DNAF	ETD-77285	DNA does not respect the DNAF_PARAMS configuration parameter.	Low
DNAF	ETD-77908	DNA does not respect the DNAF_PARAMS configuration parameter	Low
DNAF, FISR	ETD-77824	FISR overreacting to voltage violations on one section of feeder	Medium
E-OMS	ETD-78133	Missing Orders in ADMS/SA	High
E-OMS, OMS, OMS-Incidents	ETD-77869	Send Email and Send Text should be removed	Medium
FISR	ETD-74603	DNAF getting crashed on 3.14	High

Component	Issue Number	Description	Severity
FISR	ETD-74867	FISR Fails to Place a Fault when the normal open tie switch has old "set" fault indicator measurements	High
FISR	ETD-77628	Protection setting Phase Trip value discrepancy in FISR result file and tabular display	Medium
FISR	ETD-77692	FISR doesn't take default Problem Formulation	Medium
FISR	ETD-78195	FISR doesn't consider all open ties for plan generation.	Medium
Grid Tabular	ETD-76485	3.15 comment field on anno summary duplicating comments	Medium
Grid Tabular	ETD-77515	Columns on grid tabular reorder themselves	Medium
History Correction	ETD-75136	Data Correction Not Saving when Hitting OK	Medium
IDMS UI	ETD-4444	Permanent Jumper Label management	Medium
IDMS UI, Tagging	ETD-76796	Tag creation on switch (which is part of COMPSW) creating three tag placements instead of one on schematic display	High
Interfaces	ETD-77502	Below Minimum Meter Voltage Threshold value does not create any outage in OMS.	Medium
Modeling	ETD-76040	Customers Appear de-energized if model built with 3.14.0.10224 or newer converter	High
NRT, OMS	ETD-75558	ArchiveApp Crash due to Integrity Check - Requires Manual Intervention	High
OMS	ETD-74269	Orders Worksheet ETR field is populated with ITR when should be empty per TC	Low
OMS	ETD-75696	Playing SWO in Study Mode can Send commands to RT DNOM	High
OMS	ETD-76622	Restoration Orders with 0 Customers Affected on them	Medium
OMS	ETD-76992	Cannot remove Jumper to no-where from Mobile Subs	Medium
OMS	ETD-76994	Updating the working status manually does not fire the case note change	Medium
OMS	ETD-77189	History Correction Issue After OMS Failover	High
OMS	ETD-77230	System allowed to create predicted outage even though there is a confirmed order on device	Medium
OMS	ETD-77304	Pending Clear Order on a De-energized line with multiple outages on the same breaker	High
OMS	ETD-77544	CLOSE task type doesn't exist on Task type dropdown.	Low
OMS	ETD-77850	Customer Binary Loads causes Old Calls to reactivate	High
OMS	ETD-77851	WorkModel Custom Fields Not Syncing to Standby Servers	Critical
OMS	ETD-77892	Called Customers	Medium
OMS	ETD-78255	OMS Server abort when WFM updates incident that no longer exists	High
OMS-Incidents	ETD-77204	Structure navigation on incidents goes to substation instead of the Pole	High

Component	Issue Number	Description	Severity
OMS-Incidents	ETD-77442	OMS - Smoke Test - In ADMS 3.12 (TEST2), duplicate Incidents are being created on Devices (Fuse, Recloser, Substation Breaker) with Inactive Customers underneath	High
OMS-Incidents	ETD-77609	OMS - Multiple incidents are created when a predicted fuse is confirmed out on a multi phase feeder	High
SA	ETD-77632	SA 2.1 - In Storm Assist, user is able to open and close SCADA devices (ex. breakers, reclosers)	High
SA	SA-6232	Create single customer incident from map creates order task as confirmed	Medium
Tools-Modeling	ETD-67798	Incorrect Message displayed for converter. The power flow was not solved.,An unknown Transformer Type Type6SimpleX was assigned to Transformer	Low
Tools-Modeling	ETD-75890	Converter crashes when "Create Station Details File" option is set to True	High
UI	ETD-75575	Some fields should be exempted from /RIGHTTOLEFTLAYOUT	Low
UI	ETD-75635	OMS.RT.WORKMODEL not displaying all orders	High
UI	ETD-77215	Four icons on Data Correction tabular call up empty sub-tabular	Medium
Work Model	ETD-77877	Order Worksheet Error : Order status Can't be Changed to Cancelled from RMB after entering Dispatcher Comment	Medium

32. Known Issues

This chapter identifies important issues that are yet to be resolved at the time of this release.

32.1. Priority Not Retained After Merging Confirmed Outages

Identified in ADMS 16.25.1

When two confirmed outages with different priority levels are merged, the highest priority is not retained. (ETD-84411)

32.2. Model Export Does Not Populate CommunicationId

Identified in ADMS 16.25.1

Model Export does not populate the CommunicationId field in the exported data. (ETD-84408)

32.3. Order Name Changes After Convert and Merge

Identified in ADMS 16.25.1

When a non-outage order is converted and merged into a predicted order, the order name changes. (ETD-84404)

32.4. Customer Compiler Output (BPCL) Misaligned with Duplicate or Missing Data

Identified in ADMS 16.25.1

The BPCL customer compiler output is misaligned and contains duplicate or missing data. (ETD-84391)

32.5. Incorrect Transformer Display After ServicePoint Configuration Change

Identified in ADMS 16.25.1

After modifying ServicePoint settings in the VCE (clearing Permanent Non-Main Hide and setting Visible Threshold Zoom to 4.5), navigating to a distribution transformer from a tabular view may result in incorrect Toolbar data, blank transformer attributes, and incorrect transformer icon display. (ETD-84376)

32.6. Event Scriptor Does Not Create Calls Under Correct Conditions

Identified in ADMS 16.25.1

Event Scriptor does not create calls when the configured conditions are met. (ETD-84374)

32.7. Model Manager Reprocess Does Not Update Station Status

Identified in ADMS 16.25.1

The Model Manager Reprocess feature does not update the station status as expected. (ETD-84371)

32.8. Assign Crew Pop-Up List Truncated

Identified in ADMS 16.25.1

When assigning a crew from the Order Worksheet, the crew selection list may appear truncated. In Outage Assist (OA), the behavior depends on the zoom level, and in ADMS, it depends on the position of the Assign Crew pop-up on the screen. (ETD-84366)

32.9. Predicted Order Recreated After Repair of Merged Order

Identified in ADMS 16.25.1

When repairing a merged order, ADMS recreates a predicted order on the device that was originally merged. (ETD-84365)

32.10. OA Orders Table and Filters Tab Inconsistency

Identified in ADMS 16.25.1

When a saved filter is edited and the browser is refreshed, the Filters tab does not reflect the

updated selection, while the Orders table remains filtered based on the modified criteria. (ETD-84363)

32.11. Fuse Repair Causes Lateral De-Energization and Missing Switching Step

Identified in ADMS 16.25.1

When repairing a fuse, the lateral may become de-energized instead of being restored, and the expected switching step is not generated. (ETD-84361)

32.12. Cut Phase Position Always Displays as Closed in OA

Identified in ADMS 16.25.1

In Outage Assist, the cut phase position is not displayed correctly and always appears as Closed, regardless of its actual state. (ETD-84358)

32.13. Vehicle Icon Not Displayed When Toggling Show/Hide Vehicle

Identified in ADMS 16.25.1

When the Show/Hide Vehicle toolbar button is used, the vehicle icon is not displayed on the map as expected. (ETD-84352)

32.14. OMS Restoration Not Completing During Downstream Confirm

Identified in ADMS 16.25.1

OMS does not complete restoration when performing a downstream confirm. (ETD-84302)

32.15. History/Data Correction Counts Not Updating in OMS Display Plugin

Identified in ADMS 16.25.1

History/Data Correction counts are not updated automatically on the side menu of the OMS Display Plugin. (ETD-84301)

32.16. Archive Upgrade Script Errors

Identified in ADMS 16.25.1

Errors occur while running the archive upgrade script. (ETD-84295)

32.17. AMI Events on 3-Phase Underground Model Create Duplicate Devices

Identified in ADMS 16.25.1

AMI events on a 3-phase underground device modeled as three single-phase devices create two fuses and multiple service points. (ETD-84287)

32.18. Customer Compiler Internal Error with High Memory Usage

Identified in ADMS 16.25.1

The Customer Compiler may throw an "Array dimensions exceeded supported range" internal error when memory usage becomes high. (ETD-84229)

32.19. Mobility Gateway Sends Customer Address with Extra Whitespace

Identified in ADMS 16.25.1

The Mobility Gateway sends customer address values with extra whitespace. (ETD-84192)

32.20. Incident Lock Timeout During Storm Conditions

Identified in ADMS 16.25.0

During storm conditions, incident locks may not be acquired successfully. The application may return timeout errors when attempting to obtain an incident lock. Restarting the OIS service temporarily resolves the issue. (ETD-84279)

32.21. CEA Message Clears Data After Populating Text-Based Incident Custom Field

Identified in ADMS 16.25.0

When a CEA message is processed after a text-based incident custom field is populated, the existing data may be cleared. (ETD-84276)

32.22. Voltage Not Displayed in Call Ping Popup

Identified in ADMS 16.25.0

Voltage values are not displayed in the Call Ping popup. Although the popup for single-customer and multiple-customer call pings includes voltage columns and the API supports returning voltage data, the popup does not refresh or display the voltage values correctly. (ETD-84267)

32.23. .NET Exception When Moving Up in Order List

Identified in ADMS 16.25.0

A .NET exception may occur when attempting to use the Move Up action from the order list. (ETD-84235)

32.24. OMS Main Crash with Dump Files Generated

Identified in ADMS 16.25.0

The OMS Main service may crash and generate dump files during operation. (ETD-84232)

32.25. Non-Outage Call Stuck After Merge into Confirmed Outage

Identified in ADMS 16.25.0

A non-outage call may remain in the CE payload affected customers list after being merged into a confirmed outage incident. (ETD-84230)

32.26. Auto-Merging Between Customer Outage and Non-Outage Calls

Identified in ADMS 16.25.0

When an outage customer call and a non-outage customer call exist for the same customer, both calls may automatically merge into the non-outage call incident or order, while the outage call incident or order remains active. (ETD-84208)

32.27. Incidents Missing in Archive Database

Identified in ADMS 16.25.0

Some incidents may exist in the archive journal file but may not be present in the archive database. (ETD-84160)

32.28. Duplicate Restoration and Extra Switching Steps During Partial Switching

Identified in ADMS 16.25.0

Partial switching operations may result in duplicate restoration steps and additional switching steps being created. (ETD-84069)

32.29. Model Export JSON Omits Static Loads and Adds LOAD_ Suffix to Transformer ID

Identified in ADMS 16.25.0

The Model Export JSON may not include static loads, and the transformer ID may be exported with an unintended LOAD_ suffix. (ETD-84002)

32.30. End Time Error on Order Worksheet for Emergency Switch

Identified in ADMS 16.25.0

An end time validation error may occur on the Order worksheet when performing a switch in Emergency mode. (ETD-83959)

32.31. Workorder Database Sync

Identified in ADMS 16.25.0

After switching to a new server, the previously enabled Workorder database synchronization may throw a foreign key (FK) constraint exception. (ETD-83957)

32.32. Outage Assist Cannot Merge Non-Outage into Predicted Outage

Identified in ADMS 16.24.0

Outage Assist prevents merging a non-outage incident into a predicted outage and displays no master eligible incidents, even though the action is supported in ADMS. (ETD-84152)

32.33. Unable to Navigate to Meter from Ping Requests List

Identified in ADMS 16.24.0

Navigation to a meter fails when initiated from the Ping Requests list accessed through the Incident Customer tab. (ETD-84124)

32.34. Archive PURGE Fails Due to FK_INCIDENT_DEVICE Constraint

Identified in ADMS 16.24.0

The Archive PURGE command fails because of a FK_INCIDENT_DEVICE foreign key constraint violation. (ETD-84061)

32.35. Incorrect Task Creation When Confirming Predicted Phases

Identified in ADMS 16.20.0

In some cases, when confirming open for a subset of predicted phases on a device, an OPEN task is created on the device instead of a check prediction task, and no REPAIR task is created. (BUG-1068171)

Workaround: Close the device from the map. This creates a CLOSE task on the order, which allows

the order to be completed and closed.

32.36. ETR Increment and Decrement Buttons Incorrectly Enabled and Disabled

Identified in ADMS 16.16.2

The + and - buttons associated with the ETR may become incorrectly enabled or disabled on the order worksheet. It may be necessary to update the ETR by typing the correct date and time or opening the popup to set the date and time values.

32.37. Updating Crew Response Times from the Order Worksheet

Identified in ADMS 16.16.0

If an initial attempt is made to update the dispatch, en route, arrival or release time for a crew on the Order Worksheet or Crew Details panel and the entered time is found to be invalid, the invalid value is discarded and the original value remains on the user interface. If a second attempt is then made to perform such an update with the same invalid value, however, the validation failure message is not displayed and the invalid value is displayed on the user interface. The value is not saved, however, and refreshing or reopening the dialog shows the previous valid value.

32.38. Creating or Editing Tasks with a Completion Time from the Order Worksheet

Identified in ADMS 16.16.0

When creating or editing a task with the Create or Edit Task dialogs from the Order Worksheet, if an invalid completion time is entered validation runs and prevents the value from being saved. If the date/time control is opened again with the same invalid value however, pressing "Apply" does not enforce the completion time validation rules and the invalid completion time may be saved.

32.39. Order Ownership Issue in Map View

Identified in ADMS 16.4.0

Taking Ownership of an order doesn't work from Map View, but it works from Tabular view. The workaround is to set ShowOrdersInsteadOfIncidents=true in Netconf_localhost.xml file for the client.

32.40. LDAP Filter Issue in Internal Notifications

Identification in ADMS 16.2.0

Internal notifications are not recognizing users as authorized, and the LDAP filter for Microsoft Active Directory seems to be malfunctioning.

32.41. Availability of Internal Notifications

Identified in ADMS 3.14 or earlier

As of ADMS 3.14, the Internal Notifications application is not highly available. High availability will be supported in a future release.

32.42. Software Development Kit

Identified in ADMS 3.14 or earlier

The ADMS Software Development Kit (ADMS_Interfaces_SDK.msi) contains coding samples to enable integration with third-party products and enterprise applications. The coding samples are intended to provide an overview on how ADMS APIs can be used to create an interface application.

Important

The delivered coding samples are not ready for production and are not supported. It is the utility's responsibility to make sure that applications developed based on ADMS SDK samples are ready for production before using the applications.

33. Limitations

This chapter identifies important restrictions and exclusions.

33.1. Call Server and Crew Server TCP/IP Interfaces

Using the Call Server and Crew Server HTTP interfaces instead of the legacy TCP/IP interfaces is recommended for communicating with external systems, such as External Interactive Voice Response (IVR), Automated Metering Infrastructure (AMI), Customer Information System (CIS), and Workforce Management System (WFM).

The legacy Call Server and Crew Server TCP/IP interfaces are only supported with security enabled. For more information about enabling the TCP/IP interface security, see section Security in the *ADMS Outage Management Integration Programmer Guide*.

33.2. Multiple Order Worksheets

When using order-based workflows, multiple Order Worksheets can be opened at the same time in either popup or tabbed mode. Although there is currently no coded limit to the number of worksheets that can be opened at once, the CPU, memory, and database size can impose constraints on how many can be managed without adversely affecting client performance. It is therefore currently recommended that no more than 2 or 3 worksheets be left open on a client session at one time.

33.3. Optimized OMS Modeling

OMS is designed to efficiently handle scenarios where each distribution transformer serves only a single downstream service point. To achieve optimal performance, it is recommended to model the system following this principle.

33.4. Outage Prediction for Transformers with a Single Customer

When resuming outage prediction, incidents for transformers that serve only one customer are not converted into predicted incidents. This behavior affects scenarios where a single-customer incident is expected to transition automatically during the resume process. Although this does not impact the overall outage prediction workflow, it is a current limitation for single-customer transformer scenarios.

33.5. Multi-Site Configuration Support

Internal Notifications is not site-aware and does not support multi-site configuration.

33.6. Sealights Installation on ADMS Service Machines

Sealights must not be installed on machines where ADMS services are deployed, as the instrumentation overhead can impact system performance and affect testing results.

33.7. Handling of Related and Follow-Up Maintenance Orders

For Related Maintenance Orders and Follow-Up Maintenance Orders, the system creates a new incident rather than associating the orders with the existing incident.

Additionally, the order name may not follow the incident name because a group name or parent order name is assigned to subsequent orders. This allows follow-up orders to be grouped together for easier identification and tracking.

33.8. Security

33.8.1. Enabling HTTPS Mode

By default, ADMS servers are configured to use HTTP. GE strongly recommends that mutual TLS with HTTPS is enabled before using ADMS in production environment. For more information about enabling HTTPS, refer to the *ADMS Software Installation and Maintenance Guide*.

33.8.2. Using Certificate Authority Certificates in HTTPS Mode

When running web services in HTTPS mode in production, SSL certificates issued by a certificate authority must be used. (Security vulnerabilities appear when using self-signed certificates.)

34. Security Notes

34.1. Common Vulnerabilities and Exposures

A Common Vulnerabilities and Exposures (CVE) Scan was completed for this release. The following table describes the High or Critical severity CVEs that were identified and whether or not they affect this release.

If they do not affect the release, these details are provided for your information only.

ADMS Component	CVE Number	Description	Affect This Release?	Comments
ADMS preferences_service.jar	CVE-2025-66566	yawkat LZ4 Java provides LZ4 compression for Java. Insufficient clearing of the output buffer in Java-based decompressor implementations in lz4-java 1.10.0 and earlier allows remote attackers to read previous buffer contents via crafted compressed input.	No	Component Details- org.elasticsearch:elasticsearch
	CVE-2025-37731			Version- 7.17.29 Fix versions- 8.19.8, 9.1.8, 9.2.2 Impact Path ADMS_server.msi_unarchived.tar.gz//preferences_service.jar_unarchived/BOOT-INF/lib/elasticsearch-7.17.29.jar
Internal Notification UI	CVE-2025-15284	Improper Input Validation vulnerability in qs (parse modules) allows HTTP DoS.	No	Component Details: qs; Version: 6.14.0; Fix version: 6.14.1
	CVE-2025-61729			
	CVE-2025-66412			Component Details- github.com/golang/go; Version- 1.23.12; Fix versions- 1.24.11, 1.25.5
	CVE-2025-61725			Component Details- @angular/compiler; Version- 9.0.0; Fix versions- 19.2.17, 20.3.15, 21.0.2 Component Details- github.com/golang/go; Version- 1.23.12; Fix versions- 1.24.8, 1.25.2

ADMS Component	CVE Number	Description	Affect This Release?	Comments
Work Model	CVE-2025-5914 CVE-2025-24928 CVE-2025-6965	A vulnerability has been identified in the libarchive library, specifically within the archive_read_format_rar_seek_data() function.	No	Components Details- 8:libarchive Version- 0:3.3.3-5.el8 Fix versions- 0:3.5.3- 2.el9_0.1 (RHSA- 2025:14141), 0:3.5.3- 4.el9_4.1 (RHSA- 2025:14142), 0:3.5.3- 5.el9_2 (RHSA- 2025:15024), 0:3.5.3- 6.el9_6 (RHSA- 2025:14130) Impact Paths- adms- docker_adms-xray -manifest_docker -images-scan → sha256_ 3d25156be0060f7c8cb 24181b6a1 c906d4c619fb95046f1 ba55cb00d78f0953d → 9:libarchive 9:libarchive:0:3.5.3- 4.el9 sha256_3d25156be00 60f 7c8cb24181b6a1c906d 4c61 9fb95046f1ba55cb00d 78f0953d Component Details: 9:libarchive Version0:3.5.3-4.el9 Fix versions0:3.5.3- 2.el9_0.1 (RHSA- 2025:14141), 0:3.5.3- 4.el9_4.1 (RHSA- 2025:14142), 0:3.5.3- 5.el9_2 (RHSA- 2025:15024), 0:3.5.3- 6.el9_6 (RHSA- 2025:14130) Component Details- 9:libxml2 Version0:2.9.13- 6.el9_5.1 Fix versions0:2.9.13- 1.el9_0.4 (RHSA- 2025:2483), 0:2.9.13- 3.el9_2.6 (RHSA- 2025:2482), 0:2.9.13- 6.el9_5.2 (RHSA- 2025:2679), 0:2.9.13- 9.el9_4 (RHSA-

ADMS Component	CVE Number	Description	Affect This Release?	Comments
Outage Assist	CVE-2025-59375	libexpat in Expat before 2.7.2 allows attackers to trigger large dynamic memory allocations via a small document that is submitted for parsing.	No	Component Details- 8:expat Version- 0:2.2.5-13.el8_10 Fix version- 0:2.5.0-1.el8_10 (RHSA-2025:21776)

34.2. Cyber Security Change Control

In order to meet the NERC CIP Requirement: CIP-003 Cyber Security — Security Management Controls, Table below shows the Critical Cyber Assets that have changed in this release.

For a list of ports used by ADMS, refer to the *ADMS Software Installation and Maintenance Guide*.

Table 1. Changes to Critical Cyber Assets

Critical Cyber Asset	Change	Comment
Mobility Gateway App	Add	Added port 8443

35. System Requirements and Software Compatibility

This chapter lists the system requirements for this release and provides information about the software compatible with the release along with the minimum version and patch/build level.

35.1. What Do "Supported" and "Certified" Mean?

At GE, we state that a particular system or configuration is "unsupported", "supported", or "certified". These terms usually refer to operating system and compiler configurations, but they can also be used to refer to other scenarios.

Below are GE's definitions for each of these terms.

- An *unsupported* configuration is one that has not been tested at all and is not guaranteed to work. Just because a configuration is unsupported, it does not necessarily mean that it will not work — but there are no guarantees. GE strongly recommends against using unsupported configurations, especially in an operational environment. If a customer does choose to use an unsupported configuration and runs into difficulties, any time spent supporting that configuration will be charged directly to the specific customer requiring support.
- A *supported* configuration is one that we believe will work without any problems, but on which we have performed only basic testing. Due to the lower level of testing, it is possible that some unexpected behavior may occur. However, since it is a supported configuration, if any issues do arise, we will work quickly to get them resolved. Note that, if a customer chooses to use a supported configuration, additional testing may be required by the customer, to compensate for the reduced level of testing performed by GE.
- A *certified* configuration is one that GE has set up in our labs and performed significant product testing on. Problems with a certified configuration are less likely due to our extensive testing; however, if any problems do occur, we will work quickly to get them resolved.

The following sections describe the supported software and hardware configurations, as well as the certified configurations.

35.2. Hardware Requirements

35.2.1. ADMS

The system used for certification testing in GE's development lab was based on the following hardware:

- Servers:
 - **CPU:** 4 Intel Xeon X7560 @ 2.27 GHz processors with 8 cores each (with hyper-threading)

enabled, 64 threads can run in parallel).

- **RAM:** 64 GB
- **Clients:**
 - **CPU:** 2 @ 2.27 GHz processors with 4 cores each.
 - **RAM:** 16 GB
 - **Storage:** 5 GB
 - **Display:** Due to the intense nature of the displays of the distribution networks, higher-performance video cards that support multiple high-resolution monitors are recommended.
 - **Scaling:** The certified size scaling for text, applications, and other items on the screen is 100%. Display layout issues may appear at higher size scaling.
 - **Resolution:** The certified display resolution is 1920x1080. Ultra-high display resolutions (for example, 3840 x 2160) are not certified. Display layout issues may appear.

For more information, refer to the *ADMS Software Installation and Maintenance Guide*.

Note

ADMS offers flexible deployment options, such as core applications that can be deployed together on the same machine or deployed separately across several machines. It is recommended that you consult with a solution architect for individual project hardware configurations.

35.2.2. EOMS

35.2.2.1. Deployment in Kubernetes

To successfully install and use Enterprise OMS in Kubernetes, the minimum hardware requirements are as follows:

- **Kubernetes Server:** A high availability topology-capable (HA) cluster setup is highly recommended. Use at least one node with the following properties:
 - **Memory:** 16 GB per node.
 - **CPU:** 4 cores per node.
 - **Storage:** At least 20 GB per node.
 - **Operating System:** Red Hat Enterprise Linux 7.5 or later or CentOS 7.x.
- **Deployment Server:** A server specifically for deployment purposes.
 - **Memory:** 16 GB.
 - **CPU:** 4 cores.
 - **Storage:** 200 GB.

- **Operating System:** Microsoft Windows Server 2022 (Windows 11), Red Hat Enterprise Linux 7.5 or later, or CentOS 7.x.
- **Clients:** The Enterprise OMS user interface is embedded into the ADMS client. The following requirements apply to ADMS clients.
 - **CPU:** 2 @ 2.27 GHz processors with 4 cores each.
 - **Memory:** 16 GB.
 - **Storage:** 5 GB.
 - **Display:** Due to the intense nature of the displays of the distribution networks, higher-performance video cards that support multiple high-resolution monitors are recommended.
 - **Monitor:** A monitor with full HD (1920x1080) display resolution is recommended.
 - **Scaling:** The certified size scaling for text, applications, and other items on the screen is 100%. Display layout issues may appear at higher size scaling.
 - **Resolution:** The certified display resolution is 1920x1080. Ultra-high display resolutions (for example, 3840 x 2160) are not certified. Display layout issues may appear.
 - **Operating System:** Windows 11.

For ADMS server and client requirements, refer to [ADMS](#).

It is recommended that you consult with GE regarding your project-specific hardware and software requirements.

35.2.2.2. Installing in Windows

To successfully install and use Enterprise OMS in Windows, the minimum hardware requirements are as follows:

- **Enterprise OMS Server:**
 - **CPU:** 4 cores.
 - **Memory:** 16 GB of RAM.
 - **Operating System:** Windows Server 2022 or Windows Server 2019.
- **Clients:** The Enterprise OMS user interface is embedded into the ADMS client. The following requirements apply to ADMS clients.
 - **CPU:** 2 @ 2.27 GHz processors with 4 cores each.
 - **Memory:** 16 GB.
 - **Storage:** 5 GB.
 - **Display:** Due to the intense nature of the displays of the distribution networks, higher-performance video cards that support multiple high-resolution monitors are recommended.
 - **Monitor:** A monitor with full HD (1920x1080) display resolution is recommended.
 - **Scaling:** The certified size scaling for text, applications, and other items on the screen is

100%. Display layout issues may appear at higher size scaling.

- **Resolution:** The certified display resolution is 1920x1080. Ultra-high display resolutions (for example, 3840 x 2160) are not certified. Display layout issues may appear.
- **Operating System:** Windows 11.

35.2.3. Outage Assist

The Outage Assist hardware requirements are driven by the Foundation hardware requirements.

To successfully install and use Outage Assist with the demo system (Fantasy Island), the minimum Kubernetes Foundation hardware requirements are as follows:

- **Kubernetes Servers:** For performance reasons, use at least four nodes with the following properties:
 - **Memory:** 16 GB per node
 - **CPU:** 4 cores per node
 - **Storage:** At least 200 GB per node
 - **Operating System:** Foundation Base is supported on Red Hat Enterprise Linux (RHEL) 8.3 64-bit, using the Minimal Install option, with the latest updates and access to RHEL RPM repositories.

Note

A high availability (HA) topology-capable cluster setup is highly recommended. The hardware configuration will vary depending on the container replica set required in the HA topology. To support a basic HA topology, use at least five nodes with the above hardware configuration, ensuring that:

- Three of the nodes are configured as "stacked" master nodes (master plus agent)
- The remaining nodes configured as "agent" nodes

For greater workloads and to support more concurrent clients, the resources assigned to each Storm Assist worker node can be increased.

- **Deployment Server:** A server specifically for deployment purposes.
 - **Memory:** 16 GB
 - **CPU:** 4 cores
 - **Storage:** 200 GB
 - **Operating System:** Red Hat Enterprise Linux (RHEL) 8.7 64 bit
- **Read-Only Servers:** Used to house NetviewerAdapter Read-Only services and OMS Read-Only server to provide EOMS data for Outage Assist. These servers should have the following properties:

- **Memory:** 64 GB
- **CPU:** 16 cores
- **Storage:** 200 GB
- **Operating System:** Microsoft Windows Server 2019
- **Outage Assist Clients:**
 - **Memory:** 4 GB
 - **CPU:** 2 cores
 - **Display:** Using a monitor with full HD (1920x1080) display resolution is recommended

The above sizing numbers reflect minimal resource requirements for Foundation layers; the actual sizing should be determined for a specific solution and set of workloads, with realistic data sets (and load test).

It is recommended that you consult with GE regarding your project-specific hardware and software requirements.

35.3. Operating System

35.3.1. ADMS

This section describes which operating systems are certified and supported for each ADMS component.

Note

Red Hat Enterprise Linux 7.5 or later is supported only for Internal Notifications and Enterprise OMS.

Table below presents the supported (S), unsupported (-), and certified (C) platforms for each ADMS component.

Table 2. Certified and Supported Platforms

ADMS Component	Windows 10 64-Bit	Window 11 64-Bit	Windows Server 2019 64-Bit	Windows Server 2022 64-Bit
ADMS DMS Server (NetSrv)	S	C	S	C
ADMS OMS Server (NetServer)	S	C	S	C
ADMS DNAF Server (DnafSrv)	S	C	S	C
ADMS Client	S	C	S	C
ADMS Tools	S	C	S	C
ADMS File Updater Server	S	C	S	C

ADMS Component	Windows 10 64-Bit	Window 11 64-Bit	Windows Server 2019 64-Bit	Windows Server 2022 64-Bit
ADMS File Updater Client	S	C	S	C
ADMS Documentation Suite	S	C	S	C
Other ADMS Components	S	C	S	C

Note

GE recommends using Windows Server 2022. For more details, contact your GE project team.

35.3.2. EOMS

This section describes which operating systems are compatible with EOMS.

- **Server Operating System:**
 - **Deployment in Kubernetes:** Red Hat Enterprise Linux 7.5 and later or CentOS 7.x.
 - **Deployment in Windows:** Windows Server 2022 or Windows Server 2019.
- **Client Operating System:** Windows 11.

35.3.3. Outage Assist

This section provides compatibility details for operating systems.

- **Kubernetes Server Operating System:** Red Hat Enterprise Linux (RHEL) 8.7 64 bit
- **Deployment Server Operating System:** Red Hat Enterprise Linux (RHEL) 8.7 64 bit
- **Client Operating System:** Outage Assist clients are platform-independent.

35.4. Compatible GE Software

ADMS requires and works with a number of other GE products to form a comprehensive and advanced distribution management system.

The tables in this section list the GE products and components (along with the required minimum version and patch) that are compatible with ADMS 16.25.1. Each of these products is delivered with release notes. After you have read this document, GE encourages you to review the release notes for all other products that you plan to install.

Table 3. Certified and Supported GE Products

Product	Version	Patch	Supported / Certified	Comments
Habitat	5.12.2		Certified	

Product	Version	Patch	Supported / Certified	Comments
Scada	3.6.2		Certified	
	3.5		Supported	
Advanced Presentation Framework	4.4 SP2		Certified	Advanced Presentation Framework is required for running Browser.
Browser	4.4 SP2		Certified	Only Windows 10 (x64) clients are certified.
	4.4		Supported	

Table 4. Other Compatible GE Products and Components

Product	Version	Patch/B uild Level	Comments
Display Builder	6.4 SP1 (for Browser 4.4 SP2)		Display Builder is used to modify, create, and compile Browser displays. Substation Editor uses the output of Display Builder element set (ESET) files and creates a DDL file for Scada displays. When using Display Builder, you must build displays that are compatible with Display Builder 3.6, which is used to build ADMS displays.
	6.4		
Forecasting	3.11.0 & 16.25.1 (certified)		Forecasting enables utilities to forecast demand for electric energy and production of renewable energy (photovoltaic, wind, concentrated solar, etc.) while factoring in weather, historical data, and holidays (calendar) for a given network or grid.
	3.5 (supported)		
Gateway	2.9.1 & 16.25.1 (Certified)		The Gateway solution bundle includes the IEEE 2030.5 Gateway and the DER Registration and Lifecycle Manager components. Gateway enables offline and online management of Distributed Energy Resources (DERs) in the utility operational context.
	2.3 (supported)		
Foundation	25R13		GE's common Kubernetes-based platform, which provides security, storage, notification, and monitoring functionality for GE products that are deployed in Kubernetes.
RKE2 Kubernetes cluster	v1.24.7+rke2r1		Deployed using GE's Kubernetes PDI.

In addition to the products listed in the tables above, other GE products (for example, Source, Historian, and Control) are compatible but not required for ADMS. The specific versions of other GE products are documented in the release notes for those products.

35.5. Third-Party Software

35.5.1. ADMS

To be fully functional, ADMS requires the third-party products listed in the following tables. For details about third-party product licenses, refer to the `ThirdPartyLicenses.txt` file, which is provided with the Release Notes.

Table 5. Applications and Libraries for the Common Functionality

Product Name	Certified Version	Required?	Installed by ADMS?	Dependent ADMS Components
Amazon Corretto	17.0.8.8.1	For the Group Control application and Enterprise OMS. Ensure Environment Variable CORRETTO_HOME is set to the Corretto 17 installation directory.	No	Client
Boost	1.82.0	Yes	Portions	Server, File Updater Server, and File Updater Client
colamd/symamd	2.1	Yes	Portions	Server, Client, and Model Export
convertUTF.c	October 2004	Yes	Portions	Server, File Updater Server, and File Updater Client
FO.NET	1	For exporting switch orders to PDF.	Portions	Server, Client, and Software Development Kit (Notification Publisher)
NuGet Packages	N/A	Multiple	Portions	Server, Client, Tools, and Interfaces
i Note A list of required NuGet packages can be provided upon request.				
Microsoft .NET Framework	4.8	Yes	No	Server, Client, Configuration Tools, and Modeling Tools
.NET 8.0.3 Runtime (part of Microsoft .NET Framework 4.8.0)	8.0.3	Yes	No	Process Automation
ASP.NET Core Runtime Hosting Bundle (part of Microsoft .NET Framework 4.8.0)	8.0.3	Yes	No	Interfaces
.NET Desktop Runtime (part of Microsoft .NET Framework 4.8.0)	8.0.3	Yes	No	Interfaces

Product Name	Certified Version	Required?	Installed by ADMS?	Dependent ADMS Components
Microsoft SQL Server	2022 RTM (certified) 2019 RTM (certified) 2014 SP3 CU4 (supported) 2012 SP4 (supported)	For Archive DB on SQL Server.	No	Archive
Microsoft SQL Server Reporting Services	2022	For Reports.	No	Reports
Microsoft Visual C++ 2015-2022 Redistributable Package	x64 & x86 (14.40.33810)	Yes	No	DNOM2HDB, Server, Client, and Configuration Tools
OpenStreetMaps	N/A	Optional	No	Client (web maps)
Oracle Data Provider for .NET (ODP.NET)	19.21.0	For Archive app with Archive DB on Oracle.	No	Archive
Oracle Database	19c (certified) 11gR2 (supported) 12cR2 (supported)	For Archive DB on Oracle.	No	Archive
VanDyke SecureFX	9.3.0	For SFTP mode on Windows.	No	File Updater Client and File Updater Server (for SFTP mode on Windows)
VanDyke VShell Server for Windows	4.8.0	For SFTP mode on Windows.	No	File Updater Server (for SFTP mode on Windows)
Xerces	3.2.4	For Safety Doc, Orphan Tags, DNOM2HDB, Model Export, Model File Exporter, and DNOM Unique ID Checker.	Portions	Server, Client, Tools, Model Export, and DNOM2HDB
zlib	1.2.13	For Server, Model Export, Model File Exporter, File Updater, and DNOM Unique ID Checker.	Yes	Server, Client, File Updater Server, File Updater Client, and Model Export
gRPC	1.51.1	Server	Yes	Server
jquery	3.5.1	For HTML displays.	Yes	Client
NewtonSoft.Json	13.0.3	Json serialization	Yes	Server, Client, Tools and Interfaces
jqueryui	3.5.1	For HTML displays.	Yes	Client

Product Name	Certified Version	Required?	Installed by ADMS?	Dependent ADMS Components
moment-timezone	0.5.41	For the Order Worksheet web pages. For the Crew Select dialog.	Yes	Client
moment.js	2.29.4	For the Order Worksheet web pages. For the Crew Select dialog.	Yes	Client
moment.js	2.26.0	For HTML displays.	Yes	Client
Chart.js	3.3.2	For HTML displays.	Yes	Client
D3	6.7.0	For LVM Monitor.	Yes	Client
font-awesome	4.7.0	For LVM Monitor.	Yes	Server and Client
@angular/animations	15.2.10	For the Group Control application.	Yes	Client
@angular/animations	11.2.11	For the Switching Order Gantt Chart.	Yes	Client
@angular/animations	20.2.3	For the Order Worksheet web pages.	Yes	Client
@angular/animations	18.2.6	For the Crew Select dialog. For LVM Monitor.	Yes	Client
@angular/cdk	11.0.7	For the Switching Order Gantt Chart.	Yes	Client
@angular/cdk	20.2.1	For the Order Worksheet web pages.	Yes	Client
@angular/cdk	19.2.3	For the Crew Select dialog.	Yes	Client
@angular/cli	11.2.8	For the Switching Order Gantt Chart.	Yes	Client
@angular/cli	20.2.1	For the Order Worksheet web pages.	Yes	Client
@angular/common	15.2.10	For the Group Control application.	Yes	Client
@angular/common	11.0.7	For the Switching Order Gantt Chart.	Yes	Client
@angular/common	20.2.3	For the Order Worksheet web pages.	Yes	Client
@angular/common	19.2.0	For the Crew Select dialog.	Yes	Client
@angular/common	18.2.6	For LVM Monitor.	Yes	Client
@angular/compiler	15.2.10	For the Group Control application.	Yes	Client
@angular/compiler	11.0.7	For the Switching Order Gantt Chart.	Yes	Client
@angular/compiler	20.2.3	For the Order Worksheet web pages.	Yes	Client
@angular/compiler	19.2.0	For the Crew Select dialog.	Yes	Client

Product Name	Certified Version	Required?	Installed by ADMS?	Dependent ADMS Components
@angular/compiler	18.2.6	For LVM Monitor.	Yes	Client
@angular/core	15.2.10	For the Group Control application.	Yes	Client
@angular/core	11.0.7	For the Switching Order Gantt Chart.	Yes	Client
@angular/core	20.2.3	For the Order Worksheet web pages.	Yes	Client
@angular/core	19.2.0	For the Crew Select dialog.	Yes	Client
@angular/core	18.2.6	For LVM Monitor.	Yes	Client
@angular/forms	15.2.10	For the Group Control application.	Yes	Client
@angular/forms	11.0.7	For the Switching Order Gantt Chart.	Yes	Client
@angular/forms	11.2.14	For QSTS Monitor.	Yes	Client
@angular/forms	20.2.3	For the Order Worksheet web pages.	Yes	Client
@angular/forms	19.2.0	For the Crew Select dialog.	Yes	Client
@angular/forms	18.2.6	For LVM Monitor.	Yes	Client
@angular/localize	11.0.7	For the Switching Order Gantt Chart.	Yes	Client
@angular/localize	20.2.3	For the Order Worksheet web pages.	Yes	Client
@angular/localize	18.2.6	For the Crew Select dialog.	Yes	Client
@angular/material	11.0.7	For the Switching Order Gantt Chart.	Yes	Client
@angular/material	20.2.3	For the Order Worksheet web pages.	Yes	Client
@angular/material	19.2.3	For the Crew Select dialog.	Yes	Client
@angular/material-moment-adapter	18.2.6	For the Order Worksheet web pages. For the Crew Select dialog.	Yes	Client
@angular/platform-browser	15.2.10	For the Group Control application.	Yes	Client
@angular/platform-browser	11.0.7	For the Switching Order Gantt Chart.	Yes	Client
@angular/platform-browser	20.2.3	For the Order Worksheet web pages.	Yes	Client
@angular/platform-browser	19.2.0	For the Crew Select dialog.	Yes	Client
@angular/platform-browser	18.2.6	For LVM Monitor.	Yes	Client
@angular/platform-browser-dynamic	15.2.10	For the Group Control application.	Yes	Client

Product Name	Certified Version	Required?	Installed by ADMS?	Dependent ADMS Components
@angular/platform-browser-dynamic	11.0.7	For the Switching Order Gantt Chart.	Yes	Client
@angular/platform-browser-dynamic	20.2.3	For the Order Worksheet web pages.	Yes	Client
@angular/platform-browser-dynamic	19.2.0	For the Crew Select dialog.	Yes	Client
@angular/platform-browser-dynamic	18.2.6	For LVM Monitor.	Yes	Client
@angular/router	15.2.10	For the Group Control application.	Yes	Client
@angular/router	11.0.7	For the Switching Order Gantt Chart.	Yes	Client
@angular/router	20.2.3	For the Order Worksheet web pages.	Yes	Client
@angular/router	19.2.0	For the Crew Select dialog.	Yes	Client
@angular/router	18.2.6	For LVM Monitor.	Yes	Client
@ngx-translate/core	12.1.2	For the Switching Order Gantt Chart.	Yes	Client
@ngx-translate/http-loader	4.0.0	For the Switching Order Gantt Chart.	Yes	Client
@ngx-translate/i18n-polyfill	1.0.0	For the Switching Order Gantt Chart. For the Order Worksheet. For the Crew Select dialog.	Yes	Client
ngx-mat-select-search	7.0.6	For the Order Worksheet web pages. For the Crew Select dialog.	Yes	Client
angular-i18next	18.0.0	For LVM Monitor.	Yes	Client
angular-resizable-element	7.0.2	For the Order Worksheet web pages. For the Crew Select dialog.	Yes	Client
core-js	2.6.11	For the Switching Order Gantt Chart.	Yes	Server, Client
core-js	3.10.0	For LVM Monitor.	Yes	Server, Client
rxjs	7.8.1	For the Order Worksheet web pages. For the Crew Select dialog.	Yes	Client
rxjs	6.5.5	For LVM Monitor. For the Switching Order Gantt Chart. For the Group Control application.	Yes	Server, Client

Product Name	Certified Version	Required?	Installed by ADMS?	Dependent ADMS Components
zone.js	0.10.2	For LVM Monitor. For the Switching Order Gantt Chart. For the Group Control application.	Yes	Server, Client
zone.js	0.15.1	For the Order Worksheet web pages.	Yes	Client
zone.js	0.14.4	For the Crew Select dialog.	Yes	Client
i18next-browser-languagedetector	6.1.1	For LVM Monitor. For the Order Worksheet. For the Crew Select dialog.	Yes	Client
i18next-xhr-backend	3.2.2	For LVM Monitor.	Yes	Client
tslib	2.0.0	For the Order Worksheet web pages. For the Crew Select dialog. For the Group Control application.	Yes	Client
tslib	2.1.0	For LVM Monitor. For the Switching Order Gantt Chart.	Yes	Client
uuid	8.3.1	For the Order Worksheet web pages. For the Crew Select dialog.	Yes	Client
cpprestsdk	2.10.7	Yes	Yes	Server
brotili	1.3.2	For the Group Control application.	Yes	Client
browserify-fs	1.0.0	For the Group Control application.	Yes	Client
Apache HTTP Server	2.4.46	For the Group Control application.	No	Server, Client
PostgreSQL	14	For the Group Control application.	No	Server, Client
Crypto++	8.2	For the AesKeyMgr tool. For encrypting station models.	Yes	Server, Tools
Google.Protobuf	3.15.8	Yes	Yes	Server, Client, Tools, Interfaces
Apache FOP	2.7	Optionally or switching order PDF export.	Yes	Server

Table 6. Third-Party Products for the Internal Notifications Application

Product Name	Version	Required?	Installed by ADMS?	Dependent ADMS Components
font-awesome	4.7.0	For the Internal Notifications UI	Yes	Internal Notifications Application
Red Hat Docker	docker-1.13.1-91.git07f3374.el7.x86_64 or later	For the Internal Notifications deployment.	No	Internal Notifications Application

Product Name	Version	Required?	Installed by ADMS?	Dependent ADMS Components
Docker CE	18.06.1-ce or later	For the Internal Notifications deployment.	No	Internal Notifications Application
Kubernetes	1.19–1.24	For the Internal Notifications deployment.	No	Internal Notifications Application
Helm	2.14.3 or later	For the Internal Notifications deployment.	No	Internal Notifications Application
Google Chrome	102 or later	For displaying Internal Notifications web pages on the client.	No	Internal Notifications Application
PostgreSQL Database	15	For the Internal Notifications database.	No	Internal Notifications Application
core-js	2.6.12	For the Internal Notifications UI.	Yes	Internal Notifications Application
express	4.17.1	For the Internal Notifications UI.	Yes	Internal Notifications Application
rxjs-compat	6.6.7	For the Internal Notifications UI.	Yes	Internal Notifications Application
jquery	3.6.0	For the Internal Notifications UI.	Yes	Internal Notifications Application
font-awesome	4.7.0	For the Internal Notifications UI.	Yes	Internal Notifications Application
ag-grid-community	31.3.4	For the Internal Notifications UI.	Yes	Internal Notifications Application
rxjs	7.8.1	For the Internal Notifications UI.	Yes	Internal Notifications Application
zone.js	0.14.10	For the Internal Notifications UI.	Yes	Server, Client, and Internal Notifications Application
tslib	2.0.0	For the Internal Notifications UI.	Yes	Server, Client, and Internal Notifications Application
@angular/cdk	20.2.4	For the Internal Notifications UI.	Yes	Internal Notifications Application
@angular/common	20.3.1	For the Internal Notifications UI.	Yes	Internal Notifications Application
@angular/compiler	20.3.1	For the Internal Notifications UI.	Yes	Internal Notifications Application
@angular/core	20.3.1	For the Internal Notifications UI.	Yes	Internal Notifications Application
@angular/forms	20.3.1	For the Internal Notifications UI.	Yes	Internal Notifications Application
@angular/material	20.2.4	For the Internal Notifications UI.	Yes	Internal Notifications Application

Product Name	Version	Required?	Installed by ADMS?	Dependent ADMS Components
@babel/runtime	7.20.1	For the Internal Notifications UI.	Yes	Internal Notifications Application
@angular/platform-browser	20.3.1	For the Internal Notifications UI.	Yes	Internal Notifications Application
@angular/platform-browser-dynamic	20.3.1	For the Internal Notifications UI.	Yes	Internal Notifications Application
@angular/platform-server	20.3.1	For the Internal Notifications UI.	Yes	Internal Notifications Application
@angular/router	20.3.1	For the Internal Notifications UI.	Yes	Internal Notifications Application
angular-tree-component	8.5.6	For the Internal Notifications UI.	Yes	Internal Notifications Application
angular2-text-mask	9.0.0	For the Internal Notifications UI.	Yes	Internal Notifications Application
ng2-search-filter	0.5.1	For the Internal Notifications UI.	Yes	Internal Notifications Application
@types/lodash	4.14.158	For the Internal Notifications UI.	Yes	Internal Notifications Application
ngx-toastr	15.2.2	For the Internal Notifications UI.	Yes	Internal Notifications Application
ngx-cookie	5.0.2	For the Internal Notifications UI.	Yes	Internal Notifications Application
ngx-cookie-service	1.0.10	For the Internal Notifications UI.	Yes	Internal Notifications Application
ngx-spinner	16.0.2	For the Internal Notifications UI.	Yes	Internal Notifications Application
bootstrap	4.6.1	For the Internal Notifications UI.	Yes	Internal Notifications Application
toastr	2.1.4	For the Internal Notifications UI.	Yes	Internal Notifications Application
zone.js	0.15.0	For the Internal Notifications UI	Yes	Internal Notifications Application
url-parse	1.5.9	For the Internal Notifications UI	Yes	Internal Notifications Application
unset-value	2.0.1	For the Internal Notifications UI	Yes	Internal Notifications Application
tslib	2.0.0	For the Internal Notifications UI	Yes	Internal Notifications Application
rxjs-compat	6.6.7	For the Internal Notifications UI	Yes	Internal Notifications Application
Minimist	1.2.6	For the Internal Notifications UI	Yes	Internal Notifications Application

35.5.2. Outage Assist

This section lists the third-party software associated with Outage Assist.

Note

[Third-Party Licenses](#) provides license information for the third-party products that require including a copy of the license.

Table 7. Third-Party Software for Outage Assist

Product Name	Certified Version	Required?	Supplied with Outage Assist?	Dependent Outage Assist Components
Docker	20.10.10	Yes	No	Deployment
RKE2 Kubernetes	v1.24.7+rke2r1	Yes	Yes	Deployment
Google Chrome x64	107 or later with an enabled JavaScript engine	Yes	No	For displaying Foundation and Outage Assist web pages on the client
@angular/animations	20.0.3	Yes	Yes	User Interface
@angular/common	20.0.3	Yes	Yes	User Interface
@angular/core	20.0.3	Yes	Yes	User Interface
@angular/forms	20.0.3	Yes	Yes	User Interface
@angular/localize	20.0.3	Yes	Yes	User Interface
@angular/platform-browser	20.0.3	Yes	Yes	User Interface
@angular/platform-browser-dynamic	20.0.3	Yes	Yes	User Interface
@angular/router	20.0.3	Yes	Yes	User Interface
@angular/material	20.0.1	Yes	Yes	User Interface
lodash	4.17.21	Yes	Yes	User Interface
rxjs	7.8.1	Yes	Yes	User Interface
tslib	2.4.1	Yes	Yes	User Interface
xliff	5.5.3	Yes	Yes	User Interface
zone.js	0.15.1	Yes	Yes	User Interface

36. Installation Notes

36.1. ADMS

This section provides information that pertains to the installation of ADMS components. The material in this section augments the detailed installation instructions found in the software installation and maintenance guides.

36.1.1. Software Delivery Media

The ADMS delivery media include all required software to run a standard configuration of ADMS. The kits include documentation and the FantasyIsland sample data set, including an updated SCADA database for the standard EMP60 Platform model.

For advanced configurations, additional third-party software installations may be required. The required software is described in other chapters of this document, as well as in the *ADMS Software Installation and Maintenance Guide*.

This ADMS release is made available as modular and separate installation kits and archives:

File Name	Description
ADMS_server.msi	ADMS Server: For the real-time operations control.
ADMS_client.msi	ADMS Client: Provides an advanced geographical distribution user interface for the operators and analysts.
ADMS_tools.msi	ADMS Configuration Tools: For the analysts and support engineers to configure the ADMS server and client for OMS, DMS, and Simulator modules, and for the distribution modeling engineers to model both the distribution network and the components that overlap with the SCADA subsystem.
ADMS_fileupdaterserver.msi ADMS_fileupdaterclient.msi	ADMS File Updater Server and Client: For updating the models and configuration files available on the ADMS server and ADMS client machines.
ADMS_docs.msi	ADMS Documentation Suite: Documentation for end users to become familiar with and use the GE distribution management system solution.
ADMS_FantasyIsland_EP.msi	ADMS Fantasy Island Expansion Pack: For installing an additional configuration and model for development and testing.
ADMS_modelexport.msi	ADMS Model Export tool for exporting substation model data into a common format.
ADMS_symbols_server.msi	Contains PDB files for the Server kits. These files contain method signatures that are used for debugging purposes, such as creating crash dumps.

File Name	Description
ADMS_symbols_client_tools_interface.msi	Contains PDB files for the Interface, Client, and Tools kits. These files contain method signatures that are used for debugging purposes, such as creating crash dumps. NOTE: The Symbols kit must be installed on each project so that crash dumps could be created and provided to the GE project team in case of any issues.
ADMS_sample_reports.msi	ADMS outage management reports based on Microsoft SQL Server Reporting Services.
ADMS_Interfaces_SDK.msi	ADMS Software Development Kit: Contains coding samples to enable integration with third-party products and enterprise applications.
ADMS_interfaces.msi	ADMS Interfaces Kit: Contains NetViewer Adapter, DERM Kafka Adapter, Work Order Adapter, Core Data Capture Server, and other applications. Required to support the Enterprise OMS and Internal Notifications functionality and to integrate with Storm Assist and DER Gateway. This kit can also install the OMS Interface Server, OMS Adapter, Call Server, Crew Server, and DMS Interface Server applications, which are typically installed by the Server and Client kits.
ADMS_dnom2hdb.msi	ADMS DNOM2HDB utility for archiving.
ADMS_CallEntry.msi	ADMS standalone Call Entry tool for entering customer trouble calls.
ADMS_CredentialEditor.msi	ADMS Credential Editor.
ADMS_relational_database.msi	ADMS relational database setup and upgrade.
ADMS_ProcessAutomation.msi	ADMS Process Automation tool for initial crew assignment and dispatch based on configured rules.
adms_InternalNotification_install.zip	Contains Internal Notifications application artifacts.

36.1.1.1. Supported Installation Method on Windows

Using the Windows installer packages with an administrator-privileged account is the only supported installation method.

Warning

When deploying upgraded software to many client machines, simply copying the program files may cause unexpected results and is not supported. It is recommended that you uninstall the old version and re-install the new one.

36.1.2. Prerequisites

This section describes the prerequisites or dependencies for installing the application on each of the supported operating systems. The prerequisites for the ADMS installation are:

- Habitat, Scada, Browser, and APF (see section [Compatible GE Software](#)).
- Third-party software (see section [Third-Party Software](#)).

Since ADMS links to external third-party libraries, there are the following prerequisites for ADMS installation:

- Windows C++ runtime
- .NetFramework and .NetCore
- Java 17 (required to run Group Control)
- 16.11.0 and later
 - Windows C++ runtime (Microsoft Visual C++ - 2015 – 2022 redistributable (x64) & (x86) - 14.40.33810)
 - .NetFramework 4.8.0, .Net Core 8 and .Net runtime 8.0.3
- 16.6.0 to 16.10.0
 - Windows C++ runtime (Microsoft Visual C++ - 2015 – 2022 redistributable x64 & x86)
 - .NetFramework 4.8.0, .Net Core 8 and .Net runtime 8.0.3
- 16.0.0 to 16.5.0.
 - Windows C++ runtime (Microsoft Visual C++ - 2015 – 2022 redistributable x64 & x86)
 - .NetFramework 4.8.0, .Net Core 6 and .Net runtime 6.0.14 or later
- 3.15 and earlier.
 - Windows C++ runtime (Microsoft Visual C++ - 2015 – 2022 redistributable (x64 & x86)
 - .NetFramework 4.7.2 and .NetCore 5.0.400 and .NetCore 2.1.x for ADMS 3.10

36.1.3. Product Documentation

This section provides a summary of the application’s documentation set. The following table lists the documents that are included in this release.

i Note

The ADMS Documentation Index page is available from the Windows Start menu by navigating to Eterra > Distribution > Documentation.

Title	File Name	PDF Document	Online Help Document	Other
Introductory and Overview				
ADMS 16.25.1 Release Notes	adms-16-25-1-release_notes	•		
ADMS 16.0.0 Third-Party Licenses	ThirdPartyLicenses.txt			•
ADMS 16.0.0 GDPR Letter of Applicability	adms-16-gdpr-letter-of-applicability	•		

Title	File Name	PDF Document	Online Help Document	Other
ADMS 16.25.1 Architecture Overview	adms-16-25-1-architecture-overview	•		
ADMS Glossary of Terms	adms-glossary-of-terms	•	•	
ADMS 16.25.1 System Overview	adms-16-25-1-system-overview	•	•	
Operators and Analysts				
ADMS 16.25.1 Archive User Guide	adms-16-25-1-archive-user-guide	•		
ADMS 16.25.1 Display Reference Manual	adms-16-25-1-display-reference-manual	•	•	
ADMS 16.25.1 Distribution Management System User Guide	adms-16-25-1-distribution-management-system-dms-user-guide	•	•	
ADMS 16.25.1 Distribution Network Analysis Functions (DNAF) User Guide	adms-16-25-1-dnaf-user-guide	•	•	
ADMS 16.25.1 Geographic Network View Symbolology Reference Guide	adms-16-25-1-geographic-network-view-symbolology-reference-guide	•	•	
ADMS 16.25.1 Group Control User Guide	adms-16-25-1-group-control-user-guide	•		
ADMS 16.25.1 Internal Notifications User Guide	adms-16-25-1-internal-notifications-user-guide	•	•	
Outage Management System Call Entry User Guide	oms-call-entry-user-guide	•	•	
Outage Management System User Guide	oms-user-guide	•	•	
ADMS 16.25.1 Reports User Guide	adms-16-25-1-reports-user-guide	•		
ADMS 16.25.1 Snapshot Viewer User Guide	adms-16-25-1-snapshot-viewer-user-guide	•	•	
ADMS 16.25.1 Switching Operations User Guide	adms-16-25-1-switching-operations-user-guide	•	•	
ADMS 16.25.1 Switching Order Client Configuration Editor User Guide	adms-16-25-1-switching-order-client-configuration-editor-user-guide	•	•	
ADMS 16.25.1 Switching Order Server Configuration Editor User Guide	adms-16-25-1-switching-order-server-configuration-editor-user-guide	•	•	
Trainers and Testers				
ADMS 16.25.1 Distribution Operator Training Simulator (DOTS) User Guide	adms-16-25-1-distribution-operator-training-simulator-dots-user-guide	•	•	
System Administrators - Installation and Maintenance				

Title	File Name	PDF Document	Online Help Document	Other
ADMS 16.25.1 DNOM2HDB Software Installation and Maintenance Guide	adms-16-25-1-dnom2hdb-installation-guide	•	•	
ADMS 16.25.1 Relational Database Software Installation and Maintenance Guide	adms-16-25-1-relational-database-installation-guide	•	•	
ADMS 16.25.1 Software Installation and Maintenance Guide	adms-16-25-1-installation-guide	•	•	
ADMS 16.25.1 File Updater Software Installation and Maintenance Guide	adms-16-25-1-file-updater-installation-guide	•	•	
ADMS 16.25.1 Internal Notifications Software Installation and Maintenance Guide	adms-16-25-1-internal-notifications-installation-guide	•	•	
ADMS 16.25.1 Model Export Software Installation and Maintenance Guide	adms-16-25-1-model-export-installation-guide	•	•	
ADMS 16.25.1 Model Manager Software Installation and Maintenance Guide	adms-16-25-1-model-manager-installation-guide	•	•	
ADMS 16.25.1 Reports Software Installation and Maintenance Guide	adms-16-25-1-reports-installation-guide	•		
ADMS 16.25.1 Substation Editor Software Installation and Maintenance Guide	adms-16-25-1-substation-editor-installation-guide	•	•	
System Administrators - Setup and Configuration				
ADMS Alarmable Events	adms-alarmable-events.xlsx			•
ADMS 16.25.1 AMI Server – Landis and Gyr Command Center Interface Integration Guide	adms-16-25-1-ami-server-landis-and-gyr-command-center-interface-integration-guide	•		
ADMS 16.25.1 Annotation Importer/Exporter User Guide	adms-16-25-1-annotation-importer-exporter-user-guide	•	•	
ADMS 16.25.1 Coded Translation Editor User Guide	adms-16-25-1-coded-translation-editor-user-guide	•	•	
ADMS 16.25.1 Distribution Network Analysis Functions (DNAF) Advanced Applications Analyst Guide	adms-16-25-1-dnaf-advanced-applications-analyst-guide	•	•	
ADMS 16.25.1 Distribution Network Analysis Functions (DNAF) Standard Applications Analyst Guide	adms-16-25-1-dnaf-standard-applications-analyst-guide	•	•	

Title	File Name	PDF Document	Online Help Document	Other
ADMS 16.25.1 DNOM2HDB User Guide	adms-16-25-1-dnom2hdb-user-guide	•	•	
ADMS 16.25.1 Grid Tabular Builder User Guide	adms-16-25-1-grid-tabular-builder-user-guide	•	•	
ADMS 16.25.1 Net Dynamics Configuration Editor User Guide	adms-16-25-1-net-dynamics-configuration-editor-user-guide	•	•	
Outage Management System Client Configuration Editor User Guide	oms-client-configuration-editor-user-guide	•	•	
Outage Management System Server Configuration Editor User Guide	oms-server-configuration-editor-user-guide	•	•	
ADMS 16.25.1 Permission Management Plug-In User Guide	adms-16-25-1-permission-management-plug-in-user-guide	•	•	
ADMS 16.25.1 Permit Builder User Guide	adms-16-25-1-permit-builder-user-guide	•	•	
Process Automation User Guide	oms-process-automation-user-guide	•	•	
ADMS 16.25.1 Resource Editor User Guide	adms-16-25-1-resource-editor-user-guide	•	•	
ADMS 16.25.1 Viewer Configuration Editor User Guide	adms-16-25-1-viewer-configuration-editor-user-guide	•	•	
Model Managers and GIS Integrators				
Crew Modeling User Guide	oms-crew-modeling-user-guide	•	•	
Customer Field Definition Mapping	adms-customer-field-definition-mapping.xlsx			•
ADMS 16.25.1 Customer List Compiler User Guide	adms-16-25-1-customer-list-compiler-user-guide	•	•	
DNOM Converter Error Message Debugging Guide	adms-converter-error-message-debugging-guide.xml			•
 Note This XML file can be viewed in Microsoft Excel.				
ADMS 16.25.1 Geographic Background Compiler User Guide	adms-16-25-1-geographic-background-compiler-user-guide	•	•	
ADMS 16.25.1 Model Manager User Guide	adms-16-25-1-model-manager-user-guide	•	•	
ADMS 16.25.1 Modeling Overview and Converter User Guide	adms-16-25-1-modeling-overview-and-converter-user-guide	•	•	
ADMS 16.25.1 Modeling Reference Guide	adms-16-25-1-modeling-reference-guide	•	•	

Title	File Name	PDF Document	Online Help Document	Other
ADMS Modeling Reference Tables	adms-modeling-reference-tables.xlsx			•
ADMS Outage Management System Query Fields	adms-oms-query-fields.xlsx			•
ADMS 16.25.1 Substation Builder User Guide	adms-16-25-1-substation-builder-user-guide	•	•	
ADMS 16.25.1 Substation Editor User Guide	adms-16-25-1-substation-editor-user-guide	•	•	
ADMS 16.25.1 Substation File Builder Lite User Guide	adms-16-25-1-substation-file-builder-lite-sfblite-user-guide	•	•	
Programmers and Report Writers				
ADMS 16.25.1 Distribution Management Interface Server Programmer Guide	adms-16-25-1-distribution-management-system-dms-interface-server-programmer-guide	•	•	
Outage Management Integration Programmer Guide	oms-integration-programmer-guide	•	•	
Outage Management Interface Server Programmer Guide	oms-interface-server-programmer-guide	•	•	
Outage Management Read-Only Interface Programmer Guide	oms-read-only-interface-programmer-guide	•	•	
ADMS 16.25.1 Software Development Kit (SDK) Programmer Guide	adms-16-25-1-software-development-kit-sdk-programmer-guide	•	•	
Mobile Enterprise Order Web Service	mobile-enterprise-order-web-service.html			•
Mobile Enterprise Order Web Service	mobile-enterprise-order-web-service.json			•

36.2. EOMS

This section provides information that pertains to the installation of EOMS components. The material in this section augments the detailed installation instructions found in the *Enterprise OMS Software Installation and Maintenance Guide* published with this release.

Note

Installations are HTTP-enabled by default. GE strongly recommends enabling mutual Transport Layer Security (TLS) with HTTPS before using the software in production.

36.2.1. Software Delivery Media

The delivery media includes the software to deploy Enterprise OMS. Note that third-party software is also required. The required software is described in other chapters of this document, as well as in the *Enterprise OMS Software Installation and Maintenance Guide*.

This Enterprise OMS release is delivered as a set of files and archives as listed in Table below.

Note

GE has reviewed the critical Common Vulnerabilities and Exposures (CVEs) found in the EOMS cache container and our applications are not impacted by these CVEs.

Table 8. Enterprise OMS Deliverables

File Name	Description
work-model-cache-1.5.0.tar	Cache container image with artifacts needed for Enterprise OMS deployment. Used for deployment in environments without connection to the GE Docker registry, DTR.
foundation-base-cache-Q2-2023.tar	Cache container image with artifacts needed for Foundation Base Q2 2023 deployment. Used for deployment in environments without connection to the GE Docker registry, DTR.
eoms-manifest.yaml	A sample input manifest file to define the Enterprise OMS deployment.
foundation-manifest.yaml	A sample input manifest file to define the Foundation Base deployment.
bsfdeploy-<VERSION>.zip	The Bsfdeploy client tool zip archive (for Windows).
BundleDeployBsfEnv.tar.gz	Installation package for installing Nexus and Bsfdeploy client in Linux.
ADMS_Enterprise_OMS_Module.msi	Installer for the Enterprise OMS deployment in Windows.

36.2.2. Supported Install Method

36.2.2.1. Deployment in Kubernetes

The Enterprise OMS deployment in Kubernetes is performed using the Bsfdeploy solution deployment engine, which is a simplified means of running Product Deploy Images to deploy products to a Kubernetes cluster. For more details, refer to the *Enterprise OMS Software Installation and Maintenance Guide*.

36.2.2.2. Installing in Windows

The Enterprise OMS installation in Windows is performed using the ADMS_Enterprise_OMS_Module.msi installer, which contains Java executable files to run Enterprise OMS services.

36.2.3. Prerequisites

This section describes the prerequisites or dependencies for installing the application on each of the supported operating systems.

The prerequisites for the EOMS installation are:

- ADMS.
- Third-party software (see section [Third-Party Software](#)).

36.2.4. Process Restart Recommendations

Configure Habitat to restart ADMS core applications (for example, DMS Main Server, DNAF Server, OMS Server, NetDataServer) at least once in the event of a crash before failing over to a standby system. This approach accounts for the multi-threaded design of these applications, where certain crash conditions might not recur.

36.2.5. Product Documentation

Table below lists the documents that are included with this release.

Table 9. Documentation Summary

Title	File Name	PDF Document
Enterprise OMS Software Installation and Maintenance Guide	enterprise-oms-installation-guide	•
Enterprise OMS Programmer Guide	enterprise-oms-programmer-guide	•
Enterprise OMS Data Correction Interface User Guide	enterprise-oms-data-correction-interface-user-guide	•
Enterprise OMS User Guide	enterprise-oms-user-guide	•

36.3. Outage Assist

This section provides information that you should be aware of when preparing to install Outage Assist.

For complete installation and configuration instructions for Outage Assist, refer to the Outage Assist Software Installation and Maintenance Guide published with this release.

36.3.1. Delivery Media

This section outlines the Outage Assist delivery media.

The delivery media includes the software to deploy Outage Assist. Note that the third-party software is also required. The required software is described in other chapters of this document, as well as in the Outage Assist Software Installation and Maintenance Guide.

This Outage Assist release is delivered as a set of files and archives:

File Name	Description
sa-cache-16.25.1.tar.gz	Cache container image with artifacts needed for Outage Assist deployment (without Foundation services). Used for deployment in environments without connection to the GE Docker registry, DTR.
k8s-base-manifest.yaml	A sample input manifest file to define the rancher PDI deployment.
foundation-base.yaml	A sample input manifest file to define the foundation-base Helm chart deployment.
foundation-osb.yaml	A sample input manifest file to define the foundation-osb Helm chart deployment.
foundation-ui.yaml	A sample input manifest file to define the foundation-ui Helm chart deployment.
outage-assist.yaml	A sample input manifest file to define the outage-assist Helm chart deployment.

36.3.2. Prerequisites

This section describes the prerequisites or dependencies for installing Outage Assist.

The prerequisites for the Outage Assist installation are:

- RKE2 Kubernetes cluster and Foundation services (see [GE Software](#)).
- Third-party software (see [Third-Party Software](#)).

36.3.3. Outage Assist Recommended Deployment

This section describes the standardized deployment method for Outage Assist. This method ensures consistency, streamlining the installation and support process to enhance overall system efficiency. Although variations in deployment may have been used by customers in the past, GE strongly encourages the use of this standard deployment in order to effectively support active installations.

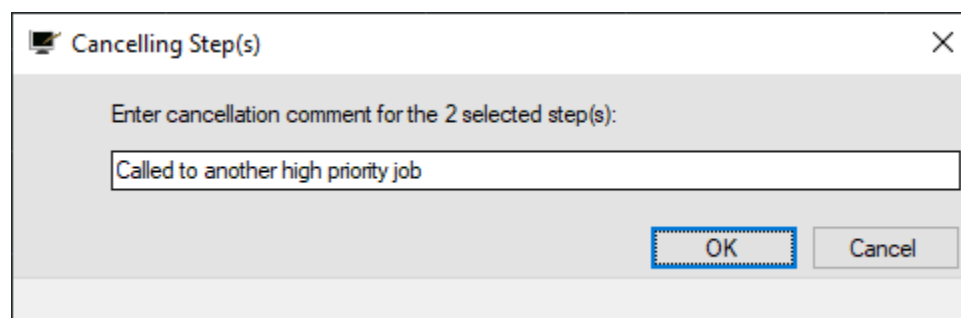


Figure 1. Outage Assist/ADMS Deployment Architecture

The figure above shows the recommended deployment architecture, mainly designed for Outage

Assist. Only the relevant components for Outage Assist are shown in the ADMS section. The current foundation architecture supports two RKE2 clusters in an active/standby mode, while ADMS can support four nodes. In this setup, ADMS nodes A and B run on the same site as Cluster A, and nodes C and D operate where Cluster B is located. The main highlights of this deployment architecture are:

1. Planned Failover Coordination:

- Crucial for coordinated planned failovers between ADMS and Outage Assist clusters.
- In the case of unplanned failovers, the system continues operation with limited performance. Relocating to the same cluster is highly advised.

2. GTM (Global Traffic Manager) Limitation:

- Currently, the foundation does not support GTM.
- In case of failovers, users need an alternate shortcut to log in to the new cluster.
- If a project chooses to implement its own GTM, it is crucial to replicate session information stored in Infinispan to the secondary cluster for a smooth transition. Note that this support is not anticipated in the foundation's future development roadmap.

3. LTM (Local Traffic Manager) Limitation:

- The cluster currently does not support LTM.
- To achieve scalability and availability within the cluster, projects should deploy the recommended architecture outlined in the Kubernetes documents provided here. The foundation plans to include this support in the roadmap for future development.

4. Database Replication using PostGRES:

- The Foundation supports database replication between sites using PostGRES.
- The secondary cluster's database instance is set to read-only.
- No user authentication is possible via the secondary cluster, even when all services are running.

5. ADMS Node Tasks in Secondary Site:

The following tasks are run in active mode to avoid any delay after site transfer:

- **NetviewerAdapter:** Publishes tiles to the secondary cluster.
- **Read-Only NetServer:** Resolves queries from client sessions.
- **NetDataReplicator:** Replicates journals to the secondary site.
- **File Updater Client:** Moves all model files to the secondary site.
- **DbSync** (EOMS Procman deployment mode): Copies EOMS database from the Primary site to the Secondary site.

All other ADMS tasks are fast to start or transition from Standby to Enabled.

36.3.4. Product Documentation

This section provides a summary of the application's documentation set. The following table lists the documents that are included with this release.

Table 10. Outage Assist Documentation

Title	File Name
Outage Assist Software Installation and Maintenance Guide	outage-assist-software-installation-and-maintenance-guide
Outage Assist User Guide	outage-assist-user-guide

37. Test Information

37.1. ADMS Performance Testing Guidelines

37.1.1. Load (Storm) Simulation

This section outlines the parameters for simulating load (storm) scenarios in the ADMS environment.

Simulate Customer Calls

- Simulate 60K customer calls within an hour.
- Simulate 120K customer calls within two hours.
- Simulate 100K customer calls and device interactions, utilizing around 500-1000 breakers for de-energization.

37.1.2. Test Process

This section provides details about the testing process followed during performance testing.

ADMS Deployment

- ADMS is deployed on a four-machine topology consisting of OMS, DMS, DOTS, and SCADA.

Hardware Requirements

All machines are AWS based.

Server Types	AWS Instance Type
OMS	m6a.16xlarge
DMS	m5a.8xlarge
INTERFACE	m6a.12xlarge
SIM	m5a.12xlarge
SCADA	m5a.8xlarge
RO	m5a.8xlarge
CLIENT	t2.2xlarge

Static and Dynamic Data Storage

Below files should be stored/available on the local machine

- Static files

- DMS station model (.mod)

Note

During performance testing, ADMS distribution models were partitioned at the station level and compacted into binary files to optimize loading performance. The platform is optimized for station model sizes between 1 MB and 20 MB. Stations exceeding 20 MB may experience slower network view loading and reduced responsiveness during panning, zooming, and other UI interactions.

- Customer model
- Crew model
- Dynamic files
 - DMS Dynamics
 - OMS Journal and Snapshots

Benchmarking

- Benchmarking is based on the following model size:
 - Number of Stations - 637
 - Number of Customers - 6.09 millions
 - Crew Sizing - 4000

Testing Iterations

- Each simulation is executed at least three times to ensure consistent test results.

Monitoring and Evaluation

- System health and integrity are monitored for a duration of 6-12 hours after load generation.
- Functional tests are conducted concurrently with load simulations.

37.1.3. Test Verification

This section outlines the verification criteria used to assess the performance testing results.

System Stability

- System stability is maintained throughout testing.
- No system crashes occur.
- No memory leaks are detected.

Response Time

Action/No. of Incidents	1	1K	5K	10K	20K	30K
OMSDisplayPlugin initial load time	5 sec	4 sec	4 sec	9 sec	25 sec	38 sec
New incident to appear in INCIDENT TAB in OMSDisplayPlugin	6 sec	6 sec	6 sec	9 sec	9 sec	11 sec
New Incident to appear on GEOMAP	6 sec	8 sec	8 sec	8 sec	12 sec	14 sec
Open update incident from INCIDENT TAB in OMSDisplayPlugin	4 sec	4 sec	4 sec	4 sec	4 sec	4 sec
New order to appear in ORDER TAB in OMSDisplayPlugin	5 sec	4 sec	7 sec	9 sec	12 sec	42 sec
New order to appear on GEOMAP	5 sec	7 sec	7 sec	10 sec	12 sec	45 sec
Open the order worksheet from ORDER TAB in OMSDisplayPlugin	4 sec	5 sec	5 sec	5 sec	6 sec	6 sec
Open order worksheet from GEOMAP	4 sec	4 sec	4 sec	4 sec	4 sec	4 sec
Open STATION LIST	1 sec	1 sec	1 sec	1 sec	1 sec	1 sec
Click on the globe to load a new geographic from INCIDENT TAB in OMSDisplayPlugin	3 sec	3 sec	3 sec	4 sec	4 sec	5 sec
Click on the globe to load a new geographic from STATION LIST	2 sec	4 sec	4 sec	4 sec	5 sec	5 sec

Action/No. of Incidents	1	1K	5K	10K	20K	30K
Navigate to SCADA display from STATION LIST	1 sec	1 sec	1 sec	1 sec	1 sec	1 sec
Open STATION OVERVIEW	1 sec	1 sec	1 sec	1 sec	1 sec	1 sec
Open a new geographic from STATION OVERVIEW	3 sec	3 sec	3 sec	3 sec	3 sec	3 sec
Select magnifying glass from tool bar	1 sec	1 sec	1 sec	1 sec	1 sec	1 sec
Search a device Results are presented	3 sec	3 sec	3 sec	3 sec	3 sec	3 sec
Select the device from the search list. Device located in GEOMAP	3 sec	4 sec	4 sec	4 sec	4 sec	5 sec
Run the trace	1 sec	1 sec	1 sec	1 sec	1 sec	1 sec
Clear trace	1 sec	1 sec	1 sec	1 sec	1 sec	1 sec
Zoom In	1 sec	1 sec	1 sec	1 sec	1 sec	1 sec
Zoom out	1 sec	1 sec	1 sec	1 sec	1 sec	1 sec
show/hide customers	1 sec	1 sec	1 sec	1 sec	1 sec	1 sec
show/hide laterals	1 sec	1 sec	1 sec	1 sec	1 sec	1 sec
Show/hide labels	1 sec	1 sec	1 sec	1 sec	1 sec	1 sec

Advance Filtering

Steps/No of Incidents/Orders	1K	10K	20K	30K	40K
Open Filter panel	<1	<1	<1	<2	<2
Navigate to Advance filter TAB	<1	<1	<1	<2	<3
Create filters (nested/grouped) Response time for selection	<1	<1	<1	<3	<3

Steps/No of Incidents/Orders	1K	10K	20K	30K	40K
Save the created filter	<1	<1	<1	<2	<2
SaveAs any new created new filter	<1	<1	<1	<2	<2
Delete any old saved filter	<1	<1	<1	<2	<3
Navigate to Advance Saved TAB	<1	<1	<1	<2	<2
Open any saved filter (with Auto Apply)	<1	<2	<4	<6	<12
Check/Uncheck Auto Apply (In Filter TAB)	<1	<1	<2	<3	<3
Open any saved filter (without Auto Apply)	<1	<1	<1	<2	<3
Apply filter	<1	<1	<1	<2	<2
Close the Advance Filter window	<1	<1	<1	<1	<2

Order Life-Cycle

No of Incidents	1	1K	5K	10K	20K	30K
Order appeared on geo map	51	58	59	66	67	76
Orders appeared in oms display plugin	49	57	58	65	67	75
Open order worksheet	4	4	4	4	4	4
Assign crew - crew selection window	1	1	1	1	1	1
Select crew dropdown	1	1	1	1	1	1
Assign crew (OW)	4	4	4	3	4	4
Unassign crew (OW)	1	1	1	1	1	1
Assign crew (Tabular)	8	7	8	7	9	9

No of Incidents	1	1K	5K	10K	20K	30K
Assign crew (Map)	6	6	9	12	12	14
Assign another crew(OW)	4	4	4	3	4	4
Change crew status -Dispatch crew	1	1	1	1	1	1
Change crew status - En-route	1	1	1	1	1	2
Change crew status - Arrived	1	1	1	1	1	2
Perform task - Confirm open	1	1	1	1	1	1
Open ETR window popup	1	1	1	1	1	1
Change ETR time	1	1	1	1	1	2
Perform repair task	1	1	1	1	1	1
Save crew comment	1	1	1	1	1	1
Save dispatcher comments	1	1	1	1	1	1
Resolution cause - equipment failure	1	1	1	1	1	1
Change oder status to close	1	1	1	1	1	1

Geographic View Responsiveness

- The geographic view remains responsive and interactive, with possible slight delays in response time at higher incident counts.

Resource Utilization Metrics

- Resource utilization metrics adhere to the following guidelines:

Component/ Incidents or Orders		1K	5K	10K	20K	30K
APF	%CPU	10	7	9	2	9
	Memory (MB)	9936.99	10225.65	10621.57	12010.34	10621.57

Component/ Incidents or Orders		1K	5K	10K	20K	30K
CefSharp.Brows erSubprocess	%CPU	5	7	12	13	12
	Memory (MB)	32.59	32.59	32.59	32.59	32.59
CefSharp.Brows erSubprocess#1	%CPU	5	7	12	13	12
	Memory (MB)	23.97	24.07	23.97	23.97	23.97
netsrv	%CPU	1	1	1	1	1
	Memory (MB)	16315.34	16315.39	16321.52	16315.62	16321.52
SwitchOrderSer ver	%CPU	2	0	1	0	1
	Memory (MB)	87.7	88.12	88.39	88.55	88.39
NetServerSim	%CPU	2	2	3	1	2
	Memory (MB)	28185.96	27698.04	26141.67	26171.36	26056.79
NetServer	%CPU	0	14	0	6	0
	Memory (MB)	26553.51	27019.93	27462.08	28876.64	27462.08
WorkOrderAdap ter	%CPU	14	0	1	2	1
	Memory (MB)	5832.16	5933.81	6061.07	6378.73	6061.07
NetDataServer	%CPU	0	1	1	2	1
	Memory (MB)	7907.62	7980.73	8097.3	8563.21	8097.3
WM Service	%CPU	1	0	1	5	1
	Memory (MB)	1677.82	1710.1	1711.02	1711.3	1711.02
WM EventHandler	%CPU	1	0	1	5	1
	Memory (MB)	1536.34	1576.37	1557.04	1575.36	1557.04
WM Notification	%CPU	1	0	1	5	1
	Memory (MB)	491.74	494.66	494.92	495.35	494.92
CallServer	%CPU	1	1	1	1	1
	Memory (MB)	4013.44	4075.61	4041.25	4378.4	4041.25
RelayServer	%CPU	1	1	1	1	1
	Memory (MB)	3277.99	3469.7	3466.31	3652.25	3466.31
CrewServer	%CPU	0	2	4	4	4
	Memory (MB)	65.75	66.36	67.81	67.93	67.81
OmsInterfaceSe rver	%CPU	2	2	1	4	1
	Memory (MB)	72.59	90.01	107.66	117.32	107.66
DmsInterfaceSe rver	%CPU	1	2	4	1	4
	Memory (MB)	3376.7	3368.06	3367.88	3369.93	3367.88
DataCorrectionI nterface	%CPU	2	4	1	2	1
	Memory (MB)	38.34	41.94	42.34	42.45	42.34

Component/ Incidents or Orders		1K	5K	10K	20K	30K
DnafSrv	%CPU	25	10	25	30	25
	Memory (MB)	20410.31	20553.14	20485.05	20065	20485.05

Time for Readiness of Components

Below metrics highlights the time for the components to become ready after restart.

Module	Time to get ready
NetServer	1 min 42 secs
Netdatasrv	3 mins 15 secs
NetSrv	15 sec
SWO	1 min 9 secs
Call Server	1 mins 51 secs
Crew Server	51 secs

Peak Load Testing – 1

Test Description

Executed 55K customer call scripts over a duration of 3 hours, generating approximately 44K orders. After the calls completed, the system was kept in a steady state for an additional 2 hours to observe potential memory-related issues.

Purpose

Ensure the system remains stable and meets performance benchmarks when operating at the highest anticipated load.

Test Details

- Test Type - Storm calls (customer calls only)
- Total Number of Calls - 55254
- Total Test Duration - 5 hours
- Call Duration - 3 hours
- System Idle Time After Last Call - 2 hours

Observations

- Incidents Created - 43717
- NetServer Settling Period After Last Call - 15 minutes
- OmsCallsReceivedOrSent - 55254
- AffectedCustomersCount - 97490

Average Process Metrics (5 Hours)

Process	CPU	Private Bytes
CefSharp.BrowserSubprocess#1	0.00 %	23.98 MB
CefSharp.BrowserSubprocess	0.00 %	32.52 MB
Apf	12.53 %	10312.81 MB
RO CefSharp.BrowserSubprocess#1	0.00 %	24.08 MB
RO CefSharp.BrowserSubprocess	0.00 %	33.10 MB
RO Apf	5.64 %	8125.53 MB
SwitchOrderServer	0.86 %	154.27 MB
netsrv	1.85 %	16329.29 MB
DnafSrv	21.57 %	19652.42 MB
OmsInterfaceServer	0.08 %	145.73 MB
DmsInterfaceServer	1.40 %	3380.65 MB
DataCorrectionInterface	0.01 %	0.88 MB
CrewServer	0.15 %	76.93 MB
RelayServer	0.17 %	3640.85 MB
CallServer	0.28 %	4043.41 MB
WorkOrderAdapter	0.10 %	6777.69 MB
NetServer	7.99 %	30554.27 MB
NetDataServer	1.10 %	8875.59 MB
Work-model-service	0.19 %	1608.76 MB
Work-model-eventhandler	0.08 %	1545.49 MB
Work-model-notifications	0.07 %	1468.14 MB
NetServerRO	13.49 %	31932.46 MB
NetDataReplicator	0.08 %	851.58 MB
SCADA	0.14 %	300.66 MB
NetServerSim	9.18 %	26973.43 MB

.NET Application Memory Metrics (5-Hour Average)

Application	Bytes in All Heaps	Pinned Objects	Gen 0 Heap	Gen 1 Heap	Gen 2 Heap	Large Object Heap
SwitchOrderServer	83.10 MB	10	4.03 MB	0.53 MB	70.70 MB	11.86 MB
NetServer	11933.85 MB	3	1381.47 MB	22.58 MB	8533.67 MB	3377.60 MB
NetDataServer	8635.37 MB	10	6.00 MB	0.67 MB	6514.70 MB	2119.99 MB
NetServerRO	13054.45 MB	1	260.98 MB	32.25 MB	8988.18 MB	4034.03 MB
NetDataReplicator	114.26 MB	1	236.00 MB	9.16 MB	33.60 MB	71.50 MB

Disk I/O Metrics (5-Hour Average)

Disk Metric	Client	DMS	Dnaf	Interface	OMS	RO	SCADA	SIM
Avg. Disk Queue Length	0.01	0.01	0.24	0.80	0.04	0.01	0.15	0.02
Disk Reads/sec	2.92	0.16	1.89	0.26	0.28	0.14	0.19	0.20
Disk Writes/sec	2.93	8.05	248.65	853.57	39.13	8.26	149.17	12.36
Physical Disk Queue Length	0.01	0.01	0.24	0.80	0.04	0.01	0.15	0.02
Disk Bytes/sec	0.83 MB	0.14 MB	3.39 MB	6.04 MB	1.29 MB	0.34 MB	2.34 MB	1.51 MB
Physical Disk Reads/sec	2.92	0.16	1.89	0.26	0.28	0.14	0.19	0.20

Network Interface Metrics (5-Hour Average)

Metric	Client	DMS	Dnaf	Interface	OMS	RO	SCADA	SIM
Bytes Received/sec	0.13 MB	0.01 MB	0.04 MB	0.14 MB	0.03 MB	0.39 MB	0.00 MB	0.01 MB
Bytes Sent/sec	0.02 MB	0.03 MB	0.40 MB	0.02 MB	0.45 MB	0.00 MB	0.00 MB	0.04 MB
Bytes Total/sec	0.14 MB	0.03 MB	0.43 MB	0.16 MB	0.49 MB	0.39 MB	0.01 MB	0.05 MB
Current Bandwidth	1 Gbps	5 Gbps	5 Gbps	5 Gbps	5 Gbps	5 Gbps	5 Gbps	5 Gbps
Output Queue Length	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Remarks

- No resource utilization issues observed.
- No application crashes detected.

Peak Load Testing - 2

Test Description

Ran 5K call scripts and 50K switching operation scripts over a duration of 11 hours. This generated 12K orders. After the scripts finished, the system was left in a steady state for 4 hours to observe potential memory-related issues.

Purpose

To ensure the system remains stable and meets performance benchmarks even when fully utilized at its highest anticipated load.

Test Details

- Test Type: Storm calls (Storm calls + Switching)
- Total number of calls made - 5K
- Total number of switches operated - 50K
- Total test duration - 15 hours
- Call + Switching duration - 11 hours
- System left idle after last call - 4 hours

Observations

- Incidents created - 11580
- NetServer settling period after last call - 15 min
- OmsCallsReceivedOrSent - 201172
- AffectedCustomersCount -1800422
- OmsAMIMessagesReceivedOrSent - 382686

Average Resource Usage (15-hour Average by Process)

Process	CPU	Private Bytes
Main CefSharp.BrowserSubprocess#1	0.00 %	24.33 MB
Main CefSharp.BrowserSubprocess	0.00 %	32.44 MB
Apf Main	7.29 %	8558.76 MB
RO CefSharp.BrowserSubprocess#1	0.00 %	24.10 MB
RO CefSharp.BrowserSubprocess	0.00 %	32.99 MB
RO Apf	3.05 %	5709.78 MB
SwitchOrderServer	1.47 %	155.94 MB
netsrv	1.68 %	16339.75 MB
DnafSrv	18.27 %	19750.39 MB
OmsInterfaceServer	0.09 %	101.92 MB
DmsInterfaceServer	1.34 %	4749.10 MB
DataCorrectionInterface	0.00 %	0.85 MB
CrewServer	0.14 %	85.67 MB
RelayServer	0.18 %	3706.47 MB
CallServer	0.27 %	4052.15 MB
WorkOrderAdapter	0.39 %	7152.61 MB
NetServer	6.58 %	28876.89 MB
NetDataServer	1.87 %	10141.21 MB
Work-model-service	0.12 %	1730.99 MB
Work-model-eventhandler	0.06 %	1567.21 MB
Work-model-notifications	0.05 %	1506.67 MB
NetServerRO	13.55 %	36070.23 MB
NetDataReplicator	0.09 %	1077.65 MB
SCADA	0.16 %	298.59 MB
NetServerSim	9.65 %	28801.71 MB

.NET CLR Memory Usage (15-hour Average)

Application	Bytes in all Heaps	Pinned Objects	Gen 0 Heap	Gen 1 Heap	Gen 2 Heap	Large Object Heap
SwitchOrderServer	82.30 MB	11 count	4.01 MB	0.40 MB	0.40 MB	0.40 MB
NetServer	10079.83 MB	3 count	1320.13 MB	10.19 MB	7528.34 MB	2541.29 MB
NetDataServer	9823.48 MB	11 count	6.00 MB	0.72 MB	7665.44 MB	2541.29 MB
NetServerRO	14258.63 MB	1 count	199.78 MB	14.11 MB	9908.85 MB	4335.66 MB
NetDataReplicator	147.70 MB	1 count	500.20 MB	17.58 MB	22.02 MB	108.11 MB

Disk I/O (15-hour Average by Server)

Metric	Client	DMS	Dnaf	Interface	OMS	RO	SCADA	SIM
LogicalDisk Avg. Disk Queue Length	0.01	0.01	0.23	0.81	0.03	0.01	0.15	0.02
LogicalDisk Disk Reads/sec	0.36	0.22	0.23	1.03	0.37	0.20	0.24	0.35
LogicalDisk Disk Writes/sec	2.72	7.54	240.99	859.35	34.15	8.31	154.99	14.80
PhysicalDisk Avg. Disk Queue Length	0.01	0.01	0.23	0.80	0.03	0.01	0.15	0.02
PhysicalDisk Disk Bytes/sec	0.70 MB	0.11 MB	3.81 MB	5.79 MB	1.45 MB	0.37 MB	2.35 MB	2.00 MB
PhysicalDisk Disk Reads/sec	0.36	0.22	0.23	1.03	0.37	0.20	0.24	0.35

Network Interface Usage (15-hour Average)

Metric	Client	DMS	Dnaf	Interface	OMS	RO	SCADA	SIM
Bytes Received/sec	0.04 MB	0.01 MB	0.01 MB	0.26 MB	0.03 MB	0.80 MB	0.00 MB	0.00 MB
Bytes Sent/sec	0.01 MB	0.07 MB	0.81 MB	0.02 MB	0.12 MB	0.00 MB	0.00 MB	0.17 MB
Bytes Total/sec	0.05 MB	0.07 MB	0.82 MB	0.28 MB	0.15 MB	0.80 MB	0.00 MB	0.17 MB
Current Bandwidth	1 Gbps	5 Gbps	5 Gbps	5 Gbps	5 Gbps	5 Gbps	5 Gbps	5 Gbps

Metric	Client	DMS	Dnaf	Interface	OMS	RO	SCADA	SIM
Output Queue Length	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Remarks

- No resource utilization issues were found.
- No crashes were observed.

Endurance Test

Test Description

Ran 55K call scripts over a duration of 3 hours, generating approximately 44K orders. The system was then left in a steady state for 2 hours to observe potential memory-related issues.

Purpose

To detect memory leaks, resource exhaustion, or performance degradation that may only appear after prolonged and continuous system usage.

Test Details

- Test type - Customers Calls + Switching + Incidents/Orders operated by simulator and manual operation.
- Call + Switching Duration - 30 hours
- Total Test Duration - 100 hour approx. (4 days+)
- Total Calls Generated - 300
- Total Switches Operated - 700
- All Incidents Closed/Cleared/Cancelled - Within 24 hours (via simulator or manual operation)
- System Idle After Final Incident/Order Clearance - 80 hours

Observations

- Incidents Created and Cleared - 1300
- OmsCallsReceivedOrSent - 13347
- OmsAMIMessagesReceivedOrSent - 43147

Note

This test was performed using the FantasyIsland data model.

NetServer (Day-wise Average)

Date	Bytes in All Heaps	Gen 2 Heap	Large Object Heap	CPU	Private Bytes
18-02-2026	868.14 MB	615.70 MB	249.59 MB	39.16 %	2732.31 MB
19-02-2026	815.08 MB	574.49 MB	238.00 MB	44.97 %	2697.17 MB
20-02-2026	708.28 MB	531.17 MB	173.80 MB	9.40 %	2600.12 MB
21-02-2026	710.07 MB	532.93 MB	174.31 MB	9.89 %	2601.95 MB
22-02-2026	710.80 MB	533.64 MB	174.36 MB	10.09 %	2602.80 MB

Work Order Adapter (WOA)

Date	CPU	Private Bytes
18-02-2026	1.06 %	718.02 MB
19-02-2026	1.00 %	709.15 MB
20-02-2026	0.42 %	674.43 MB
21-02-2026	0.43 %	675.18 MB
22-02-2026	0.44 %	675.21 MB

NetDataServer

Date	Bytes in All Heaps	Gen 2 Heap	Large Object Heap	CPU	Private Bytes
18-02-2026	881.11 MB	623.11 MB	257.23 MB	9.03 %	989.67 MB
19-02-2026	1005.55 MB	749.64 MB	255.27 MB	9.26 %	1118.71 MB
20-02-2026	987.63 MB	731.38 MB	255.49 MB	1.20 %	1101.21 MB
21-02-2026	984.72 MB	728.88 MB	255.07 MB	1.23 %	1098.31 MB
22-02-2026	984.95 MB	729.11 MB	255.05 MB	1.24 %	1098.48 MB

Switch Order Server (SWO)

Date	Bytes in All Heaps	Gen 2 Heap	Large Object Heap	CPU	Private Bytes
18-02-2026	36.17 MB	29.13 MB	6.60 MB	9.34 %	87.52 MB
19-02-2026	37.53 MB	30.12 MB	7.02 MB	18.73 %	89.55 MB
20-02-2026	38.87 MB	30.67 MB	7.80 MB	44.16 %	93.21 MB
21-02-2026	38.86 MB	30.67 MB	7.80 MB	45.38 %	93.31 MB
22-02-2026	38.87 MB	30.67 MB	7.80 MB	45.74 %	93.33 MB

Other Process Metrics (Day-wise Average)

NetSrv

Date	CPU	Private Bytes
18-02-2026	0.85 %	964.84 MB

Date	CPU	Private Bytes
19-02-2026	0.99 %	977.11 MB
20-02-2026	0.79 %	977.58 MB
21-02-2026	0.98 %	978.93 MB
22-02-2026	1.07 %	978.94 MB

DNAF

Date	CPU	Private Bytes
18-02-2026	15.47 %	1399.06 MB
19-02-2026	21.91 %	1391.23 MB
20-02-2026	2.88 %	1337.26 MB
21-02-2026	3.25 %	1372.92 MB
22-02-2026	3.33 %	1349.03 MB

Call Server

Date	CPU	Private Bytes
18-02-2026	1.31 %	522.36 MB
19-02-2026	1.49 %	574.91 MB
20-02-2026	2.42 %	985.02 MB
21-02-2026	2.52 %	987.42 MB
22-02-2026	2.57 %	990.54 MB

Crew Server

Date	CPU	Private Bytes
18-02-2026	1.93 %	80.55 MB
19-02-2026	1.19 %	94.34 MB
20-02-2026	0.92 %	95.95 MB
21-02-2026	0.97 %	95.66 MB
22-02-2026	0.99 %	95.39 MB

DMS Interface Server

Date	CPU	Private Bytes
18-02-2026	0.28 %	377.99 MB
19-02-2026	0.29 %	381.39 MB
20-02-2026	0.28 %	366.60 MB
21-02-2026	0.34 %	368.41 MB
22-02-2026	0.35 %	370.12 MB

OMS Interface Server

Date	CPU	Private Bytes
18-02-2026	0.83 %	89.55 MB
19-02-2026	0.46 %	109.57 MB
20-02-2026	0.01 %	105.63 MB
21-02-2026	0.02 %	105.70 MB
22-02-2026	0.02 %	105.73 MB

Data Correction

Date	CPU	Private Bytes
18-02-2026	0.04 %	55.06 MB
19-02-2026	0.04 %	57.27 MB
20-02-2026	0.06 %	58.01 MB
21-02-2026	0.06 %	57.11 MB
22-02-2026	0.07 %	57.61 MB

SCADA

Date	CPU	Private Bytes
18-02-2026	0.21 %	29.12 MB
19-02-2026	0.22 %	61.33 MB
20-02-2026	0.21 %	211.89 MB
21-02-2026	0.22 %	212.04 MB
22-02-2026	0.24 %	212.13 MB

APF

Date	CPU	Private Bytes
18-02-2026	8.96 %	1709.33 MB
19-02-2026	11.13 %	1970.40 MB
20-02-2026	5.58 %	2924.56 MB
21-02-2026	6.14 %	2973.19 MB
22-02-2026	6.15 %	2993.10 MB

CefSharp.BrowserSubprocess

Date	CPU	Private Bytes
18-02-2026	0.31 %	32.77 MB
19-02-2026	1.87 %	33.03 MB
20-02-2026	0.32 %	33.43 MB

Date	CPU	Private Bytes
21-02-2026	0.32 %	33.43 MB
22-02-2026	0.31 %	33.43 MB

CefSharp.BrowserSubprocess#1

Date	CPU	Private Bytes
18-02-2026	0.62 %	24.05 MB
19-02-2026	1.26 %	24.54 MB
20-02-2026	0.63 %	26.26 MB
21-02-2026	0.95 %	26.27 MB
22-02-2026	1.56 %	26.27 MB

Remarks

- No abnormal resource utilization was observed.
- No memory leaks were detected during the test duration.

37.2. Outage Assist Performance Test

37.2.1. Summary

- No performance degradation was observed compared to previous versions.
- No errors were detected during the JMeter test run.
- User experience while handling Tabular views, Order Worksheet (OW), and Map view remained within acceptable limits.

37.2.2. Goal

- Measure the performance of the Outage Assist (OA) – ADMS integrated system.
- Verify that the system can support 600 concurrent users with up to 23,000 incidents/orders loaded in the system.
- Ensure the response error rate remains below 5%.
- Verify that users can open the Order Worksheet (OW) and perform updates while the system is under load from 600 concurrent users.

37.2.3. Hardware and Environment

37.2.3.1. Data Model

- Customer Size – 491,201
- Crew Size - 4,000
- Number of Stations - 640

37.2.3.2. Environment Preparation

- AWS-based machines/nodes were used to create the environment using a Jenkins pipeline.
- EventScripter was used to generate incidents and orders in ADMS.
- JMeter was used to drive load against OA.

JMeter scripts repository: <https://github.com/software-governova/grid-operate/oa-automation-test/tree/master/SA-2.0/performance-test/jmeter-jmx>

37.2.3.3. Client Display Requirements

The Outage Assist user interface was validated using the following display configuration during performance testing:

- Minimum supported screen resolution: 1690 x 1031

37.2.3.4. Hardware

37.2.3.4.1. ADMS Servers

#	Server Type	Instance Type	Sizing (vCPU / RAM / Storage)	Network Bandwidth	Components Deployed
1	OMS	m5a.16xlarge	64 vCPU / 251 GB / 300 GB	Up to 10 Gbps	NetServer, NetDataReplicator, NetDataServer, WorkModel Services, NVA
2	DMS	m5a.8xlarge	32 vCPU / 128 GB / 300 GB	Up to 10 Gbps	NetSrv
3	DNAF	m5a.8xlarge	32 vCPU / 128 GB / 300 GB	Up to 10 Gbps	DNAF
4	SIM	m5a.12xlarge	48 vCPU / 189 GB / 300 GB	Up to 10 Gbps	DOTS

#	Server Type	Instance Type	Sizing (vCPU / RAM / Storage)	Network Bandwidth	Components Deployed
5	INTERF	m5a.12xlarge	48 vCPU / 189 GB / 300 GB	Up to 10 Gbps	Interfaces
6	RO-A	m5a.8xlarge	32 vCPU / 128 GB / 300 GB	Up to 10 Gbps	NetServer-RO, NetDataReplicator-RO, NVA-RO
7	RO-B	m5a.8xlarge	32 vCPU / 128 GB / 300 GB	Up to 10 Gbps	NetServer-RO, NetDataReplicator-RO, NVA-RO

37.2.3.4.2. OA Cluster

#	Node Role	Instance Type	Sizing (vCPU / RAM / Storage)	Network Bandwidth
1	Master / Control Plane	m5.8xlarge	32 vCPU / 128 GiB / 300 GB	Up to 10 Gbps
2	Agent	m5.8xlarge	32 vCPU / 128 GiB / 300 GB	Up to 10 Gbps
3	Agent	m5.8xlarge	32 vCPU / 128 GiB / 300 GB	Up to 10 Gbps
4	Agent	m5.8xlarge	32 vCPU / 128 GiB / 300 GB	Up to 10 Gbps

37.2.3.4.3. JMeter Clients

#	Server	Instance Type	Sizing (vCPU / RAM / Storage)	Network Bandwidth	Purpose
1	JMeter Client	m5a.4xlarge	16 vCPU / 64 GiB / 300 GB	Up to 10 Gbps	Drive load against the system
2	JMeter Client	m5a.4xlarge	16 vCPU / 64 GiB / 300 GB	Up to 10 Gbps	Drive load against the system

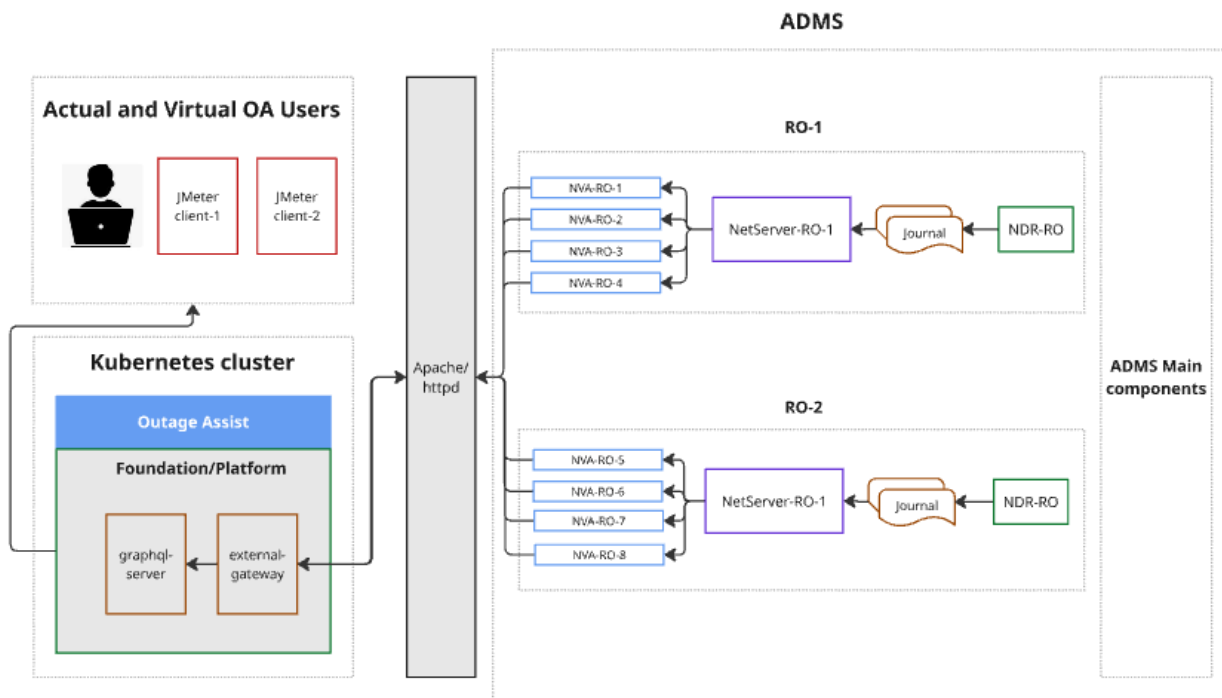
Note

All AWS instance types use SSD-backed storage.

37.2.3.5. Deployment Architecture

- ADMS was deployed using the standard distributed topology described in the Hardware section.
- For incident and order data retrieval, Read-Only (RO) components were configured to distribute traffic and enable horizontal scaling.

- Each RO server contains:
 - One NetServer-RO
 - One NetDataReplicator-RO (NDR-RO)
 - Four NetviewerAdapter-RO (NVA-RO)
- Four RO server instances were configured in the environment.
- Apache httpd was used as a load balancer to distribute traffic to the NVA-RO services.
- Two client machines were used to generate JMeter traffic against the system.



37.2.4. Test Configuration

37.2.4.1. Data Preparation

37.2.4.1.1. Users and AOR Distribution

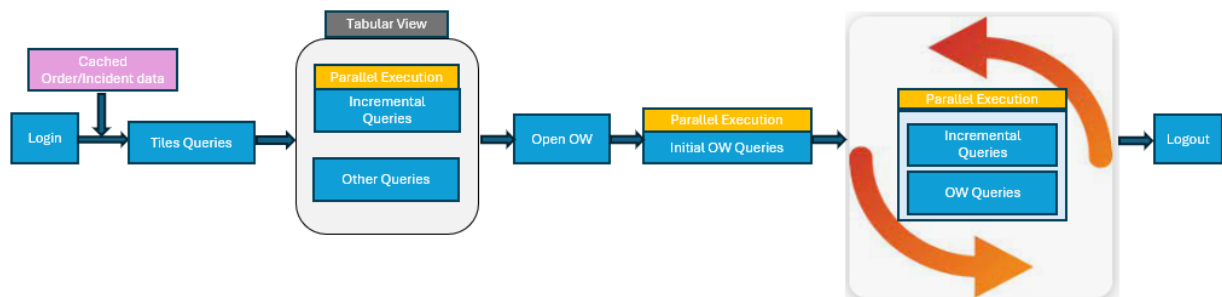
- More than 600 test users were distributed across five AORs.
- Each user was configured to access 300 to 2,000 incidents or orders, rather than all 25,000 orders.
- These users were added to the system during deployment.
- Approximately 5% of users were granted access to all AORs, allowing access to the entire set of 25,000 orders.

37.2.4.1.2. Incident and Order Generation

- A total of 25,000 incidents and orders were generated using customer calls and AMI outage events.
- EventScripter was used to execute the load generation scripts.

37.2.4.2. Test Script Architecture

The test scripts were designed to simulate concurrent user activity against the OA-ADMS integrated system using JMeter.



37.2.5. Test Run

- Total concurrent users - 600
- Ramp-up duration - 50 minutes
- All 600 users were placed in a steady-state load using loops for 3 hours.
- Total loops executed - 100
- Pause between loops - 70 seconds
- Mutation queries - Configured to represent 10% of the total loop count

During the steady-state phase, the following functional verifications were performed:

- Open Order Worksheet (OW)
- Assign and unassign crews
- Update crew comments
- Restore orders

System health was monitored using the following tools:

- Performance Counters
- Lens
- Apache Balance Manager

Query performance was measured using:

- JMeter logs
- Apache Load Balancer logs
- ADMS logs

37.2.6. Test Metrics

37.2.6.1. Resource Utilization

No significant resource utilization issues were observed on the ADMS side during the test execution.

Application	%CPU		Private Bytes (GB)	
	Avg.	Max	Avg.	Max
NetServer	8.5	18	33.4	38.5
NVA	6.2	43	22	40
WOA	<1	1	7.6	8
NetServer-RO	16.5	28	31.8	35
NVA-RO	<1	20.5	5.8	15

37.2.7. Cluster

No significant resource utilization issues were observed on the cluster side during the test execution.

37.2.7.1. Overall Query Response Time

The following table shows the number of queries and their response times for different user groups.

Latency Range (ms)	1-200 Threads	201-400 Threads	401-600 Threads
0-999	72454	183296	979398
1000-1999	660	1214	4788
2000-2999	60	182	1338
3000-3999	164	206	620
4000-4999	42	28	204
5000-5999	94	8	94
6000-6999	78	26	44
7000-7999	18	62	28

Latency Range (ms)	1–200 Threads	201–400 Threads	401–600 Threads
8000–8999	6	72	8
9000–9999	10	28	44
10000–10999	0	12	88
11000–11999	18	22	78
12000–12999	16	26	28
13000–13999	10	8	10
14000–14999	12	8	8
15000–15999	4	6	22
16000–16999	2	0	0
17000–17999	0	2	0

37.2.7.2. Query Response Time by Query Type

The following table shows the average latency for each query type.

Query Type	Average Latency (ms)
AssignCrews	989.181
codedTranslation - ActionTaken	4.563333333
codedTranslation - ActionType	4.87
codedTranslation - ARMSSTATUS	5.01
codedTranslation - AssetSubType	5.023333333
codedTranslation - AssetType	4.753333333
codedTranslation - CallType	8.933333333
codedTranslation - CaseNote	4.98
codedTranslation - Condition_1	22.47333333
codedTranslation - Condition_2	5.163333333
codedTranslation - CrewAvailabilityStatus	5.26
codedTranslation - CrewShiftStatus	4.633333333
codedTranslation - CriticalType	4.58
codedTranslation - CriticalType2	5.366666667
codedTranslation - CriticalType3	4.526666667
codedTranslation - CriticalType4	4.793333333
codedTranslation - CriticalType5	4.706666667
codedTranslation - Description	10.81333333
codedTranslation - DeviceType	5.676666667
codedTranslation - FailureMode	4.49
codedTranslation - FaultLocationSource	5.246666667

Query Type	Average Latency (ms)
codedTranslation - IncidentWorkingStatus	4.933333333
codedTranslation - Manufacturer	4.63
codedTranslation - Material	5.283333333
codedTranslation - MobileCommunicationStatus	5.04
codedTranslation - OutageIncidentType	5.076666667
codedTranslation - OutageType	4.72
codedTranslation - PowerQualityType	4.516666667
codedTranslation - PRIORITY_CODE	5.676666667
codedTranslation - Reason	10.63666667
codedTranslation - ReasonGroup	5.196666667
codedTranslation - SuppressReason	5.44
codedTranslation - SystemAffected	7.616666667
codedTranslation - Weather	4.62
codedTranslation - WorkModelOrderStatus	4.6
codedTranslation - WorkModelOrderType	8.816666667
codedTranslation - WorkOrderCrewStatus	4.703333333
codedTranslation - WorkOrderSuspendedReasons	4.55
codedTranslation - WorkOrderTaskStatus	4.9
codedTranslation - WorkOrderTaskType	5.82
codedTranslation - WorkOrderTaskTypeResults	5.206666667
codedTranslation - WorkOrganizations	5.473333333
Condition for Mutation query	0
Get Crews	4480.638333
Get Device	145.8579104
Get Order	70.19508251
Get SessionAor (required for zitadel workflow)	467.35
Get Userinfo (required for zitadel workflow)	8.326666667
IncidentCustomFieldDefinitions	169.31
Incremental Query - WORKORDERLIST - Details	105.4941333
Incremental Query (Full Refresh) - WORKORDERLIST - Details	13045.08889
Incremental Query - CREWS	75.83353333
Incremental Query - UNASSOCCALLS	75.6975
Incremental Query - WORKORDERLIST - DMS	102.3095
Incremental Query (Full Refresh) - CREWS	375.95
Incremental Query (Full Refresh) - UNASSOCCALLS	480.9003322

Query Type	Average Latency (ms)
Incremental Query (Full Refresh) - WORKORDERLIST - DMS	2592.210291
Incremental Query (Full Refresh) - WORKORDERLIST - OMS	1788.27
OrderNextStatus	27.90653465
Structure	13.08778878
Tasks Query	32.46481848
TaskTypes	49.04333333
UnassignCrews	606.046
UpdateIncidentFields	90.29533333
UpdateOrder	320.5173333

37.2.8. UI Response Time

Orders	Initial Page Load Time	Order Worksheet Open Time	Crew Assignment Time	Map Search
100	5 sec	2 sec	2 sec	1 sec
23000	30 sec	2 sec	2 sec	1 sec

These guidelines ensure the robustness and reliability of the ADMS system under varying loads and conditions.

38. Third-Party Licenses

38.1. ADMS

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38.1.1. Spring Framework

Version

6.0.11

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Version 2.0, January 2004

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Reference

<https://mvnrepository.com/artifact/org.springframework/spring-context/>

38.1.2. Spring Boot

Version

3.1.2

License

Apache License 2.0

For more information, refer to [Spring Framework](#).

Reference

<https://mvnrepository.com/artifact/org.springframework.boot/spring-boot/>

38.1.3. Spring Cloud

Version

2021.0.7

License

Apache License 2.0

For more information, refer to [Spring Framework](#).

Reference

<https://mvnrepository.com/artifact/org.springframework.cloud/spring-cloud-commons/>

38.1.4. Spring Integration

Version

6.0.6

License

Apache License 2.0

For more information, refer to [Spring Framework](#).

Reference

<https://mvnrepository.com/artifact/org.springframework.integration/spring-integration-stream/>

38.1.5. Hibernate Core

Version

6.1.5.Final

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Version 2.1, February 1999

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Reference

<https://mvnrepository.com/artifact/org.hibernate/hibernate-core/>

38.1.6. Hibernate Validator

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6.1.5.Final

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<https://mvnrepository.com/artifact/org.hibernate.validator/hibernate-validator/>

38.1.7. Hazelcast

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38.1.8. Hazelcast Kubernetes

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2.2

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38.1.10. Apache HttpClient

Version

4.5.13

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<https://mvnrepository.com/artifact/org.apache.httpcomponents/httpclient/>

38.1.11. Bouncy Castle

Version

1.67 & 1.69

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38.1.12. FasterXML Jackson Dataformat

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2.15.2

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<https://mvnrepository.com/artifact/com.fasterxml.jackson.dataformat/jackson-dataformat-xml/>

38.1.13. FasterXML Jackson Databind

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2.15.2

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<https://mvnrepository.com/artifact/com.fasterxml.jackson.core/jackson-databind/>

38.1.14. PostgreSQL Database

Version

15

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<https://www.postgresql.org/about/licence/>

38.1.15. PostgreSQL JDBC Driver

Version

42.3.7

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Reference

<https://mvnrepository.com/artifact/org.postgresql/postgresql/>

38.1.16. Reactor IO Project

Version

3.4.29

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Reference

<https://mvnrepository.com/artifact/io.projectreactor/reactor-core>

38.1.17. IO Micrometer Core

Version

1.9.10

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Reference

<https://mvnrepository.com/artifact/io.micrometer/micrometer-core>

38.1.18. IO Micrometer Prometheus

Version

1.11.0

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<https://mvnrepository.com/artifact/io.micrometer/micrometer-registry-prometheus>

38.1.19. IO Netty

Version

4.1.91.Final

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<https://mvnrepository.com/artifact/io.netty/netty-all>

38.1.20. Apache Commons

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3.8.1

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38.1.21. Apache Commons Logging

Version

1.2

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Version

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@angular/common 20.0.3
@angular/core 20.0.3
@angular/forms 20.0.3
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Reference

<https://github.com/angular/angular/blob/master/LICENSE>

38.2.2. rxjs

Version

7.8.1

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Reference

<https://github.com/ReactiveX/rxjs/blob/master/LICENSE.txt>

38.2.3. zone.js

Version

0.15.1

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38.2.4. tslib

Version

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38.2.5. xliff

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